Topology and Differential Geometry in Physics

Study Group Notes

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Abstract. This course offers a semi-rigorous introduction to the mathematical structures underlying theoretical physics. By semi-rigorous, we mean that all definitions and theorems will be stated with mathematical precision, but not all will be proven in full detail. Instead, the focus will be on developing the mathematical intuition and thinking style necessary for understanding and contributing to modern theoretical physics.
We aim to build everything from scratch no prior background in advanced mathematics or physics is assumed. The only requirement is curiosity and a willingness to learn. Where we skip formal proofs, we'll compensate with rich and illuminating examples drawn from physics and geometry. The ultimate goal is to equip participants with a rigorous mathematical lens and the foundational tools needed to independently approach advanced topics in mathematical physics. .

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1 What is Physics

One of the oldest sciences in the world is physics. A "physicist" could conduct both theoretical and practical scientific experiments at the beginning of this field of study. Isaac Newton, who established the laws of motion and discovered gravity, is a prime example. However, contemporary physics has become extremely difficult, to the point where physicists concentrate either on the theoretical or experimental aspects of the subject. In this section, we define theoretical physics along with its framework, theory, and models.

1.1 Definition of Physics

Definition 1.1: Theoretical physics

Physics is the scientific study of physical systems in the universe. Depending on the scale, different **frameworks** are employed to construct **theories** and **models** of these systems.

A physical system can range from something as small as an elementary particle to something as vast as the entire universe, depending on the field of study.

In the next section we are going to define and highlight the difference between what is framework, theory and model. These the definitions of those concepts can be subjective and different from a physicist to another but in our case we will follow the definition of Alex Flournoy.

1.2 Framework

1.2.1 Definition of framework

Definition 1.2: Framework

A **framework** in physics is a collection of mathematical and conceptual data that provides a general structure for describing physical systems. It typically includes:

- A **spacetime manifold** M, which encodes the geometric and causal structure of the universe.
- A **space of states** S, where each element (a state) represents a complete description of the degrees of freedom of the system.
- A **space of observables** O, representing measurable quantities or operations that can act on states.
- A **law of evolution**, which determines how states and/or observables change over time, often dictated by equations of motion or dynamical rules.
- A **measurement process**, which specifies how observables are used to extract information from states, possibly affecting them in the process.

Spacetime is where you physics happens, states

1.2.2 Examples of framework

Example 1.: Examples of Frameworks

- **Classical Mechanics (Newtonian)**:
  - $M = \mathbb{R}^3 \times \mathbb{R}$: Euclidean 3-space over time.
  - S: positions and velocities of particles.
  - O: functions like kinetic energy, momentum.
  - Evolution: Newton's second law $F = ma$.
  - Measurement: position, velocity via classical instruments.
- **Special Relativity**:
  - $M = \mathbb{R}^{1,3}$: Minkowski spacetime.
  - S: four-velocity and four-momentum of particles.
  - O: relativistic energy, invariant mass.
  - Evolution: Lorentz-invariant equations of motion.
  - Measurement: proper time intervals, relativistic Doppler effect.
- **Quantum Mechanics**:
  - $M = \mathbb{R}$: time axis.
  - $S = \mathcal{H}$: a complex Hilbert space.
  - O: self-adjoint operators on $\mathcal{H}$.
  - Evolution: Schrödinger equation $i\hbar \frac{d}{dt}|\psi\rangle = H|\psi\rangle$.
  - Measurement: Born rule, collapse postulate.

Some frameworks are more fundamental than others

- For example, with respect to the **velocity** of the physical object or process, **special relativity** is more fundamental than classical mechanics. Special relativity is valid at any speed, while classical mechanics is only valid when the speed is significantly lower than the speed of light. In fact, classical mechanics can be recovered from special relativity in the low-speed approximation.
- Another example involves the **size** of the system. **Quantum mechanics** is more fundamental than classical mechanics because it applies at all scales: every physical system is, in principle, a quantum system. Classical mechanics, by contrast, is only valid for macroscopic systems where quantum effects are negligible.

Question: What if you want to study a system where both relativistic and quantum effects are relevant, such as a particle in an accelerator? In this case, you need to incorporate both quantum mechanics and special relativity. The appropriate framework

for such systems is called **quantum field theory**. **Quantum field theory** is the most powerful framework in physics. It has applications ranging from the microscopic scale, such as particle physics, to the largest scales, such as cosmology.

Even though it is a framework, it is called a *theory*. The reason is mostly historical and aesthetic *Quantum Field Theory* (QFT) sounds better than *Quantum Field Framework* (QFF). All jokes aside, the distinction between a framework and a theory is somewhat subjective and varies from one physicist to another. Some physicists consider it more of a theory than a framework. In the next section, we highlight the existence of this ambiguity when we define what is a *theory*.

1.2.3 Alternative formulation of frameworks

A framework can have different formulations that describe the same physics, each with its own definitions of state space, observable space, evolution law, and measurement method. For example, classical mechanics can be formulated using Newtonian, Lagrangian, or Hamiltonian formalisms.

Lagrangian Mechanics

- **State Space:** Configuration space — the space of generalized coordinates q^i.
- **Observables:** Functions of position and velocity, typically written as the Lagrangian $L(q, \dot{q})$.
- **Evolution Law:** Euler–Lagrange equations:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}^i}\right) = \frac{\partial L}{\partial q^i}$$

- **Measurement and Interpretation:** Motion arises by extremizing the action functional:

$$S[q] = \int L(q, \dot{q})\, dt$$

  This formulation emphasizes a variational principle: nature selects the path that minimizes (or extremizes) the action.

Hamiltonian Mechanics

- **State Space:** Phase space — the cotangent bundle T^*Q, with coordinates (q^i, p_i).
- **Observables:** Functions on phase space, typically the Hamiltonian $H(q, p)$, which often represents the total energy.
- **Evolution Law:** Hamilton's equations:

$$\dot{q}^i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q^i}$$

- **Measurement and Interpretation:** The dynamics are viewed geometrically as flows on a symplectic manifold. Time evolution is governed by the Hamiltonian vector field associated with H.

1.2.4 Power of Frameworks: The Case of Spin

Some frameworks in physics are more powerful than others—not only because they apply in a broader range of physical situations, but because they reveal deeper structures and symmetries in nature.

One striking example is the concept of *spin*:

- **Classical Mechanics:** Spin does not exist in classical mechanics. Particles are treated as point masses, and angular momentum arises only from motion in space ($\vec{L} = \vec{r} \times \vec{p}$). There is no concept of intrinsic angular momentum.

- **Non-Relativistic Quantum Mechanics:** Spin is introduced as an extra degree of freedom—an internal quantum number that has no classical analogue. It is added by hand, typically through Pauli matrices and two-component spinors, and leads to phenomena like Zeeman splitting and spin-orbit coupling.

- **Quantum Field Theory (QFT):** Spin emerges naturally as a consequence of spacetime symmetries. In particular, QFT respects the Poincaré symmetry of special relativity, and spin labels irreducible representations of the Poincaré group. Furthermore, results like the *spin-statistics theorem*—which links integer spin to bosons and half-integer spin to fermions—are only visible in this framework.

This progression illustrates how more powerful frameworks not only expand the applicability of physics but also deepen our understanding of its fundamental structure.

1.3 Theory

1.3.1 Definition of theory and physical systems

Definition 1.3: Theory

A **theory** is a framework that is applied to describe a specific physical system or interaction.

Here is a list of physical systems:

Examples of Physical Systems

The following is a non-exhaustive list of physical systems that physicists study across different frameworks:

- Elementary particles
- Atoms and molecules
- Solids
- Gases and plasmas
- Fluids and liquids
- Biological systems (e.g., plants, cells)
- Stars
- Galaxies
- Black holes
- The universe

And we also have four fundamental forces in nature:

The Four Fundamental Forces

In physics, all interactions in the universe can be described by four fundamental forces:

- **Gravitational Force** – Governs the attraction between objects with mass; dominant at astronomical scales.
- **Electromagnetic Force** – Acts between electrically charged particles; responsible for electricity, magnetism, and light.
- **Weak Nuclear Force** – Responsible for radioactive decay and neutrino interactions; crucial in nuclear processes like fusion.
- **Strong Nuclear Force** – Binds quarks together to form protons and neutrons, and holds atomic nuclei together.

1.3.2 Example of theories

Theories Applied to Physical Systems

We illustrate how various frameworks are applied to different physical systems. In many cases, classical mechanics serves as a starting point, but deeper understanding requires quantum mechanics, special relativity, or quantum field theory.

- **Solid-State Systems**
  - **Classical Mechanics:** Models lattice vibrations and thermal behavior using continuum mechanics.
  - **Quantum Mechanics:** Explains phenomena like conductivity, band structure, and quantized heat capacity (e.g., via phonons or specific heat at low temperature).
- **Plasmas**
  - **Classical Electromagnetism:** Describes plasma dynamics through Maxwell's equations and fluid models (e.g., magnetohydrodynamics).
  - **Quantum Field Theory:** Needed in high-energy plasmas, such as those in the early universe or inside stars.
- **Black Holes**
  - **General Relativity:** Essential for understanding spacetime curvature, event horizons, and gravitational collapse.
  - **Quantum Field Theory in Curved Spacetime:** Needed to derive Hawking radiation and explore quantum aspects of gravity.
- **Biological Systems**
  - **Classical Physics:** Used in biomechanics, thermodynamics of cells, and modeling organismal motion.
  - **Quantum Mechanics:** Plays a role in phenomena like photosynthesis (e.g., quantum coherence in energy transport) and olfaction.

Is quantum field theory and general relativity a theory, theories, or frameworks?

You can view them as both frameworks and theories. Quantum field theory (QFT), for example, can serve as a framework for studying particle physics and condensed matter physics. At the same time, it is still considered an incomplete framework, as it lacks a fully rigorous mathematical formulation unlike quantum mechanics or general relativity. Physicists and mathematicians are still investigating what a "field" really is. In this sense, they apply quantum mechanics and special relativity (or general relativity, if studying curved spacetime) to understand the physics of fields. General relativity (GR), on the

other hand, tells a different story. GR has a well-defined mathematical structure formulated through differential geometry. You can think of it as the application of classical field theory to the study of the gravitational force.

1.4 Model

1.4.1 Model and effective theories

Definition 1.4: Model

A **model** is an effective theory that requires some input parameters which are not predicted by the theory itself, but are chosen to match the desired behavior of the system.

Definition 1.4: Effective Theory

An **effective theory** is a scientific theory that provides an accurate description of physical phenomena within a limited range of scales (such as energy, length, or time), while approximating or ignoring effects outside that range. It is not considered a complete or fundamental theory, but rather a useful approximation valid within its domain.

1.4.2 Examples of models

Example 1.1: Fermi's Theory of Beta Decay

Fermi's theory of beta decay is a prototypical example of an **effective theory**. It models the weak interaction responsible for neutron decay as a contact four-fermion interaction. While it provides accurate predictions at low energies, it fails at higher energies where the full electroweak theory (developed by Glashow, Weinberg, and Salam) becomes necessary. Thus, Fermi's theory is valid only as a low-energy approximation of the more fundamental electroweak framework.

Example 1.2: Thomas–Fermi Model

The Thomas–Fermi model is an approximate, statistical description of the distribution of electrons in atoms. It treats the electron density as a continuous function and simplifies quantum many-body problems.

Parameters: The atomic number Z and the total number of electrons must be specified. These determine the boundary conditions for the model's differential equation.

Example 1.3: Standard Model

The Standard Model is a quantum field theory that describes electromagnetic, weak, and strong interactions. It encompasses quarks, leptons, gauge bosons, and the Higgs boson. While it predicts many physical processes with high accuracy, its parameters are not derived within the theory.

Parameters: Masses of elementary particles (e.g., m_e, m_u, m_d, m_t), gauge coupling constants (e.g., g, g', g_s), the CKM matrix elements, and the Higgs vacuum expectation value v are all input from experimental measurements.

Example 1.4: Ising Model

The Ising model describes a system of spins on a lattice, each taking values ± 1, and interacting with neighboring spins. It's a cornerstone model for understanding phase transitions in statistical mechanics.

Parameters: The interaction strength J, the external magnetic field h, and the temperature T are inputs that determine the behavior of the system, including whether it exhibits spontaneous magnetization.

Remark 1.1

Theory vs Model: String Theory vs Standard Model String Theory and the Standard Model both attempt to describe the fundamental forces of nature, but they play different roles.

- **String Theory** is considered a **theory** because it aims to explain all physical constants and phenomena from first principles. In principle, no external input is needed once the theory is fully specified — all particle properties and interactions should be derived. However, string theory is not yet experimentally confirmed or mathematically complete.
- **The Standard Model** is a **model**: although it makes extremely accurate predictions, it depends on about 19 parameters (like masses, coupling constants, and mixing angles) that must be determined from experiments. It does not explain the values of these parameters.

This distinction illustrates a deeper philosophical divide in physics: *theories aim to answer "why", while models aim to describe "what".*

2 Set theory

In the following sections, we introduce the foundational languages of mathematics: set theory.

2.1 Review of Set Theory

2.1.1 Sets

Most mathematical objects are collections of other mathematical objects. For example, vector spaces are collections of vectors, topological spaces are collections of sets called open sets, and the real numbers form a collection that includes both rational and irrational numbers. To study these structures rigorously, we begin by recalling the formal notion of a set.

Definition 2.1: Set

A **set** is a well-defined collection of distinct objects, called *elements* or *members*.

Remark 2.

The order of elements and repetition do not matter in a set. For example:

$$\{1, 2, 3\} = \{3, 1, 2\} = \{1, 1, 2, 3, 3\}$$

Definition 2.2: Set Operations

Let A and B be subsets of a universal set. We define the following standard operations:

- **Union** $A \cup B$: the set of all elements that are in A, in B, or in both.
- **Intersection** $A \cap B$: the set of all elements that are in both A and B.
- **Set Difference** $A \setminus B$: the set of all elements that are in A but not in B.
- **Complement** A^c: the set of all elements not in A, taken relative to the universal set.

While doing mathematics, we often want to work with sets whose elements satisfy specific properties. Let x be a mathematical object of interest and $P(x)$ a logical statement representing the property we desire. Then the set of all x for which $P(x)$ is true is called the **truth set** of P.

Definition 2.3: Truth Set

Given a property represented by a logical statement $P(x)$, the **truth set** of P is the set:

$$\{x \mid P(x) \text{ is true}\}.$$

Here are some examples:

- **Even numbers:** $\{x \in \mathbb{Z} \mid x \equiv 0 \mod 2\}$
- **Odd numbers:** $\{x \in \mathbb{Z} \mid x \equiv 1 \mod 2\}$

- **Prime numbers:** $\{x \in \mathbb{N} \mid x > 1 \text{ and divisible only by 1 and } x\}$
- **Special relativity:** $\{p \in \mathbb{R}^4 \mid p^2 - m^2c^2 \geq 0\}$ the set of all 4-momenta with real mass.
- **Analytic geometry (circle):** $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = r^2\}$
- **Algebraic solution set:** $\{x \in \mathbb{R} \mid x^2 + 2x - 1 = 0\}$
- **Differential equation:** $\{f \in C^1(\mathbb{R}) \mid f'(x) + 2f(x) = 0\}$

In many mathematical and physical applications, we need to deal with a collection of sets that are indexed by another set, often representing time, space, particle labels, or energy levels.

Definition 2.4: Indexed Elements

Let A be a set and I a set called the **index set**. An **indexed family of elements** of A is a collection $\{a_i\}_{i \in I}$, where each $a_i \in A$. The subscript i identifies each element with a label from the index set.

2.1.2 Relations

Let H be the set of all humans on Earth. This set by itself is not particularly interesting, but what makes humans interesting is our relationships with each other, which allow us to form friendships, families, enemies, and communities.

We begin by considering a set H, representing all humans on Earth. Within this set, we can identify various meaningful groupings based on relationships or shared interests.

- The **friendship set** consists of pairs of people who are friends. We write $x \; F \; y$ to indicate that person x is a friend of person y.
- The **family set** consists of pairs of people who are related by blood or legal family ties. We write $x \; R \; y$ to indicate that x is a family member of y.
- The **community set**, for example the community of physics fans, consists of people who share a common interest. We write $x \; C \; y$ to indicate that x and y belong to the same physics community.

Definition 2.4: Relation

A **relation** R from a set A to a set B is a rule that associates to some elements $x \in A$ and $y \in B$ the statement $x \; R \; y$, meaning "x is related to y by R."

Here a few examples of relations that is common mathematics:

Example 2.2: Examples from number theory

- Let $A = B = \mathbb{N}$, and define xRy by $x < y$:

$$R_< = \{1 < 2,\ 1 < 3,\ 2 < 3,\ \ldots\} = \{xRy \mid x \text{ is strictly smaller than } y\}$$

- Let $A = B = \mathbb{Z}$, and define xRy by $x = y$:

$$R_= = \{0 = 0,\ -1 = -1,\ 42 = 42,\ \ldots\} = \{xRy \mid x \text{ equals } y\}$$

- Let $A = B = \mathbb{N}$, and define xRy if both x and y are even:

$$R = \{xRy \mid x \equiv y \equiv 0 \pmod 2\}$$

- Let $A = B = \mathbb{N}$, and define xRy if both x and y are odd:

$$R = \{xRy \mid x \equiv y \equiv 1 \pmod 2\}$$

- Let $A = B = \mathbb{Z}_{>0}$, and define xRy if $\gcd(x, y) = 1$:

$$R = \{xRy \mid x \text{ and } y \text{ are coprime}\}$$

Example 2.3: Triangles

Let $A = B = \mathcal{T}$ be the set of all triangles in the plane with side lengths in $\mathbb{R}_{>0}$. We define a relation xRy on $\mathcal{T}$ by:

$$xRy \quad \text{if and only if the sorted side lengths of } x \text{ and } y \text{ are equal.}$$

That is, if triangle x has side lengths $\{a, b, c\}$ with $a \leq b \leq c$, and triangle y has side lengths $\{a', b', c'\}$ with $a' \leq b' \leq c'$, then

$$xRy \iff (a, b, c) = (a', b', c').$$

This relation captures triangle congruence by side lengths (SSS criterion) and ignores rotation or position.

$$R = \{xRy \mid \text{triangles } x \text{ and } y \text{ have the same sorted side lengths}\}.$$

Example 2.4: Relation Based on Equal Determinants

Let $A = B = M_{n\times n}(\mathbb{R})$, the set of all $n \times n$ real matrices. Define the relation xRy by:

$$xRy \quad \text{if and only if} \quad \det(x) = \det(y).$$

That is, two matrices are related if they have the same determinant. This relation partitions the set of matrices into equivalence classes of constant determinant.

$$R = \{xRy \mid \text{matrix } x \text{ and matrix } y \text{ have the same determinant}\}.$$

Example 2.5: Relation Based on Trace Comparison

Let $A = B = M_{n\times n}(\mathbb{R})$, the set of all $n \times n$ real matrices. Define the relation xRy by:

$$xRy \quad \text{if and only if} \quad \operatorname{tr}(x) \leq \operatorname{tr}(y).$$

That is, one matrix is related to another if the trace (the sum of the diagonal elements) of the first is less than or equal to that of the second.

$$R = \{xRy \mid \operatorname{tr}(x) \leq \operatorname{tr}(y)\}.$$

Example 2.6: Relation Based on the Equation $x^2 + y^2 = 1$

Let $A = B = \mathbb{R}$. Define the relation xRy by:

$$xRy \quad \text{if and only if} \quad x^2 + y^2 = 1.$$

This relation captures pairs of real numbers whose coordinates lie on the unit circle in $\mathbb{R}^2$.

$$R = \left\{xRy \mid x^2 + y^2 = 1\right\}.$$

Examples:

$$0R1, \quad 1R0, \quad \tfrac{\sqrt{2}}{2}R\tfrac{\sqrt{2}}{2}, \quad -1R0.$$

Example 2.7: Relation Based on the Number of Elements in Sets

Let $A = B = \mathcal{P}(\{1, 2, 3, 4\})$, the power set of a 4-element set. Define the relation xRy by:

$$xRy \quad \text{if and only if} \quad |x| = |y|.$$

This relation groups subsets according to their cardinality.

$$R = \{xRy \mid x, y \subseteq \{1, 2, 3, 4\}, \text{ and } |x| = |y|\}.$$

Examples:

$$\{1, 2\}R\{3, 4\}, \quad \emptyset R\{\}, \quad \{1\}R\{2\}, \quad \{1, 2, 3\}R\{2, 3, 4\}.$$

Definition 2.4: Important Properties of Relations

Let R be a relation on a set A. We define the following properties using the notation xRy to mean that $(x, y) \in R$:

- **Reflexive:** R is reflexive if every element is related to itself:

$$\forall x \in A, \quad xRx$$

- **Irreflexive:** R is irreflexive if no element is related to itself:

$$\forall x \in A, \quad \neg(xRx)$$

- **Symmetric:** R is symmetric if xRy implies yRx:

$$\forall x, y \in A, \quad xRy \Rightarrow yRx$$

- **Antisymmetric:** R is antisymmetric if xRy and yRx together imply $x = y$:

$$\forall x, y \in A, \quad (xRy \wedge yRx) \Rightarrow x = y$$

- **Asymmetric:** R is asymmetric if xRy implies $\neg(yRx)$:

$$\forall x, y \in A, \quad xRy \Rightarrow \neg(yRx)$$

- **Transitive:** R is transitive if xRy and yRz together imply xRz:

$$\forall x, y, z \in A, \quad (xRy \wedge yRz) \Rightarrow xRz$$

Example 2.6: Same First Letter of Last Name

Let A be the set of USTHB students participating in this class.

$$A = \{\text{Amir H, Mohamed H, Zakaria H, Nadia A, Anis G, Mohamed S, Roufaida G, Iness G, Asma A, Soulef B, Abir G, Abedldjalil M, Ibrahim C, Abdelmouen Z, Asma H, Rania B, Sirine C, Redouane A, Nada A, Lydia B, Haithem B}\}$$

Define the relation R on A by:

$$xRy \iff x \text{ and } y \text{ have the same first letter of their last names}$$

Then,

$$R = \{xRy \mid x \text{ and } y \text{ have the same first letter of their last names}\}$$

We analyze the properties of this relation:

- **Reflexive:** Yes, every person shares the same initial with themselves.
- **Irreflexive:** No, because it's reflexive.
- **Symmetric:** Yes, if xRy, then yRx.
- **Antisymmetric:** No, for example, Amir H and Mohamed H satisfy xRy and yRx, but they are different people.
- **Transitive:** Yes, if Amir H is related to Zakaria H and Zakaria H is related to Mohamed H, then Amir H is related to Mohamed H.

Example 2.: Relation: $x < y$ on $\mathbb{N}$

Properties: Irreflexive, Asymmetric, Transitive.

Example 2.: Relation: $x = y$ on any set

Properties: Reflexive, Symmetric, Antisymmetric, Transitive.

Example 2.: Relation: $x^2 + y^2 = 1$ on $\mathbb{R}$

Properties: Symmetric.

Example 2.: Relation: $x \geq y$ on $\mathbb{N}$

Properties: Reflexive, Antisymmetric, Transitive.

Example 2.: Relation: Triangles with equal sorted side lengths (Congruence)

Properties: Reflexive, Symmetric, Transitive.

Example 2.: Relation: Sets with equal number of elements

Properties: Reflexive, Symmetric, Transitive.

Example 2.: Relation: Natural numbers with same parity

Properties: Reflexive, Symmetric, Transitive.

Example 2.: Relation: $\gcd(x, y) = 1$ on $\mathbb{N}$

Properties: Symmetric.

Example 2.: Relation: Matrices with equal determinant

Properties: Reflexive, Symmetric, Transitive.

Example 2.: Relation: $\mathrm{tr}(A) \leq \mathrm{tr}(B)$ on square matrices

Properties: Reflexive, Antisymmetric, Transitive.

Observation 2.2

Equivalence vs Order Relations We observe that:

- Relations representing **equality, equivalence, or similarity** (such as $x = y$, congruent triangles, sets with same number of elements, etc.) are always:

  Reflexive, Symmetric, Transitive

- Relations representing **order or hierarchy** (such as $x \geq y$, $\mathrm{tr}(A) \leq \mathrm{tr}(B)$) are always:

  Reflexive, Antisymmetric, Transitive

These are the two most important types of relations in mathematics.

Definition 2.3: Equivalence Relation

A relation R on a set A is called an **equivalence relation** if it satisfies the following properties:

- **Reflexive:** xRx for all $x \in A$
- **Symmetric:** If xRy, then yRx
- **Transitive:** If xRy and yRz, then xRz

Definition 2.4: Partial Order

A relation R on a set A is called a **partial order** if it satisfies:

- **Reflexive:** xRx for all $x \in A$
- **Antisymmetric:** If xRy and yRx, then $x = y$
- **Transitive:** If xRy and yRz, then xRz

Definition 2.5: Total Order

A partial order R on A is called a **total order** (or **linear order**) if every pair of elements is comparable:

$$\forall x, y \in A, \quad xRy \text{ or } yRx$$

That is, any two elements can be compared under the order.

2.1.3 Functions

In the previous section, we encountered important types of relations such as equivalence relations and partial orders. These relations can associate an element in the domain with multiple elements in the codomain. For example, the relation $x < y$ on $\mathbb{N}$ links a number to all larger numbers. Similarly, the relation $x = y$ includes pairs like $(1, 1)$, $(2, 2)$, and so on, but does not restrict to unique outputs.

However, in many real-world situations, we want to assign **exactly one value** to each input. For example:

- A thermometer assigns a unique temperature reading to each time of day.
- The height of a person is a single value associated with their name.
- The distance from a city to the capital is a specific number for each city.
- A physics experiment might associate each time with the position of a particle.

In these examples, for each input, there is one and only one output. This kind of relation is so common and so useful that it has a special name in mathematics: a **function**.

Definition 2.1: Function

A **function** from a set A to a set B is a relation that assigns to each element $x \in A$ a **unique** element $y \in B$. If f is a function, we write:

$$f : A \to B$$

and for $x \in A$, we denote the unique element $y \in B$ assigned to x by:

$$f(x) = y.$$

This definition formalizes a key idea: each input from the domain A has exactly one output in the codomain B. We move away from the general notation xRy and instead write $f(x) = y$, which is the standard function notation used throughout mathematics and science.

Definition 2.2: Domain, Codomain, and Range

Let $f : A \to B$ be a function.

- The set A is called the **domain** of the function. It consists of all inputs for which $f(x)$ is defined.
- The set B is called the **codomain** of the function. It is the set that contains all possible outputs.
- The **range** of the function, also called the **image** of f, is the set:

$$\text{range}(f) = \{f(x) \mid x \in A\} \subseteq B.$$

Definition 2.3: Image and Preimage

Let $f : A \to B$ be a function.

- If $x \in A$, then the **image** of x under f is $f(x) \in B$.
- If $Y \subseteq B$, the **preimage** (or inverse image) of Y under f is:

$$f^{-1}(Y) = \{x \in A \mid f(x) \in Y\}.$$

- In particular, if $y \in B$, then

$$f^{-1}(\{y\}) = \{x \in A \mid f(x) = y\}.$$

Non-Example 2.1: Relations That Are NOT Functions

The following examples show relations that fail to be functions because they violate the "unique output" requirement:

Example 1: Square Root Relation Let $A = B = \mathbb{R}_{\geq 0}$ and define the relation R by:

$$xRy \iff y^2 = x$$

This relation is not a function because for $x = 4$, we have both $4R2$ and $4R(-2)$, since $2^2 = 4$ and $(-2)^2 = 4$. One input maps to multiple outputs.

Example 2: Circle Relation Let $A = B = \mathbb{R}$ and define R by:

$$xRy \iff x^2 + y^2 = 1$$

For $x = 0$, we have both $0R1$ and $0R(-1)$, so this is not a function.

Example 3: Age and Grade Relation Let A be the set of students in a school and $B = \{1, 2, 3, ..., 12\}$ be grade levels. Define:

$$xRy \iff \text{student } x \text{ is } y \text{ years old}$$

This fails to be a function because students of the same age can be in different grades.

Example 4: Matrix Determinant Relation Let $A = B = M_{2\times 2}(\mathbb{R})$ and define:

$$ARB \iff \det(A) = \det(B)$$

This is not a function because many different matrices can have the same determinant.

Example 2.1: Functions on Different Sets

Here are examples of functions with diverse domains and codomains:

Example 1: Polynomial Function

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = x^3 - 2x^2 + x - 1$$

Example 2: Modular Arithmetic

$$g : \mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}, \quad g(n) = n \bmod 5$$

Example 3: Set Cardinality

$$h : \mathcal{P}(\{1, 2, 3, 4\}) \to \mathbb{N}, \quad h(S) = |S|$$

For example: $h(\{1, 3\}) = 2$, $h(\emptyset) = 0$, $h(\{1, 2, 3, 4\}) = 4$.

Example 4: Matrix Trace

$$\mathrm{tr} : M_{n\times n}(\mathbb{R}) \to \mathbb{R}, \quad \mathrm{tr}(A) = \sum_{i=1}^{n} a_{ii}$$

Example 5: Complex Conjugation

$$\overline{(\cdot)} : \mathbb{C} \to \mathbb{C}, \quad \overline{a + bi} = a - bi$$

Example 6: First Coordinate Projection

$$\pi_1 : \mathbb{R}^3 \to \mathbb{R}, \quad \pi_1(x, y, z) = x$$

Example 7: Vector Length in Physics

$$\|\cdot\| : \mathbb{R}^3 \to \mathbb{R}_{\geq 0}, \quad \|(x, y, z)\| = \sqrt{x^2 + y^2 + z^2}$$

Example 8: Group Homomorphism

$$\phi : (\mathbb{Z}, +) \to (\mathbb{Z}/n\mathbb{Z}, +), \quad \phi(k) = k \bmod n$$

Example 9: Inclusion Function Let $A \subseteq B$. The inclusion function is:

$$\iota : A \to B, \quad \iota(x) = x$$

Example 10: Characteristic Function For a subset $S \subseteq X$:

$$\chi_S : X \to \{0, 1\}, \quad \chi_S(x) = \begin{cases} 1 & \text{if } x \in S \\ 0 & \text{if } x \notin S \end{cases}$$

Example 2.2: Functions from Physics

Example 1: Position as Function of Time

$$x : \mathbb{R}_{\geq 0} \to \mathbb{R}^3, \quad x(t) = \text{position of particle at time } t$$

Example 2: Energy-Momentum Relation

$$E : \mathbb{R}^3 \to \mathbb{R}_{\geq 0}, \quad E(\vec{p}) = \sqrt{|\vec{p}|^2 c^2 + m^2 c^4}$$

Example 3: Temperature Field

$$T : \mathbb{R}^3 \times \mathbb{R}_{\geq 0} \to \mathbb{R}, \quad T(\vec{r}, t) = \text{temperature at position } \vec{r} \text{ and time } t$$

Example 4: Lorentz Factor

$$\gamma : [0, c) \to [1, \infty), \quad \gamma(v) = \frac{1}{\sqrt{1 - v^2/c^2}}$$

Example 2.3: Functions in Computer Science

Example 1: Hash Function

$$h : \text{Strings} \to \mathbb{Z}/p\mathbb{Z}, \quad h(s) = \text{hash value of string } s$$

Example 2: Boolean Function

$$f : \{0, 1\}^n \to \{0, 1\}, \quad f(x_1, x_2, ..., x_n) = \text{output bit}$$

Example 3: Database Query

$$\text{query} : \text{Database} \to \text{ResultSet}, \quad \text{query}(db) = \text{matching records}$$

Example 2.4: Functions in Everyday Life

Example 1: Student ID Assignment

$$\text{ID} : \text{Students} \to \mathbb{N}, \quad \text{ID}(\text{student}) = \text{unique ID number}$$

Example 2: Price Function

$$\text{price} : \text{Products} \to \mathbb{R}_{>0}, \quad \text{price}(\text{item}) = \text{cost in dollars}$$

Example 3: Birthday Function

$$\text{birthday} : \text{People} \to \text{Dates}, \quad \text{birthday}(\text{person}) = \text{birth date}$$

Example 4: Grade Assignment

$$\text{grade} : \text{Students} \times \text{Courses} \to \{A, B, C, D, F\}$$

Types of Functions

When studying functions, we often care about special properties that relate the structure of the domain to the structure of the codomain. Three fundamental properties help us classify functions based on how they map elements between sets.

Definition 2.1: Injective Function (One-to-One)

A function $f : A \to B$ is called **injective** (or **one-to-one**) if different elements in the domain map to different elements in the codomain. Formally:

$$f \text{ is injective} \iff \forall x_1, x_2 \in A, \quad f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

Equivalently:

$$f \text{ is injective} \iff \forall x_1, x_2 \in A, \quad x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$$

Definition 2.2: Surjective Function (Onto)

A function $f : A \to B$ is called **surjective** (or **onto**) if every element in the codomain is the image of at least one element in the domain. Formally:

$$f \text{ is surjective} \iff \forall y \in B, \quad \exists x \in A \text{ such that } f(x) = y$$

Equivalently, f is surjective if and only if $\text{range}(f) = B$.

Definition 2.3: Bijective Function (One-to-One and Onto)

A function $f : A \to B$ is called **bijective** if it is both injective and surjective. That is:

$$f \text{ is bijective} \iff f \text{ is injective and } f \text{ is surjective}$$

Example 2.1: Examples of Injective Functions

Example 1: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 3$ This is injective because if $f(x_1) = f(x_2)$, then $2x_1 + 3 = 2x_2 + 3$, which implies $x_1 = x_2$.
Example 2: $g : \mathbb{N} \to \mathbb{N}$, $g(n) = 2n$ This maps each natural number to its double. Different inputs give different outputs.
Example 3: $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^3$ The cubing function is injective because if $x_1^3 = x_2^3$, then $x_1 = x_2$.
Example 4: Inclusion function $\iota : A \to B$ where $A \subseteq B$, $\iota(x) = x$ This is always injective.

Non-Example 2.1: Examples of Non-Injective Functions

Example 1: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ Not injective because $f(2) = f(-2) = 4$, but $2 \neq -2$.
Example 2: $g : \mathbb{R} \to [-1, 1]$, $g(x) = \sin(x)$ Not injective because $\sin(0) = \sin(\pi) = 0$.
Example 3: Constant function $c : A \to B$, $c(x) = b_0$ for some fixed $b_0 \in B$ Not injective (unless $|A| \leq 1$).

Example 2.2: Examples of Surjective Functions

Example 1: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3$ This is surjective because for any $y \in \mathbb{R}$, we can find $x = \sqrt[3]{y}$ such that $f(x) = y$.
Example 2: $g : \mathbb{R} \to [0, \infty)$, $g(x) = x^2$ This is surjective because for any $y \geq 0$, we have $g(\sqrt{y}) = y$.
Example 3: $h : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$, $h(k) = k \bmod n$ This is surjective because every residue class is represented.

Non-Example 2.2: Examples of Non-Surjective Functions

Example 1: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ Not surjective because negative numbers are not in the range.
Example 2: $g : \mathbb{N} \to \mathbb{N}$, $g(n) = 2n$ Not surjective because odd numbers are not in the range.
Example 3: $h : \mathbb{R} \to \mathbb{R}$, $h(x) = e^x$ Not surjective because the range is $(0, \infty)$, not all of $\mathbb{R}$.

Example 2.3: Examples of Bijective Functions

Example 1: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 3$ This is both injective and surjective, hence bijective.
Example 2: $g : \mathbb{R} \to \mathbb{R}$, $g(x) = x^3$ This is both injective and surjective, hence bijective.
Example 3: $h : (0, \infty) \to \mathbb{R}$, $h(x) = \ln(x)$ The natural logarithm is bijective from positive reals to all reals.
Example 4: Identity function $\mathrm{id}_A : A \to A$, $\mathrm{id}_A(x) = x$ Always bijective.
Example 5: $f : [0, \pi/2] \to [0, 1]$, $f(x) = \sin(x)$ Bijective when restricted to appropriate domains.

Function Composition

Before we study special types of functions, we need to understand how functions can be combined. One of the most important ways to create new functions from existing ones is through composition.

Definition 2.1: Function Composition

Let $f : A \to B$ and $g : B \to C$ be functions such that the codomain of f is the domain of g. The **composition** of g and f, denoted $g \circ f$ (read as "g composed with f" or "g after f"), is the function:

$$g \circ f : A \to C$$

defined by:

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in A$$

Remark 2.1

Note the order: $(g \circ f)(x) = g(f(x))$ means we apply f first, then apply g to the result. The function written on the right is applied first.

Example 2.1: Example of Function Composition

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2 + 1$, and let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x) = 2x - 3$.
Computing $g \circ f$:

$$(g \circ f)(x) = g(f(x)) = g(x^2 + 1) = 2(x^2 + 1) - 3 = 2x^2 + 2 - 3 = 2x^2 - 1$$

Computing $f \circ g$:

$$(f \circ g)(x) = f(g(x)) = f(2x - 3) = (2x - 3)^2 + 1 = 4x^2 - 12x + 9 + 1 = 4x^2 - 12x + 10$$

Verification with a specific value: Let $x = 2$. - $(g \circ f)(2) = 2(2)^2 - 1 = 8 - 1 = 7$ - Direct computation: $f(2) = 4 + 1 = 5$, then $g(5) = 2(5) - 3 = 7$ ✓- $(f \circ g)(2) = 4(2)^2 - 12(2) + 10 = 16 - 24 + 10 = 2$ - Direct computation: $g(2) = 4 - 3 = 1$, then $f(1) = 1 + 1 = 2$ ✓
Notice that $g \circ f \neq f \circ g$ in general!

Proposition 2.1: Properties of Function Composition

Let $f : A \to B$, $g : B \to C$, and $h : C \to D$ be functions. Then:

1. **Associativity:** $(h \circ g) \circ f = h \circ (g \circ f)$
2. **Identity Elements:** For the identity function $\mathrm{id}_A : A \to A$ defined by $\mathrm{id}_A(x) = x$:
$$f \circ \mathrm{id}_A = f \quad \text{and} \quad \mathrm{id}_B \circ f = f$$
3. **Non-Commutativity:** In general, $g \circ f \neq f \circ g$ (when both compositions are defined)
4. **Preservation of Properties:**
   - If f and g are both injective, then $g \circ f$ is injective
   - If f and g are both surjective, then $g \circ f$ is surjective
   - If f and g are both bijective, then $g \circ f$ is bijective

Example 2.2: Examples from Physics and Mathematics

Example 1: Unit Conversions Let $C : \mathbb{R} \to \mathbb{R}$ convert Celsius to Kelvin: $C(x) = x + 273.15$ Let $F : \mathbb{R} \to \mathbb{R}$ convert Kelvin to Fahrenheit: $F(x) = \frac{9}{5}x - 459.67$ The composition $(F \circ C)(x) = F(C(x)) = F(x + 273.15) = \frac{9}{5}(x + 273.15) - 459.67 = \frac{9}{5}x + 32$ directly converts Celsius to Fahrenheit.
Example 2: Coordinate Transformations Let $p : \mathbb{R}^2 \to \mathbb{R}^2$ be a rotation by 45ř: $p(x, y) = \left(\frac{x-y}{\sqrt{2}}, \frac{x+y}{\sqrt{2}}\right)$ Let $s : \mathbb{R}^2 \to \mathbb{R}^2$ be scaling by factor 2: $s(x, y) = (2x, 2y)$ Then $(s \circ p)(x, y)$ first rotates, then scales.
Example 3: Mathematical Functions Let $f : \mathbb{R}_{>0} \to \mathbb{R}$, $f(x) = \ln(x)$ Let $g : \mathbb{R} \to \mathbb{R}_{>0}$, $g(x) = e^x$
Then $(f \circ g)(x) = f(g(x)) = f(e^x) = \ln(e^x) = x$ and $(g \circ f)(x) = g(f(x)) = g(\ln(x)) = e^{\ln(x)} = x$.
This shows $f \circ g = \mathrm{id}_{\mathbb{R}}$ and $g \circ f = \mathrm{id}_{\mathbb{R}_{>0}}$, so f and g are inverses!

Inverse Functions

The concept of an inverse function is fundamental in mathematics and physics. It allows us to "undo" the action of a function, returning from the output back to the original input.

Definition 2.1: Inverse Function

Let $f : A \to B$ be a function. A function $g : B \to A$ is called the **inverse function** of f, denoted f^{-1}, if:

$$g(f(x)) = x \quad \text{for all } x \in A$$

and

$$f(g(y)) = y \quad \text{for all } y \in B$$

In other words, $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.

Remark 2.1

The notation f^{-1} for the inverse function should not be confused with $f^{-1}(Y)$ for the preimage of a set Y. The context usually makes the meaning clear.

Lemma 2.1: Uniqueness of Inverse

If a function $f : A \to B$ has an inverse, then the inverse is unique.

Example 2.1: Examples of Inverse Functions

Example 1: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 3$ The inverse is $f^{-1}(y) = \frac{y-3}{2}$. Check: $f^{-1}(f(x)) = f^{-1}(2x+3) = \frac{(2x+3)-3}{2} = x$
Example 2: $g : \mathbb{R} \to \mathbb{R}$, $g(x) = x^3$ The inverse is $g^{-1}(y) = \sqrt[3]{y}$.
Example 3: $h : (0, \infty) \to \mathbb{R}$, $h(x) = \ln(x)$ The inverse is $h^{-1}(y) = e^y$.
Example 4: Temperature conversion $F : \mathbb{R} \to \mathbb{R}$, $F(C) = \frac{9}{5}C + 32$ The inverse is $F^{-1}(F) = \frac{5}{9}(F - 32)$.

Non-Example 2.1: Functions Without Inverses

Example 1: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ No inverse exists because f is not injective. We cannot determine whether the preimage of 4 is 2 or -2.
Example 2: $g : \mathbb{R} \to [0, \infty)$, $g(x) = x^2$ No inverse exists because g is not surjective from $\mathbb{R}$ to $\mathbb{R}$.
Example 3: $h : \mathbb{R} \to \mathbb{R}$, $h(x) = \sin(x)$ No inverse exists because h is neither injective nor surjective.

Theorem 2.1: Characterization of Invertible Functions

Let $f : A \to B$ be a function. Then f has an inverse function if and only if f is bijective.
Proof: ($\Rightarrow$) Suppose f has an inverse $f^{-1} : B \to A$.
Injectivity: If $f(x_1) = f(x_2)$, then applying f^{-1} to both sides:

$$f^{-1}(f(x_1)) = f^{-1}(f(x_2)) \Rightarrow x_1 = x_2$$

So f is injective.
Surjectivity: For any $y \in B$, let $x = f^{-1}(y) \in A$. Then:

$$f(x) = f(f^{-1}(y)) = y$$

So every element of B is in the range of f, making f surjective.
($\Leftarrow$) Suppose f is bijective. We construct $f^{-1} : B \to A$ as follows: for each $y \in B$, since f is surjective, there exists $x \in A$ such that $f(x) = y$. Since f is injective, this x is unique. Define $f^{-1}(y) = x$.
By construction, $f^{-1}(f(x)) = x$ for all $x \in A$ and $f(f^{-1}(y)) = y$ for all $y \in B$. $\square$

Observation 2.1

Consequences of the Characterization Theorem The characterization theorem tells us that:

- A function has an inverse if and only if it is bijective
- To find an inverse, we must ensure both injectivity and surjectivity
- Many functions can be made invertible by restricting their domain or codomain appropriately

Example 2.2: Making Functions Invertible by Restriction

Example 1: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ This is not bijective, but we can restrict: - $f : [0, \infty) \to [0, \infty)$, $f(x) = x^2$ is bijective with inverse $f^{-1}(y) = \sqrt{y}$
Example 2: $g : \mathbb{R} \to \mathbb{R}$, $g(x) = \sin(x)$ This is not bijective, but we can restrict: - $g : [-\pi/2, \pi/2] \to [-1, 1]$, $g(x) = \sin(x)$ is bijective with inverse $g^{-1}(y) = \arcsin(y)$
Example 3: $h : \mathbb{R} \to \mathbb{R}$, $h(x) = e^x$ This is not surjective, but we can adjust the codomain: - $h : \mathbb{R} \to (0, \infty)$, $h(x) = e^x$ is bijective with inverse $h^{-1}(y) = \ln(y)$

Proposition 2.1: Properties of Inverse Functions

Let $f : A \to B$ and $g : B \to C$ be bijective functions. Then:

1. $(f^{-1})^{-1} = f$
2. $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$
3. f^{-1} is also bijective

2.2 Creating New Spaces from Old Ones.

A common theme in mathematics is to build new spaces from old ones. These constructions begin in set theory and are essential across all areas of math and physics. Importantly, we define these constructions **as maps** so that key structures (such as linearity, continuity, or smoothness) can be transferred to the new spaces.

Why maps? Because maps are the most important objects when studying structure:

- In vector spaces, we study **linear maps**.
- In topological spaces, we study **continuous maps**.
- In manifolds, we study **smooth maps**.

To ensure these types of maps can be preserved or induced on new constructions, we must define the constructions themselves via maps.

2.2.1 Subset (as a Map)

Given a set X and a subset $A \subseteq X$, we define the inclusion map:

$$\iota : A \hookrightarrow X, \quad \iota(a) = a$$

This inclusion map allows us to treat A as a part of X, preserving any structure on X that restricts to A.

Example 2.1: Examples of Subsets and Inclusion Maps

Example 1: Integers in Real Numbers Let $X = \mathbb{R}$ (the set of real numbers) and $A = \mathbb{Z}$ (the set of integers). The inclusion map is $\iota : \mathbb{Z} \hookrightarrow \mathbb{R}$, where $\iota(n) = n$. This allows us to view integers as a subset of real numbers, preserving operations like addition and multiplication when restricted to integers.

Example 2: Open Interval in Real Numbers Let $X = \mathbb{R}$ and $A = (0, 1)$ (the open interval). The inclusion map is $\iota : (0, 1) \hookrightarrow \mathbb{R}$, where $\iota(x) = x$. This means that an open interval, when considered as a subset of $\mathbb{R}$, inherits properties like distance and order from $\mathbb{R}$.

Example 3: Subspace of a Vector Space Let $X = \mathbb{R}^3$ (a 3D vector space) and $A = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$ (the xy-plane, a 2D subspace). The inclusion map is $\iota : A \hookrightarrow \mathbb{R}^3$, where $\iota(x, y, 0) = (x, y, 0)$. The xy-plane inherits the vector space structure (addition, scalar multiplication) from $\mathbb{R}^3$.

2.2.2 Product (as a Map)

Given two sets X and Y, their product is:

$$X \times Y = \{(x, y) \mid x \in X, y \in Y\}$$

with projection maps:

$$\pi_X : X \times Y \to X, \quad \pi_Y : X \times Y \to Y$$

These maps let us recover the original components of each point in the product.

Example 2.1: Examples of Product Sets and Projection Maps

Example 1: Cartesian Plane Let $X = \mathbb{R}$ and $Y = \mathbb{R}$. Their product is $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$, the Cartesian plane. The projection maps are:

$$\pi_X : \mathbb{R}^2 \to \mathbb{R}, \quad \pi_X(x, y) = x$$

$$\pi_Y : \mathbb{R}^2 \to \mathbb{R}, \quad \pi_Y(x, y) = y$$

These maps allow us to isolate the x-coordinate and y-coordinate of any point in the plane.

Example 2: Position and Time in Classical Mechanics Let $X = \mathbb{R}^3$ (position space) and $Y = \mathbb{R}_{\geq 0}$ (time). The product $\mathbb{R}^3 \times \mathbb{R}_{\geq 0}$ represents the state space of a particle (position and time). A point $(\vec{r}, t)$ in this space has projections $\pi_{\text{pos}}(\vec{r}, t) = \vec{r}$ and $\pi_{\text{time}}(\vec{r}, t) = t$.

Example 3: Configuration Space of a Double Pendulum The configuration of a double pendulum can be described by two angles, θ_1 and θ_2. The configuration space is $S^1 \times S^1$, which is a torus. The projection maps $\pi_1 : S^1 \times S^1 \to S^1$ and $\pi_2 : S^1 \times S^1 \to S^1$ recover the individual angles.

Example 4: Product of Sample Spaces (Probability) If we roll a die and flip a coin, the set of outcomes for the die is $D = \{1, 2, 3, 4, 5, 6\}$ and for the coin is $C = \{\text{Heads, Tails}\}$. The combined sample space is the product $D \times C = \{(d, c) \mid d \in D, c \in C\}$. The projection maps allow us to determine the die roll or the coin flip independently from a combined outcome.

2.2.3 Disjoint Union (as a Map)

Given sets X and Y, their disjoint union is:

$$X \sqcup Y = \{(x, 0) \mid x \in X\} \cup \{(y, 1) \mid y \in Y\}$$

with canonical inclusions:

$$\iota_X : X \to X \sqcup Y, \quad x \mapsto (x, 0), \quad \text{and} \quad \iota_Y : Y \to X \sqcup Y, \quad y \mapsto (y, 1)$$

Example 2.1: Examples of Disjoint Unions and Inclusion Maps

Example 1: Combining Categories of Objects Suppose we have a set of "Fruits" $F = \{\text{Apple}, \text{Banana}\}$ and a set of "Vegetables" $V = \{\text{Carrot}, \text{Broccoli}\}$. Their disjoint union $F \sqcup V$ would be $\{(\text{Apple}, 0), (\text{Banana}, 0), (\text{Carrot}, 1), (\text{Broccoli}, 1)\}$. The canonical inclusions ensure that an Apple from F is distinct from any item potentially named "Apple" if it were in V (which it isn't here, but illustrates the point).

Example 2: Energy Levels in Quantum Mechanics Consider a system that can be in one of two distinct sets of energy states, say $E_1 = \{e_{1,1}, e_{1,2}\}$ for one type of particle and $E_2 = \{e_{2,1}, e_{2,2}, e_{2,3}\}$ for another. The total set of possible energy levels for a system composed of these two non-interacting types would be $E_1 \sqcup E_2$. The inclusions map $e_{1,j}$ to $(e_{1,j}, 0)$ and $e_{2,k}$ to $(e_{2,k}, 1)$, distinguishing them even if numerical values might coincide.

Example 3: Building a "Two-Sheeted" Space Imagine two copies of the real line, $X = \mathbb{R}_A$ and $Y = \mathbb{R}_B$. Their disjoint union $\mathbb{R}_A \sqcup \mathbb{R}_B$ is like having two parallel number lines where, for instance, 0_A is distinct from 0_B. This construction is crucial in building spaces that are locally Euclidean but globally more complex, such as Riemann surfaces where different "sheets" of a function are distinct.

2.2.4 Quotient (as a Map)

Given a set X and an equivalence relation $\sim$, the quotient set is:

$$X/\sim = \{[x] \mid x \in X\}$$

with the canonical projection map:

$$q : X \to X/\sim, \quad x \mapsto [x]$$

This map collapses elements of X that are related by $\sim$ into a single point.

Definition 2.1: Equivalence Class

Let $\sim$ be an equivalence relation on a set X. For any element $x \in X$, the **equivalence class** of x, denoted by $[x]$, is the set of all elements in X that are related to x:

$$[x] = \{y \in X \mid y \sim x\}$$

The elements of the quotient set $X/\sim$ are precisely these equivalence classes.

Example 2.1: Examples of Equivalence Classes

Example 1: Even and Odd Numbers Let $X = \mathbb{Z}$ (the set of integers). Define the equivalence relation $a \sim b \iff a \equiv b \pmod 2$ (i.e., a and b have the same parity). There are two equivalence classes:

- $[0] = \{\ldots, -4, -2, 0, 2, 4, \ldots\}$ (the set of even numbers)
- $[1] = \{\ldots, -3, -1, 1, 3, 5, \ldots\}$ (the set of odd numbers)

The quotient set $\mathbb{Z}/\sim$ is $\{[0], [1]\}$.

Example 2: Rational Numbers Let $X = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ (pairs of integers where the second is non-zero). Define the equivalence relation $(a, b) \sim (c, d) \iff ad = bc$. This relation captures the idea of equivalent fractions. An equivalence class $[(a, b)]$ represents a rational number. For example:

- $[(1, 2)] = \{\ldots, (-2, -4), (-1, -2), (1, 2), (2, 4), \ldots\}$ represents the rational number $1/2$.
- $[(3, 1)] = \{\ldots, (-6, -2), (-3, -1), (3, 1), (6, 2), \ldots\}$ represents the rational number 3.

The set of all such equivalence classes is the set of rational numbers, $\mathbb{Q}$.

Example 3: People with the Same Birthday Let X be the set of all people. Define the equivalence relation $P_1 \sim P_2 \iff P_1$ and P_2 have the same birthday. An equivalence class consists of all people who share a specific birthday. For example:

- [Person born on Jan 1] = {all people born on January 1st}
- [Person born on Dec 31] = {all people born on December 31st}

The quotient set $X/\sim$ would be the set of all possible birthdays in a year.

Example 4: Geometric Congruence Let X be the set of all triangles in a plane. Define the equivalence relation $T_1 \sim T_2 \iff T_1$ is congruent to T_2. An equivalence class consists of all triangles that are congruent to each other. For example:

- [Equilateral triangle with side length 1] = {all equilateral triangles with side length 1}
- [Right triangle with sides 3,4,5] = {all right triangles with sides 3,4,5}

The quotient set $X/\sim$ is the set of all possible "shapes and sizes" of triangles, ignoring their position and orientation in the plane.

Given a set X and an equivalence relation $\sim$, the quotient set is:

$$X/\sim = \{[x] \mid x \in X\}$$

with the canonical projection map:

$$q : X \to X/\sim, \quad x \mapsto [x]$$

This map collapses elements of X that are related by $\sim$ into a single point.

Example 2.1: Examples of Quotient Sets and Projection Maps

Example 1: Integers Modulo n Let $X = \mathbb{Z}$ and the equivalence relation be $a \sim b \iff a \equiv b \pmod{n}$. The quotient set $\mathbb{Z}/\sim$ is typically denoted $\mathbb{Z}_n$ or $\mathbb{Z}/n\mathbb{Z}$, the set of integers modulo n. The projection map $q : \mathbb{Z} \to \mathbb{Z}_n$, $q(k) = [k]_n$, sends each integer to its residue class modulo n. For $n = 2$, $\mathbb{Z}_2 = \{[0], [1]\}$, where $[0]$ contains all even numbers and $[1]$ contains all odd numbers.

Example 2: Identifying Points on a Line to Form a Circle Let $X = \mathbb{R}$ and define $x \sim y \iff x - y \in \mathbb{Z}$ (i.e., x and y differ by an integer). The quotient set $\mathbb{R}/\mathbb{Z}$ is topologically equivalent to a circle S^1. The projection map $q : \mathbb{R} \to S^1$ effectively "wraps" the real line around the circle, with points $0, 1, 2, \ldots$ all mapping to the same point on the circle.

Example 3: Projective Space Let $X = \mathbb{R}^{n+1} \backslash \{0\}$ and define $\vec{x} \sim \vec{y} \iff \vec{y} = k\vec{x}$ for some $k \in \mathbb{R}, k \neq 0$. This means two non-zero vectors are equivalent if they are scalar multiples of each other (i.e., they lie on the same line through the origin). The quotient set $X/\sim$ is the real projective space $\mathbb{R}P^n$. The projection map $q : \mathbb{R}^{n+1} \setminus \{0\} \to \mathbb{R}P^n$ maps each non-zero vector to the unique line passing through the origin that it lies on. This is fundamental in geometry and physics (e.g., describing directions).

Example 4: Quotienting a Plane to Form a Cylinder or Torus Consider a plane $X = \mathbb{R}^2$. - To form a cylinder: Define $(x_1, y_1) \sim (x_2, y_2)$ if $x_1 = x_2$ and $y_1 - y_2 \in \mathbb{Z}$. This identifies points vertically. The quotient $\mathbb{R}^2/\sim$ is a cylinder. - To form a torus: Define $(x_1, y_1) \sim (x_2, y_2)$ if $x_1 - x_2 \in \mathbb{Z}$ and $y_1 - y_2 \in \mathbb{Z}$. This identifies points both horizontally and vertically. The quotient $\mathbb{R}^2/\sim$ is a torus.

2.2.5 Pushout (as a Map)

The **pushout** is a universal way to glue two sets (or spaces) together along a common part.

Suppose we are given sets and functions:

$$A \xrightarrow{f} X, \quad A \xrightarrow{g} Y$$

The pushout $P = X \sqcup_A Y$ is formed by taking the disjoint union $X \sqcup Y$ and identifying $f(a) \sim g(a)$ for each $a \in A$. Formally:

$$P = (X \sqcup Y)/\sim \quad \text{where } f(a) \sim g(a)$$

We get a commutative diagram:

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "g"'] & X \arrow[d, "\bar{f}"] \\ Y \arrow[r, "\bar{g}"] & X \sqcup_A Y \end{tikzcd}$$

The pushout comes with canonical maps:

$$\bar{f} : X \to X \sqcup_A Y, \quad \bar{g} : Y \to X \sqcup_A Y$$

such that $\bar{f} \circ f(a) = \bar{g} \circ g(a)$. This is the universal space where X and Y are glued along the image of A.

Remark 2.1

Pushouts allow us to construct new spaces by "gluing" along shared substructures. They are fundamental in topology (attaching cells), geometry (gluing manifolds), and category theory.

Example 2.1: Examples of Pushouts

Example 1: Gluing Two Disks Along a Common Boundary Let X and Y be two closed disks (e.g., $D^2 = \{(x, y) \mid x^2 + y^2 \leq 1\}$). Let A be a circle (their common boundary, $S^1 = \{(x, y) \mid x^2 + y^2 = 1\}$). Let $f : S^1 \to D^2$ and $g : S^1 \to D^2$ be the inclusion maps of the boundary circle into each disk. The pushout $D^2 \sqcup_{S^1} D^2$ is topologically equivalent to a sphere (S^2). We are literally gluing two hemispheres along their common equator.

Example 2: Attaching a Handle to a Sphere (Torus Construction) Start with a sphere S^2. We want to attach a "handle" (a cylinder $S^1 \times I$). Let $X = S^2$ and $Y = S^1 \times I$. To glue them, we need to choose two disjoint disks on the sphere, say $D_1, D_2 \subset S^2$, and two disjoint boundary circles on the cylinder, say $S^1 \times \{0\}$ and $S^1 \times \{1\}$. Let $A = S^1 \sqcup S^1$ (two disjoint circles). Let $f : A \to X$ embed the two circles as the boundaries of D_1 and D_2 on the sphere. Let $g : A \to Y$ embed the two circles as the boundaries of the cylinder. The pushout is $S^2 \sqcup_A (S^1 \times I)$, which results in a torus with a sphere "bulge", or more generally, can construct higher genus surfaces.

Example 3: The Cone Construction Let X be any set (or topological space) and $Y = \{*\}$ be a single point. Let $A = X$. Let $f : A \to X$ be the identity map ($f = \mathrm{id}_X$). Let $g : A \to Y$ be the unique map from X to the single point Y (i.e., $g(x) = *$ for all $x \in X$). The pushout $X \sqcup_X \{*\}$ is the **cone** over X. It's formed by taking X and collapsing all its points to a single point, forming an apex, with lines connecting each point of X to the apex. For example, the cone over a circle S^1 is a disk D^2.

Example 4: Suspension of a Space The suspension of a space X, denoted SX, can be defined using a pushout. Let $X \times I$ be the product of X with the unit interval $I = [0, 1]$. We want to collapse $X \times \{0\}$ to one point (south pole) and $X \times \{1\}$ to another point (north pole). Let $A = (X \times \{0\}) \sqcup (X \times \{1\})$. Let $f : A \to X \times I$ be the inclusion map. Let $Y = \{*, *'\}$ (two distinct points, representing the poles). Let $g : A \to Y$ map all points in $X \times \{0\}$ to $*$ and all points in $X \times \{1\}$ to $*'$. The pushout $SX = (X \times I) \sqcup_A \{*, *'\}$ is the suspension of X. For example, the suspension of a sphere S^2 is S^3.

2.3 Cardinality and Comparing Sizes of Sets

2.3.1 Classification in Mathematics

In our discussion of equivalence relations, we mentioned that one of their uses is to show equivalence between mathematical objects. One of the main questions in different branches of mathematics is about classification, and we can describe this problem in terms of equivalence relations and functions that preserve the relevant mathematical structure. One of the main requirements for the latter is for your function to be bijective. Now we are going to show that bijective functions form an equivalence relation.

Proposition 2.1: Bijective Functions Form an Equivalence Relation

Let $\mathcal{S}$ be a collection of sets. Define a relation $\sim$ on $\mathcal{S}$ such that for any two sets $A, B \in \mathcal{S}$,

$$A \sim B \iff \text{there exists a bijection } f : A \to B$$

Then $\sim$ is an equivalence relation.

Proof. To show that $\sim$ is an equivalence relation, we must demonstrate that it satisfies reflexivity, symmetry, and transitivity.

1. **Reflexivity:** For any set $A \in \mathcal{S}$, we need to show that $A \sim A$. Consider the identity function $\mathrm{id}_A : A \to A$ defined by $\mathrm{id}_A(x) = x$ for all $x \in A$. We know that the identity function is always bijective (it is both injective and surjective). Therefore, there exists a bijection from A to A, so $A \sim A$.

2. **Symmetry:** For any two sets $A, B \in \mathcal{S}$, we need to show that if $A \sim B$, then $B \sim A$. Assume $A \sim B$. By definition, there exists a bijection $f : A \to B$. Since f is bijective, its inverse function $f^{-1} : B \to A$ exists and is also bijective (as shown in the properties of inverse functions). Therefore, there exists a bijection from B to A, so $B \sim A$.

3. **Transitivity:** For any three sets $A, B, C \in \mathcal{S}$, we need to show that if $A \sim B$ and $B \sim C$, then $A \sim C$. Assume $A \sim B$ and $B \sim C$. By definition, there exists a bijection $f : A \to B$ and a bijection $g : B \to C$. Consider the composition $g \circ f : A \to C$. As shown in the properties of function composition, if two functions are bijective, their composition is also bijective. Therefore, there exists a bijection from A to C, so $A \sim C$.

Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation. $\square$

Definition 2.1: Cardinality

For any set A, its **cardinality**, denoted $|A|$, is the number of elements in the set.

Example 2.1: Finite Sets with the Same Cardinality

Let's consider some simple finite sets. Two finite sets have the same cardinality if and only if there exists a bijection between them, which intuitively means they have the "same number of elements."

Example 1: Set of Days in a Week and First Seven Natural Numbers Let $A = \{$Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday$\}$ Let $B = \{1, 2, 3, 4, 5, 6, 7\}$ We can define a bijection $f : A \to B$ by mapping Monday to 1, Tuesday to 2, and so on. Since such a bijection exists, $|A| = |B|$, which is 7.

Example 2: Vowels and First Five Positive Even Numbers Let $V = \{$a, e, i, o, u$\}$ Let $E = \{2, 4, 6, 8, 10\}$ We can define a bijection $g : V \to E$, for instance, $g(\text{a}) = 2$, $g(\text{e}) = 4$, $g(\text{i}) = 6$, $g(\text{o}) = 8$, $g(\text{u}) = 10$. Since a bijection exists, $|V| = |E|$, which is 5.

Example 3: Standard Deck Suits and Standard Card Ranks Let $S = \{$Clubs, Diamonds, Hearts, Spades$\}$ Let $R = \{$Ace, King, Queen, Jack$\}$ We can define a bijection $h : S \to R$. For example, $h(\text{Clubs}) = \text{Ace}$, $h(\text{Diamonds}) = \text{King}$, $h(\text{Hearts}) = \text{Queen}$, $h(\text{Spades}) = \text{Jack}$. Both sets have cardinality 4.

Example 4: Planets (prior to 2006) and Musical Notes Let $P = \{$Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune, Pluto$\}$ (classical planets including Pluto) Let $M = \{$C, D, E, F, G, A, B, C'$\}$ (octave of musical notes, where C' is higher C) Wait, P has 9 elements and M has 8. These two sets do *not* have the same cardinality. A bijection between them cannot be formed. This illustrates that if no bijection exists, the cardinalities are different.

Observation 2.1

The Significance of Bijective Correspondence The proposition and examples highlight a profound idea:

- When we say two finite sets have the "same number of elements," we are implicitly asserting that there exists a bijection between them. This is how counting works: by establishing a one-to-one correspondence between the set of objects and a set of natural numbers $\{1, 2, \ldots, n\}$.
- The only structure inherent to a set itself is its cardinality (the "number" of its elements). Bijective functions are precisely the maps that preserve this structure. Hence, in the context of sets, bijections are the fundamental structure-preserving maps. This property is what allows us to classify sets based on their "size."

This equivalence relation forms the basis for defining cardinality, a cornerstone of set theory.

2.3.2 Comparing Infinite Sets

Beyond finite sets, the concept of cardinality allows us to compare the "sizes" of infinite sets. This comparison is precisely defined by the existence of bijections.

Definition 2.1: Same Cardinality

Two sets A and B have the same cardinality, $|A| = |B|$, if there exists a bijection $f : A \to B$.

Definition 2.2: Finite, Countable, and Uncountable Sets

- A set is **finite** if it has exactly $n \in \mathbb{N}$ elements.
- A set is **countably infinite** (or simply **countable**) if there exists a bijection with $\mathbb{N}$, i.e., $|A| = |\mathbb{N}|$.
- A set is **uncountable** if it is not countable.

Example 2.1: Examples of Set Cardinalities

- The set $\mathbb{N}$ of natural numbers is countable.
  - Its elements can be listed: $1, 2, 3, 4, \ldots$
- The set $\mathbb{Z}$ of integers is countable.
  - A bijection $f : \mathbb{N} \to \mathbb{Z}$ can be constructed as $f(n) = (-1)^n \lceil n/2 \rceil$.
  - Listing: $0, 1, -1, 2, -2, 3, -3, \ldots$
- The set $\mathbb{Q}$ of rational numbers is countable.
  - This is a non-trivial result, typically shown by "zigzagging" through a grid of positive and negative fractions.
- The set $\mathbb{R}$ of real numbers is uncountable.
  - This is proven by Cantor's diagonalization argument.

Example 2.2: Further Examples of Countable and Uncountable Sets

Example 1: Even Natural Numbers The set of even natural numbers, $E = \{2, 4, 6, \ldots\}$, is countable. A bijection $f : \mathbb{N} \to E$ is given by $f(n) = 2n$. This demonstrates that a proper subset of an infinite set can have the same cardinality as the original set.

Example 2: Intervals of Real Numbers Any open interval (a, b) where $a < b$ has the same cardinality as $\mathbb{R}$, and is thus uncountable. A bijection $f : (a, b) \to \mathbb{R}$ exists, for example, using a combination of linear scaling and the tangent function. This means that even a "small" interval contains "as many" points as the entire real line.

Example 3: Points in a Plane The set of points in the Cartesian plane $\mathbb{R}^2$ has the same cardinality as $\mathbb{R}$, and is therefore uncountable. This can be shown by constructing a bijection between $\mathbb{R}^2$ and $\mathbb{R}$, for instance, by interleaving decimal expansions. This is a surprising result, as $\mathbb{R}^2$ seems intuitively "larger" than $\mathbb{R}$.

Theorem 2.1: Cantor's Theorem

The power set $\mathcal{P}(A)$ of any set A has strictly greater cardinality than A itself. In particular:

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| > |\mathbb{N}|$$

This proves that $\mathbb{R}$ is uncountable.

Theorem 2.2: Countable Union of Countable Sets

A countable union of countable sets is countable.

Remark 2.1

While all finite sets have well-defined sizes, comparing infinite sets requires understanding bijections and functions. Surprisingly, some infinite sets are "larger" than others!

2.4 Geometry vs Algebra from Set-Theoretic Constructions

In physical theories, we often distinguish between two fundamental components: the **static** and **dynamic** aspects of a system.

- The **static** aspect refers to the *mathematical space* in which a physical system is defined. This is the structural background or *arena*. For instance:
  - In classical mechanics, motion unfolds in Euclidean space.
  - In special relativity, events are situated in Minkowski space.
  - In quantum mechanics, states live in Hilbert space.
- The **dynamic** aspect is described by evolution laws that dictate how the state of a system changes over time within this space.

These mathematical arenas can be broadly classified into two overlapping types:

- **Algebraic structures:** These involve sets equipped with operations (often binary), such as addition or multiplication. They include structures like groups, vector spaces, algebras, and rings. Algebra is concerned with the formal manipulation of elements using these rules.
- **Geometric structures:** These are concerned with shapes, spaces, and configurations. A geometric structure is typically described in terms of a **specified collection of subsets** of S (like open sets, neighborhoods, or manifolds), often arising from topological or metric properties. They are often more visual or spatial in nature.

Remark 2.1

In practice, most physical and mathematical frameworks blend both algebraic and geometric features. For example, differential geometry (used in general relativity) combines smooth manifolds (geometric) with tensor calculus (algebraic). Similarly, quantum field theory unites operator algebras with fields over spacetime. The interplay between algebra and geometry is central to modern theoretical physics.

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Let's first formalize what we mean by algebraic structures:

2.4.1 Algebraic Structures

An **algebraic structure** consists of a set (or sets) equipped with one or more binary operations, often referred to as laws of composition.

Definition 2.1: Law of Composition

Let S be a set. A **binary law of composition** on S is a map:

$$\varphi : S \times S \to S.$$

Given any $a, b \in S$, the image $\varphi(a, b) \in S$ is called the **product** of a and b. It is often denoted by ab, or other symbols like $a \cdot b$, $a + b$, $a \times b$, $[a, b]$, or $a \circ b$, depending on context.

- Most algebraic structures are built by equipping a set S with one or more such binary operations.
- Sometimes multiple sets are involved. For instance, in a vector space, we have a set of vectors and a set of scalars, with a law of composition:

$$\varphi : \text{Scalars} \times \text{Vectors} \to \text{Vectors}.$$

Remark 2.2

Higher-arity operations ($n > 2$) can also be considered, but many algebraic theories primarily focus on binary operations, as other operations can often be decomposed or expressed in terms of families of binary ones.

Key Properties of Binary Operations

- **Commutativity:** A binary operation φ is commutative if

$$ab = ba \quad \forall a, b \in S.$$

  This is usually assumed for addition: $a + b = b + a$.

- **Associativity:** The operation is associative if

$$a(bc) = (ab)c.$$

  Examples include:

  - Matrix multiplication
  - Function composition: $f \circ (g \circ h) = (f \circ g) \circ h$

  A non-example is the vector (cross) product in $\mathbb{R}^3$, where:

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \neq (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}.$$

  Instead, we have the identity:

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$$

—

2.4.2 Geometric Structures

A **geometric structure** on a set S is a way of designating certain subsets of S as "geometrically acceptable," according to rules that govern how they intersect and combine. Formally, a geometric structure $\mathcal{G}$ is typically defined as a specified collection of subsets of S, satisfying certain closure or compatibility properties.

Remark 2.1

Geometric structures are often visual in nature and provide the *arena* in which geometry or physical theories take place. Unlike algebraic structures, which are focused on composition and operations, geometric structures are about configuration, shape, and continuity.

We now explore two foundational examples:

Euclidean Geometry

In Euclidean geometry, the underlying set S is typically $\mathbb{R}^n$, and the distinguished subsets include points, lines, circles, planes, and more generally, geometric figures defined using notions of distance and angle.

- Acceptable subsets include: points, line segments, triangles, circles, polygons, etc.
- The structure is governed by axioms (e.g., Euclid's postulates) that define relationships such as parallelism, congruence, and the behavior of angles.
- Metric notions like length, area, and angle are defined, allowing for geometric reasoning.

Topology

In topology, the geometric structure $\mathcal{T} \subseteq 2^S$ is called a **topology**, and it satisfies the following conditions:

- The empty set and the whole set are in $\mathcal{T}$
- Any union of sets in $\mathcal{T}$ is also in $\mathcal{T}$
- Any finite intersection of sets in $\mathcal{T}$ is also in $\mathcal{T}$

The elements of $\mathcal{T}$ are called **open sets**, and they generalize the idea of "neighborhoods" or "continuous deformation."

Example 2.1: Standard Topology on the Real Line

Let $S = \mathbb{R}$, and define $\mathcal{T}$ to be the collection of all open intervals and arbitrary unions of them. This gives the standard topology on $\mathbb{R}$, where continuity, connectedness, and compactness can be studied.

Remark 2.2

Both Euclidean geometry and topology serve as geometric frameworks in physics. Euclidean geometry underlies classical mechanics and rigid body motion, while topology plays a central role in modern physics—especially in areas like condensed matter, quantum field theory, and general relativity.

3 Real analysis

3.1 Real Numbers

3.1.1 The Incompleteness of Rational Numbers

In physics, we have a good intuition about what numbers are. We understand their arithmetic, how to compare them, and how to work with functions involving them. From this foundation, the concept of calculus was developed.

However, in mathematics, we must build the notion of a number from scratch in order to eliminate any ambiguity in our definitions and theorems.

Our focus is to define the real numbers, but we will give a brief overview of how to construct the other number systems.

Constructing Number Systems and Their Limitations

- **Natural Numbers** $\mathbb{N}$:
  - **Construction:**Natural numbers are about counting. To define them rigorously, we use the Peano axioms, which characterize the idea of counting by defining a "successor" function. For example, we start from 0 and define 1 as the successor of 0, then 2 as the successor of 1, and so on.
  - **Limitation:** Natural numbers are incomplete they do not include negative numbers, so we cannot solve equations like $x + 2 = 1$ in $\mathbb{N}$.
- **Integers** $\mathbb{Z}$:
  - **Construction:**Natural Integers generalize natural numbers by incorporating the concept of differences. We construct them as equivalence classes of ordered pairs of natural numbers, such as (a, b) representing the integer $a - b$.
  - **Limitation:** While integers allow subtraction, they still cannot handle division problems like $2x = 1$.
- **Rational Numbers** $\mathbb{Q}$:
  - **Construction:** Rational numbers represent ratios of integers, defined as equivalence classes of ordered pairs (a, b) with $b \neq 0$, under the relation $(a, b) \sim (c, d)$ if $ad = bc$.

We will explore the limitation of the rationals in more detail.

Proposition 3.:

The equation $x^2 = 2$ has no solution in $\mathbb{Q}$.

Scratch Work and Strategy: We are about to write our first proof in this course. A key component of any proof is understanding the *logical structure* of the statement.

This proposition asserts that:

$$\forall x \in \mathbb{Q}, \quad x^2 \neq 2$$

In other words, it's a universal negative statement. A standard strategy in such cases is to assume the opposite that there *is* a rational solution and derive a contradiction.

Proof. Assume, for contradiction, that there exists a rational number $x \in \mathbb{Q}$ such that $x^2 = 2$. Since x is rational, we can write $x = \frac{m}{n}$, where $m, n \in \mathbb{Z}$, $n \neq 0$, and the fraction is in lowest terms (i.e., $\gcd(m, n) = 1$).

Then:

$$x^2 = \left(\frac{m}{n}\right)^2 = \frac{m^2}{n^2} = 2 \Rightarrow m^2 = 2n^2$$

This implies that m^2 is even, so m must also be even (since the square of an odd number is odd). Thus, write $m = 2k$ for some integer k. Substituting:

$$(2k)^2 = 4k^2 = 2n^2 \Rightarrow 2k^2 = n^2$$

So n^2 is even, hence n is also even. But this contradicts the assumption that $\frac{m}{n}$ is in lowest terms both m and n are even, so they share a common factor of 2.

Therefore, our assumption must be false. There is no rational number x such that $x^2 = 2$. □

The above proposition shows that the rational numbers have gaps in them, and therefore are incomplete. We express these gaps in terms of sets and sequences, which ultimately leads to the construction of the real numbers. Before we proceed with that construction, however, we will explore some important properties of the rational numbers that will help us define the real numbers precisely.

The Gaps in Rational Numbers in Terms of the Least Upper Bound Property

Definition 3.: Field

A *field* F is a set equipped with two operations, addition and multiplication, such that:

- $(F, +)$ is an abelian group with identity element 0,
- $F \setminus \{0\}$ is an abelian group under multiplication with identity element 1,
- Multiplication distributes over addition: for all $a, b, c \in F$, we have

$$a(b + c) = ab + ac.$$

The set of rational numbers $\mathbb{Q}$ is a field. You are encouraged to verify the axioms yourself.

Definition 3.: Upper Bound

Let $S \subseteq F$. An element $u \in F$ is called an *upper bound* of S if

$$\forall s \in S, \quad s \leq u.$$

Example 3.: L

et $S = \{x \in \mathbb{Q} \mid x < 2\}$. Then $3, 5, 8 \in \mathbb{Q}$ are upper bounds. Among all upper bounds, 2 is the smallest one, so it is the *least upper bound*.

Example 3.: L

et $T = \{x \in \mathbb{Q} \mid x < \frac{1}{2}\}$. Then $\frac{3}{5}, 1, 100$ are upper bounds. The least upper bound is $\frac{1}{2}$.

Definition 3.: Least Upper Bound Property

A field F is said to have the **least upper bound property** if every nonempty subset $S \subseteq F$ that is bounded above has a least upper bound (supremum) in F. That is, there exists some $\alpha \in F$ such that:

(i) $\forall s \in S$, $s \leq \alpha$ (so α is an upper bound of S),

(ii) If $\beta < \alpha$, then β is not an upper bound of S.

Remark 3.

This property tells us whether a field has "no gaps." If a set has a supremum outside the field, then the field is missing something and that's precisely what happens in $\mathbb{Q}$.

Example 3.: An upper bound with a least upper bound

Let $A = \{x \in \mathbb{Q} \mid x < 2\}$. The number 2 is clearly the least upper bound of A, and since $2 \in \mathbb{Q}$, this set has a least upper bound in $\mathbb{Q}$.

Non-Example 3.: A set without a least upper bound

Let $B = \{x \in \mathbb{Q} \mid x^2 < 2\}$. This set is bounded above in $\mathbb{Q}$; for example, 2 is an upper bound. However, the least upper bound of B is $\sqrt{2}$, which is irrational and not in $\mathbb{Q}$. Therefore, $\mathbb{Q}$ does not contain the supremum of this set.

Remark 3.

This example shows that $\mathbb{Q}$ is *incomplete*. The field of real numbers $\mathbb{R}$ was constructed specifically to fill these kinds of gaps. One way to construct $\mathbb{R}$ is to add all missing suprema using Dedekind cuts.

Proposition 3.: The field $\mathbb{Q}$ does not have the LUB property

There exists a subset of $\mathbb{Q}$ that is bounded above but has no least upper bound in $\mathbb{Q}$. Hence, $\mathbb{Q}$ fails the least upper bound property.

We have expressed the gaps in the rational numbers as bounded subsets of $\mathbb{Q}$ that do not possess a least upper bound.

Gaps in the Rationals via Cauchy Sequences

Definition 3.: Absolute Value and Distance

Let $x \in \mathbb{Q}$. The **absolute value** of x, denoted $|x|$, is defined as:

$$|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

Let $x, y \in \mathbb{Q}$. The **distance** between x and y, denoted $d(x, y)$, is defined by:

$$d(x, y) := |x - y|.$$

For instance, $d(2, 7) = |2 - 7| = 5$.

Proposition 3.: Basic Properties of Absolute Value

Let $x, y \in \mathbb{Q}$. Then:

(a) $|x| \geq 0$, with equality if and only if $x = 0$,

(b) $|-x| = |x|$,

(c) $|xy| = |x||y|$,

(d) $-|x| \leq x \leq |x|$,

(e) $|x + y| \leq |x| + |y|$ (Triangle inequality).

Proposition 3.: Properties of Distance

Let $x, y, z \in \mathbb{Q}$. Then:

(a) $d(x, y) = 0$ if and only if $x = y$,

(b) $d(x, y) = d(y, x)$ (Symmetry),

(c) $d(x, z) \leq d(x, y) + d(y, z)$ (Triangle inequality).

Definition 3.: ε-Closeness

Let $\varepsilon > 0$, and let $x, y \in \mathbb{Q}$. We say that y is **ε-close** to x if

$$d(y, x) := |y - x| \leq \varepsilon.$$

Example 3.: Understanding ε-Closeness

- The numbers 3.01 and 3.02 are 0.02-close, since $|3.01 - 3.02| = 0.01 \leq 0.02$.
- The numbers 5 and 4.95 are 0.1-close, but not 0.04-close, because $|5 - 4.95| = 0.05$.
- Any number is ε-close to itself for all $\varepsilon > 0$; for example, 7 is ε-close to 7 for any positive ε.
- The numbers 1 and 1.00000001 are 10^{-7}-close but not 10^{-9}-close.

Proposition 3.: Properties of ε-Closeness

Let $x, y, z, w \in \mathbb{Q}$ and $\varepsilon, \delta > 0$. Then:

(a) If $x = y$, then x is ε-close to y for every $\varepsilon > 0$. Conversely, if x is ε-close to y for every $\varepsilon > 0$, then $x = y$.

(b) If x is ε-close to y, then y is ε-close to x. (Symmetry)

(c) If x is ε-close to y, and y is δ-close to z, then x and z are $(\varepsilon + \delta)$-close. (Triangle inequality)

(d) If x and y are ε-close, and z and w are δ-close, then:

$$x + z \text{ and } y + w \text{ are } (\varepsilon + \delta)\text{-close}, \quad \text{and} \quad x - z \text{ and } y - w \text{ are } (\varepsilon + \delta)\text{-close}.$$

(e) If x and y are ε-close, then they are also ε'-close for every $\varepsilon' > \varepsilon$. (Monotonicity)

(f) If y and z are both ε-close to x, and w lies between y and z (i.e., $y \leq w \leq z$ or $z \leq w \leq y$), then w is also ε-close to x.

(g) If x and y are ε-close, and $z \neq 0$, then xz and yz are $\varepsilon|z|$-close.

(h) If x and y are ε-close, and z and w are δ-close, then xz and yw are $(\varepsilon|z| + \delta|x| + \varepsilon\delta)$-close.

Remark 3.

The ε-closeness relation $x \sim_\varepsilon y \iff |x - y| \leq \varepsilon$ behaves similarly to a relation:

- It is **reflexive**: every number is ε-close to itself.
- It is **symmetric**: if x is ε-close to y, then y is ε-close to x.
- It is *not transitive* in the usual sense, but satisfies a fuzzy transitivity: if $x \sim_\varepsilon y$ and $y \sim_\varepsilon z$, then $x \sim_{2\varepsilon} z$.

Thus, ε-closeness is not a true equivalence relation, but it shares many structural features with one.

Proposition 3.: Approximating $\sqrt{2}$ from Below

For every rational number $\varepsilon > 0$, there exists a nonnegative rational number x such that

$$x^2 < 2 < (x + \varepsilon)^2.$$

Proof (by contradiction). Let $\varepsilon > 0$ be a rational number.

Suppose for contradiction that there is *no* nonnegative rational number x such that

$$x^2 < 2 < (x + \varepsilon)^2.$$

Then for every nonnegative rational number x satisfying $x^2 < 2$, we must also have $(x + \varepsilon)^2 < 2$. (Note: $(x + \varepsilon)^2 = 2$ is impossible, as $\sqrt{2} \notin \mathbb{Q}$.)

Let's apply this to $x = 0$, which clearly satisfies $0^2 = 0 < 2$. Then we have:

$$(\varepsilon)^2 < 2.$$

Now let's try $x = 2\varepsilon$. Again,

$$(2\varepsilon)^2 = 4\varepsilon^2 < 2 \quad \text{(since } \varepsilon^2 < 2\text{, and 4 is fixed).}$$

We continue inductively. Suppose that $x = n\varepsilon$, then:

$$(n\varepsilon)^2 = n^2\varepsilon^2 < 2 \quad \text{for all } n \in \mathbb{N}.$$

But this leads to a contradiction. By the Archimedean property (see Proposition **??**), there exists an integer n such that

$$n > \frac{2}{\varepsilon} \quad \Rightarrow \quad n\varepsilon > 2 \quad \Rightarrow \quad (n\varepsilon)^2 > 4 > 2,$$

contradicting the claim that $(n\varepsilon)^2 < 2$ for all n. Therefore, our initial assumption must be false.

Hence, for every rational $\varepsilon > 0$, there exists a rational $x \geq 0$ such that

$$x^2 < 2 < (x + \varepsilon)^2.$$

$\square$

Example 3.: Approximating $\sqrt{2}$ with $\varepsilon = 0.001$

Let $\varepsilon = 0.001$, and take $x = 1.414$. Then:

$$x^2 = (1.414)^2 = 1.999396 < 2 < 2.002225 = (1.415)^2 = (x + \varepsilon)^2.$$

Here we view 1.414 as the rational number $\frac{1414}{1000}$. (We will discuss terminating decimals and their rational representation in a separate supplemental handout.)

Remark 3.

This proposition shows that although $\sqrt{2} \notin \mathbb{Q}$, we can find rational numbers that get arbitrarily close to it from below. Consider the sequence:

$$1.4,\ 1.41,\ 1.414,\ 1.4142,\ 1.41421,\ \ldots$$

Each term is rational, and their squares are:

$$1.96,\ 1.9881,\ 1.99396,\ 1.99996164,\ 1.9999899241,\ \ldots$$

These values approach 2, suggesting that the sequence gets closer and closer to $\sqrt{2}$. This motivates constructing the real numbers $\mathbb{R}$ as limits of Cauchy sequences in $\mathbb{Q}$.

Remark 3.

We have now seen two distinct ways in which the rational numbers $\mathbb{Q}$ are incomplete:

- There exist subsets of $\mathbb{Q}$ that are bounded above but have no least upper bound in $\mathbb{Q}$, such as the set $\{x \in \mathbb{Q} \mid x^2 < 2\}$.
- There are sequences of rational numbers that get arbitrarily close to values like $\sqrt{2}$, yet these limits do not belong to $\mathbb{Q}$.

These two perspectives reveal the "gaps" in the rational number system. To construct the real numbers $\mathbb{R}$, we will now explore two elegant and mathematically rigorous solutions that fill these gaps:

1. The **Dedekind cut construction**, which builds real numbers as cuts (or partitions) of $\mathbb{Q}$, using the idea of least upper bounds.
2. The **Cauchy sequence construction**, which defines real numbers as limits of sequences of rational numbers that become arbitrarily close together.

Both constructions produce the same number system $\mathbb{R}$, but offer different insights. In this course, we will focus primarily on the Cauchy sequence approach, while leaving the Dedekind cut method as optional supplemental reading.

3.1.2 Real Numbers

Dedekind Cuts

Definition 3.: Dedekind Cut

A **Dedekind cut** is a subset $A \subseteq \mathbb{Q}$ satisfying the following three properties:

1. A is nonempty and not equal to all of $\mathbb{Q}$,
2. If $r \in A$ and $s < r$, then $s \in A$ (i.e., A is closed downward),
3. A has no greatest element: for every $r \in A$, there exists $s \in A$ such that $r < s$.

Each cut corresponds intuitively to a real number: the set of all rationals less than that number.

Example 3.: Dedekind Cuts

- The rational number 2 corresponds to the cut:

$$2 = \{q \in \mathbb{Q} \mid q < 2\}.$$

- The number -1 corresponds to the cut:

$$-1 = \{q \in \mathbb{Q} \mid q < -1\}.$$

- The irrational number $\sqrt{3}$ corresponds to the cut:

$$\sqrt{3} = \{q \in \mathbb{Q} \mid q^2 < 3 \text{ or } q < 0\}.$$

- The cube root of 2 corresponds to the cut:

$$\sqrt[3]{2} = \{q \in \mathbb{Q} \mid q^3 < 2\}.$$

Theorem 3.: Completeness of $\mathbb{R}$ (Dedekind Version)

Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$.

Remark 3.

This is the famous **least upper bound property**, and it characterizes the completeness of the real number system.

Proof Sketch. We define $\mathbb{R}$ as the set of Dedekind cuts $A \subset \mathbb{Q}$. To prove completeness, suppose we have a nonempty collection $\mathcal{A}$ of cuts that is bounded above. We want to show that $\sup \mathcal{A}$ exists in $\mathbb{R}$.

Main idea: We will construct the least upper bound cut by taking the union of all the cuts in $\mathcal{A}$. We must then verify that this union satisfies the three properties of a cut and that it is the least upper bound.

Tools used: Properties of rational numbers, unions of downward-closed sets, and careful verification of the Dedekind axioms. □

Proof. Let $\mathcal{A}$ be a nonempty collection of Dedekind cuts that is bounded above. Let

$$S := \bigcup_{A \in \mathcal{A}} A.$$

We claim that S is a Dedekind cut:

1. $S \subsetneq \mathbb{Q}$: Since each $A \in \mathcal{A}$ omits some rational, so does the union.

2. S is nonempty: Because $\mathcal{A}$ is nonempty and each $A \in \mathcal{A}$ is nonempty.

3. S is downward-closed: If $r \in S$, then $r \in A$ for some $A \in \mathcal{A}$, so any $s < r$ lies in $A \subseteq S$.

4. S has no greatest element: Given $r \in S$, it lies in some A, which has no greatest element, so we can find $s > r$ in $A \subseteq S$.

Thus $S \in \mathbb{R}$ and serves as an upper bound. It is the least such upper bound because any smaller cut would omit some rational in some $A \in \mathcal{A}$.

Hence, $\mathbb{R}$ is complete. $\square$

Remark 3.

Using the definition of Dedekind cuts, one can define:

- Addition and multiplication of real numbers,
- A total order on $\mathbb{R}$,

making $\mathbb{R}$ into a **complete ordered field**. These constructions are rigorous but require careful definitions. We will not go into them here in detail.

Cauchy Sequences

We previously saw that we can approximate irrational numbers such as $\sqrt{2}$ using rational numbers that get arbitrarily close. Now we aim to build the machinery that formalizes this idea the notion of a **Cauchy sequence**.

Remark 3.

For every $\varepsilon > 0$, we can find a rational number x such that $x^2 < 2 < (x+\varepsilon)^2$. This gave us a list of rational numbers that appear to "approach" $\sqrt{2}$. To make this rigorous, we need to formalize the idea of a *sequence*.

Definition 3.: Sequence of Rational Numbers

Let $m \in \mathbb{Z}$. A **sequence** $(a_n)_{n=m}^{\infty}$ of rational numbers is a function:

$$a : \{n \in \mathbb{Z} \mid n \geq m\} \to \mathbb{Q}.$$

More informally, it is a list of rational numbers $a_m, a_{m+1}, a_{m+2}, \ldots$, one for each integer $n \geq m$.

Example 3.: Examples of Sequences

- The sequence $(n^2)_{n=0}^{\infty} = 0, 1, 4, 9, 16, \ldots$.
- The constant sequence $(3)_{n=0}^{\infty} = 3, 3, 3, 3, \ldots$.
- The sequence $(n^2)_{n=3}^{\infty} = 9, 16, 25, \ldots$.

Our goal is to define the real numbers as limits of certain rational sequences. But not all sequences "converge." We need a criterion to distinguish the ones that are trying to settle down (like approaching $\sqrt{2}$) from those that do not.

Example 3.: Converging and Non-Converging Sequences

The following sequences appear to **converge**:

- $1.4,\ 1.41,\ 1.414,\ 1.4142,\ 1.41421,\ \ldots$ (approaching $\sqrt{2}$)
- $0.1,\ 0.01,\ 0.001,\ 0.0001,\ \ldots$ (approaching 0)

The following sequences clearly do **not** converge:

- $1,\ 2,\ 4,\ 8,\ 16,\ \ldots$ (diverges to infinity),
- $1,\ 0,\ 1,\ 0,\ 1,\ \ldots$ (oscillates, never settling).

Cauchy Sequences

To define real numbers using sequences, we return to the idea of *closeness*. Recall:

Definition 3.: ε-Closeness (recall)

Two rational numbers x, y are said to be ε-**close** if

$$d(x, y) := |x - y| < \varepsilon.$$

Now we use this idea to define what it means for a sequence to become "steady" or "stable" within a certain range.

Definition 3.: ε-Steady Sequence

Let $\varepsilon > 0$. A sequence $(a_n)_{n=0}^{\infty}$ of rational numbers is said to be ε-**steady** if:

$$d(a_j, a_k) \leq \varepsilon \quad \text{for all } j, k \in \mathbb{N}.$$

That is, all the terms in the sequence are ε-close to each other.
More generally, a sequence $(a_n)_{n=N}^{\infty}$ is ε-steady if:

$$d(a_j, a_k) \leq \varepsilon \quad \text{for all } j, k \geq N.$$

Example 3.: Examples

- $1, 0, 1, 0, 1, \ldots$ is 1-steady but not $\frac{1}{2}$-steady.
- $0.1, 0.01, 0.001, 0.0001, \ldots$ is 0.1-steady, but not 0.01-steady. (Why?)
- $1, 2, 4, 8, 16, \ldots$ is not ε-steady for *any* $\varepsilon > 0$. (Why?)
- $2, 2, 2, 2, \ldots$ is ε-steady for *every* $\varepsilon > 0$.

However, ε-steadiness is not quite what we need: it is too sensitive to the early terms of a sequence. For example:

Example 3.: Too Sensitive: A Problem with Early Terms

The sequence $10, 0, 0, 0, 0, \ldots$ is 10-steady but not ε-steady for any $\varepsilon < 10$, even though it clearly converges quickly to 0.
What we want is a notion that *eventually* the terms settle down.

Definition 3.: Eventually ε-Steady

A sequence $(a_n)_{n=0}^{\infty}$ is **eventually ε-steady** if there exists an integer $N \geq 0$ such that:

$$d(a_j, a_k) \leq \varepsilon \quad \text{for all } j, k \geq N.$$

Example 3.: Eventually Steady

- Let $a_n = \frac{1}{n}$. Then:

$$1,\ \tfrac{1}{2},\ \tfrac{1}{3},\ \ldots \text{ is not 0.1-steady, but it is eventually 0.1-steady.}$$

- The sequence $10, 0, 0, 0, 0, \ldots$ is eventually ε-steady for *every* $\varepsilon > 0$. (Why?)

We now give the correct definition that captures the intuitive idea of "wanting to converge":

Definition 3.: Cauchy Sequence

A sequence $(a_n)_{n=0}^{\infty}$ of rational numbers is a **Cauchy sequence** if for every $\varepsilon > 0$, there exists an integer $N \geq 0$ such that:

$$d(a_j, a_k) = |a_j - a_k| \leq \varepsilon \quad \text{for all } j, k \geq N.$$

Example 3.: Informal Intuition

The sequence

$$1.4,\ 1.41,\ 1.414,\ 1.4142,\ 1.41421,\ \ldots$$

seems to converge to $\sqrt{2}$. It's already 1-steady. Discarding the first term gives:

$$1.41,\ 1.414,\ 1.4142,\ \ldots$$

which is 0.1-steady. Going further gives 0.01-steady, 0.001-steady, and so on. So it seems this sequence becomes ε-steady for every $\varepsilon > 0$, and is therefore a Cauchy sequence.
Of course, to be rigorous, we would need to define the sequence formally not just using decimals which we will do in the next subsection.

Example 3.: A Rigorous Example of a Cauchy Sequence

Let $a_n := \frac{1}{n}$. Then the sequence

$$(a_n) = \left(1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4},\ \ldots\right)$$

is a Cauchy sequence in $\mathbb{Q}$.
Proof: Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $\frac{1}{N} < \frac{\varepsilon}{2}$. Then for all $j, k \geq N$, we have:

$$|a_j - a_k| = \left|\frac{1}{j} - \frac{1}{k}\right| \leq \left|\frac{1}{N}\right| < \frac{\varepsilon}{2} < \varepsilon.$$

Thus, the terms eventually get arbitrarily close to each other. So (a_n) is a Cauchy sequence.

Non-Example 3.: A Cauchy Sequence that Does Not Converge in $\mathbb{Q}$

Let $a_n \in \mathbb{Q}$ be the decimal truncation of $\sqrt{2}$ to n decimal places, e.g.:

$$a_1 = 1.4, \quad a_2 = 1.41, \quad a_3 = 1.414, \quad a_4 = 1.4142, \quad \ldots$$

Formally, define each $a_n = \left\lfloor 10^n \cdot \sqrt{2} \right\rfloor / 10^n \in \mathbb{Q}$.
Then (a_n) is a Cauchy sequence in $\mathbb{Q}$, since:

$$|a_j - a_k| \leq \frac{1}{10^{\min(j,k)}} \to 0 \quad \text{as } j, k \to \infty.$$

But: (a_n) does not converge in $\mathbb{Q}$, because $\lim a_n = \sqrt{2} \notin \mathbb{Q}$.
Conclusion: $\mathbb{Q}$ is not complete it contains Cauchy sequences that do not converge within $\mathbb{Q}$. This motivates the construction of $\mathbb{R}$ as a **completion** of $\mathbb{Q}$.

Remark 3.

As we saw earlier, verifying that a sequence is Cauchy from first principles that is, directly from the definition without using limit machinery requires some care. Even for a simple sequence like $a_n = \frac{1}{n}$, we had to think in reverse: given an $\varepsilon > 0$, what condition on N guarantees that the tail of the sequence becomes ε-steady? Later, we will develop limit laws that make these verifications much easier. For now, we explore some basic properties of Cauchy sequences beginning with boundedness.

Definition 3.: Bounded Sequence

Let $M \geq 0$. A finite sequence $a_1, a_2, \ldots, a_n$ is said to be **bounded by** M if:

$$|a_i| \leq M \quad \text{for all } 1 \leq i \leq n.$$

An infinite sequence $(a_n)_{n=1}^{\infty}$ is **bounded by** M if:

$$|a_i| \leq M \quad \text{for all } i \geq 1.$$

A sequence is said to be **bounded** if it is bounded by some $M \geq 0$.

Example 3.: Examples

- The finite sequence $1, -2, 3, -4$ is bounded by 4 (and also by any number ≥ 4).
- The infinite sequence $1, -2, 3, -4, 5, -6, \ldots$ is **unbounded**. (Can you prove this using the Archimedean property?)
- The sequence $1, -1, 1, -1, \ldots$ is bounded (e.g., by 1), but is **not** a Cauchy sequence.
- Note: a sequence is bounded by M if and only if it is M-close to the zero sequence $0, 0, 0, \ldots$. (Why?)

Lemma 3.: Every Finite Sequence is Bounded

Every finite sequence $a_1, a_2, \ldots, a_n$ of rational numbers is bounded.

Proof. We proceed by induction on n.

Base case: When $n = 1$, the sequence is just a_1, which is clearly bounded by $M := |a_1|$.

Inductive step: Suppose the lemma holds for some $n \geq 1$; that is, the sequence $a_1, a_2, \ldots, a_n$ is bounded by some $M \geq 0$. Then consider the sequence $a_1, a_2, \ldots, a_n, a_{n+1}$. The number $|a_{n+1}| \leq M + |a_{n+1}|$, and all previous terms are also bounded by $M + |a_{n+1}|$. Hence, the extended sequence is bounded.

By induction, every finite sequence is bounded. $\square$

Remark 3.

While this argument guarantees boundedness of *finite* sequences no matter how long it does not apply to *infinite* sequences. (Infinity is not a natural number!) For that, we prove the following key lemma.

Lemma 3.: Cauchy Sequences are Bounded

Every Cauchy sequence $(a_n)_{n=1}^{\infty}$ is bounded.

Remark 3.

The sequence

$$1.1,\ -1.1,\ 1.1,\ -1.1,\ 1.1,\ \ldots$$

has all its terms 0.1-close to each other, yet it is **not** 0.1-steady it oscillates and does not settle. This motivates defining closeness *between* two sequences.

Definition 3.: Eventually ε-Close Sequences

Let $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ be two sequences of rational numbers, and let $\varepsilon > 0$. We say that (a_n) is **eventually ε-close** to (b_n) if there exists an integer $N \geq 0$ such that:

$$|a_n - b_n| \leq \varepsilon \quad \text{for all } n \geq N.$$

Example 3.: Eventually Close Sequences

Consider the sequences:

$$a_n = 1.1,\ 1.01,\ 1.001,\ 1.0001,\ \ldots$$

$$b_n = 0.9,\ 0.99,\ 0.999,\ 0.9999,\ \ldots$$

These are not 0.1-close at the start (e.g., $|1.1 - 0.9| = 0.2 > 0.1$), but from the second term onward, they become 0.1-close. In fact, by discarding more terms, they become 0.01-close, then 0.001-close, etc. Thus, they are eventually ε-close for all $\varepsilon > 0$.

Definition 3.: Equivalent Sequences

Two sequences $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ are said to be **equivalent** if for every $\varepsilon > 0$, they are eventually ε-close. That is:

$$\forall \varepsilon > 0,\ \exists N \geq 0 \text{ such that } |a_n - b_n| \leq \varepsilon \quad \text{for all } n \geq N.$$

Remark 3.

Equivalence of sequences captures the idea that two sequences "converge to the same number," even if they look different initially. This idea will allow us to define a real number as an entire class of equivalent Cauchy sequences.

Proposition 3.: Claim

Let $a_n = 1 + 10^{-n}$ and $b_n = 1 - 10^{-n}$. Then (a_n) and (b_n) are equivalent sequences.

Proof. Fix $\varepsilon > 0$. We want to find an integer N such that for all $n \geq N$, we have:

$$|a_n - b_n| = |(1 + 10^{-n}) - (1 - 10^{-n})| = |2 \cdot 10^{-n}| = 2 \cdot 10^{-n} \leq \varepsilon.$$

Let $N \in \mathbb{N}$ be such that $10^{-N} \leq \frac{\varepsilon}{2}$, which is possible by the Archimedean property. Then for all $n \geq N$, we have:

$$|a_n - b_n| = 2 \cdot 10^{-n} \leq 2 \cdot 10^{-N} \leq \varepsilon.$$

Thus, the two sequences are eventually ε-close for every $\varepsilon > 0$, hence they are equivalent. $\square$

Constructing the Real Numbers via Cauchy Sequences

We are now ready to construct the real numbers.

We introduce a new symbolic notation $\mathrm{LIM}_{n\to\infty} a_n$, similar in spirit to the earlier notations for formal subtraction $(-)$ and division $(//)$. As the notation suggests, this object will eventually match the familiar limit operation but for now, it simply denotes an equivalence class of Cauchy sequences.

Definition 3.: Real Numbers via Cauchy Sequences

A **real number** is defined to be a formal expression of the form

$$\mathrm{LIM}_{n\to\infty} a_n,$$

where $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence of rational numbers.
Two such expressions $\mathrm{LIM}_{n\to\infty} a_n$ and $\mathrm{LIM}_{n\to\infty} b_n$ are said to be **equal** if the sequences (a_n) and (b_n) are equivalent that is, for every $\varepsilon > 0$, there exists $N \geq 0$ such that:

$$|a_n - b_n| \leq \varepsilon \quad \text{for all } n \geq N.$$

The set of all such formal limits is denoted by $\mathbb{R}$.

Example 3.: Equivalent Cauchy Sequences Representing the Same Real Number

Let (a_n) and (b_n) be the following sequences:

$$a_n = 1.4,\ 1.41,\ 1.414,\ 1.4142,\ 1.41421,\ \ldots$$

$$b_n = 1.5,\ 1.42,\ 1.415,\ 1.4143,\ 1.41422,\ \ldots$$

These are both Cauchy sequences that approximate $\sqrt{2}$. Though they differ in early terms, they become arbitrarily close that is, they are equivalent. Hence:

$$\mathrm{LIM}_{n\to\infty} a_n = \mathrm{LIM}_{n\to\infty} b_n \in \mathbb{R}.$$

Remark 3.

We refer to $\mathrm{LIM}_{n\to\infty} a_n$ as the **formal limit** of the sequence (a_n). Later, we will define the familiar notion of limit, and prove that every Cauchy sequence in $\mathbb{Q}$ has a unique formal limit in $\mathbb{R}$, which coincides with its actual limit in $\mathbb{R}$. At that point, we will no longer need the formal LIM notation.

To ensure this is a valid construction of a number system, we must verify that the notion of equality we've defined satisfies the standard properties of equality:

Proposition 3.: Basic Properties of Equality in $\mathbb{R}$

Let $x = \mathrm{LIM}_{n\to\infty} a_n$, $y = \mathrm{LIM}_{n\to\infty} b_n$, and $z = \mathrm{LIM}_{n\to\infty} c_n$ be real numbers. Then:

(i) $x = x$ (Reflexivity),

(ii) If $x = y$, then $y = x$ (Symmetry),

(iii) If $x = y$ and $y = z$, then $x = z$ (Transitivity).

Proof. Each of these follows directly from the definition of equivalence of sequences:

(i) Reflexivity: For any Cauchy sequence (a_n), we have $|a_n - a_n| = 0 \leq \varepsilon$ for all $\varepsilon > 0$, so $(a_n) \sim (a_n)$.

(ii) Symmetry: If $(a_n) \sim (b_n)$, then $|a_n - b_n| \leq \varepsilon \Rightarrow |b_n - a_n| \leq \varepsilon$, so $(b_n) \sim (a_n)$.

(iii) Transitivity: If $(a_n) \sim (b_n)$ and $(b_n) \sim (c_n)$, then for every $\varepsilon > 0$, there exist N_1 and N_2 such that for $n \geq \max(N_1, N_2)$:

$$|a_n - b_n| \leq \tfrac{\varepsilon}{2}, \quad |b_n - c_n| \leq \tfrac{\varepsilon}{2} \Rightarrow |a_n - c_n| \leq \varepsilon,$$

by the triangle inequality. So $(a_n) \sim (c_n)$. $\square$

Overview: Arithmetic, Order, and Completeness of $\mathbb{R}$

Arithmetic and Order in $\mathbb{R}$. Once we have defined real numbers as equivalence classes of Cauchy sequences of rational numbers, we can define arithmetic operations and order on $\mathbb{R}$ by performing those operations term-by-term on representatives.

Let $x = \mathrm{LIM}\, a_n$ and $y = \mathrm{LIM}\, b_n$ be two real numbers (i.e., two equivalence classes of Cauchy sequences).

- **Addition:** $x + y := \mathrm{LIM}(a_n + b_n)$
- **Multiplication:** $x \cdot y := \mathrm{LIM}(a_n \cdot b_n)$
- **Negation:** $-x := \mathrm{LIM}(-a_n)$
- **Order:** $x < y$ if there exists $\varepsilon > 0$ such that $a_n < b_n - \varepsilon$ for all sufficiently large n

One must verify that these operations are well-defined i.e., they do not depend on the choice of representative sequence within an equivalence class and that they satisfy the axioms of a field and total order.

Completeness of $\mathbb{R}$. The field $\mathbb{R}$, as constructed above, satisfies the following crucial property:

Theorem 3.: Completeness of $\mathbb{R}$

Every Cauchy sequence of real numbers converges to a real number.

Proof Sketch. Every real number $x = \mathrm{LIM}\, a_n$ is, by definition, the limit of a Cauchy sequence in $\mathbb{Q}$. If (x_n) is a Cauchy sequence in $\mathbb{R}$, each $x_n = \mathrm{LIM}\, a_n^{(n)}$, for some Cauchy sequence $(a_n^{(n)}) \subset \mathbb{Q}$.

One can then "diagonalize" to construct a new sequence (b_n) in $\mathbb{Q}$ that approximates x_n closely enough for large n, and show that (b_n) is itself a Cauchy sequence in $\mathbb{Q}$, whose formal limit $\mathrm{LIM}\, b_n \in \mathbb{R}$ is the limit of (x_n).

Thus, $\mathbb{R}$ is **complete**. $\square$

Contrast: $\mathbb{Q}$ is Not Complete. The rationals $\mathbb{Q}$ are not complete. That is, there exist Cauchy sequences of rational numbers that do not converge in $\mathbb{Q}$. A classic example is:

$$a_n := \text{the decimal approximation of } \sqrt{2} \text{ to } n \text{ digits}$$

This sequence is Cauchy in $\mathbb{Q}$, but it does not converge to any rational number since $\sqrt{2} \notin \mathbb{Q}$.

Thus, while $\mathbb{Q}$ is dense and algebraically rich, it lacks limits of some of its own Cauchy sequences. This incompleteness is exactly what our construction of $\mathbb{R}$ remedies.

Further Description and Properties of $\mathbb{R}$

Now that we have defined the real numbers $\mathbb{R}$ as equivalence classes of Cauchy sequences of rational numbers, and shown that they form a complete ordered field, we explore some foundational properties of $\mathbb{R}$. These include how we generalize ε-closeness to real numbers, how rationals are distributed within $\mathbb{R}$, and a basic yet powerful structural axiom known as the Archimedean property.

Definition 3.: ε-Closeness in $\mathbb{R}$

Let $x, y \in \mathbb{R}$, and let $\varepsilon > 0$ be a rational number. We say that x is **ε-close** to y if:

$$|x - y| < \varepsilon.$$

Proposition 3.: Basic Properties of ε-Closeness in $\mathbb{R}$

Let $x, y, z, w \in \mathbb{R}$, and let $\varepsilon, \delta > 0$. Then:

(a) If $x = y$, then x is ε-close to y for all $\varepsilon > 0$. Conversely, if x is ε-close to y for all $\varepsilon > 0$, then $x = y$.

(b) ε-closeness is symmetric: if x is ε-close to y, then y is ε-close to x.

(c) Triangle inequality: if x is ε-close to y, and y is δ-close to z, then x is $(\varepsilon + \delta)$-close to z.

(d) Addition and subtraction: if x is ε-close to y, and z is δ-close to w, then:

$$x + z \text{ is } (\varepsilon + \delta)\text{-close to } y + w, \quad \text{and} \quad x - z \text{ is } (\varepsilon + \delta)\text{-close to } y - w.$$

(e) Monotonicity: if x is ε-close to y, then it is also ε'-close to y for all $\varepsilon' > \varepsilon$.

(f) Intermediate point: if y and z are both ε-close to x, and w lies between y and z, then w is also ε-close to x.

(g) Multiplication: if x and y are ε-close, and $z \in \mathbb{R} \setminus \{0\}$, then xz and yz are $\varepsilon|z|$-close.

(h) Mixed multiplication: if x is ε-close to y, and z is δ-close to w, then:

$$xz \text{ is } (\varepsilon|z| + \delta|x| + \varepsilon\delta)\text{-close to } yw.$$

Theorem 3.: Density of $\mathbb{Q}$ in $\mathbb{R}$

Between any two real numbers $x < y$, there exists a rational number $q \in \mathbb{Q}$ such that:

$$x < q < y.$$

Idea. Since $\mathbb{Q}$ is dense in itself, and every real number is a limit of rational numbers, one

can construct a rational approximation within any interval (x, y) by choosing a rational with enough decimal precision. $\square$

Remark 3.

This means $\mathbb{Q} \subset \mathbb{R}$ is a dense subfield. Despite not being complete, the rationals are "everywhere" in $\mathbb{R}$.

Theorem 3.: Archimedean Property

For any $x \in \mathbb{R}$, there exists a natural number $n \in \mathbb{N}$ such that:

$$n > x.$$

Equivalently: for any positive real number $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that:

$$\frac{1}{n} < \varepsilon.$$

Remark 3.

This property ensures there are no "infinitely large" or "infinitesimally small" real numbers all reals are comparable in size to the natural numbers. This is essential in analysis, especially in proving limits and convergence.

Building New Spaces from $\mathbb{R}$

We now explore how to build new geometric spaces out of the real numbers. This leads to Euclidean space $\mathbb{R}^m$, the dot product, and many important concepts such as norms, convexity, and inner product spaces.

Definition 3.: Open and Closed Intervals

Let $a, b \in \mathbb{R}$ with $a < b$. Then:

- The **open interval** $(a, b) := \{x \in \mathbb{R} \mid a < x < b\}$,
- The **closed interval** $[a, b] := \{x \in \mathbb{R} \mid a \leq x \leq b\}$,
- Half-open intervals: $(a, b], [a, b)$ defined similarly.

Definition 3.: Euclidean Space

For any positive integer m, the Euclidean space $\mathbb{R}^m$ is the Cartesian product:

$$\mathbb{R}^m := \mathbb{R} \times \cdots \times \mathbb{R} = \{(x_1, \ldots, x_m) \mid x_i \in \mathbb{R}\}.$$

Each element $\vec{x} = (x_1, \ldots, x_m)$ is called a **vector**.

Example 3.: Examples in Low Dimensions

- $\mathbb{R}^1$: the real line.
- $\mathbb{R}^2$: the Euclidean plane points like $(3, 5)$, $(0, -1.2)$.
- $\mathbb{R}^3$: 3D space common in physics.

Definition 3.: Dot Product

Given vectors $\vec{x} = (x_1, \ldots, x_m)$, $\vec{y} = (y_1, \ldots, y_m)$ in $\mathbb{R}^m$, their **dot product** is:

$$\vec{x} \cdot \vec{y} := \sum_{i=1}^{m} x_i y_i.$$

Definition 3.: Length of a Vector (Norm)

The **length** or **norm** of a vector $\vec{x} \in \mathbb{R}^m$ is:

$$\|\vec{x}\| := \sqrt{\vec{x} \cdot \vec{x}} = \sqrt{\sum_{i=1}^{m} x_i^2}.$$

Proposition 3.: Cauchy–Schwarz Inequality

For all $\vec{x}, \vec{y} \in \mathbb{R}^m$,

$$|\vec{x} \cdot \vec{y}| \leq \|\vec{x}\| \cdot \|\vec{y}\|.$$

Equality holds if and only if $\vec{x}, \vec{y}$ are linearly dependent.

Proposition 3.: Triangle Inequality

For all $\vec{x}, \vec{y} \in \mathbb{R}^m$,

$$\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|.$$

Definition 3.: Unit Ball and Unit Sphere

- The **unit ball** in $\mathbb{R}^m$ is:

$$B^m := \{\vec{x} \in \mathbb{R}^m \mid \|\vec{x}\| < 1\}.$$

- The **unit sphere** in $\mathbb{R}^m$ is:

$$S^{m-1} := \{\vec{x} \in \mathbb{R}^m \mid \|\vec{x}\| = 1\}.$$

Definition 3.: Convex Subset

A subset $C \subseteq \mathbb{R}^m$ is called **convex** if for all $\vec{x}, \vec{y} \in C$ and all $t \in [0, 1]$,

$$(1-t)\vec{x} + t\vec{y} \in C.$$

Example 3.: Examples of Convex Sets

- The unit ball B^m is convex.
- Any affine subspace (e.g. a line or plane) is convex.
- The union of two disjoint balls is *not* convex.

Definition 3.: Inner Product Space

A vector space V over $\mathbb{R}$ is called an **inner product space** if there is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying:

- $\langle \vec{x}, \vec{y} \rangle = \langle \vec{y}, \vec{x} \rangle$ (symmetry),
- $\langle \lambda\vec{x}, \vec{y} \rangle = \lambda \langle \vec{x}, \vec{y} \rangle$,
- $\langle \vec{x} + \vec{z}, \vec{y} \rangle = \langle \vec{x}, \vec{y} \rangle + \langle \vec{z}, \vec{y} \rangle$,
- $\langle \vec{x}, \vec{x} \rangle \geq 0$, with equality iff $\vec{x} = 0$.

Example 3.: Inner Product on $\mathbb{R}^n$

The dot product $\vec{x} \cdot \vec{y} = \sum x_i y_i$ defines an inner product on $\mathbb{R}^n$. Hence, $\mathbb{R}^n$ is an inner product space.

Definition 3.: Norm

A function $\| \cdot \| : V \to \mathbb{R}$ is called a **norm** if for all $\vec{x}, \vec{y} \in V$, $\lambda \in \mathbb{R}$:

- $\|\vec{x}\| \geq 0$, with equality iff $\vec{x} = 0$,
- $\|\lambda\vec{x}\| = |\lambda| \cdot \|\vec{x}\|$,
- $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$ (triangle inequality).

Example 3.: Examples of Norms

- Euclidean norm: $\|\vec{x}\|_2 = \sqrt{x_1^2 + \cdots + x_n^2}$,
- Taxicab (Manhattan) norm: $\|\vec{x}\|_1 = |x_1| + \cdots + |x_n|$,
- Maximum norm: $\|\vec{x}\|_\infty = \max_i |x_i|$.

Proposition 3.: Norm from Inner Product

Given an inner product $\langle \cdot, \cdot \rangle$, one can define a norm by:

$$\|\vec{x}\| := \sqrt{\langle \vec{x}, \vec{x} \rangle}.$$

3.1.3 Functions on the Reals

We now turn our attention to functions whose domain and codomain are subsets of the real numbers. These functions will serve as a foundation for much of calculus and analysis. Our goal is to motivate the precise definition of continuity through examples both intuitive and pathological.

Definition 3.: Function Between Sets

Let X and Y be sets. A function f from X to Y is a rule that assigns to each element $x \in X$ a unique element $f(x) \in Y$. We write this as:

$$f : X \to Y.$$

Definition 3.: Graph of a Function

Given a function $f : X \to \mathbb{R}$, the **graph** of f is the set of ordered pairs:

$$\{(x, f(x)) \mid x \in X\} \subseteq \mathbb{R}^2.$$

Example 3.: Well-behaved Functions (Intuitively "Nice")

- $f(x) = x^2$ on $\mathbb{R}$. Its graph is a smooth parabola; small changes in x yield small changes in $f(x)$.
- $f(x) = \sin x$. This function oscillates gently in a predictable, bounded way.
- $f(x) = \dfrac{1}{1 + x^2}$. This function decays smoothly toward 0 as $|x| \to \infty$, without any jumps or wild behavior.

Non-Example 3.: Functions With Pathological or Interesting Behavior

- **Discontinuous at a Point:**

$$f(x) = \begin{cases} x^2 & \text{if } x \neq 2, \\ 5 & \text{if } x = 2. \end{cases}$$

  This function looks smooth, except for the single "wrong" value at $x = 2$. It does not behave like its neighbors at that point.

- **Jump Discontinuity:**

$$f(x) = \begin{cases} 1 & \text{if } x < 0, \\ 0 & \text{if } x \geq 0. \end{cases}$$

  This function has an abrupt jump at $x = 0$.

- **Oscillatory Behavior:**

$$f(x) = \sin\left(\frac{1}{x}\right) \quad \text{for } x \neq 0, \quad f(0) := 0.$$

  Near $x = 0$, the function oscillates faster and faster, without settling down. Its graph looks extremely erratic near the origin.

- **Wildly Oscillating Step Function:**

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

  This function, known as the **Dirichlet function**, is "jumping" at every point it's completely unapproachable.

- **Another Weird Limit:**

$$f(x) = x \cdot \sin\left(\frac{1}{x}\right), \quad f(0) := 0.$$

  This function oscillates like $\sin(1/x)$, but its amplitude goes to zero as $x \to 0$. It behaves better than $\sin(1/x)$, but still has lots of wiggles near 0.

We observed in previous examples that, as we approach certain points from the left or right, the function's output seems to tend toward a particular value. This behavior is what we call the **limit** of the function at that point. When this expected limiting behavior agrees with the actual value of the function at that point, we informally consider the function to be "well-behaved" or **continuous** there.

We now make these ideas precise.

Closeness of Functions: Toward Limits

Just as we used the notions of ε-closeness and eventual ε-closeness to rigorously define limits of sequences, we now develop similar ideas to handle limits of functions.

Definition 3.: ε-Closeness of a Function

Let $X \subseteq \mathbb{R}$, and let $f : X \to \mathbb{R}$ be a function. Fix a real number $L \in \mathbb{R}$ and $\varepsilon > 0$. We say that the function f is **ε-close to** L on X if:

$$|f(x) - L| \leq \varepsilon \quad \text{for all } x \in X.$$

Example 3.: Examples

- Let $f(x) = x^2$. On the interval $[1, 3]$, we have:

  $$1 \leq f(x) \leq 9 \quad \Rightarrow \quad |f(x) - 4| \leq 5.$$

  So f is 5-close to 4 on $[1, 3]$.

- On the smaller interval $[1.9, 2.1]$, we compute:

  $$f(x) = x^2 \in [3.61, 4.41] \quad \Rightarrow \quad |f(x) - 4| \leq 0.41,$$

  so f is 0.41-close to 4 on $[1.9, 2.1]$.

- Let $f(x) = \sin x$. On $[0, \pi/2]$, we have:

  $$0 \leq \sin x \leq 1 \Rightarrow |f(x) - 0.5| \leq 0.5,$$

  so f is 0.5-close to 0.5 on that interval.

Non-Example 3.: Non-Examples

- Let $f(x) = \frac{1}{x}$ on $(0, 1]$, and try to claim that $f(x)$ is 5-close to 0 on this domain.

  This is false, because $f(x) \to \infty$ as $x \to 0$. So $|f(x) - 0|$ is not bounded by 5 in fact, for $x = 0.01$, we already have $|f(x) - 0| = 100 \gg 5$.

- Let $f(x) = x^2$, and consider the whole real line $\mathbb{R}$. Then $f(x)$ is not 1000-close to 0, because $f(100) = 10{,}000$, and $|10{,}000 - 0| = 10{,}000 \gg 1000$.

  So no matter how large ε is, if the domain is unbounded, $f(x)$ might not be ε-close to a fixed L.

3.1.4 Limits and ε-Closeness for Functions

Just as we used the notions of ε-closeness and eventual ε-closeness to deal with limits of sequences, we now extend this idea to functions. This will help us develop the formal

notion of limits and continuity.

Definition 3.: Epsilon-Closeness of a Function

Let $X \subseteq \mathbb{R}$, $f : X \to \mathbb{R}$ a function, $L \in \mathbb{R}$, and $\varepsilon > 0$. We say that f **is ε-close to** L iff

$$\forall x \in X, \quad |f(x) - L| \leq \varepsilon.$$

Example 3.: Examples

- Let $f(x) := x^2$, restricted to $X = [1, 3]$. Then f is 5-close to 4 since for all $x \in [1, 3]$, we have $1 \leq f(x) \leq 9$, so $|f(x) - 4| \leq 5$.
- If we restrict $f(x) := x^2$ to $X = [1.9, 2.1]$, then f is 0.41-close to 4 since for all $x \in [1.9, 2.1]$, we have $3.61 \leq f(x) \leq 4.41$, so $|f(x) - 4| \leq 0.41$.

Definition 3.: Epsilon-Closeness Near a Point

Let $X \subseteq \mathbb{R}$, $f : X \to \mathbb{R}$, $L \in \mathbb{R}$, and x_0 be an adherent point of X. We say that f **is ε-close to L near** x_0 iff

$$\exists \delta > 0 \text{ such that } \forall x \in X,\ 0 < |x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon.$$

Example 3.: Examples

- Let $f(x) := x^2$, restricted to $[1, 3]$. Then f is 0.1-close to 4 near $x_0 = 2$, since for $x \in (1.99, 2.01)$, we have $|f(x) - 4| < 0.1$.
- The same function $f(x) = x^2$ is 0.1-close to 9 near $x_0 = 3$, since on the interval $(2.99, 3]$, we have $8.9401 < f(x) \leq 9$, so $|f(x) - 9| < 0.1$.
- Let $f(x) := \frac{\sin x}{x}$ for $x \in (0, 1]$. Then f is ε-close to 1 near $x_0 = 0$, since $\lim_{x \to 0^+} \frac{\sin x}{x} = 1$.

Non-Example 3.: Non-Examples

Non-Examples

- Define $f(x) := \begin{cases} 1 & x \in \mathbb{Q}, \\ 0 & x \notin \mathbb{Q}. \end{cases}$ Then f is not ε-close to any real number near any x_0, because in any interval around x_0, f takes both 0 and 1, so it fails all bounds.
- Let $f(x) := \sin\left(\frac{1}{x}\right)$ on $(0, 1]$. Then f is not ε-close to any real number near 0, since it oscillates between -1 and 1 infinitely often as $x \to 0$, with no consistent limiting behavior.

3.1.5 Continuity of a Function

We now introduce one of the most fundamental concepts in the theory of real functions: continuity.

Definition 3.: Continuity at a Point

Let $X \subseteq \mathbb{R}$, and let $f : X \to \mathbb{R}$ be a function. Let $x_0 \in X$. We say that f is **continuous at** x_0 if

$$\lim_{x \to x_0;\, x \in X} f(x) = f(x_0).$$

In other words, the limit of $f(x)$ as $x \to x_0$ (within the domain X) exists and equals the function's value at that point.

Definition 3.: Continuity on a Set

A function $f : X \to \mathbb{R}$ is said to be **continuous on** X (or simply *continuous*) if it is continuous at every point $x_0 \in X$.
If f is not continuous at $x_0 \in X$, then we say that f is **discontinuous at** x_0.

3.1.6 Examples and Non-Examples of Continuous Functions

Example 3.: Constant Function

Let $c \in \mathbb{R}$, and define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) := c$. Then for every $x_0 \in \mathbb{R}$, we have

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0} c = c = f(x_0),$$

so f is continuous at every point $x_0 \in \mathbb{R}$. Thus, f is continuous on $\mathbb{R}$.

Example 3.: Identity Function

Let $f : \mathbb{R} \to \mathbb{R}$ be the identity function $f(x) := x$. Then for every $x_0 \in \mathbb{R}$, we have

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0} x = x_0 = f(x_0),$$

hence f is continuous on all of $\mathbb{R}$.

Example 3.: Sign Function (Discontinuous at 0)

Let $\operatorname{sgn} : \mathbb{R} \to \mathbb{R}$ be defined by

$$\operatorname{sgn}(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ -1 & \text{if } x < 0. \end{cases}$$

Then $\operatorname{sgn}(x)$ is continuous at every point $x \neq 0$. For example, at $x = 1$,

$$\lim_{x \to 1} \operatorname{sgn}(x) = \lim_{x \to 1} 1 = 1 = \operatorname{sgn}(1).$$

However, $\operatorname{sgn}(x)$ is *not* continuous at $x = 0$, since the left-hand and right-hand limits disagree:

$$\lim_{x \to 0^-} \operatorname{sgn}(x) = -1, \quad \lim_{x \to 0^+} \operatorname{sgn}(x) = 1.$$

Example 3.: Dirichlet Function (Nowhere Continuous)

Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) := \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Then f is discontinuous at *every* point $x_0 \in \mathbb{R}$. Intuitively, in every neighborhood of x_0, no matter how small, there are both rational and irrational numbers and the function jumps from 0 to 1 with no approach.

Example 3.: Step Function

Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) := \begin{cases} 1 & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

This function is discontinuous at $x = 0$, because

$$\lim_{x \to 0^-} f(x) = 0 \neq 1 = \lim_{x \to 0^+} f(x).$$

However, f is continuous at every $x \neq 0$. Moreover, if we restrict the domain to $[0, \infty)$, then the function becomes continuous on this restricted domain.

3.1.7 Properties of Continuous Functions

We now explore key properties of continuous functions, focusing on how their behavior is controlled when defined on closed intervals.

Proposition 3.: Composition of Continuous Functions

Let $f : X \to \mathbb{R}$ and $g : Y \to \mathbb{R}$, where $f(X) \subseteq Y$. If f is continuous at $x_0 \in X$, and g is continuous at $f(x_0)$, then the composition $g \circ f$ is continuous at x_0.

Theorem 3.: Boundedness Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous. Then $f([a, b])$ is bounded in $\mathbb{R}$; that is, there exist $m, M \in \mathbb{R}$ such that

$$m \leq f(x) \leq M \quad \text{for all } x \in [a, b].$$

Theorem 3.: Extreme Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous. Then there exist $x_0, x_1 \in [a, b]$ such that

$$f(x_0) \leq f(x) \leq f(x_1) \quad \text{for all } x \in [a, b].$$

That is, f achieves its absolute minimum and maximum values on $[a, b]$.

Theorem 3.: Intermediate Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous. Suppose $f(a) = \alpha$, $f(b) = \beta$, and let γ be any real number between α and β (i.e., $\min(\alpha, \beta) \leq \gamma \leq \max(\alpha, \beta)$). Then there exists $c \in [a, b]$ such that

$$f(c) = \gamma.$$

Definition 3.: Uniform Continuity

Let $f : X \to \mathbb{R}$. We say f is **uniformly continuous** on X if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in X$,

$$|x - y| < \delta \quad \Rightarrow \quad |f(x) - f(y)| < \varepsilon.$$

This is stronger than pointwise continuity, as δ depends only on ε, not on the point $x \in X$.

Example 3.: Examples of Uniform Continuity

- The function $f(x) = x^2$ is **not** uniformly continuous on $\mathbb{R}$, but it is uniformly continuous on any closed interval $[a, b]$.
- The function $f(x) = \sin(x)$ is uniformly continuous on all of $\mathbb{R}$.
- The function $f(x) = \frac{1}{x}$ is not uniformly continuous on $(0, 1)$, because it blows up near 0.

Theorem 3.: Fundamental Theorem of Continuous Functions

Every continuous function $f : [a, b] \to \mathbb{R}$ satisfies:

- f is bounded (Boundedness Theorem),
- f attains its minimum and maximum values (Extreme Value Theorem),
- f attains all intermediate values (Intermediate Value Theorem),
- f is uniformly continuous.

How Discontinuous Functions Fail Fundamental Theorems

We now illustrate how the key theorems about continuous functions can fail if the function is not continuous.

Example 3.: Failure of Boundedness Theorem

Theorem: A continuous function on a closed interval is bounded.
Counterexample: Let

$$f(x) := \frac{1}{x}, \quad \text{defined on } (0, 1].$$

This function is not defined at $x = 0$ and diverges as $x \to 0^+$. If we try to extend it to $[0, 1]$, it becomes discontinuous at $x = 0$ and is unbounded on that interval. Thus, the function is not continuous on $[0, 1]$ and does not satisfy the conclusion of the boundedness theorem.

Example 3.: Failure of Extreme Value Theorem

Theorem: A continuous function on a closed interval attains its minimum and maximum values.
Counterexample: Define

$$f(x) := \begin{cases} x & \text{if } 0 < x < 1, \\ 2 & \text{if } x = 1. \end{cases}$$

This function is not continuous at $x = 1$. While f is bounded on $(0, 1]$, it does not attain its supremum (which is 1) within the open interval $(0, 1)$. Therefore, the extreme value theorem fails.

Example 3.: Failure of Intermediate Value Theorem

Theorem: A continuous function on an interval takes all intermediate values.
Counterexample: Let

$$f(x) := \begin{cases} 1 & \text{if } x < 0, \\ 3 & \text{if } x \geq 0. \end{cases}$$

This function has a jump discontinuity at $x = 0$. For example, $f(-1) = 1$ and $f(1) = 3$, but there is no x such that $f(x) = 2$. Hence, the intermediate value theorem fails.

Example 3.: Failure of Uniform Continuity

Theorem: A continuous function on a closed and bounded interval is uniformly continuous.
Counterexample: Let

$$f(x) := \frac{1}{x}, \quad \text{defined on } (0, 1).$$

This function is continuous on $(0, 1)$ but not uniformly continuous: as $x \to 0^+$, the function becomes arbitrarily steep. For small differences in x, the corresponding values of $f(x)$ can differ greatly. Thus, uniform continuity fails even though the function is continuous.

3.2 Metric Spaces and Topology

We now generalize the notions of convergence and continuity from the real numbers to more abstract spaces.

Review: Sequences and Continuity in $\mathbb{R}$

A sequence (x_n) of real numbers converges to $x \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have

$$|x_n - x| < \varepsilon.$$

A function $f : \mathbb{R} \to \mathbb{R}$ is continuous at a point $x_0 \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon.$$

Definition: Metric Space

A **metric space** is a pair (X, d), where X is a set and $d : X \times X \to \mathbb{R}$ is a function (called a **metric** or **distance function**) satisfying for all $x, y, z \in X$:

(i) $d(x, y) \geq 0$, and $d(x, y) = 0 \iff x = y$ (Non-negativity and identity)

(ii) $d(x, y) = d(y, x)$ (Symmetry)

(iii) $d(x, z) \leq d(x, y) + d(y, z)$ (Triangle inequality)

Examples of Metric Spaces

- $(\mathbb{R}, d(x,y) := |x-y|)$
- $(\mathbb{R}^n, d(x,y) := \sqrt{(x_1-y_1)^2 + \cdots + (x_n-y_n)^2})$ Euclidean space
- $(\mathbb{R}^n, d(x,y) := \max\{|x_i - y_i|\})$ Maximum or sup metric
- $(C([a,b]), d(f,g) := \sup_{x\in[a,b]} |f(x)-g(x)|)$ Space of continuous functions on an interval

Convergence of Sequences in Metric Spaces

Let (X,d) be a metric space. A sequence (x_n) in X **converges** to a point $x \in X$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$d(x_n, x) < \varepsilon.$$

Continuity in Metric Spaces

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. We say that f is **continuous at a point** $x_0 \in X$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \in X$,

$$d_X(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon.$$

Sequential characterization: The function f is continuous at $x_0 \in X$ if whenever $x_n \to x_0$ in X, we have $f(x_n) \to f(x_0)$ in Y.

Theorem 11: Equivalent Definitions of Continuity

Let $f : M \to N$ be a function between metric spaces. The following are equivalent:

(i) (ε, δ)**-Condition:** For every $x \in M$ and every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$d_M(x,y) < \delta \quad \Rightarrow \quad d_N(f(x), f(y)) < \varepsilon.$$

(ii) **Sequential Continuity:** For every sequence (x_n) in M with $x_n \to x$, we have $f(x_n) \to f(x)$.

(iii) **Closed Set Condition:** For every closed set $C \subseteq N$, the preimage $f^{-1}(C) \subseteq M$ is closed.

(iv) **Open Set Condition:** For every open set $U \subseteq N$, the preimage $f^{-1}(U) \subseteq M$ is open.

Examples in $\mathbb{R}$

- **Example 1 (Identity Function):** Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x$. This function is continuous because it satisfies all four equivalent conditions.
- **Example 2 (Constant Function):** Let $f(x) = 3$. Then for any closed set $C \subseteq \mathbb{R}$, either $f^{-1}(C) = \mathbb{R}$ (if $3 \in C$) or $\emptyset$ (if $3 \notin C$). Both are closed, so f is continuous.
- **Example 3 (Step Function):** Define

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

  This function is not continuous at $x = 0$: it fails the (ε, δ)-condition, and the preimage of the open set $(0.5, 1.5)$ is not open in $\mathbb{R}$. Hence it fails all four conditions at that point.

3.3 Derivatives

3.3.1 Definition of the Derivative

Let $f : D \to \mathbb{R}$ be a function defined on a subset $D \subseteq \mathbb{R}$, and let $x_0 \in D$ be an interior point. We say that f is **differentiable at** x_0 if the following limit exists:

$$f'(x_0) := \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

In this case, $f'(x_0)$ is called the **derivative of** f **at** x_0.

Formal ε–δ Definition. We say that f is differentiable at x_0 with derivative L if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$0 < |x - x_0| < \delta \quad \Rightarrow \quad \left| \frac{f(x) - f(x_0)}{x - x_0} - L \right| < \varepsilon.$$

This is equivalent to the limit definition:

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = L.$$

Geometric Interpretation. The derivative $f'(x_0)$ represents the slope of the **tangent line** to the graph of f at the point $(x_0, f(x_0))$. It describes the instantaneous rate of change of the function at that point. If the graph is smooth at x_0, the tangent line just touches the curve there without crossing it.

Examples.

- **Constant Function.** Let $f(x) = c$ for all $x \in \mathbb{R}$. Then

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{c - c}{x - x_0} = 0 \quad \Rightarrow \quad f'(x_0) = 0.$$

  Every constant function is differentiable with derivative 0 everywhere.

- **Identity Function.** Let $f(x) = x$. Then

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{x - x_0}{x - x_0} = 1 \quad \Rightarrow \quad f'(x_0) = 1.$$

- **Quadratic Function.** Let $f(x) = x^2$. Then

$$\frac{f(x) - f(x_0)}{x - x_0} = \frac{x^2 - x_0^2}{x - x_0} = \frac{(x - x_0)(x + x_0)}{x - x_0} = x + x_0.$$

  Taking the limit as $x \to x_0$, we get $f'(x_0) = 2x_0$.

- **Non-Differentiable Function.** Let $f(x) = |x|$. Then:

$$\frac{f(x) - f(0)}{x - 0} = \frac{|x|}{x} = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}.$$

  This limit does not exist at $x = 0$, so f is not differentiable at 0.

3.3.2 Rules of Differentiation

Once we know that a function is differentiable, we can compute its derivative more easily using the following fundamental rules:

Proposition 3.: Basic Rules of Differentiation

Let f and g be differentiable functions on an open interval, and let $c \in \mathbb{R}$. Then:

(a) **Linearity Rules**

- $(f+g)' = f' + g'$
- $(cf)' = cf'$

(b) **Product Rule**

$$(fg)' = f'g + fg'$$

(c) **Quotient Rule** (provided $g(x) \neq 0$)

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

(d) **Chain Rule**

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

Examples.

- If $f(x) = x^3 + 2x$, then $f'(x) = 3x^2 + 2$. (Linearity Rule)
- If $f(x) = x \cdot \sin(x)$, then

$$f'(x) = 1 \cdot \sin(x) + x \cdot \cos(x) = \sin(x) + x\cos(x).$$

(Product Rule)

- If $f(x) = \frac{\ln x}{x}$, then

$$f'(x) = \frac{(1/x) \cdot x - \ln x \cdot 1}{x^2} = \frac{1 - \ln x}{x^2}.$$

(Quotient Rule)

- If $f(x) = \cos(x^2)$, then

$$f'(x) = -\sin(x^2) \cdot 2x = -2x\sin(x^2).$$

(Chain Rule)

3.3.3 Properties of the Derivative

Differentiability Implies Continuity. If a function f is differentiable at a point x_0, then it is continuous at x_0.

Proof. Assume f is differentiable at x_0, then

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0).$$

Since this limit exists and is finite, it follows that

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Hence, f is continuous at x_0. $\square$

Mean Value Theorem (MVT). If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Corollary: Global Lipschitz Condition. If f is differentiable on (a, b) and $|f'(x)| \leq M$ for all $x \in (a, b)$, then for all $t, x \in (a, b)$,

$$|f(t) - f(x)| \leq M|t - x|.$$

In particular, if $f'(x) = 0$ for all $x \in (a, b)$, then $f(x)$ is constant on (a, b).

Ratio Mean Value Theorem. Suppose f and g are continuous on $[a, b]$ and differentiable on (a, b). Then there exists $\theta \in (a, b)$ such that

$$\Delta f \cdot g'(\theta) = \Delta g \cdot f'(\theta),$$

or equivalently,

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\theta)}{g'(\theta)}.$$

L'Hôpital's Rule. Let f and g be differentiable on (a, b), with $f(x) \to 0$ and $g(x) \to 0$ as $x \to b$, and assume $g(x), g'(x) \neq 0$. If

$$\lim_{x \to b} \frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x \to b} \frac{f(x)}{g(x)} = L.$$

Intermediate Value Property of Derivatives. If f is differentiable on (a, b), then f' has the intermediate value property: if $f'(a) < \lambda < f'(b)$, then there exists $c \in (a, b)$ such that $f'(c) = \lambda$.

Corollary: Derivatives Have No Jump Discontinuities. The derivative of a differentiable function cannot have a jump discontinuity. That is, if f is differentiable, then f' is not allowed to jump at any point in its domain.

Pathological Example. Consider the function

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & \text{if } x > 0, \\ 0, & \text{if } x \leq 0. \end{cases}$$

This function is differentiable everywhere, including at $x = 0$, where $f'(0) = 0$. But for $x > 0$, its derivative is

$$f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x},$$

which oscillates increasingly rapidly as $x \to 0^+$. Thus, f' is discontinuous at $x = 0$. However, this is not a jump discontinuity; the discontinuity is due to wild oscillation. The function f remains smooth.

This example illustrates that while the derivative of a differentiable function can never jump, it may still fail to be continuous.

example-f-prime-discontinuity.png

3.3.4 Higher Derivatives and Smoothness Classes

The derivative of f', if it exists, is called the **second derivative** of f:

$$f''(x) = (f')'(x) = \lim_{t \to x} \frac{f'(t) - f'(x)}{t - x}.$$

Definition (Higher Derivatives). Higher-order derivatives are defined inductively:

$$f^{(r)} := \left(f^{(r-1)}\right)', \quad \text{with } f^{(0)}(x) := f(x).$$

If $f^{(r)}(x)$ exists, we say that f is **r-th order differentiable** at x. If $f^{(r)}(x)$ exists for all $x \in (a, b)$, then f is r-times differentiable on (a, b). If all derivatives $f^{(r)}(x)$ exist for all $r \in \mathbb{N}$ and $x \in (a, b)$, then f is said to be **smooth** or **infinitely differentiable**.

Theorem 3.: Continuity of Lower Derivatives

If f is r-times differentiable for $r \geq 1$, then $f^{(r-1)}$ is continuous on (a, b).

Smoothness Classes C^r

Definition. Let $r \in \mathbb{N} \cup \{0\}$. A function f is said to be of class C^r on an interval (a, b) if: - All derivatives up to order r exist on (a, b), and - $f^{(r)}$ is continuous on (a, b).

We define: - C^0: The class of continuous functions, - C^1: The class of functions with continuous first derivatives, - C^2: Continuous second derivatives, - $\cdots$, - $C^\infty = \bigcap_{r \in \mathbb{N}} C^r$: The class of smooth functions.

Hierarchy of Smoothness.

$$C^0 \supset C^1 \supset C^2 \supset \cdots \supset C^\infty$$

Each inclusion is proper: there are functions in C^r that are not in C^{r+1}.

Examples

- $f(x) = |x|$: This function is continuous but not differentiable at $x = 0$, hence $f \in C^0 \setminus C^1$.
- $f(x) = x|x|$: Differentiable with continuous first derivative, but the second derivative is not continuous at $x = 0$, so $f \in C^1 \setminus C^2$.
- $f(x) = |x|^3$: Twice differentiable with continuous second derivative, but the third derivative is not continuous at $x = 0$, so $f \in C^2 \setminus C^3$.

3.3.5 Analytic Functions and Taylor Approximation

Definition (Analytic Function). Let $f : \mathbb{R} \to \mathbb{R}$ be a function. We say that f is **analytic at a point** $x_0 \in \mathbb{R}$ if there exists a power series

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n$$

which converges to $f(x)$ for all x in some neighborhood of x_0, and such that

$$a_n = \frac{f^{(n)}(x_0)}{n!}.$$

That is, f is analytic at x_0 if it equals its Taylor series in a neighborhood of x_0.

If f is analytic at every point of an open interval $I \subset \mathbb{R}$, we say that f is **real analytic on** I.

Example (Smooth but Not Analytic). Let

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0 \\ 0, & x = 0. \end{cases}$$

This function is **infinitely differentiable** (i.e., smooth) on $\mathbb{R}$, and all derivatives vanish at $x = 0$: $f^{(n)}(0) = 0$ for all $n \geq 0$. Hence, the Taylor series of f at 0 is identically 0, but $f(x) \neq 0$ for $x \neq 0$. So $f \notin C^\omega(\mathbb{R})$, the class of analytic functions.

Taylor Approximation

Theorem (Taylor's Theorem). Let f be a function with $(r+1)$-continuous derivatives on an interval containing x_0 and x. Then

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{f^{(r)}(x_0)}{r!}(x - x_0)^r + R_{r+1}(x),$$

where the remainder term is given by

$$R_{r+1}(x) = \frac{f^{(r+1)}(\theta)}{(r+1)!}(x - x_0)^{r+1}$$

for some θ between x_0 and x.

Examples.

- $f(x) = e^x$, about $x_0 = 0$:
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$
- $f(x) = \sin x$, about $x_0 = 0$:
$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$
- $f(x) = \ln(1 + x)$, about $x_0 = 0$, for $|x| < 1$:
$$\ln(1 + x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$$

Inverse Function Theorem in Dimension One

Theorem (Inverse Function Theorem in $\mathbb{R}$). Let $f : \mathbb{R} \to \mathbb{R}$ be a continuously differentiable function. Suppose $f'(x_0) \neq 0$. Then there exists an open interval I containing x_0 and an open interval J containing $f(x_0)$ such that:

- f is a bijection from I to J,
- The inverse function $f^{-1} : J \to I$ exists and is continuously differentiable,
- The derivative of the inverse is given by

$$(f^{-1})'(f(x_0)) = \frac{1}{f'(x_0)}.$$

3.4 Integration

3.4.1 Definitions and Techniques

We will define the Riemann integral and discuss criteria for integrability.

4 Topology: Definition and Construction

4.1 Motivation and Origins

Motivation. As we saw in the previous sections, **continuous functions** are among the most well-behaved functions in analysis. They satisfy key theorems such as:

- **Intermediate Value Theorem (IVT)**;
- **Extreme Value Theorem**;
- **Uniform Continuity and Compactness**.

Continuity is necessary for differentiability and integrability. To generalize these ideas, we need a setting that captures convergence and continuity without relying on a metric — this leads us to **topology**.

Hierarchy of Convergence and Continuity. We explain how these concepts evolve:

1. In $\mathbb{R}$: via ε-δ definitions.
2. In metric spaces: via general metrics.
3. Using open balls: $B_r(x) = \{y \mid d(x, y) < r\}$.

From Open Balls to Open Sets. Open balls lead us to a more abstract concept:

- $U \subset M$ is open if $\forall x \in U, \exists r > 0$ such that $B_r(x) \subset U$.
- The collection of all such U's forms a topology.

This leads naturally to the idea of a **topological space**, abstracting all distance-based structure into a framework based on open sets.

4.2 Continuity in Topological Spaces

Continuity in Metric Spaces via Open Balls

Let $f : X \to Y$ be a function between metric spaces. Then f is **continuous at a point** $x_0 \in X$ if:

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } x \in B_X(x_0, \delta) \Rightarrow f(x) \in B_Y(f(x_0), \epsilon)$$

This means that the image of every open ball around x_0 is contained in some open ball around $f(x_0)$.

Open Sets from Open Balls

Definition 4.: Open Sets in Metric Spaces

Let (M, d) be a metric space. A subset $U \subseteq M$ is called **open** if for every point $p \in U$, there exists an $r > 0$ such that the open ball

$$B_r(p) := \{q \in M : d(p, q) < r\}$$

is contained in U, i.e., $B_r(p) \subseteq U$.

Theorem 4.: Properties of Open Sets in Metric Spaces

Let (M, d) be a metric space. Then:

1. The empty set $\emptyset$ and the entire space M are open.
2. Arbitrary unions of open sets are open.
3. Finite intersections of open sets are open.
4. Every open ball $B_r(p)$ is an open set.

Reformulating Continuity Using Open Sets

Definition 4.: Topological Continuity

Let $f : X \to Y$ be a function between topological spaces. We say f is **continuous** if for every open set $V \subseteq Y$, the preimage $f^{-1}(V) \subseteq X$ is open.

Forward Direction. Assume f is continuous in the metric sense. Let $V \subseteq Y$ be open and $x \in f^{-1}(V)$. Since V is open, there exists $\epsilon > 0$ such that:

$$B_Y(f(x), \epsilon) \subseteq V.$$

By continuity, there is $\delta > 0$ such that:

$$f(B_X(x,\delta)) \subseteq B_Y(f(x),\epsilon) \Rightarrow B_X(x,\delta) \subseteq f^{-1}(V).$$

So $f^{-1}(V)$ is open.

Backward Direction. Assume that for every open $V \subseteq Y$, the set $f^{-1}(V) \subseteq X$ is open. Fix $x_0 \in X$, and let $\epsilon > 0$. The ball $B_Y(f(x_0),\epsilon)$ is open, so $f^{-1}(B_Y(f(x_0),\epsilon))$ is open and contains x_0. Thus, there exists $\delta > 0$ such that:

$$B_X(x_0,\delta) \subseteq f^{-1}(B_Y(f(x_0),\epsilon)).$$

This is the ε–δ condition.

Topological Continuity without Metrics

Definition 4.: Topology

A **topology** on a set X is a collection $\mathcal{T}$ of subsets of X (called *open sets*) satisfying:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.
2. If $\{U_\alpha\}_{\alpha\in A}$ is a family of sets in $\mathcal{T}$, then $\bigcup_{\alpha\in A} U_\alpha \in \mathcal{T}$.
3. If $U_1,\ldots,U_n \in \mathcal{T}$, then $\bigcap_{i=1}^n U_i \in \mathcal{T}$.

The pair $(X,\mathcal{T})$ is called a **topological space**.

Remark 4.

This generalizes the notion of open sets from metric spaces. Every metric space defines a topology via its open balls.

Summary Table.

Context	Definition of Continuity
Metric spaces	ε–δ using open balls
Topology (from metric)	Preimages of open sets are open
General topology	Same preimage condition (no distances)

4.3 Explicit Examples of Topologies

Topologies on a Finite Set

Example 4.: Three Topologies on $\{1, 2, 3\}$

Let $X = \{1, 2, 3\}$. We present three distinct topologies on X:

(1) Indiscrete Topology:

$$\mathcal{T}_1 = \{\emptyset, \{1, 2, 3\}\}$$

Only the empty set and the whole space are open.

(2) A Non-Trivial Topology:

$$\mathcal{T}_2 = \{\emptyset, \{1\}, \{1, 2, 3\}\}$$

(3) Discrete Topology:

$$\mathcal{T}_3 = \mathcal{P}(X) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Topologies on $\mathbb{R}$

Example 4.: Standard Topology

The basis consists of all open intervals $(a, b) \subset \mathbb{R}$, which generate the standard topology.

Example 4.: Lower Limit Topology

The basis is formed by all intervals of the form $[a, b)$, which produce a strictly finer topology than the standard one.

Topology on the Rational Numbers $\mathbb{Q}$

Example 4.: The p-adic Topology on $\mathbb{Q}$

Fix a prime number p. Define a metric:

$$d_p(x, y) = |x - y|_p$$

where the p-adic absolute value is:

$$|x|_p = \begin{cases} p^{-v_p(x)} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

This gives a topology where open sets are defined by p-adic balls:

$$B_r(x) = \{y \in \mathbb{Q} \mid |x - y|_p < r\}.$$

4.4 Abstract Topologies

Example 4.: Discrete Topology

Let X be any set. The *discrete topology* on X is:

$$\mathcal{T}_{\text{disc}} = \{U \subseteq X\}$$

Every subset is open. This is the finest topology on X.

Example 4.: Indiscrete Topology

Let X be any set. The *indiscrete topology* on X is:

$$\mathcal{T}_{\text{indisc}} = \{\emptyset, X\}$$

Only the empty set and the whole space are open. This is the coarsest topology on X.

Example 4.: Cofinite Topology

Let X be any set. The *cofinite topology* on X is:

$$\mathcal{T}_{\text{cof}} = \{U \subseteq X \mid X \setminus U \text{ is finite}\} \cup \{\emptyset\}$$

A subset is open if its complement is finite or if it is empty.

Example 4.: Cocountable Topology

Let X be any set. The *cocountable topology* on X is:

$$\mathcal{T}_{\text{coc}} = \{U \subseteq X \mid X \setminus U \text{ is countable}\} \cup \{\emptyset\}$$

A subset is open if its complement is countable or if it is empty.

Example 4.: Zariski Topology on $\mathbb{R}$

Let $X = \mathbb{R}$. The *Zariski topology* is defined by declaring the closed sets to be finite subsets of $\mathbb{R}$ and $\mathbb{R}$ itself. Hence, the open sets are:

$$\mathcal{T}_{\text{Zar}} = \{\mathbb{R} \setminus Z \mid Z \subseteq \mathbb{R} \text{ is finite}\}.$$

4.5 Non-Examples of Topologies

Non-Example 4.: Fails Union Condition

Let $X = \{1, 2, 3\}$, and define $\mathcal{T} = \{\emptyset, \{1\}, \{2\}, \{1, 2, 3\}\}$. This is **not** a topology because it is not closed under unions:

$$\{1\} \cup \{2\} = \{1, 2\} \notin \mathcal{T}.$$

Non-Example 4.: Fails Intersection Condition

Let $X = \{1, 2, 3\}$, and define $\mathcal{T} = \{\emptyset, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\}$. This is **not** a topology because it is not closed under finite intersections:

$$\{1, 2\} \cap \{2, 3\} = \{2\} \notin \mathcal{T}.$$

Non-Example 4.: Fails Identity Conditions

Let $X = \{1, 2, 3\}$, and define $\mathcal{T} = \{\{1\}, \{2\}, \{3\}\}$. This is **not** a topology because:

$$\emptyset \notin \mathcal{T}, \quad X \notin \mathcal{T}.$$

Both must be present in any valid topology.

4.6 Bases for Topologies

Definition 4.: Basis

A **basis** $\mathcal{B}$ for a topology on a set X is a collection of subsets of X (called *basis elements*) such that:

1. For every $x \in X$, there exists at least one $B \in \mathcal{B}$ with $x \in B$.
2. If $x \in B_1 \cap B_2$ for some $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Theorem 4.: Characterization of Open Sets from a Basis

Let $\mathcal{B}$ be a basis for a topology on a set X. A subset $U \subseteq X$ is open (i.e., belongs to the topology generated by $\mathcal{B}$) if and only if for every $x \in U$, there exists $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Definition 4.: Topology Generated by a Basis

The **topology generated by a basis** $\mathcal{B}$ is the collection:

$$\mathcal{T} = \{U \subseteq X \mid \forall x \in U, \exists B \in \mathcal{B} \text{ with } x \in B \subseteq U\}.$$

That is, every open set is a union of basis elements.

Examples and Non-Examples of Bases

Example 4.: Standard Basis on $\mathbb{R}$

Let $\mathcal{B} = \{(a, b) \mid a < b,\ a, b \in \mathbb{R}\}$. Then $\mathcal{B}$ forms a basis for the *standard topology* on $\mathbb{R}$.

Example 4.: Lower Limit Topology on $\mathbb{R}$

Let $\mathcal{B} = \{[a, b) \mid a < b\}$. This forms a basis for the lower limit topology on $\mathbb{R}$, which is strictly finer than the standard topology.

Example 4.: Finite Complement Basis on $\mathbb{R}$

Let $\mathcal{B} = \{\mathbb{R} \setminus F \mid F \subset \mathbb{R} \text{ finite}\}$. The topology generated by this basis is the *cofinite topology*.

Non-Example 4.: Invalid Basis

Let $\mathcal{B} = \{(0, 1), (1, 2)\} \subset \mathbb{R}$. This is **not** a basis for a topology on $\mathbb{R}$, because not every point of $\mathbb{R}$ is contained in one of these sets.

4.7 Continuous Functions

Definition 4.: Continuous Function

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. A function $f : X \to Y$ is said to be **continuous** if for every open set $V \in \mathcal{T}_Y$, the preimage $f^{-1}(V) \in \mathcal{T}_X$; that is,

$$\forall V \in \mathcal{T}_Y, \quad f^{-1}(V) \in \mathcal{T}_X.$$

Examples and Non-Examples

Example 4.: Standard Continuous Function

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$. This function is continuous with respect to the standard topology since the preimage of any open interval is a union of open intervals.

Example 4.: Identity Function

Let X be any topological space. The identity map $\mathrm{id}_X : X \to X$ is continuous because the preimage of an open set is itself.

Non-Example 4.: Non-Continuous Function

Let $f : \mathbb{R}_{\text{std}} \to \mathbb{R}_{\text{disc}}$ be the identity map. Then f is **not** continuous, since the preimage of a singleton (which is open in the discrete topology) is not open in the standard topology.

Remark 4.

This definition generalizes the ε-δ definition from calculus and the open-ball version from metric spaces. In general topology, continuity is expressed entirely in terms of open sets.

4.8 Properties of Continuous Functions

Proposition 4.: Composition of Continuous Functions

Let $f : X \to Y$ and $g : Y \to Z$ be continuous maps between topological spaces. Then the composition $g \circ f : X \to Z$ is also continuous.

Proof. Let $U \subseteq Z$ be open. Since g is continuous, $g^{-1}(U)$ is open in Y. Since f is continuous, $f^{-1}(g^{-1}(U)) = (g \circ f)^{-1}(U)$ is open in X. $\square$

Example 4.: Composition

Let $f(x) = \sin x$ and $g(x) = x^2$. Then $g \circ f(x) = \sin^2 x$ is continuous since both f and g are continuous.

Proposition 4.: Restriction of a Continuous Function

Let $f : X \to Y$ be continuous and let $A \subseteq X$. Then the restriction $f|_A : A \to Y$ is continuous with respect to the subspace topology on A.

Proof. Let $V \subseteq Y$ be open. Then $f^{-1}(V)$ is open in X, so $f^{-1}(V) \cap A = (f|_A)^{-1}(V)$ is open in the subspace topology on A. □

Proposition 4.: Gluing Lemma

Let $X = U \cup V$, where U and V are open subsets of X. Suppose $f_U : U \to Y$ and $f_V : V \to Y$ are continuous functions such that $f_U|_{U\cap V} = f_V|_{U\cap V}$. Then the glued function

$$f(x) = \begin{cases} f_U(x) & \text{if } x \in U \\ f_V(x) & \text{if } x \in V \end{cases}$$

defines a continuous function $f : X \to Y$.

Proof. Let $W \subseteq Y$ be open. Then $f^{-1}(W) = f_U^{-1}(W) \cup f_V^{-1}(W)$, which is open in X because U and V are open. □

Remark 4.

These properties make the category of topological spaces and continuous maps well-behaved. Continuity is preserved under natural operations like composition, restriction, and gluing.

Proposition 4.: Identity and Inclusions are Continuous

Let X be a topological space.

- The identity map $\mathrm{id}_X : X \to X$ is continuous.
- If $U \subseteq X$ is open, the inclusion map $\iota : U \hookrightarrow X$ is continuous.

Proof. **(Identity)** Let $V \subseteq X$ be open. Then $\mathrm{id}_X^{-1}(V) = V$, which is open.

(Inclusion) Let $V \subseteq X$ be open. Then $\iota^{-1}(V) = U \cap V$, which is open in the subspace topology. □

Proposition 4.: Locality of Continuity

Let $f : X \to Y$ be a map between topological spaces. Suppose that for every point $x \in X$, there exists an open neighborhood $U_x \subseteq X$ containing x such that the restriction

$$f|_{U_x} : U_x \to Y$$

is continuous. Then f is continuous on all of X.

Proof. Let $V \subseteq Y$ be open. Then for each $x \in X$, $f^{-1}(V) \cap U_x$ is open in U_x, and hence in X. Since $X = \bigcup_x U_x$, we have:

$$f^{-1}(V) = \bigcup_x (f^{-1}(V) \cap U_x),$$

which is open. □

4.9 Examples of Continuous Maps

(1) Addition. The map

$$\mathrm{add}\colon \mathbb{R}^2 \to \mathbb{R}, \quad (x, y) \mapsto x + y$$

is continuous.

(2) Multiplication. The map

$$\mathrm{mult}\colon \mathbb{R}^2 \to \mathbb{R}, \quad (x, y) \mapsto xy$$

is continuous.

(3) Projections. Each coordinate projection

$$\mathrm{pr}_i\colon \mathbb{R}^m \to \mathbb{R}, \quad (x_1, \ldots, x_m) \mapsto x_i$$

is continuous.

(4) Products. Let $F\colon \mathbb{R}^a \to \mathbb{R}^{b+c} = \mathbb{R}^b \times \mathbb{R}^c$. If both $\mathrm{pr}_1 \circ F$ and $\mathrm{pr}_2 \circ F$ are continuous, then F is continuous.

(5) Metric Function. For a metric space (X, d), the distance function

$$d\colon X \times X \to \mathbb{R}, \quad (x, y) \mapsto d(x, y)$$

is continuous with respect to the product topology on $X \times X$.

Proposition 4.: Corollary

Let $m > 0$. Then any polynomial function

$$p\colon \mathbb{R}^m \to \mathbb{R}$$

is continuous.

Proposition 4.: Corollary

Let $m, n > 0$. Let

$$P\colon \mathbb{R}^m \to \mathbb{R}^n, \quad x \mapsto (p_1(x), \ldots, p_n(x)),$$

where each p_j is a polynomial. Then P is continuous.

Proposition 4.: Corollary

The following standard functions are continuous:

- $\cos(x) : \mathbb{R} \to \mathbb{R}$
- $\sin(x) : \mathbb{R} \to \mathbb{R}$
- $e^x : \mathbb{R} \to \mathbb{R}$
- $\log(x) : (0, \infty) \to \mathbb{R}$
- $\tan(x) : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$
- $\sqrt[p]{\cdot} : (0, \infty) \to \mathbb{R}$

- **Diagonal map:**
$$\mathrm{diag}\colon \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^n, \quad x \mapsto (x, x)$$

- **Squaring map:**
$$\mathrm{sq}\colon \mathbb{R} \to \mathbb{R}, \quad x \mapsto x^2$$
This is the composition of the diagonal and multiplication maps:
$$x \xrightarrow{\text{diag}} (x, x) \xrightarrow{\text{mult}} x^2.$$

- **Mixed coordinate map:**
$$f\colon \mathbb{R}^2 \to \mathbb{R}^2, \quad (x, y) \mapsto (xy, y + 2)$$
The first coordinate is a product, and the second is a composition involving projections and addition.

Remark 4.

The class of continuous maps is broad. Many functions in analysis and geometry are built from continuous pieces using operations that preserve continuity.

5 Topology: Constructing new spaces from old ones

5.1 Comparing Topologies and Effects on Continuity

Definition 5.: Finer and Coarser Topologies

Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on the same set X. We say:

- $\mathcal{T}_1$ is **finer** than $\mathcal{T}_2$ if $\mathcal{T}_2 \subseteq \mathcal{T}_1$,
- $\mathcal{T}_1$ is **coarser** than $\mathcal{T}_2$ if $\mathcal{T}_1 \subseteq \mathcal{T}_2$.

If the inclusion is strict, we say "strictly finer" or "strictly coarser."

Proposition 5.: Effect on Continuity

Let $f\colon X \to Y$ be a function.

- Making the **domain topology finer** makes it *harder* for f to be continuous.
- Making the **codomain topology coarser** makes it *easier* for f to be continuous.

Formally:

1. If $\mathcal{T}'_X \supseteq \mathcal{T}_X$, and $f\colon (X, \mathcal{T}'_X) \to (Y, \mathcal{T}_Y)$ is continuous, then so is $f\colon (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$.
2. If $\mathcal{T}'_Y \subseteq \mathcal{T}_Y$, and $f\colon (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$ is continuous, then so is $f\colon (X, \mathcal{T}_X) \to (Y, \mathcal{T}'_Y)$.

Definition 5.: The Discrete and Indiscrete Topologies

Let X be any set.

- The **discrete topology** on X is the finest possible topology:

$$\mathcal{T}_{\text{disc}} = \mathcal{P}(X)$$

- The **indiscrete topology** on X is the coarsest possible topology:

$$\mathcal{T}_{\text{indisc}} = \{\emptyset, X\}$$

Proposition 5.: Universal Continuity for Discrete and Indiscrete Topologies

Let $f : X \to Y$ be any function.

1. If X has the discrete topology, then every function $f : X \to Y$ is continuous.
2. If Y has the indiscrete topology, then every function $f : X \to Y$ is continuous.

Proof. **(1)** Every subset of X is open, so any preimage $f^{-1}(U)$ is open.

(2) The only open sets in Y are $\emptyset$ and Y. Their preimages are always open in any topology on X. □

5.2 Subspace Topology

Motivation. Recall that a topological space is a set X equipped with a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ satisfying the open set axioms. Given a subset $A \subseteq X$, we can always construct a topology on A by selecting subsets of A. However, we want a topology on A that is *inherited* from X in a canonical way.

In particular, we want to define a topology on A such that the inclusion map

$$i : A \hookrightarrow X$$

is continuous. This choice ensures good behavior under composition: for instance, if $f : X \to Y$ is continuous and $A \subseteq X$, then $f \circ i : A \to Y$ is also continuous. To guarantee this, we define a topology on A that makes i continuous—and that is the *coarsest possible topology* on A with this property.

Definition 5.: Subspace Topology

Let $(X, \mathcal{T}_X)$ be a topological space, and let $A \subseteq X$. The **subspace topology** on A, denoted $\mathcal{T}_A$, is the coarsest topology on A such that the inclusion map $i : A \to X$ is continuous.
Explicitly, this topology is given by:

$$\mathcal{T}_A = \{i^{-1}(U) \mid U \in \mathcal{T}_X\} = \{U \cap A \mid U \in \mathcal{T}_X\}.$$

That is, a subset $V \subseteq A$ is open in the subspace topology if and only if there exists an open set $U \subseteq X$ such that $V = U \cap A$.

Proposition 5.: Subspace Topology is a Topology

Let $(X, \mathcal{T}_X)$ be a topological space, and let $A \subseteq X$. Then the collection

$$\mathcal{T}_A = \{U \cap A \mid U \in \mathcal{T}_X\}$$

defines a topology on A; that is, $\mathcal{T}_A$ satisfies the axioms of a topological space.

Proof. We must check that $\mathcal{T}_A$ satisfies the three topology axioms on the set A:

(1) $\emptyset \in \mathcal{T}_A$ and $A \in \mathcal{T}_A$: We have $\emptyset = \emptyset \cap A$, and $\emptyset \in \mathcal{T}_X$, so $\emptyset \in \mathcal{T}_A$. Also, $A = X \cap A$, and since $X \in \mathcal{T}_X$, we have $A \in \mathcal{T}_A$.

(2) Arbitrary unions: Let $\{V_\alpha\}_{\alpha \in \Lambda} \subseteq \mathcal{T}_A$. Then for each α, there exists $U_\alpha \in \mathcal{T}_X$ such that $V_\alpha = U_\alpha \cap A$. Then:

$$\bigcup_{\alpha \in \Lambda} V_\alpha = \bigcup_{\alpha \in \Lambda} (U_\alpha \cap A) = \left(\bigcup_{\alpha \in \Lambda} U_\alpha \right) \cap A,$$

and since $\bigcup_\alpha U_\alpha \in \mathcal{T}_X$, the union is in $\mathcal{T}_A$.

(3) Finite intersections: Let $V_1 = U_1 \cap A$, $V_2 = U_2 \cap A$ with $U_1, U_2 \in \mathcal{T}_X$. Then:

$$V_1 \cap V_2 = (U_1 \cap A) \cap (U_2 \cap A) = (U_1 \cap U_2) \cap A,$$

and since $U_1 \cap U_2 \in \mathcal{T}_X$, it follows that $V_1 \cap V_2 \in \mathcal{T}_A$.

Thus, $\mathcal{T}_A$ satisfies all the axioms of a topology on A. □

Examples and Non-Examples of Subspace Topology

Example 5.: Open Intervals as Subspaces of $\mathbb{R}$

Let $X = \mathbb{R}$ with the standard topology, and let $A = (1, 3) \subset \mathbb{R}$. Then A inherits the subspace topology from $\mathbb{R}$.
A set like $(1.5, 2.5) \subseteq A$ is open in A because it is the intersection of A with an open set in $\mathbb{R}$:

$$(1.5, 2.5) = (1.5, 2.5) \cap (1, 3).$$

More generally, any open interval fully contained in $(1, 3)$ is open in the subspace topology.

Example 5.: Closed Interval as a Subspace

Let $A = [0, 1] \subset \mathbb{R}$ with the subspace topology.
Important distinction: In $\mathbb{R}$, the interval $(0, 1)$ is open. But in the subspace $[0, 1]$, the set $(0, 1)$ is not open, because it is not the intersection of $[0, 1]$ with an open set of $\mathbb{R}$. However, $(0, 1) = [0, 1] \cap (0, 1)$, and since $(0, 1) \subset \mathbb{R}$ is open, this shows that $(0, 1) \subset [0, 1]$ is open in the subspace topology.
In contrast, the half-open interval $[0, 0.5)$ is open in the subspace topology since

$$[0, 0.5) = [0, 1] \cap (-0.5, 0.5),$$

but the singleton $\{0\}$ is *not* open in $[0, 1]$, even though it's a boundary point of an open interval in $\mathbb{R}$.

Example 5.: The Circle as a Subspace of $\mathbb{R}^2$

Let $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$. This defines the unit circle in the plane as a subspace of $\mathbb{R}^2$.
The topology on S^1 is inherited from the standard topology on $\mathbb{R}^2$. That is, a set $U \subseteq S^1$ is open if there exists an open set $V \subseteq \mathbb{R}^2$ such that

$$U = S^1 \cap V.$$

This allows us to study S^1 locally using the open neighborhoods of $\mathbb{R}^2$, while respecting the circular geometry of S^1.

Example 5.: Subspace Definition of the Torus

1. Torus as a Subspace of $\mathbb{R}^3$

The torus T^2 can be defined as a subset of $\mathbb{R}^3$ via the parametric equations:

$$T^2 = \{(x, y, z) \in \mathbb{R}^3 : x = (R + r\cos\theta)\cos\phi,\ y = (R + r\cos\theta)\sin\phi,\ z = r\sin\theta\},$$

where:

- $R > r > 0$: R is the distance from the center of the torus to the center of the tube, and r is the radius of the tube.
- $\theta \in [0, 2\pi)$: Angular coordinate around the tube.
- $\phi \in [0, 2\pi)$: Angular coordinate around the central axis of the torus.

The torus T^2 inherits its topology as a subspace of $\mathbb{R}^3$.
Comment: While this gives a concrete realization of the torus, it is analytically complex. Later, we will see a much simpler and elegant description of the torus using quotient topology:

$$T^2 = \mathbb{R}^2/\mathbb{Z}^2,$$

which makes its structure as a compact manifold more transparent.

Proposition 5.: Open Subsets and Subspace Topology

Let $(X, \mathcal{T})$ be a topological space, and let $U \subseteq X$ be an open subset. Then the subspace topology on U coincides with the topology $\mathcal{T}_U$ inherited from X; in other words, U becomes a topological space in its own right, with the same open sets as in $\mathcal{T}$ restricted to U.
Moreover, if $V \subseteq U$, then V is open in U (in the subspace topology) if and only if $V \in \mathcal{T}$.

Proof. Since $U \subseteq X$, the subspace topology on U is:

$$\mathcal{T}_U = \{V \cap U \mid V \in \mathcal{T}\}.$$

But if $U \in \mathcal{T}$, then for any $V \subseteq U$, the condition $V = W \cap U$ for some $W \in \mathcal{T}$ is equivalent to $V \in \mathcal{T}$, since U is already open and subsets of open sets that are also open in the whole space are open in the subset.

So open subsets inherit the same topology they would have as sets in X, but now viewed as a topological space in their own right. □

5.3 Product Topology

Definition 5.: Product Topology

Let $\{X_\alpha\}_{\alpha \in A}$ be a family of topological spaces. The **product space** is

$$\prod_{\alpha \in A} X_\alpha,$$

and the **product topology** is the coarsest topology that makes all the projection maps

$$\pi_\beta : \prod_{\alpha \in A} X_\alpha \to X_\beta$$

continuous.
Explicitly, a set $U \subseteq \prod_{\alpha \in A} X_\alpha$ is open if and only if it is a union of basic open sets — finite intersections of sets of the form $\pi_\alpha^{-1}(U_\alpha)$, where $U_\alpha \subseteq X_\alpha$ is open.

Subbasis and Basis Description for $X \times Y$

Let X and Y be topological spaces. The collection

$$\mathcal{S} = \{\pi_1^{-1}(U) \mid U \text{ open in } X\} \cup \{\pi_2^{-1}(V) \mid V \text{ open in } Y\}$$

is a subbasis for the product topology on $X \times Y$. The finite intersections

$$\pi_1^{-1}(U) \cap \pi_2^{-1}(V) = U \times V$$

form a basis.

Thus, the product topology on $X \times Y$ is generated by all sets of the form $U \times V$, where $U \subseteq X$ and $V \subseteq Y$ are open.

Universal Property of the Product Topology

Let $\{X_\alpha\}_{\alpha \in A}$ be a family of topological spaces, and let $\prod_{\alpha \in A} X_\alpha$ be their product with the product topology. Then:

- Each projection map $\pi_\alpha : \prod X_\alpha \to X_\alpha$ is continuous.
- For any topological space Y, and any family of continuous maps $f_\alpha : Y \to X_\alpha$, there exists a unique continuous map

$$f : Y \to \prod_{\alpha \in A} X_\alpha, \quad f(y) = (f_\alpha(y))_{\alpha \in A}$$

such that $\pi_\alpha \circ f = f_\alpha$ for all α.

5.4 Disjoint Union Topology

Definition 5.: Disjoint Union Topology

Let $\{X_\alpha\}_{\alpha \in A}$ be a family of topological spaces. The **disjoint union set** is

$$\coprod_{\alpha \in A} X_\alpha = \{(x, \alpha) \mid x \in X_\alpha,\ \alpha \in A\}.$$

Each X_α is embedded as the subset

$$X_\alpha^* = \{(x, \alpha) \mid x \in X_\alpha\}.$$

The **disjoint union topology** is the finest topology on $\coprod X_\alpha$ such that each inclusion map

$$\iota_\alpha : X_\alpha \hookrightarrow \coprod X_\alpha, \quad x \mapsto (x, \alpha)$$

is continuous.
Explicitly, a subset $U \subseteq \coprod X_\alpha$ is open if and only if

$$\iota_\alpha^{-1}(U) = \{x \in X_\alpha \mid (x, \alpha) \in U\}$$

is open in X_α for all $\alpha \in A$.

Examples of Disjoint Unions

Example 5.: Finite Disjoint Union

If $A = \{1, 2\}$, then

$$X_1 \amalg X_2 := \coprod_{\alpha \in \{1,2\}} X_\alpha = (X_1 \times \{1\}) \cup (X_2 \times \{2\}).$$

Each X_α is topologically embedded as a separate "layer" in the disjoint union.

Remark 5.

Disjoint union topology models spaces made by keeping distinct components totally separated. It's commonly used when classifying disconnected components or when analyzing maps that behave differently on different domains.

5.5 Quotient Topology

Definition 5.: Quotient Topology

Let X be a topological space and let $\sim$ be an equivalence relation on X. Define the quotient set $X/\sim$ as the set of equivalence classes:

$$X/\sim = \{[x] \mid x \in X\}.$$

Let $\pi : X \to X/\sim$ be the canonical projection map given by $\pi(x) = [x]$. The **quotient topology** on $X/\sim$ is the finest topology that makes π continuous. Explicitly, a subset $U \subseteq X/\sim$ is open if and only if

$$\pi^{-1}(U) \text{ is open in } X.$$

Universal Property of the Quotient Topology

Let X be a topological space and $\pi : X \to X/\sim$ the projection map. Then the quotient topology is characterized by:

- π is continuous.
- For any topological space Y, and any continuous map $f : X \to Y$ such that

  $$x \sim x' \quad \Rightarrow \quad f(x) = f(x'),$$

  there exists a unique continuous map $\tilde{f} : X/\sim \to Y$ such that the diagram commutes:

  $$f = \tilde{f} \circ \pi.$$

5.6 Pushout Topology

Definition 5.: Pushout (Adjunction Space)

Let X, Y, and A be topological spaces, and let $f : A \to X$, $g : A \to Y$ be continuous maps. The **pushout** (also called an *adjunction space*) is defined as the quotient of the disjoint union $X \sqcup Y$ by the identifications

$$f(a) \sim g(a) \quad \text{for all } a \in A.$$

That is,

$$X \cup_f Y := (X \sqcup Y)/\sim$$

with the quotient topology.

Remark 5.

Pushouts provide a general method for gluing topological spaces together along specified subspaces. They are fundamental in CW-complexes, surgeries, and attaching cells in algebraic topology and differential topology.

Examples of Pushouts

Example 5.: Attaching an Interval to a Circle

Let $A = \{0, 1\} \subseteq [0, 1]$, and let

$$f(0) = p, \quad f(1) = q \quad \text{(points on } S^1\text{)},$$

and let $g : A \hookrightarrow [0, 1]$ be inclusion. Then the pushout

$$S^1 \cup_f [0, 1]$$

is the space obtained by attaching the interval $[0, 1]$ to S^1, gluing its endpoints to the points p and q.

Example 5.: Wedge Sum as a Pushout

Let X and Y be spaces with chosen basepoints $x_0 \in X$ and $y_0 \in Y$. Define $A = \{*\}$, and let

$$f(*) := * \mapsto x_0 \in X, \quad g(*) := * \mapsto y_0 \in Y.$$

Then the pushout $X \cup_* Y$ is the **wedge sum** $X \vee Y$, formed by identifying the basepoints.

Example 5.: Constructing the Torus via Pushout

Start with the square $I \times I$, and impose the identifications:

$$(x, 0) \sim (x, 1), \quad (0, y) \sim (1, y).$$

This glues opposite edges of the square and forms the torus T^2. It is a pushout along the edge maps from the circle.

6 Homotopy Theory

6.1 Motivation

The problem. When are two topological spaces "the same"? Homeomorphism is the strictest answer — but often too strict. Many interesting spaces are *not* homeomorphic, yet share all the same "topological features" we care about.

The idea. Two spaces are **homotopy equivalent** if one can be continuously deformed into the other. This is weaker than homeomorphism, but captures the essential topology.

Why this matters. Homotopy groups $\pi_n(X)$ are algebraic invariants that detect "holes" of various dimensions. The fundamental group $\pi_1(X)$ detects loops that cannot be shrunk to a point. Higher homotopy groups $\pi_n(X)$ detect higher-dimensional obstructions.

6.2 Homotopy of Maps

Definition 6.1: Homotopy

Let X and Y be topological spaces, and let $f, g : X \to Y$ be continuous maps. A *homotopy* from f to g is a continuous map

$$H : X \times [0, 1] \to Y$$

such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for all $x \in X$.
We say f and g are *homotopic*, written $f \simeq g$, if such an H exists.

Remark 6.1

Think of H as a continuous family of maps $H_t : X \to Y$ (where $H_t(x) = H(x, t)$) that interpolates between $f = H_0$ and $g = H_1$.

Example 6.1: Homotopy in $\mathbb{R}^n$

Any two maps $f, g : X \to \mathbb{R}^n$ are homotopic via the straight-line homotopy:

$$H(x, t) = (1 - t)f(x) + t \cdot g(x).$$

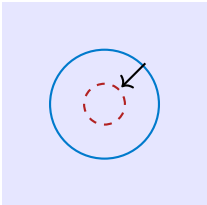
This works because $\mathbb{R}^n$ is convex.

Example 6.2: Loops in $\mathbb{R}^2$

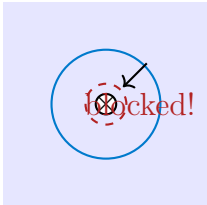
Consider two loops $\gamma_0, \gamma_1 : S^1 \to \mathbb{R}^2$ (continuous maps from the circle). They are homotopic: we can continuously deform one into the other without leaving $\mathbb{R}^2$.

Example 6.3: Loops in $\mathbb{R}^2 \setminus \{0\}$

Now consider loops in the punctured plane. A loop that winds once around the origin is **not** homotopic to a constant loop — the hole blocks the deformation!



$\mathbb{R}^2$: loops contract



$\mathbb{R}^2 \setminus \{0\}$: loops stuck

Figure 1: The puncture creates an obstruction to contracting loops.

Proposition 6.1: Homotopy is an equivalence relation

The relation $\simeq$ on the set of continuous maps $X \to Y$ is an equivalence relation:

1. **Reflexive:** $f \simeq f$ (use $H(x,t) = f(x)$).
2. **Symmetric:** $f \simeq g \Rightarrow g \simeq f$ (use $H'(x,t) = H(x, 1-t)$).
3. **Transitive:** $f \simeq g$ and $g \simeq h \Rightarrow f \simeq h$ (concatenate homotopies).

6.3 Homotopy Equivalence of Spaces

Definition 6.2: Homotopy equivalence

Spaces X and Y are *homotopy equivalent* (written $X \simeq Y$) if there exist continuous maps $f : X \to Y$ and $g : Y \to X$ such that:

$$g \circ f \simeq \mathrm{id}_X \quad \text{and} \quad f \circ g \simeq \mathrm{id}_Y.$$

The maps f and g are called *homotopy equivalences* (or *homotopy inverses*).

Remark 6.2

Homeomorphism $\Rightarrow$ homotopy equivalence, but not conversely. Homotopy equivalence is a weaker notion.

Definition 6.3: Contractible space

A space X is *contractible* if it is homotopy equivalent to a point: $X \simeq \{*\}$. Equivalently, id_X is homotopic to a constant map.

Example 6.4: $\mathbb{R}^n$ is contractible

The map $H(x,t) = (1-t)x$ contracts $\mathbb{R}^n$ to the origin. So $\mathbb{R}^n \simeq \{0\}$.

Example 6.5: D^n is contractible

The closed disk $D^n = \{x \in \mathbb{R}^n : |x| \leq 1\}$ contracts to its center.

Example 6.6: S^n is not contractible

The n-sphere S^n is **not** contractible. There is no way to continuously shrink S^n to a point while staying on S^n. (We will prove this using homotopy groups.)

Example 6.7: $\mathbb{R}^2 \setminus \{0\} \simeq S^1$

The punctured plane is homotopy equivalent to the circle! The map

$$f : \mathbb{R}^2 \setminus \{0\} \to S^1, \quad x \mapsto \frac{x}{|x|}$$

with the inclusion $g : S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$ gives a homotopy equivalence.
The homotopy $H(x,t) = (1-t)x + t \cdot \frac{x}{|x|}$ deforms each point radially onto the unit circle.

Example 6.8: $\mathbb{R}^3 \setminus \{\text{line}\} \simeq S^1$

$\mathbb{R}^3$ with a line removed is homotopy equivalent to a circle (retract onto any circle linking the line).

Observation 6.1

Homotopy equivalent spaces have:

- The same homotopy groups π_n
- The same homology groups H_n
- The same cohomology groups H^n

Homotopy equivalence preserves all "algebraic topology" invariants.

6.4 Deformation Retracts

Definition 6.4: Retraction

Let $A \subset X$ be a subspace. A *retraction* of X onto A is a continuous map $r : X \to A$ such that $r|_A = \mathrm{id}_A$ (i.e., $r(a) = a$ for all $a \in A$).

Definition 6.5: Deformation retract

A subspace $A \subset X$ is a *deformation retract* of X if there exists a homotopy $H : X \times [0,1] \to X$ such that:

1. $H(x,0) = x$ for all $x \in X$,
2. $H(x,1) \in A$ for all $x \in X$,
3. $H(a,t) = a$ for all $a \in A$ and all t.

Remark 6.3

If A is a deformation retract of X, then $X \simeq A$. The retraction $r(x) = H(x,1)$ and the inclusion $i : A \hookrightarrow X$ are homotopy inverses.

Example 6.9: S^1 is a deformation retract of $\mathbb{R}^2 \setminus \{0\}$

The homotopy $H(x,t) = (1-t)x + t \cdot \frac{x}{|x|}$ deformation retracts the punctured plane onto the unit circle.

Example 6.10: S^{n-1} is a deformation retract of $\mathbb{R}^n \setminus \{0\}$

Same idea in all dimensions: $H(x,t) = (1-t)x + t \cdot \frac{x}{|x|}$.

6.5 The Fundamental Group

Setup. Fix a basepoint $x_0 \in X$. Consider loops based at x_0: continuous maps $\gamma : [0,1] \to X$ with $\gamma(0) = \gamma(1) = x_0$.

Definition 6.6: Loop

A *loop* in X based at x_0 is a continuous map $\gamma : [0,1] \to X$ with $\gamma(0) = \gamma(1) = x_0$.

Definition 6.7: Homotopy of loops (relative to endpoints)

Two loops γ_0, γ_1 based at x_0 are *homotopic* (written $\gamma_0 \simeq \gamma_1$) if there exists a continuous map $H : [0,1] \times [0,1] \to X$ such that:

1. $H(s,0) = \gamma_0(s)$ and $H(s,1) = \gamma_1(s)$,
2. $H(0,t) = H(1,t) = x_0$ for all t (basepoint stays fixed).

Definition 6.8: Concatenation of loops

Given loops γ, η based at x_0, their *concatenation* $\gamma \cdot \eta$ is:

$$(\gamma \cdot \eta)(s) = \begin{cases} \gamma(2s) & 0 \leq s \leq 1/2, \\ \eta(2s-1) & 1/2 \leq s \leq 1. \end{cases}$$

First traverse γ, then η, at double speed each.

Definition 6.9: Fundamental group

The *fundamental group* of X based at x_0 is:

$$\pi_1(X, x_0) := \{\text{loops based at } x_0\}/\simeq$$

with the group operation $[\gamma] \cdot [\eta] = [\gamma \cdot \eta]$.

Proposition 6.2: $\pi_1(X, x_0)$ is a group

1. **Identity:** The constant loop $e(s) = x_0$ satisfies $[\gamma] \cdot [e] = [e] \cdot [\gamma] = [\gamma]$.
2. **Inverse:** The reverse loop $\bar{\gamma}(s) = \gamma(1-s)$ satisfies $[\gamma] \cdot [\bar{\gamma}] = [e]$.
3. **Associativity:** $([\alpha] \cdot [\beta]) \cdot [\gamma] = [\alpha] \cdot ([\beta] \cdot [\gamma])$.

Remark 6.4

The fundamental group is generally **non-abelian**. The order in which you traverse loops matters!

Examples of Fundamental Groups

Example 6.11: $\pi_1(\mathbb{R}^n) = 0$

$\mathbb{R}^n$ is contractible, so every loop can be shrunk to a point. The fundamental group is trivial.

Example 6.12: $\pi_1(S^1) = \mathbb{Z}$

Loops on S^1 are classified by their *winding number* — how many times they wrap around. The isomorphism is:

$$\pi_1(S^1) \xrightarrow{\sim} \mathbb{Z}, \quad [\gamma] \mapsto \text{winding number}.$$

The generator is the loop that goes around once counterclockwise.

Example 6.13: $\pi_1(S^n) = 0$ for $n \geq 2$

On higher spheres, any loop can be contracted — there's "room to move around" the loop. This requires proof (not obvious!).

Example 6.14: $\pi_1(T^2) = \mathbb{Z} \times \mathbb{Z}$

The torus has two independent loops: one around the "hole" and one around the "tube." Both commute, giving $\mathbb{Z} \times \mathbb{Z}$.

Example 6.15: $\pi_1(\mathbb{R}P^2) = \mathbb{Z}/2\mathbb{Z}$

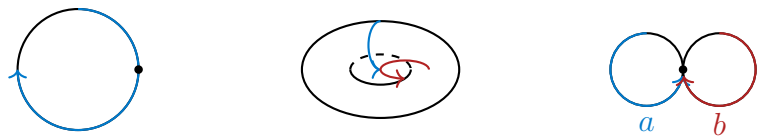
The projective plane has $\pi_1 = \mathbb{Z}/2\mathbb{Z}$. Going around the "crosscap" twice is homotopic to the identity.

Example 6.16: $\pi_1(\text{figure-8}) = F_2$

The figure-8 (wedge of two circles $S^1 \vee S^1$) has fundamental group the *free group on two generators*:

$$\pi_1(S^1 \vee S^1) = F_2 = \langle a, b \rangle.$$

This is non-abelian: $ab \neq ba$.



$\pi_1(S^1) = \mathbb{Z}$ $\quad$ $\pi_1(T^2) = \mathbb{Z}^2$ $\quad$ $\pi_1 = F_2$ (free)

Figure 2: Fundamental groups of basic spaces.

6.6 Higher Homotopy Groups

Idea. π_1 uses loops (maps $S^1 \to X$). Why not use higher spheres?

Definition 6.10: Higher homotopy groups

For $n \geq 1$, the n-th *homotopy group* of (X, x_0) is:

$$\pi_n(X, x_0) := [(S^n, *), (X, x_0)]$$

the set of homotopy classes of basepoint-preserving maps $S^n \to X$.

Remark 6.5

Equivalently, $\pi_n(X, x_0)$ is homotopy classes of maps $(I^n, \partial I^n) \to (X, x_0)$, where $I^n = [0, 1]^n$ and the entire boundary maps to the basepoint.

Proposition 6.3: π_n is abelian for $n \geq 2$

For $n \geq 2$, the group $\pi_n(X, x_0)$ is abelian. There is "room" in higher dimensions to slide maps past each other.

Observation 6.2

What π_n detects:

$\pi_0(X)$	Connected components
$\pi_1(X)$	1-dimensional holes (loops that can't contract)
$\pi_2(X)$	2-dimensional holes (spheres that can't contract)
$\pi_n(X)$	n-dimensional holes

Examples of Higher Homotopy Groups

Example 6.17: $\pi_n(\mathbb{R}^k) = 0$ for all n

Euclidean space is contractible — all homotopy groups vanish.

Example 6.18: $\pi_n(S^n) = \mathbb{Z}$

The identity map $\mathrm{id} : S^n \to S^n$ generates $\pi_n(S^n)$. Maps are classified by their *degree*.

Example 6.19: $\pi_k(S^n) = 0$ for $k < n$

A k-sphere in S^n (for $k < n$) can always be contracted — there's room to push it off itself.

Example 6.20: $\pi_3(S^2) = \mathbb{Z}$ (Hopf fibration)

This is surprising! There is a non-trivial map $S^3 \to S^2$ that cannot be deformed to a constant. This is the *Hopf fibration*:

$$S^1 \hookrightarrow S^3 \xrightarrow{\eta} S^2.$$

The generator $\eta \in \pi_3(S^2)$ is the Hopf map.

Remark 6.6

Computing $\pi_k(S^n)$ for $k > n$ is very hard! The answer is often non-trivial and is a major topic in algebraic topology.

Observation 6.3

Summary of homotopy groups of spheres (low dimensions):

	π_1	π_2	π_3	π_4	π_5
S^1	$\mathbb{Z}$	0	0	0	0
S^2	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}/2$	$\mathbb{Z}/2$
S^3	0	0	$\mathbb{Z}$	$\mathbb{Z}/2$	$\mathbb{Z}/2$
S^4	0	0	0	$\mathbb{Z}$	$\mathbb{Z}/2$

6.7 Covering Spaces

Motivation. Covering spaces provide a geometric way to understand π_1. They also let us "unwind" loops.

Definition 6.11: Covering space

A *covering space* of X is a space $\tilde{X}$ together with a map $p : \tilde{X} \to X$ such that every point $x \in X$ has a neighborhood U with:

$$p^{-1}(U) = \bigsqcup_{\alpha} V_\alpha$$

where each V_α is mapped homeomorphically onto U by p.

Example 6.21: $\mathbb{R} \to S^1$

The map $p : \mathbb{R} \to S^1$, $t \mapsto e^{2\pi i t}$, is a covering. The fiber over any point is $\mathbb{Z}$. This "unwinds" the circle into a line.

Example 6.22: $S^n \to \mathbb{R}P^n$

The quotient map $p : S^n \to \mathbb{R}P^n = S^n/(x \sim -x)$ is a 2-fold cover. Each point in $\mathbb{R}P^n$ has exactly 2 preimages.

Theorem 6.1: Lifting criterion

Let $p : \tilde{X} \to X$ be a covering, and let $f : Y \to X$ be continuous with Y connected and locally path-connected. Fix basepoints with $p(\tilde{x}_0) = x_0 = f(y_0)$.
Then f lifts to $\tilde{f} : Y \to \tilde{X}$ (with $p \circ \tilde{f} = f$ and $\tilde{f}(y_0) = \tilde{x}_0$) if and only if:

$$f_*(\pi_1(Y, y_0)) \subseteq p_*(\pi_1(\tilde{X}, \tilde{x}_0)).$$

Theorem 6.2: Classification of covering spaces

For a "nice" space X (connected, locally path-connected, semi-locally simply connected), there is a bijection:

$$\{\text{connected coverings of } X\} \longleftrightarrow \{\text{subgroups of } \pi_1(X)\}.$$

The covering $p : \tilde{X} \to X$ corresponds to the subgroup $p_*(\pi_1(\tilde{X}))$.

Example 6.23: Coverings of S^1

$\pi_1(S^1) = \mathbb{Z}$. The subgroups are $n\mathbb{Z}$ for $n \geq 0$.

- $n = 0$: trivial subgroup $\to$ universal cover $\mathbb{R} \to S^1$.
- $n \geq 1$: subgroup $n\mathbb{Z} \to n$-fold cover $S^1 \xrightarrow{z \mapsto z^n} S^1$.

Definition 6.12: Universal cover

The *universal cover* of X is a simply connected covering space $\tilde{X} \to X$. It corresponds to the trivial subgroup $\{e\} \subset \pi_1(X)$.

Observation 6.4

Key facts about the universal cover:

- The deck transformations (covering automorphisms) form a group isomorphic to $\pi_1(X)$.
- $X \cong \tilde{X}/\pi_1(X)$ (as a quotient by deck transformations).
- This gives a geometric interpretation of π_1.

6.8 Summary

Observation 6.5

Concept	What it captures
Homotopy $f \simeq g$	Maps can be deformed into each other
Homotopy equivalence $X \simeq Y$	Spaces have the same "shape"
Contractible	Homotopy equivalent to a point
$\pi_1(X)$	Loops modulo homotopy (1-holes)
$\pi_n(X)$	n-spheres modulo homotopy (n-holes)
Covering space	Geometric model of π_1
Universal cover	Simply connected version of X

7 Homology

7.1 Motivation

The problem with π_n. Homotopy groups are powerful but *extremely hard to compute.* Even $\pi_k(S^n)$ for $k > n$ requires sophisticated techniques.

The idea. Homology groups $H_n(X)$ are *computable* algebraic invariants that also detect holes. The trade-off: homology is always abelian and carries less information than homotopy, but it's much more tractable.

The construction. Instead of mapping spheres into X (homotopy), we decompose X into simple pieces (simplices) and track how they fit together algebraically. Holes appear as "cycles that aren't boundaries."

Physics connection. Homology underlies:

- Conservation laws (closed forms that aren't exact)
- Topological charges (elements of H_n)
- The Gauss-Bonnet theorem ($\chi(M) = \sum(-1)^k \dim H_k$)
- Topological field theories

7.2 Simplicial Complexes

Definition 7.1: Simplex

The *standard n-simplex* is:

$$\Delta^n = \left\{ (t_0, \ldots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, \sum_{i=0}^{n} t_i = 1 \right\}.$$

It is the convex hull of $n+1$ points in general position.

Example 7.1: Low-dimensional simplices

- Δ^0 = point
- Δ^1 = line segment (edge)
- Δ^2 = triangle (with interior)
- Δ^3 = tetrahedron (with interior)

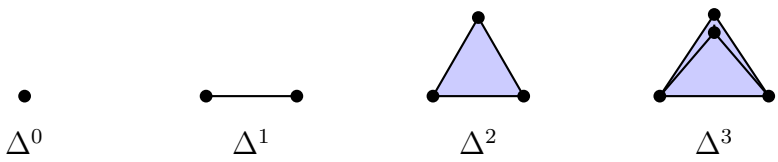


Figure 3: The standard simplices in dimensions 0, 1, 2, 3.

Definition 7.2: Face of a simplex

The *i-th face* of Δ^n is the $(n-1)$-simplex obtained by omitting the i-th vertex:

$$d_i : \Delta^{n-1} \to \Delta^n, \quad (t_0, \ldots, t_{n-1}) \mapsto (t_0, \ldots, t_{i-1}, 0, t_i, \ldots, t_{n-1}).$$

An n-simplex has $(n+1)$ faces of dimension $(n-1)$.

Definition 7.3: Simplicial complex

A *simplicial complex* K is a collection of simplices such that:

1. Every face of a simplex in K is also in K.
2. The intersection of any two simplices in K is either empty or a common face.

The *geometric realization* $|K|$ is the topological space obtained by gluing the simplices together.

Example 7.2: Simplicial complexes

- A graph is a 1-dimensional simplicial complex.
- The boundary of a tetrahedron (4 triangles) triangulates S^2.
- Any smooth manifold can be triangulated (given a simplicial structure).

7.3 Chain Complexes

The algebraic setup. We now encode the combinatorics of a simplicial complex into algebra.

Definition 7.4: Chain group

Let K be a simplicial complex. The *n-th chain group* $C_n(K; R)$ is the free R-module generated by the n-simplices of K.
An element $c \in C_n(K; R)$ is a finite formal sum:

$$c = \sum_\sigma a_\sigma \cdot \sigma, \quad a_\sigma \in R,$$

where σ ranges over n-simplices.

Remark 7.1

Common choices of coefficients:

- $R = \mathbb{Z}$ (integer coefficients) — detects torsion
- $R = \mathbb{R}$ or $\mathbb{Q}$ (field coefficients) — vector spaces, easier to compute
- $R = \mathbb{Z}/2\mathbb{Z}$ — useful for non-orientable manifolds

Definition 7.5: Boundary operator

The *boundary operator* $\partial_n : C_n(K) \to C_{n-1}(K)$ is defined on a simplex $\sigma = [v_0, \ldots, v_n]$ by:

$$\partial_n[v_0, \ldots, v_n] = \sum_{i=0}^{n} (-1)^i [v_0, \ldots, \hat{v}_i, \ldots, v_n],$$

where $\hat{v}_i$ means omit v_i. The signs ensure $\partial^2 = 0$.

Example 7.3: Boundary of a 1-simplex (edge)

For an edge $[v_0, v_1]$:

$$\partial_1[v_0, v_1] = [v_1] - [v_0].$$

The boundary of an edge is the difference of its endpoints.

Example 7.4: Boundary of a 2-simplex (triangle)

For a triangle $[v_0, v_1, v_2]$:

$$\partial_2[v_0, v_1, v_2] = [v_1, v_2] - [v_0, v_2] + [v_0, v_1].$$

The boundary is the sum of edges with alternating signs (orienting consistently).

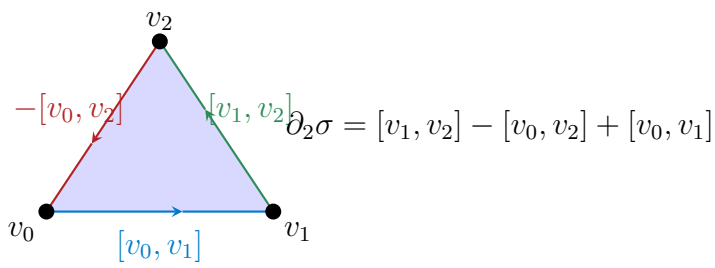


Figure 4: The boundary of a triangle is the oriented sum of its edges.

Proposition 7.1: $\partial^2 = 0$

The composition $\partial_{n-1} \circ \partial_n = 0$ for all n. In other words:

$$\partial \circ \partial = 0 \quad \text{or} \quad \partial^2 = 0.$$

Sketch. Each $(n-2)$-face of an n-simplex appears exactly twice in ∂^2, with opposite signs. □

Definition 7.6: Chain complex

A *chain complex* is a sequence of abelian groups (or modules) with maps:

$$\cdots \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} C_0 \xrightarrow{\partial_0} 0$$

such that $\partial_n \circ \partial_{n+1} = 0$ for all n.

7.4 Homology Groups

Definition 7.7: Cycles and boundaries

- The *n-cycles* are: $Z_n = \ker(\partial_n) = \{c \in C_n : \partial c = 0\}$.
- The *n-boundaries* are: $B_n = \mathrm{im}(\partial_{n+1}) = \{\partial d : d \in C_{n+1}\}$.

Remark 7.2

Since $\partial^2 = 0$, every boundary is a cycle: $B_n \subseteq Z_n$.

Definition 7.8: Homology groups

The *n-th homology group* is the quotient:

$$H_n(K; R) = \frac{Z_n}{B_n} = \frac{\ker \partial_n}{\mathrm{im}\partial_{n+1}}.$$

Elements of H_n are equivalence classes of cycles, where two cycles are equivalent if they differ by a boundary.

Observation 7.1

Interpretation:

- A *cycle* is a chain with no boundary — it "closes up."
- A *boundary* is a cycle that bounds something — it's "filled in."
- $H_n \neq 0$ means there exist cycles that don't bound anything — *holes!*

Examples

Example 7.5: H_0 counts connected components

$H_0(X)$ is the free R-module on the connected components of X.
For a connected space: $H_0(X; R) \cong R$.

Example 7.6: Homology of a point

$$H_n(\text{point}) = \begin{cases} R & n = 0, \\ 0 & n > 0. \end{cases}$$

Example 7.7: Homology of S^1

Triangulate S^1 as a triangle (3 vertices, 3 edges, 0 faces).
Chain complex: $0 \to C_1 \cong R^3 \xrightarrow{\partial} C_0 \cong R^3 \to 0$.

$$H_0(S^1) = R, \quad H_1(S^1) = R, \quad H_n(S^1) = 0 \text{ for } n \geq 2.$$

The generator of $H_1(S^1)$ is the cycle that goes around the circle once.

Example 7.8: Homology of S^n

$$H_k(S^n; R) = \begin{cases} R & k = 0 \text{ or } k = n, \\ 0 & \text{otherwise.} \end{cases}$$

The sphere has one "hole" in dimension n.

Example 7.9: Homology of the torus T^2

$$H_0(T^2) = R, \quad H_1(T^2) = R \oplus R, \quad H_2(T^2) = R.$$

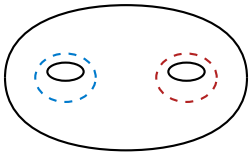
Two independent 1-cycles (around the hole and around the tube), and one 2-cycle (the torus itself).

Example 7.10: Homology of the Riemann surface Σ_g

A surface of genus g (sphere with g handles):

$$H_0(\Sigma_g) = R, \quad H_1(\Sigma_g) = R^{2g}, \quad H_2(\Sigma_g) = R.$$

Each handle contributes 2 independent 1-cycles.



Σ_2: $H_1 = R^4$

Figure 5: A genus-2 surface has 4 independent 1-cycles.

7.5 Properties of Homology

Proposition 7.2: Homotopy invariance

If $f : X \to Y$ is a homotopy equivalence, then:

$$H_n(X) \cong H_n(Y) \quad \text{for all } n.$$

Homology is a *homotopy invariant*.

Proposition 7.3: Functoriality

A continuous map $f : X \to Y$ induces homomorphisms:

$$f_* : H_n(X) \to H_n(Y) \quad \text{for all } n.$$

This respects composition: $(g \circ f)_* = g_* \circ f_*$ and $(\mathrm{id})_* = \mathrm{id}$.

Theorem 7.1: Euler characteristic

For a finite simplicial complex K:

$$\chi(K) = \sum_{n \geq 0} (-1)^n \cdot \#\{n\text{-simplices}\} = \sum_{n \geq 0} (-1)^n \dim H_n(K; \mathbb{Q}).$$

The Euler characteristic can be computed from either the combinatorics or the homology.

Example 7.11: Euler characteristic examples

Space	$V - E + F$	$\sum(-1)^n \dim H_n$	χ
Point	1	$1 - 0 + 0$	1
S^1	$3 - 3 + 0$	$1 - 1 + 0$	0
S^2	$4 - 6 + 4$	$1 - 0 + 1$	2
T^2	varies	$1 - 2 + 1$	0
Σ_g	varies	$1 - 2g + 1$	$2 - 2g$

7.6 Exact Sequences

Definition 7.9: Exact sequence

A sequence of abelian groups and homomorphisms:

$$\cdots \to A \xrightarrow{f} B \xrightarrow{g} C \to \cdots$$

is *exact* at B if $\mathrm{im}(f) = \ker(g)$.

Example 7.12: Short exact sequences

A *short exact sequence* is:

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

where f is injective, g is surjective, and $\mathrm{im}(f) = \ker(g)$.
This says: A is a subgroup of B, and $C = B/A$.

Theorem 7.2: Long exact sequence of a pair

For a pair (X, A) with $A \subset X$, there is a long exact sequence:

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots$$

Here $H_n(X, A)$ is the *relative homology* (cycles in X modulo cycles in A).

Theorem 7.3: Mayer-Vietoris sequence

If $X = U \cup V$ where U, V are open, there is a long exact sequence:

$$\cdots \to H_n(U \cap V) \to H_n(U) \oplus H_n(V) \to H_n(X) \xrightarrow{\partial} H_{n-1}(U \cap V) \to \cdots$$

This lets us compute homology by decomposing spaces.

Example 7.13: $H_*(S^n)$ via Mayer-Vietoris

Cover S^n by two hemispheres U, V (overlapping at the equator S^{n-1}).
Both U and V are contractible, so $H_k(U) = H_k(V) = 0$ for $k > 0$.
The Mayer-Vietoris sequence gives:

$$0 \to H_n(S^n) \xrightarrow{\partial} H_{n-1}(S^{n-1}) \to 0.$$

So $H_n(S^n) \cong H_{n-1}(S^{n-1}) \cong \cdots \cong H_0(S^0) = R$.

7.7 Singular Homology

Beyond simplicial complexes. Simplicial homology requires a triangulation. *Singular homology* works for any topological space.

Definition 7.10: Singular simplex

A *singular n-simplex* in X is a continuous map $\sigma : \Delta^n \to X$.

Definition 7.11: Singular chain complex

The *singular n-chains* $C_n(X)$ is the free abelian group generated by all singular n-simplices.
The boundary $\partial\sigma$ is defined by restricting to faces:

$$\partial\sigma = \sum_{i=0}^{n} (-1)^i \sigma|_{i\text{-th face}}.$$

Definition 7.12: Singular homology

The *singular homology* of X is:

$$H_n(X; R) = \frac{\ker(\partial_n : C_n \to C_{n-1})}{\operatorname{im}(\partial_{n+1} : C_{n+1} \to C_n)}.$$

Remark 7.3

Singular homology agrees with simplicial homology for triangulable spaces. The advantage: it's defined for *any* space, with no triangulation required.

7.8 Homology with Coefficients

Definition 7.13: Homology with coefficients

For any abelian group G, the *homology with coefficients in G* is:

$$H_n(X; G) = H_n(C_*(X) \otimes_{\mathbb{Z}} G).$$

Example 7.14: Coefficient choices

$H_n(X; \mathbb{Z})$	Integer homology (detects torsion)
$H_n(X; \mathbb{R})$	Real homology (vector space)
$H_n(X; \mathbb{Z}/2)$	Mod-2 homology (non-orientable OK)

Theorem 7.4: Universal coefficient theorem

There is a short exact sequence:

$$0 \to H_n(X;\mathbb{Z}) \otimes G \to H_n(X;G) \to \mathrm{Tor}(H_{n-1}(X;\mathbb{Z}), G) \to 0.$$

If G is a field (like $\mathbb{R}$ or $\mathbb{Q}$), the Tor term vanishes and $H_n(X;G) \cong H_n(X;\mathbb{Z}) \otimes G$.

7.9 Summary

Observation 7.2

Object	Meaning
C_n	Formal sums of n-dimensional pieces
∂	Boundary operator ($\partial^2 = 0$)
$Z_n = \ker \partial$	Cycles (closed chains)
$B_n = \mathrm{im}\partial$	Boundaries (chains that bound)
$H_n = Z_n/B_n$	Homology (cycles mod boundaries)
$H_n \neq 0$	Existence of n-dimensional holes

Observation 7.3

Comparison with homotopy:

	Homotopy π_n	Homology H_n
Abelian?	Only for $n \geq 2$	Always
Computable?	Very hard	Feasible
Information	More	Less
Hurewicz	$\pi_1^{\mathrm{ab}} \cong H_1$	(abelianization)

8 Cohomology

8.1 Motivation

From homology to cohomology. Homology groups $H_n(X)$ are built from chains (formal sums of simplices). Cohomology groups $H^n(X)$ are built from *cochains* — functions on simplices.

Why bother?

1. Cohomology has a **ring structure** (cup product). Homology doesn't.
2. Cohomology is **contravariant** — it pulls back under maps. This is natural for differential forms.
3. Cohomology connects directly to **de Rham theory**: $H^n_{\mathrm{dR}}(M) \cong H^n(M;\mathbb{R})$.

4. Many invariants (Chern classes, characteristic classes) live in cohomology.

The slogan. Homology measures "what's inside" (cycles). Cohomology measures "what you can integrate over" (cocycles).

8.2 Cochain Complexes

Definition 8.1: Cochain

Let C_n be a chain group. A *n-cochain* is a homomorphism:

$$\phi : C_n \to R.$$

The *n-th cochain group* is:

$$C^n(X; R) = \mathrm{Hom}(C_n(X), R).$$

Remark 8.1

If C_n is free on simplices $\{\sigma_\alpha\}$, then a cochain ϕ is determined by its values $\phi(\sigma_\alpha) \in R$ on each simplex.

Definition 8.2: Coboundary operator

The *coboundary operator* $\delta^n : C^n \to C^{n+1}$ is the dual of ∂:

$$(\delta\phi)(c) = \phi(\partial c).$$

For a cochain $\phi \in C^n$ and a chain $c \in C_{n+1}$:

$$(\delta^n\phi)(c) = \phi(\partial_{n+1}c).$$

Proposition 8.1: $\delta^2 = 0$

Since $\partial^2 = 0$, we have $\delta^2 = 0$:

$$(\delta^2\phi)(c) = (\delta\phi)(\partial c) = \phi(\partial^2 c) = \phi(0) = 0.$$

Definition 8.3: Cochain complex

A *cochain complex* is a sequence:

$$0 \to C^0 \xrightarrow{\delta^0} C^1 \xrightarrow{\delta^1} C^2 \xrightarrow{\delta^2} \cdots$$

with $\delta^{n+1} \circ \delta^n = 0$.

Remark 8.2

Notice the arrows go *up* in degree (from C^n to C^{n+1}), opposite to homology.

8.3 Cohomology Groups

Definition 8.4: Cocycles and coboundaries

- The *n-cocycles* are: $Z^n = \ker(\delta^n) = \{\phi \in C^n : \delta\phi = 0\}$.
- The *n-coboundaries* are: $B^n = \mathrm{im}(\delta^{n-1}) = \{\delta\psi : \psi \in C^{n-1}\}$.

Definition 8.5: Cohomology groups

The *n-th cohomology group* is:

$$H^n(X; R) = \frac{Z^n}{B^n} = \frac{\ker \delta^n}{\mathrm{im}\delta^{n-1}}.$$

Observation 8.1

Interpretation:

- A *cocycle* ϕ satisfies $\delta\phi = 0$ — it assigns values to cycles that are "consistent."
- A *coboundary* $\phi = \delta\psi$ is "trivial" — it can be computed from lower-dimensional data.
- $H^n \neq 0$ means there exist cocycles that carry genuine information.

Relationship to Homology

Theorem 8.1: Universal coefficient theorem for cohomology

There is a short exact sequence:

$$0 \to \mathrm{Ext}(H_{n-1}(X), R) \to H^n(X; R) \to \mathrm{Hom}(H_n(X), R) \to 0.$$

When R is a field, Ext $= 0$ and:

$$H^n(X; R) \cong \mathrm{Hom}(H_n(X), R) \cong H_n(X; R)^*.$$

Cohomology is the dual of homology (over a field).

Example 8.1: Cohomology of spheres

Over $\mathbb{R}$:

$$H^k(S^n;\mathbb{R}) \cong \begin{cases} \mathbb{R} & k = 0 \text{ or } k = n, \\ 0 & \text{otherwise.} \end{cases}$$

Same as homology, since $\mathbb{R}$ is a field.

Example 8.2: Cohomology of the torus

$$H^0(T^2) = \mathbb{R}, \quad H^1(T^2) = \mathbb{R}^2, \quad H^2(T^2) = \mathbb{R}.$$

8.4 The Cup Product

Ring structure. Cohomology has a multiplicative structure that homology lacks.

Definition 8.6: Cup product

The *cup product* is a bilinear map:

$$\smile : H^p(X) \times H^q(X) \to H^{p+q}(X).$$

On cochains, if $\phi \in C^p$ and $\psi \in C^q$, then $\phi \smile \psi \in C^{p+q}$ is defined by:

$$(\phi \smile \psi)([v_0, \ldots, v_{p+q}]) = \phi([v_0, \ldots, v_p]) \cdot \psi([v_p, \ldots, v_{p+q}]).$$

Evaluate ϕ on the "front face" and ψ on the "back face."

Proposition 8.2: Properties of the cup product

1. **Associativity:** $(\alpha \smile \beta) \smile \gamma = \alpha \smile (\beta \smile \gamma)$.
2. **Unit:** $1 \in H^0(X)$ (the constant cochain) satisfies $1 \smile \alpha = \alpha$.
3. **Graded commutativity:** $\alpha \smile \beta = (-1)^{pq} \beta \smile \alpha$ for $\alpha \in H^p$, $\beta \in H^q$.
4. **Leibniz rule:** $\delta(\phi \smile \psi) = (\delta\phi) \smile \psi + (-1)^{|\phi|} \phi \smile (\delta\psi)$.

Definition 8.7: Cohomology ring

The *cohomology ring* of X is:

$$H^*(X;R) = \bigoplus_{n \geq 0} H^n(X;R)$$

with multiplication given by the cup product. It is a graded-commutative ring.

Example 8.3: Cohomology ring of S^n

$$H^*(S^n;\mathbb{R}) = \mathbb{R}[x]/(x^2), \quad |x| = n.$$

There is a generator $x \in H^n(S^n)$ with $x^2 = 0$ (for dimension reasons).

Example 8.4: Cohomology ring of $\mathbb{R}P^n$

$$H^*(\mathbb{R}P^n;\mathbb{Z}/2) = (\mathbb{Z}/2)[x]/(x^{n+1}), \quad |x| = 1.$$

Over $\mathbb{Z}/2$, there is a generator x in degree 1, and the ring is truncated polynomial.

Example 8.5: Cohomology ring of $\mathbb{C}P^n$

$$H^*(\mathbb{C}P^n;\mathbb{Z}) = \mathbb{Z}[x]/(x^{n+1}), \quad |x| = 2.$$

The generator lives in degree 2 (since $\mathbb{C}P^n$ has cells only in even dimensions).

Example 8.6: Cohomology ring of T^n

$$H^*(T^n;\mathbb{R}) = \Lambda_{\mathbb{R}}(x_1,\ldots,x_n), \quad |x_i| = 1.$$

The cohomology is the *exterior algebra* on n generators. Each x_i corresponds to a circle factor.

Observation 8.2

The cup product carries more information than homology alone. Two spaces can have isomorphic homology groups but different cohomology rings.

8.5 Poincaré Duality

Setup. Let M be a closed (compact, no boundary), connected, oriented n-manifold.

Theorem 8.2: Poincaré duality

For a closed oriented n-manifold M:

$$H^k(M;R) \cong H_{n-k}(M;R).$$

Cohomology in degree k is dual to homology in degree $n-k$.

Remark 8.3

The isomorphism is given by the *cap product* with the fundamental class $[M] \in H_n(M)$:

$$H^k(M) \xrightarrow{\cap[M]} H_{n-k}(M).$$

Example 8.7: Poincaré duality for S^2

For S^2 ($n = 2$):

$$H^0(S^2) \cong H_2(S^2) = \mathbb{R}, \quad H^1(S^2) \cong H_1(S^2) = 0, \quad H^2(S^2) \cong H_0(S^2) = \mathbb{R}.$$

Example 8.8: Poincaré duality for Σ_g

For a genus-g surface ($n = 2$):

$$H^0 \cong H_2 = \mathbb{R}, \quad H^1 \cong H_1 = \mathbb{R}^{2g}, \quad H^2 \cong H_0 = \mathbb{R}.$$

The middle cohomology $H^1 \cong H_1$ is self-dual.

Observation 8.3

Poincaré duality implies:

- $\chi(M) = 0$ for odd-dimensional closed oriented manifolds (terms cancel in pairs).
- The Betti numbers satisfy $b_k = b_{n-k}$.
- There is a non-degenerate pairing $H^k \times H^{n-k} \to R$.

8.6 Connection to de Rham Cohomology

The bridge. On a smooth manifold, differential forms give cochains. The de Rham theorem says this gives the same cohomology.

Definition 8.8: de Rham cohomology (review)

For a smooth manifold M:

$$H^k_{\mathrm{dR}}(M) = \frac{\{\text{closed } k\text{-forms}\}}{\{\text{exact } k\text{-forms}\}} = \frac{\ker d}{\mathrm{im} d}.$$

Theorem 8.3: de Rham theorem

For a smooth manifold M:

$$H^k_{\mathrm{dR}}(M) \cong H^k(M; \mathbb{R}).$$

de Rham cohomology equals singular cohomology with real coefficients.

Remark 8.4

The isomorphism is given by integration. A closed form ω defines a cocycle by:

$$\phi_\omega : \sigma \mapsto \int_\sigma \omega.$$

Stokes' theorem ensures $\delta\phi_\omega = 0$ iff $d\omega = 0$.

Observation 8.4

This connects three worlds:

Topology	$\longleftrightarrow$	**Analysis**
Singular cochains		Differential forms
δ		d
$H^k(M;\mathbb{R})$	$\cong$	$H^k_{\mathrm{dR}}(M)$

8.7 Cohomology Operations

Beyond the cup product. There are additional operations on cohomology that carry subtle topological information.

Definition 8.9: Steenrod squares (mod 2)

The *Steenrod squares* are operations:

$$\mathrm{Sq}^i : H^n(X;\mathbb{Z}/2) \to H^{n+i}(X;\mathbb{Z}/2)$$

satisfying:

1. $\mathrm{Sq}^0 = \mathrm{id}$.
2. $\mathrm{Sq}^n(x) = x^2$ for $x \in H^n$.
3. $\mathrm{Sq}^i(x) = 0$ for $i > |x|$.
4. Cartan formula: $\mathrm{Sq}^k(xy) = \sum_{i+j=k} \mathrm{Sq}^i(x) \cdot \mathrm{Sq}^j(y)$.

Remark 8.5

Steenrod operations detect more than the ring structure. They distinguish spaces with isomorphic cohomology rings.

8.8 Applications

Example 8.9: Brouwer fixed point theorem

Claim: Every continuous map $f : D^n \to D^n$ has a fixed point.
Proof sketch: Suppose not. Then we can define a retraction $r : D^n \to S^{n-1}$ by projecting from the fixed point. But $H^{n-1}(S^{n-1}) = \mathbb{R}$ while $H^{n-1}(D^n) = 0$, and a retraction would give a contradiction via functoriality.

Example 8.10: Borsuk-Ulam theorem

Claim: For any continuous $f : S^n \to \mathbb{R}^n$, there exist antipodal points with $f(x) = f(-x)$.
Proof sketch: Uses the cohomology of $\mathbb{R}P^n$ and the structure of the cohomology ring.

Example 8.11: Intersection theory

On a closed oriented n-manifold M, the cup product corresponds to intersection of submanifolds:
If $[\alpha] \in H^p(M)$ and $[\beta] \in H^q(M)$ are Poincaré dual to submanifolds A and B, then $[\alpha] \smile [\beta]$ is Poincaré dual to $A \cap B$ (when transverse).

8.9 Summary

Observation 8.5

Object	Meaning
$C^n = \mathrm{Hom}(C_n, R)$	Cochains (functions on chains)
δ	Coboundary ($\delta^2 = 0$, dual to ∂)
$Z^n = \ker \delta$	Cocycles
$B^n = \mathrm{im}\delta$	Coboundaries
$H^n = Z^n/B^n$	Cohomology
$\smile$	Cup product (ring structure)
$H^*(X)$	Cohomology ring

Observation 8.6

Why cohomology matters in physics:

- **Electromagnetism:** $F \in \Omega^2$, $dF = 0$ means $[F] \in H^2_{\mathrm{dR}}$. Magnetic monopoles $\leftrightarrow H^2 \neq 0$.
- **Gauge theory:** Chern classes live in H^{2k}. Characteristic classes classify bundles via cohomology.
- **String theory:** Cohomology of moduli spaces, Calabi-Yau manifolds, Hodge numbers.
- **Topological field theory:** Partition functions often compute topological invariants via cohomology.

9 Smooth Manifolds

Manifolds are "nice" spaces that look locally like Euclidean space. In mathematics, they generalize curves and surfaces to arbitrary dimension and provide the natural setting for multivariable calculus beyond $\mathbb{R}^n$.

In physics, manifolds appear everywhere: spacetime in relativity, configuration spaces in classical mechanics, phase spaces in Hamiltonian mechanics, and as base spaces for bundles that carry fields (scalar fields, spinor fields, gauge fields, ...). A main goal of these notes is to introduce the minimal differential-geometric language needed to read physics smoothly, while staying honest about the underlying mathematics.

9.1 Topological manifolds

Definition 9.: Topological manifold

Let M be a topological space with topology $\mathcal{O}$. We say that M is a *topological d-manifold* if:

1. M is Hausdorff,
2. M is second countable,
3. For every $p \in M$, there exists an open set $U \in \mathcal{O}$ with $p \in U$, and a map

$$x : U \longrightarrow x(U) \subseteq \mathbb{R}^d$$

such that:

(a) x is a homeomorphism onto its image,

(b) $x(U)$ is an open subset of $\mathbb{R}^d$.

The pair (U, x) is called a *chart*. A collection of charts covering M is called an *atlas*.

Remark 9.Why Hausdorff and second countable?

These hypotheses rule out many pathologies. Hausdorffness ensures limits (when they exist) are unique, and second countability is a mild "smallness" condition that guarantees good global behavior (it implies, for instance, metrizability for manifolds and helps ensure many standard theorems apply).

We will use two running examples throughout this section: S^1 and S^2.

Example 9.: A two-chart atlas on S^1 via stereographic projection

Let

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}, \qquad N = (1, 0), \quad S = (-1, 0).$$

Define stereographic projections

$$\phi_N : S^1 \setminus \{N\} \to \mathbb{R}, \qquad \phi_N(x, y) = \frac{y}{1 - x},$$

$$\phi_S : S^1 \setminus \{S\} \to \mathbb{R}, \qquad \phi_S(x, y) = \frac{y}{1 + x}.$$

Each map is a homeomorphism with explicit inverse:

$$\phi_N^{-1}(t) = \left(\frac{t^2 - 1}{t^2 + 1}, \frac{2t}{t^2 + 1}\right), \qquad \phi_S^{-1}(t) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2}\right).$$

Since $S^1 \setminus \{N\}$ and $S^1 \setminus \{S\}$ are open in S^1 and cover S^1, the two charts

$$\big(S^1 \setminus \{N\}, \phi_N\big), \qquad \big(S^1 \setminus \{S\}, \phi_S\big)$$

form an atlas on S^1.

Example 9.: The sphere S^2 is a 2-dimensional topological manifold

Let

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\},$$

and let $N = (0, 0, 1)$ and $S = (0, 0, -1)$ be the north and south poles.
Define stereographic projections

$$\pi_N : S^2 \setminus \{N\} \longrightarrow \mathbb{R}^2, \qquad \pi_N(x, y, z) = \left(\frac{x}{1 - z}, \frac{y}{1 - z}\right),$$

$$\pi_S : S^2 \setminus \{S\} \longrightarrow \mathbb{R}^2, \qquad \pi_S(x, y, z) = \left(\frac{x}{1 + z}, \frac{y}{1 + z}\right).$$

Both maps are homeomorphisms with explicit inverses (left to the reader / given later in many references). Since $S^2 \setminus \{N\}$ and $S^2 \setminus \{S\}$ are open in S^2 and cover S^2, these two charts form an atlas and show that S^2 is a 2-dimensional topological manifold.

A point $p \in M$ may lie in the domains of two charts at the same time. On overlaps we can compare coordinates using *transition maps*. This is where differentiability will enter.

Definition 9.: Atlas

An *atlas* on M is a collection

$$\mathscr{A} := \{(U_\alpha, x_\alpha) \mid \alpha \in \mathcal{A}\}$$

of charts such that

$$\bigcup_{\alpha \in \mathcal{A}} U_\alpha = M.$$

Definition 9.: $\mathcal{C}^0$-compatible charts

Two charts (U, x) and (V, y) are said to be $\mathcal{C}^0$-*compatible* if either $U \cap V = \varnothing$, or the transition map

$$y \circ x^{-1} : x(U \cap V) \longrightarrow y(U \cap V)$$

is continuous.

Proposition 9.: $\mathcal{C}^0$-compatibility is automatic for topological manifolds

Let M be a topological d-manifold, and let (U, x) and (V, y) be two charts on M. If $U \cap V \neq \varnothing$, then the transition map

$$y \circ x^{-1} : x(U \cap V) \longrightarrow y(U \cap V)$$

is a homeomorphism. In particular, it is continuous.

Proof. Since (U, x) and (V, y) are charts, $x : U \to x(U)$ and $y : V \to y(V)$ are homeomorphisms onto open subsets. The overlap $U \cap V$ is open in M, hence $x(U \cap V)$ is open in $\mathbb{R}^d$, and x^{-1} restricts to a continuous map

$$x^{-1} : x(U \cap V) \to U \cap V.$$

Composing with the continuous map $y : U \cap V \to y(U \cap V)$ gives continuity of $y \circ x^{-1}$. Its inverse is $x \circ y^{-1}$, continuous by the same argument. □

Thus the real content begins when we require transition maps to be $\mathcal{C}^k$ (especially $\mathcal{C}^\infty$).

Proposition 9.: Well-definedness of dimension

Let M be a topological manifold. Suppose $U, V \subseteq M$ are open sets with $U \cap V \neq \varnothing$. Let

$$x : U \longrightarrow x(U) \subseteq \mathbb{R}^d, \qquad y : V \longrightarrow y(V) \subseteq \mathbb{R}^{d'}$$

be homeomorphisms onto open subsets. Then $d = d'$.

Proof. Since $U \cap V \neq \varnothing$, the transition map

$$y \circ x^{-1} : x(U \cap V) \longrightarrow y(U \cap V)$$

is a homeomorphism between nonempty open subsets of $\mathbb{R}^d$ and $\mathbb{R}^{d'}$. By invariance of domain (equivalently, topological invariance of dimension), this implies $d = d'$. $\square$

So the dimension is well-defined on each connected component of a topological manifold.

We conclude this subsection by introducing two basic types of maps.

Definition 9.: Real-valued function

Let M be a topological manifold. A *real-valued function* on M is a map

$$f : M \longrightarrow \mathbb{R}.$$

We say that f is *continuous* if it is continuous with respect to the topology of M.

Definition 9.: Curve

Let M be a topological manifold and let $I \subseteq \mathbb{R}$ be an interval. A *curve* in M is a continuous map

$$\gamma : I \longrightarrow M.$$

The curve passes through $p \in M$ at $t_0 \in I$ if $\gamma(t_0) = p$.

9.2 Construction of manifolds

In practice, manifolds are often obtained from simpler ones by standard constructions. We discuss four basic constructions: submanifolds, products, bundles, and quotients.

Submanifold

Definition 9.: Topological submanifold

Let M be a topological m-manifold and let $S \subseteq M$. We say that S is a *topological k-submanifold of M* if for every $p \in S$ there exists a chart (U, x) of M with $p \in U$ such that

$$x(U \cap S) = x(U) \cap \left(\mathbb{R}^k \times \{0\}\right) \subseteq \mathbb{R}^k \times \mathbb{R}^{m-k}.$$

In this case, S is given the subspace topology inherited from M.

Proposition 9.: I

f $S \subseteq M$ is a topological k-submanifold (in the sense above), then S is a topological k-manifold.

Proof. Fix $p \in S$ and choose (U, x) as in the definition. Then $U \cap S$ is open in S, and

$$x|_{U \cap S} : U \cap S \longrightarrow x(U) \cap (\mathbb{R}^k \times \{0\})$$

is a homeomorphism onto an open subset of $\mathbb{R}^k \times \{0\} \cong \mathbb{R}^k$. $\square$

Example 9.: $S^2 \subset \mathbb{R}^3$ as a topological submanifold

The unit sphere
$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$$
is a topological 2-submanifold of $\mathbb{R}^3$.

Product

Definition 9.: Product manifold

Let M be a topological m-manifold and N a topological n-manifold. The *product manifold* is the topological space $M \times N$ with the product topology.

Proposition 9.: I

f M is a topological m-manifold and N is a topological n-manifold, then $M \times N$ is a topological $(m + n)$-manifold.

Proof. Let (U, x) be a chart on M and (V, y) a chart on N. Then $U \times V$ is open in $M \times N$, and
$$x \times y : U \times V \to x(U) \times y(V) \subseteq \mathbb{R}^m \times \mathbb{R}^n \cong \mathbb{R}^{m+n}$$
is a homeomorphism onto an open subset. These charts cover $M \times N$. □

Example 9.: Cylinder and torus

The *cylinder* is $S^1 \times (0, 1)$ (or $S^1 \times \mathbb{R}$), a 2-manifold.
The *torus* is $T^2 = S^1 \times S^1$, also a 2-manifold.

Bundle

Definition 9.: Bundle

A *bundle* is a triple (E, π, M) where E and M are topological spaces and
$$\pi : E \to M$$
is a continuous surjective map. We call E the *total space*, M the *base space*, and π the *projection*.

Definition 9.: Fibre

Let $E \xrightarrow{\pi} M$ be a bundle and let $p \in M$. The set
$$F_p := \pi^{-1}(\{p\})$$
is called the *fibre over* p.

Definition 9.: Fibre bundle (local triviality)

Let $E \xrightarrow{\pi} M$ be a bundle and let F be a manifold. We say $E \xrightarrow{\pi} M$ is a *fibre bundle with (typical) fibre* F if for every $p \in M$ there exists an open neighborhood $U \ni p$ and a homeomorphism (a *local trivialization*)

$$\Phi : \pi^{-1}(U) \longrightarrow U \times F$$

such that $\mathrm{pr}_1 \circ \Phi = \pi$ on $\pi^{-1}(U)$.

Example 9.: Product bundle

The projection $\pi : M \times N \to M$, $\pi(m, n) = m$, defines a fibre bundle

$$M \times N \xrightarrow{\pi} M$$

with fibre N (trivial bundle).

Example 9.: Möbius strip

The Möbius strip can be viewed as a fibre bundle $E \xrightarrow{\pi} S^1$ with fibre $[0, 1]$. It is locally a product, but not globally:

$$E \ncong S^1 \times [0, 1].$$

Definition 9.: Section

Let $E \xrightarrow{\pi} M$ be a bundle. A map $\sigma : M \to E$ is called a *(cross-)section* if

$$\pi \circ \sigma = \mathrm{id}_M.$$

Remark 9.Physics translation

In physics, a *field* is often a section of a bundle over spacetime (or over configuration space). Later, vector bundles and principal bundles will be the correct setting for spinor fields and gauge fields.

Quotient

Definition 9.: Quotient space

Let X be a topological space and let $\sim$ be an equivalence relation on X. The *quotient space* $X/\!\sim$ is the set of equivalence classes with the quotient topology: a set $U \subseteq X/\!\sim$ is open iff its preimage under the quotient map $q : X \to X/\!\sim$ is open in X.

Remark 9.Warning

A quotient of a manifold need not be a manifold (it may fail to be Hausdorff, or fail to be locally Euclidean). In practice, to get a *quotient manifold* one often assumes a group action that is *free and proper*.

Example 9.: $\mathbb{RP}^1$ as a quotient manifold

Define an equivalence relation on S^1 by $p \sim -p$. The quotient space

$$\mathbb{RP}^1 := S^1/\{p \sim -p\}$$

is called the real projective line. One can show $\mathbb{RP}^1$ is a 1-dimensional topological manifold. (In fact, $\mathbb{RP}^1 \cong S^1$.)

9.3 Differentiable structure

Topological manifolds let us speak about neighborhoods and continuity. To do calculus, we need differentiability. This is achieved by requiring coordinate changes between overlapping charts to be differentiable.

Regularity classes. Let $\square \in \{C^0, C^1, C^k, C^\infty, C^\omega\}$. On open subsets of Euclidean space:

- C^0: continuous,
- C^1: once differentiable with continuous derivative,
- C^k: k-times differentiable with continuous derivatives up to order k,
- C^∞: smooth (infinitely differentiable),
- C^ω: real-analytic.

Definition 9.: $\square$-compatible charts

Let $(M, \mathcal{O})$ be a topological d-manifold. Two charts (U, x) and (V, y) are called $\square$-*compatible* if either $U \cap V = \varnothing$, or the transition map

$$y \circ x^{-1} : x(U \cap V) \longrightarrow y(U \cap V)$$

is of class $\square$.

Definition 9.: $\square$-atlas

An atlas $\mathscr{A}$ on M is called a $\square$-*atlas* if every pair of charts in $\mathscr{A}$ is $\square$-compatible.

Definition 9.: Compatibility of smooth atlases

Two C^∞-atlases $\mathscr{A}$ and $\mathscr{B}$ on M are *compatible* if $\mathscr{A} \cup \mathscr{B}$ is again a C^∞-atlas.

Definition 9.: Smooth structure / smooth manifold

A *smooth structure* on a topological manifold M is an equivalence class of pairwise compatible C^∞-atlases. A *smooth manifold* is a pair $(M, [\mathscr{A}])$.
Equivalently, starting from one smooth atlas $\mathscr{A}$, there is a unique *maximal atlas* $\mathscr{A}_{\max}$ consisting of all charts that are C^∞-compatible with $\mathscr{A}$. One may specify a smooth manifold as $(M, \mathscr{A}_{\max})$.

Remark 9.

For $\square = C^0$, compatibility is automatic on a topological manifold: transition maps between charts are always continuous. The content begins when $\square$ requires differentiability (especially C^∞).

Example 9.: Transition maps on S^1 (stereographic atlas) are smooth

Consider the stereographic atlas on S^1 from the earlier example:

$$\Big(S^1 \setminus \{N\}, \phi_N\Big), \qquad \Big(S^1 \setminus \{S\}, \phi_S\Big),$$

where $\phi_N(x, y) = \frac{y}{1-x}$ and $\phi_S(x, y) = \frac{y}{1+x}$.
On the overlap $S^1 \setminus \{N, S\}$, the transition map is

$$\phi_S \circ \phi_N^{-1} : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}.$$

Using $\phi_N^{-1}(t) = \left(\frac{t^2-1}{t^2+1}, \frac{2t}{t^2+1}\right)$, we compute

$$(\phi_S \circ \phi_N^{-1})(t) = \frac{\frac{2t}{t^2+1}}{1 + \frac{t^2-1}{t^2+1}} = \frac{\frac{2t}{t^2+1}}{\frac{2t^2}{t^2+1}} = \frac{1}{t}, \qquad t \neq 0.$$

Thus the transition maps are smooth (in fact real-analytic) on their domains.

Definition 9.: Smooth (C^∞) map

Let M and N be smooth manifolds. A function $f : M \to N$ is called *smooth* if for every $p \in M$ there exist charts (U, x) around p and (V, y) around $f(p)$ with $f(U) \subseteq V$ such that

$$y \circ f \circ x^{-1} : x(U) \longrightarrow y(V)$$

is a smooth map between open subsets of Euclidean space.

Proposition 9.: Independence of charts

Let M, N be smooth manifolds and $f : M \to N$. If $y \circ f \circ x^{-1}$ is smooth for one choice of charts (U, x) around p and (V, y) around $f(p)$ (with $f(U) \subset V$), then it is smooth for any other choice of charts around the same points (on the appropriate overlap domains).

Proof. Let (U', x') be another chart at p and (V', y') another chart at $f(p)$. On overlaps we have

$$y' \circ f \circ (x')^{-1} = (y' \circ y^{-1}) \circ (y \circ f \circ x^{-1}) \circ (x \circ (x')^{-1}),$$

a composition of smooth maps (transition maps are smooth by definition of the smooth structure). □

Definition 9.: Smooth real-valued function

Let M be a smooth d-manifold. A function $f : M \to \mathbb{R}$ is called *smooth* if for every chart (U, x), the coordinate representation

$$f \circ x^{-1} : x(U) \longrightarrow \mathbb{R}$$

is smooth.

Definition 9.: Smooth curve

Let M be a smooth manifold and let $I \subseteq \mathbb{R}$ be an interval. A *smooth curve* in M is a map $\gamma : I \to M$ such that for every chart (U, x), the map

$$x \circ \gamma : \gamma^{-1}(U) \longrightarrow \mathbb{R}^d$$

is smooth.

Definition 9.: Diffeomorphism

Let M and N be smooth manifolds. A map $f : M \to N$ is a *diffeomorphism* if it is bijective, smooth, and its inverse $f^{-1} : N \to M$ is smooth. If such a map exists, we say M and N are *diffeomorphic* and write $M \cong_{\text{diff}} N$.

Example 9.: Smooth real-valued functions on S^1

The coordinate projections

$$f_1 : S^1 \to \mathbb{R}, \qquad f_1(x, y) = x, \qquad f_2 : S^1 \to \mathbb{R}, \qquad f_2(x, y) = y,$$

are smooth.
More generally, if $F : \mathbb{R}^2 \to \mathbb{R}$ is smooth, then its restriction $F|_{S^1}$ is smooth on S^1.
For example,

$$f(x, y) = x^2 y + \sin(x)$$

is smooth on S^1.

Example 9.: Smooth curves on S^1

The standard parametrization

$$\gamma : \mathbb{R} \to S^1, \qquad \gamma(t) = (\cos t, \sin t),$$

is a smooth curve.
For any $a, b \in \mathbb{R}$, the map

$$\gamma_{a,b}(t) = (\cos(at + b), \sin(at + b))$$

defines a smooth curve on S^1.

Example 9.: Rotation as a diffeomorphism of S^1

For $\theta \in \mathbb{R}$, define

$$R_\theta : S^1 \to S^1, \qquad R_\theta(x, y) = \Big(x \cos\theta - y \sin\theta,\ x \sin\theta + y \cos\theta\Big).$$

Then R_θ is a diffeomorphism with inverse $R_{-\theta}$.

Example 9.: Stereographic charts on S^1 are diffeomorphisms

The stereographic maps $\phi_N : S^1 \setminus \{N\} \to \mathbb{R}$ and $\phi_S : S^1 \setminus \{S\} \to \mathbb{R}$ are diffeomorphisms: the maps and their inverses are given by rational formulas, hence smooth.

Example 9.: Translation on $\mathbb{R}^n$

For $a \in \mathbb{R}^n$, define

$$T_a : \mathbb{R}^n \to \mathbb{R}^n, \qquad T_a(x) = x + a.$$

Then T_a is a diffeomorphism with inverse T_{-a}.

10 Tangent space, tangent bundle and vector fields

Motivation. In calculus and multivariable calculus, tangential structures play an important role. They provide the best linear approximation of curves and surfaces at a point via the tangent line and tangent plane, respectively. These objects allow us to linearize problems and give rise to important notions such as velocity and rates of change.

Our goal is to generalize this idea to manifolds. In particular, we introduce the concept of the *tangent space* at a point, which will serve as the linear space that approximates the manifold near that point.

10.1 Review: Directional derivatives in $\mathbb{R}^2$ and $\mathbb{R}^3$

Before defining tangent vectors on manifolds, it is helpful to recall the notion of directional derivatives in Euclidean space. In $\mathbb{R}^2$ and $\mathbb{R}^3$, the directional derivative describes the rate of change of a function in a specified direction.

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function and let $v \in \mathbb{R}^n$ be a vector. The directional derivative of f at a point $p \in \mathbb{R}^n$ in the direction v is defined by

$$D_v f(p) = \left.\frac{d}{dt}\right|_{t=0} f(p + tv).$$

Since $\mathbb{R}^n$ is a vector space, it is equipped with natural notions of addition and scalar multiplication. This allows us to define directional derivatives using vectors. In particular, if $v \in \mathbb{R}^n$ and $p \in \mathbb{R}^n$, we consider the straight line

$$p + tv,$$

which represents motion starting at p in the direction v.

However, most manifolds do not come with a vector space structure. In general we cannot add points or multiply them by scalars, so expressions such as $p + tv$ do not make sense. Therefore this definition cannot be directly generalized to manifolds.

What is really important in the definition is not the straight line itself, but the fact that we move through the point p with velocity v at $t = 0$. In fact, many different curves can have the same velocity at a point.

For example, consider the curve

$$\gamma(t) = p + tv + t^2 w, \qquad w \in \mathbb{R}^n.$$

Then

$$\gamma(0) = p, \qquad \gamma'(0) = v.$$

Thus this curve passes through p and has velocity v at $t = 0$, even though it is not the straight line $p + tv$.

This observation suggests that instead of defining directions using straight lines, we can use smooth curves. Curves can be defined on any manifold, so this approach extends naturally beyond Euclidean space.

Using curves, the directional derivative of a smooth function $f : \mathbb{R}^n \to \mathbb{R}$ at a point p in the direction determined by a curve γ with $\gamma(0) = p$ can be written as

$$\left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)).$$

There is another way to describe tangent vectors. Observe that the directional derivative takes smooth functions and satisfies the product rule. This suggests an alternative definition.

Definition 10.: Derivation

Let M be a smooth manifold and $p \in \mathbb{R}^n$. A *derivation at p* is a linear map

$$D : C^\infty(M) \to \mathbb{R}$$

such that for all $f, g \in C^\infty(M)$,

$$D(fg) = f(p)D(g) + g(p)D(f).$$

We will see that derivations at p correspond exactly to tangent vectors at p. Thus the tangent space T_pM can be defined either using curves or using derivations.

10.1.1 Tangent space

We will work with the viewpoint obtained by composing a curve with a smooth function. The curve gives a natural geometric picture of a vector through its velocity at a point, while the use of smooth functions turns this geometric object into an operator, which is exactly what differentiation does.

Definition 10.: Tangent vector at a point

Let M be a smooth manifold and let $p \in M$. A *tangent vector at p* is an operator

$$v : C^\infty(M) \to \mathbb{R}$$

of the form

$$v(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)),$$

where $\gamma : (-\varepsilon, \varepsilon) \to M$ is a smooth curve such that

$$\gamma(0) = p.$$

Remark 10.

Thus a tangent vector at p measures the rate of change of smooth functions along a curve passing through p. Different curves may determine the same tangent vector if they give the same derivative on all smooth functions at p.

Definition 10.: Tangent space

Let M be a smooth manifold and let $p \in M$. The set of all tangent vectors at p is called the *tangent space at* p and is denoted by

$$T_pM.$$

10.1.2 Examples

Let us look at some concrete examples of tangent vectors on familiar manifolds. We will describe them using curves, but always interpret the tangent vector as an operator acting on smooth functions.

Example 10.: The circle S^1

Consider the circle

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

Let $p = (1, 0) \in S^1$, and consider the smooth curve

$$\gamma : (-\varepsilon, \varepsilon) \to S^1, \qquad \gamma(t) = (\cos t, \sin t).$$

Then

$$\gamma(0) = (1, 0) = p.$$

The tangent vector determined by γ at p is the operator

$$v : C^\infty(S^1) \to \mathbb{R}, \qquad v(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)).$$

Thus for every smooth function f on S^1,

$$v(f) = \left.\frac{d}{dt}\right|_{t=0} f(\cos t, \sin t).$$

This is the tangent vector to the circle at the point $(1, 0)$ in the direction of increasing angle.

Example 10.: The cylinder

Consider the cylinder

$$C = S^1 \times \mathbb{R} = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}.$$

Let $p = (1, 0, 0) \in C$.
First consider the curve

$$\gamma(t) = (\cos t, \sin t, 0),$$

which satisfies $\gamma(0) = p$. It determines a tangent vector

$$v(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)) = \left.\frac{d}{dt}\right|_{t=0} f(\cos t, \sin t, 0)$$

for every $f \in C^\infty(C)$.
Now consider the curve

$$\eta(t) = (1, 0, t),$$

which also satisfies $\eta(0) = p$. It determines another tangent vector

$$w(f) = \left.\frac{d}{dt}\right|_{t=0} f(\eta(t)) = \left.\frac{d}{dt}\right|_{t=0} f(1, 0, t).$$

These two tangent vectors represent two different directions on the cylinder: one going around the circular direction, and one going along the axis.

Example 10.: The sphere S^2

Consider the sphere

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}.$$

Let $p = (0, 0, 1) \in S^2$.
Consider the curves

$$\gamma(t) = (\sin t, 0, \cos t), \qquad \eta(t) = (0, \sin t, \cos t).$$

Both satisfy

$$\gamma(0) = \eta(0) = p.$$

They determine tangent vectors

$$v(f) = \frac{d}{dt}\bigg|_{t=0} f(\gamma(t)) = \frac{d}{dt}\bigg|_{t=0} f(\sin t, 0, \cos t),$$

and

$$w(f) = \frac{d}{dt}\bigg|_{t=0} f(\eta(t)) = \frac{d}{dt}\bigg|_{t=0} f(0, \sin t, \cos t),$$

for every $f \in C^\infty(S^2)$.
These represent two different tangent directions at the north pole.

Remark 10.

In each case, the tangent vector is not the curve itself, but the operator obtained by differentiating smooth functions along the curve at $t = 0$. This point of view extends naturally to arbitrary smooth manifolds.

Geometrically, this operator corresponds to the velocity vector $\gamma'(0)$ in the ambient Euclidean space.

Example 10.: The circle S^1

Consider the circle

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

Let $p = (1, 0) \in S^1$, and consider the smooth curve

$$\gamma : (-\varepsilon, \varepsilon) \to S^1, \qquad \gamma(t) = (\cos t, \sin t).$$

Then

$$\gamma(0) = (1, 0) = p.$$

The tangent vector determined by γ at p is the operator

$$v : C^\infty(S^1) \to \mathbb{R}, \qquad v(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)).$$

Let us evaluate this tangent vector on the concrete function

$$f(x, y) = x^2.$$

Then

$$f(\gamma(t)) = f(\cos t, \sin t) = (\cos t)^2.$$

Thus

$$v(f) = \left.\frac{d}{dt}\right|_{t=0} (\cos t)^2 = \left.\frac{d}{dt}\right|_{t=0} \cos^2 t.$$

Since

$$\frac{d}{dt} \cos^2 t = 2 \cos t(-\sin t) = -2 \cos t \sin t,$$

we obtain

$$v(f) = 0.$$

This computation illustrates how a tangent vector acts as a differential operator on smooth functions.

If instead we take

$$g(x, y) = y,$$

then

$$g(\gamma(t)) = \sin t,$$

so

$$v(g) = \left.\frac{d}{dt}\right|_{t=0} \sin t = 1.$$

10.2 T_pM as a vector space

Motivation. Manifolds are in general nonlinear and geometrically complicated objects. However, the tangent space at each point has a much simpler structure: it is a real vector

space. This is what makes tangent spaces so useful — they allow us to linearize problems locally, turning questions about a curved manifold into linear algebra.

Proposition 10.: T_pM is a vector space

Let M be a smooth d-dimensional manifold and let $p \in M$. The tangent space T_pM is a real vector space of dimension d, with operations defined as follows.

- **Addition.** Given tangent vectors $v, w \in T_pM$, their sum $v+w$ is the tangent vector defined by

$$(v+w)(f) \;=\; v(f) + w(f), \qquad f \in C^\infty(M).$$

- **Scalar multiplication.** Given $v \in T_pM$ and $\lambda \in \mathbb{R}$, the tangent vector λv is defined by

$$(\lambda v)(f) \;=\; \lambda\, v(f), \qquad f \in C^\infty(M).$$

Remark 10.

One checks that both operations are again derivations at p, so T_pM is closed under addition and scalar multiplication. The zero vector corresponds to the derivation $f \mapsto 0$. The fact that $\dim T_pM = d$ follows from the existence of a coordinate chart (U, x) around p, which provides a basis

$$\left\{ \left.\frac{\partial}{\partial x^1}\right|_p, \;\ldots,\; \left.\frac{\partial}{\partial x^d}\right|_p \right\}$$

for T_pM. We will make this precise in the next subsection.

Geometric picture. The following figures illustrate the tangent space at a point for three familiar manifolds.

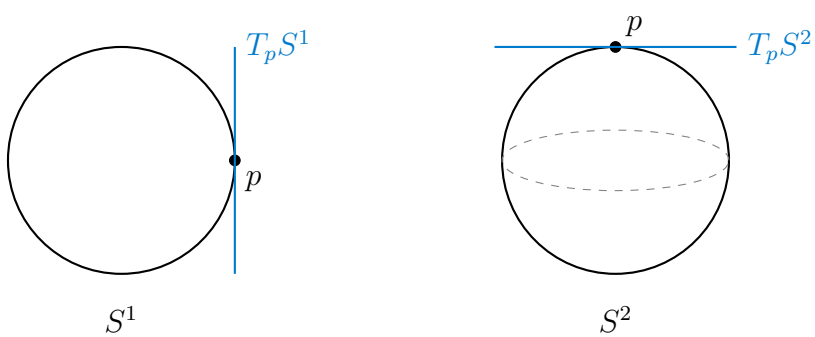


Figure 6: Left: the tangent space T_pS^1 at a point p is a copy of $\mathbb{R}$. Right: the tangent space T_pS^2 at the north pole is a copy of $\mathbb{R}^2$.

10.3 Dimension of T_pM

Since T_pM is a vector space, a natural question arises: what is its dimension, and how does it relate to the dimension of the manifold M?

Proposition 10.: $\dim T_pM = \dim M$

Let M be a smooth d-dimensional manifold and let $p \in M$. Then

$$\dim T_pM = d.$$

More precisely, if (U, x) is any chart around p, the vectors

$$\left.\frac{\partial}{\partial x^1}\right|_p, \ \ldots, \ \left.\frac{\partial}{\partial x^d}\right|_p,$$

defined by

$$\left.\frac{\partial}{\partial x^i}\right|_p (f) \ = \ \left.\frac{\partial (f \circ x^{-1})}{\partial r^i}\right|_{x(p)}, \qquad f \in C^\infty(M),$$

form a basis for T_pM.

Remark 10.

It is important to note that the dimension here refers to two different kinds of dimension, though they turn out to be equal:

- $\dim M$ is the *topological/manifold dimension* — the dimension of the Euclidean space that M locally looks like via its charts.
- $\dim T_pM$ is the *algebraic dimension* of T_pM as a real vector space — the number of elements in any basis.

The proposition says these two notions always agree. Geometrically, this means the tangent space captures exactly the degrees of freedom available at each point of M.

Examples

Let us verify this for the manifolds we have already seen.

Example 10.: The circle S^1

S^1 is a 1-dimensional manifold, so $\dim T_pS^1 = 1$ for every $p \in S^1$. The tangent space at any point is a copy of $\mathbb{R}$ — a single line. At the point $p = (1, 0)$, the basis vector is

$$\left.\frac{d}{dt}\right|_{t=0} (\cos t, \sin t) = (0, 1),$$

which points in the direction of increasing angle along the circle.

Example 10.: The cylinder $S^1 \times \mathbb{R}$

The cylinder is a 2-dimensional manifold, so $\dim T_p(S^1 \times \mathbb{R}) = 2$. At the point $p = (1, 0, 0)$, we found two tangent vectors earlier: one going around the circular direction and one going along the axis. These two vectors are linearly independent and together form a basis for $T_pM \cong \mathbb{R}^2$.

Example 10.: The sphere S^2

S^2 is a 2-dimensional manifold, so $\dim T_pS^2 = 2$ for every $p \in S^2$. The tangent space at any point is a copy of $\mathbb{R}^2$ — a plane. At the north pole $p = (0, 0, 1)$, the two basis vectors

$$v(f) = \frac{d}{dt}\bigg|_{t=0} f(\sin t, 0, \cos t), \qquad w(f) = \frac{d}{dt}\bigg|_{t=0} f(0, \sin t, \cos t)$$

span the horizontal tangent plane T_pS^2.

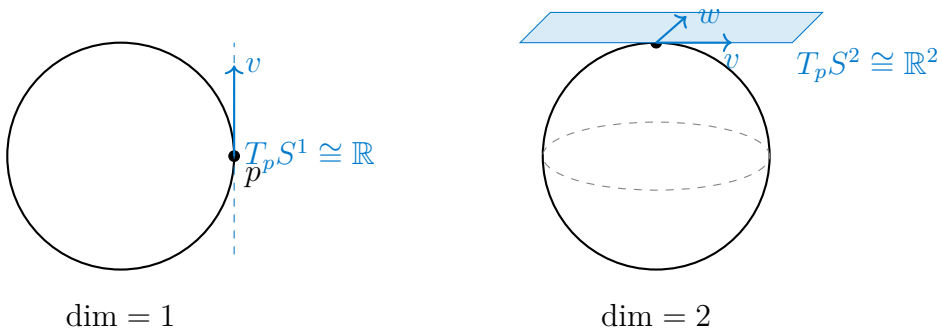


Figure 7: Left: T_pS^1 is a 1-dimensional line. Right: T_pS^2 is a 2-dimensional plane. In both cases $\dim T_pM = \dim M$.

Geometric picture.

10.4 The coordinate basis

Given a chart (U, x) around $p \in M$, with local coordinates $x = (x^1, \ldots, x^d)$, the *coordinate basis vectors* at p are the tangent vectors

$$\frac{\partial}{\partial x^i}\bigg|_p \in T_pM, \qquad i = 1, \ldots, d,$$

defined by their action on smooth functions $f \in C^\infty(M)$:

$$\frac{\partial}{\partial x^i}\bigg|_p (f) = \frac{\partial(f \circ x^{-1})}{\partial r^i}\bigg|_{x(p)}.$$

Here $r^1, \ldots, r^d$ are the standard coordinates on $\mathbb{R}^d$, so $f \circ x^{-1} : x(U) \subset \mathbb{R}^d \to \mathbb{R}$ is an ordinary function on Euclidean space and the right-hand side is a standard partial derivative.

By the proposition above, these d vectors form a basis for T_pM. Different charts around p give different bases, related by the Jacobian of the transition map. We now compute these bases explicitly in all our running examples.

10.4.1 The circle S^1

S^1 is a 1-dimensional manifold, so each tangent space T_pS^1 has a single basis vector coming from the chosen chart.

The angle atlas. Cover S^1 by the two charts

$$U_1 = S^1 \setminus \{(-1,0)\}, \quad \theta_1 : U_1 \to (-\pi, \pi), \quad \theta_1(\cos\theta, \sin\theta) = \theta,$$

$$U_2 = S^1 \setminus \{(1,0)\}, \quad \theta_2 : U_2 \to (0, 2\pi), \quad \theta_2(\cos\theta, \sin\theta) = \theta.$$

The inverse of θ_1 (and similarly θ_2) is

$$\theta_1^{-1}(t) = (\cos t, \sin t).$$

Take the point $p = (1,0) \in U_1$, so $\theta_1(p) = 0$. The coordinate basis vector at p is

$$\left.\frac{\partial}{\partial\theta_1}\right|_p (f) = \left.\frac{d(f \circ \theta_1^{-1})}{dt}\right|_{t=0} = \left.\frac{d}{dt}\right|_{t=0} f(\cos t, \sin t).$$

This is exactly the tangent vector we computed earlier using the curve $\gamma(t) = (\cos t, \sin t)$. The coordinate basis reproduces our curve-based computation.

Example 10.: Basis vector on S^1 — angle chart

Let $f(x,y) = y$. Then

$$\left.\frac{\partial}{\partial\theta_1}\right|_p (f) = \left.\frac{d}{dt}\right|_{t=0} \sin t = 1.$$

Let $g(x,y) = x^2$. Then

$$\left.\frac{\partial}{\partial\theta_1}\right|_p (g) = \left.\frac{d}{dt}\right|_{t=0} \cos^2 t = -2\cos(0)\sin(0) = 0.$$

So in the angle chart, $\left.\frac{\partial}{\partial\theta_1}\right|_p$ acts exactly like differentiation with respect to arc length along S^1 at p.

The stereographic atlas. Cover S^1 by two stereographic charts, projecting from the north pole $N = (0, 1)$ and the south pole $S = (0, -1)$:

$$U_N = S^1 \setminus \{(0,1)\}, \quad \phi_N(a,b) = \frac{a}{1-b}, \quad \phi_N^{-1}(t) = \left(\frac{2t}{t^2+1}, \frac{t^2-1}{t^2+1}\right),$$

$$U_S = S^1 \setminus \{(0,-1)\}, \quad \phi_S(a,b) = \frac{a}{1+b}, \quad \phi_S^{-1}(t) = \left(\frac{2t}{t^2+1}, \frac{1-t^2}{t^2+1}\right).$$

Take again $p = (1, 0)$. Then $\phi_N(1, 0) = 1$, so the coordinate basis vector is

$$\left.\frac{\partial}{\partial \phi_N}\right|_p (f) = \left.\frac{d(f \circ \phi_N^{-1})}{dt}\right|_{t=1} = \left.\frac{d}{dt}\right|_{t=1} f\left(\frac{2t}{t^2+1}, \frac{t^2-1}{t^2+1}\right).$$

Example 10.: Basis vector on S^1 — stereographic chart

Let $f(x, y) = x$. Then

$$f\left(\phi_N^{-1}(t)\right) = \frac{2t}{t^2+1}.$$

Differentiating:

$$\frac{d}{dt}\frac{2t}{t^2+1} = \frac{2(t^2+1) - 2t \cdot 2t}{(t^2+1)^2} = \frac{2-2t^2}{(t^2+1)^2}.$$

At $t = 1$:

$$\left.\frac{\partial}{\partial \phi_N}\right|_p (f) = \frac{2-2}{4} = 0.$$

Let $g(x, y) = y$. Then $g(\phi_N^{-1}(t)) = \frac{t^2-1}{t^2+1}$, so

$$\frac{d}{dt}\frac{t^2-1}{t^2+1} = \frac{2t(t^2+1) - (t^2-1)2t}{(t^2+1)^2} = \frac{4t}{(t^2+1)^2}.$$

At $t = 1$:

$$\left.\frac{\partial}{\partial \phi_N}\right|_p (g) = \frac{4}{4} = 1.$$

Comparing with the angle chart: both bases give the same values on f and g, confirming they represent the same tangent vector — just expressed in different coordinates. The two bases are related by the derivative of the transition map $\phi_N \circ \theta_1^{-1}$.

10.4.2 The cylinder $S^1 \times \mathbb{R}$

The cylinder $C = S^1 \times \mathbb{R}$ has dimension 2, so each T_pC has a 2-element basis. A natural chart comes from combining the angle coordinate on S^1 with the standard coordinate on $\mathbb{R}$:

$$(\theta, z) : (U_1 \times \mathbb{R}) \to (-\pi, \pi) \times \mathbb{R}, \qquad (\theta, z)(\cos\theta, \sin\theta, z) = (\theta, z).$$

The inverse is $(\theta, z)^{-1}(s, t) = (\cos s, \sin s, t)$.

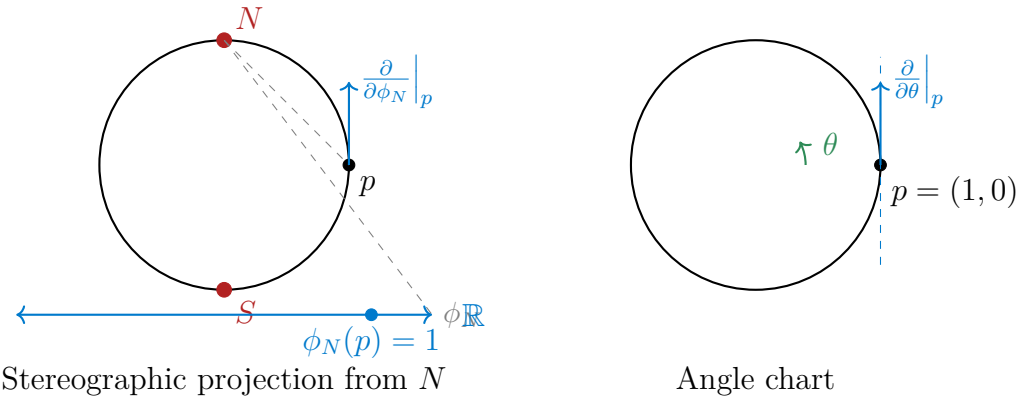


Figure 8: Two atlases on S^1, both producing the same tangent vector at $p = (1, 0)$.

Take $p = (1, 0, 0)$, so $(\theta, z)(p) = (0, 0)$. The two coordinate basis vectors at p act on $f \in C^\infty(C)$ by

$$\frac{\partial}{\partial \theta}\bigg|_p (f) = \frac{\partial (f \circ (\theta, z)^{-1})}{\partial s}\bigg|_{(0,0)} = \frac{d}{dt}\bigg|_{t=0} f(\cos t, \sin t, 0),$$

$$\frac{\partial}{\partial z}\bigg|_p (f) = \frac{\partial (f \circ (\theta, z)^{-1})}{\partial t}\bigg|_{(0,0)} = \frac{d}{dt}\bigg|_{t=0} f(1, 0, t).$$

These match exactly the two curves $\gamma(t) = (\cos t, \sin t, 0)$ and $\eta(t) = (1, 0, t)$ we used earlier. The coordinate chart simply names them.

Example 10.: Coordinate basis on the cylinder

Let $f(x, y, z) = z^2 + x$. Then:

$$\frac{\partial}{\partial \theta}\bigg|_p (f) = \frac{d}{dt}\bigg|_{t=0} \Big(\cos t + 0\Big) = -\sin(0) = 0.$$

$$\frac{\partial}{\partial z}\bigg|_p (f) = \frac{d}{dt}\bigg|_{t=0} \Big(t^2 + 1\Big) = 0.$$

Let $g(x, y, z) = y$. Then:

$$\frac{\partial}{\partial \theta}\bigg|_p (g) = \frac{d}{dt}\bigg|_{t=0} \sin t = 1, \qquad \frac{\partial}{\partial z}\bigg|_p (g) = 0.$$

The two basis vectors are independent: $\frac{\partial}{\partial \theta}\Big|_p$ detects angular motion and $\frac{\partial}{\partial z}\Big|_p$ detects axial motion.

10.4.3 The sphere S^2

S^2 is a 2-dimensional manifold. We compute the coordinate basis in two different atlases.

Spherical coordinates. The spherical coordinate chart is

$$(\theta, \varphi) : U \to (0, \pi) \times (0, 2\pi), \qquad (\theta, \varphi)^{-1}(s, t) = (\sin s \cos t,\ \sin s \sin t,\ \cos s),$$

where $U = S^2$ with the two meridians $\varphi = 0$ and $\varphi = 2\pi$ removed. Take the point $p = (0, 0, 1)$... but note p is not in this chart ($\theta = 0$ is the boundary). So take instead $p = (1, 0, 0)$, where $(\theta, \varphi)(p) = (\pi/2, 0)$.

The two coordinate basis vectors act on $f \in C^\infty(S^2)$ by

$$\left.\frac{\partial}{\partial\theta}\right|_p (f) = \left.\frac{\partial}{\partial s}\right|_{(\pi/2,0)} f(\sin s \cos 0,\ \sin s \sin 0,\ \cos s) = \left.\frac{d}{ds}\right|_{s=\pi/2} f(\sin s,\ 0,\ \cos s),$$

$$\left.\frac{\partial}{\partial\varphi}\right|_p (f) = \left.\frac{\partial}{\partial t}\right|_{(\pi/2,0)} f(\sin(\pi/2) \cos t,\ \sin(\pi/2) \sin t,\ \cos(\pi/2)) = \left.\frac{d}{dt}\right|_{t=0} f(\cos t,\ \sin t,\ 0).$$

Example 10.: Coordinate basis on S^2 — spherical chart

Let $f(x, y, z) = z$. Then:

$$\left.\frac{\partial}{\partial\theta}\right|_p (f) = \left.\frac{d}{ds}\right|_{s=\pi/2} \cos s = -\sin(\pi/2) = -1.$$

$$\left.\frac{\partial}{\partial\varphi}\right|_p (f) = \left.\frac{d}{dt}\right|_{t=0} 0 = 0.$$

Let $g(x, y, z) = y$. Then:

$$\left.\frac{\partial}{\partial\theta}\right|_p (g) = 0, \qquad \left.\frac{\partial}{\partial\varphi}\right|_p (g) = \left.\frac{d}{dt}\right|_{t=0} \sin t = 1.$$

So $\left.\frac{\partial}{\partial\theta}\right|_p$ points in the direction of decreasing latitude (towards the south pole) and $\left.\frac{\partial}{\partial\varphi}\right|_p$ points in the direction of increasing longitude.

The stereographic atlas. Cover S^2 by two charts projecting from the north pole $N = (0, 0, 1)$ and south pole $S = (0, 0, -1)$:

$$\psi_N : S^2 \setminus \{N\} \to \mathbb{R}^2, \qquad \psi_N(x, y, z) = \left(\frac{x}{1-z}, \frac{y}{1-z}\right),$$

with inverse

$$\psi_N^{-1}(u, v) = \left(\frac{2u}{u^2+v^2+1}, \frac{2v}{u^2+v^2+1}, \frac{u^2+v^2-1}{u^2+v^2+1}\right).$$

Take $p = (1, 0, 0)$, so $\psi_N(p) = (1, 0)$.

The two coordinate basis vectors at p are:

$$\left.\frac{\partial}{\partial u}\right|_p (f) = \left.\frac{\partial(f \circ \psi_N^{-1})}{\partial u}\right|_{(1,0)}, \qquad \left.\frac{\partial}{\partial v}\right|_p (f) = \left.\frac{\partial(f \circ \psi_N^{-1})}{\partial v}\right|_{(1,0)}.$$

Example 10.: Basis vector on S^2 — stereographic chart

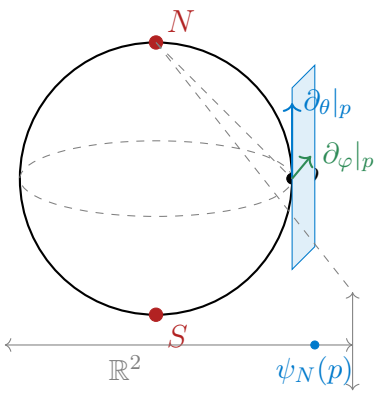
Let $f(x, y, z) = y$. Along the u-direction at $(u, v) = (1, 0)$:

$$f(\psi_N^{-1}(u, 0)) = \frac{0}{u^2 + 1} = 0 \quad \Rightarrow \quad \left.\frac{\partial}{\partial u}\right|_p (f) = 0.$$

Along the v-direction:

$$f(\psi_N^{-1}(1, v)) = \left.\frac{2v}{v^2 + 2}\right|_{v=0}, \qquad \frac{d}{dv}\left.\frac{2v}{v^2 + 2}\right|_{v=0} = \frac{2 \cdot 2}{4} = 1.$$

So $\left.\frac{\partial}{\partial v}\right|_p$ detects the y-direction, consistent with the spherical computation above.



Stereographic projection from N

Figure 9: The stereographic chart on S^2 and the two coordinate basis vectors $\partial_\theta|_p$, $\partial_\varphi|_p$ spanning T_pS^2 at $p = (1, 0, 0)$.

10.5 The pushforward

Observation 10.

Now that we have a notion of tangential structure at every point of a manifold, we can study how smooth maps between manifolds interact with these tangent spaces. Given a smooth map $f : M \to N$, we want to define a linear map that sends tangent vectors at $p \in M$ to tangent vectors at $f(p) \in N$. This is the *pushforward* of f at p.

Definition 10.: Pushforward

Let M and N be smooth manifolds, let $f : M \to N$ be a smooth map, and let $p \in M$. The *pushforward of f at p* is the linear map

$$df_p : T_pM \longrightarrow T_{f(p)}N$$

defined as follows. Given a tangent vector $v \in T_pM$, represented by a smooth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = p$, the pushforward $df_p(v) \in T_{f(p)}N$ is the tangent vector represented by the composed curve $f \circ \gamma$:

$$df_p(v)(g) \;=\; \frac{d}{dt}\bigg|_{t=0} g\big(f(\gamma(t))\big) \;=\; v(g \circ f), \qquad g \in C^\infty(N).$$

Remark 10.

The definition says: to push v forward through f, just compose the curve γ with f and differentiate. The new curve $f \circ \gamma$ lives in N, starts at $f(p)$, and its velocity at $t = 0$ is $df_p(v)$.
In terms of coordinate bases, if (U, x) is a chart near p on M and (V, y) a chart near $f(p)$ on N, the pushforward is represented by the **Jacobian matrix** of $y \circ f \circ x^{-1}$:

$$df_p\left(\frac{\partial}{\partial x^i}\bigg|_p\right) \;=\; \sum_{j=1}^{\dim N} \frac{\partial\big(y^j \circ f \circ x^{-1}\big)}{\partial r^i}\bigg|_{x(p)} \frac{\partial}{\partial y^j}\bigg|_{f(p)}.$$

The pushforward is therefore the intrinsic, coordinate-free version of the Jacobian.

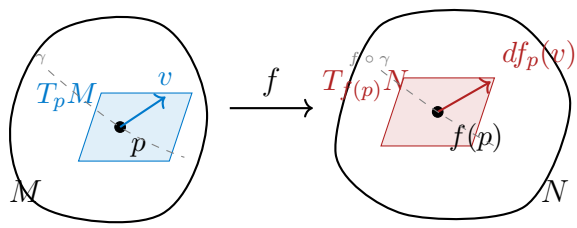


Figure 10: The pushforward df_p sends a tangent vector $v \in T_pM$ to $df_p(v) \in T_{f(p)}N$ by composing the representing curve γ with f.

Examples

1. Rotation of S^1. Let $f : S^1 \to S^1$ be the rotation by angle α:

$$f(\cos\theta, \sin\theta) = (\cos(\theta + \alpha), \sin(\theta + \alpha)).$$

Take $p = (1, 0)$ and $v = \frac{\partial}{\partial\theta}\big|_p$. The composed curve is

$$f(\gamma(t)) = (\cos(t + \alpha), \sin(t + \alpha)),$$

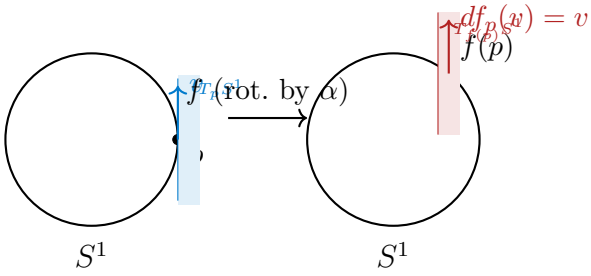


Figure 11: Rotation by α: the pushforward preserves the tangent vector.

so

$$df_p(v)(g) = \frac{d}{dt}\bigg|_{t=0} g(\cos(t+\alpha), \sin(t+\alpha)) = \frac{\partial}{\partial\theta}\bigg|_{f(p)} g.$$

Thus $df_p\left(\frac{\partial}{\partial\theta}\Big|_p\right) = \frac{\partial}{\partial\theta}\Big|_{f(p)}$. The pushforward simply *rotates the base point* while preserving the direction of the tangent vector. This makes sense: rotation is an isometry, so it preserves all geometric structure.

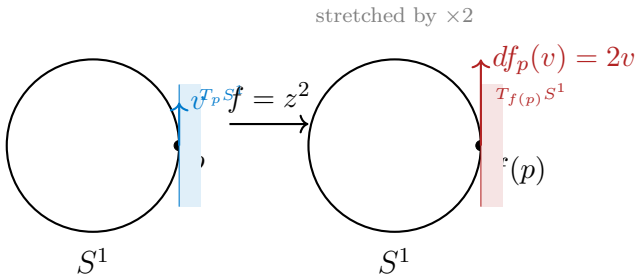


Figure 12: Double cover $z \mapsto z^2$: the pushforward stretches the tangent vector by a factor of 2.

2. The double cover $z \mapsto z^2$ on S^1. Identify $S^1 \subset \mathbb{C}$ and define $f : S^1 \to S^1$ by $f(e^{i\theta}) = e^{2i\theta}$. In real coordinates, $f(\cos\theta, \sin\theta) = (\cos 2\theta, \sin 2\theta)$. Take $p = (1, 0)$, so $f(p) = (1, 0)$ as well. The composed curve is

$$f(\gamma(t)) = (\cos 2t,\ \sin 2t),$$

so

$$df_p\left(\frac{\partial}{\partial\theta}\bigg|_p\right)(g) = \frac{d}{dt}\bigg|_{t=0} g(\cos 2t, \sin 2t) = 2\,\frac{\partial}{\partial\theta}\bigg|_{f(p)} g.$$

Hence

$$df_p\left(\frac{\partial}{\partial\theta}\bigg|_p\right) = 2\,\frac{\partial}{\partial\theta}\bigg|_{f(p)}.$$

The pushforward *stretches* the tangent vector by a factor of 2. Every point in S^1 has two preimages under f, and the factor 2 reflects the fact that f wraps the circle twice around itself.

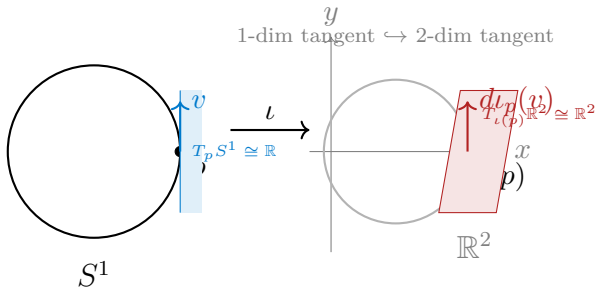


Figure 13: Inclusion $\iota : S^1 \hookrightarrow \mathbb{R}^2$: the 1-dimensional T_pS^1 embeds injectively into the 2-dimensional $T_{\iota(p)}\mathbb{R}^2$. This is the prototype of an immersion.

3. Inclusion $\iota : S^1 \hookrightarrow \mathbb{R}^2$. Let $\iota : S^1 \to \mathbb{R}^2$ be the natural inclusion $\iota(a, b) = (a, b)$. Take $p = (1, 0)$ and $v = \frac{\partial}{\partial\theta}\Big|_p$. The composed curve is $\iota(\gamma(t)) = (\cos t, \sin t)$ in $\mathbb{R}^2$, so

$$d\iota_p(v)(g) = \frac{d}{dt}\Big|_{t=0} g(\cos t, \sin t) \qquad g \in C^\infty(\mathbb{R}^2).$$

In the standard basis $\{\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\}$ of $T_p\mathbb{R}^2 \cong \mathbb{R}^2$ this reads:

$$d\iota_p(v) = -\sin(0)\,\frac{\partial}{\partial x}\Big|_p + \cos(0)\,\frac{\partial}{\partial y}\Big|_p = \frac{\partial}{\partial y}\Big|_p.$$

The pushforward *embeds* $T_pS^1 \cong \mathbb{R}$ as a subspace of $T_p\mathbb{R}^2 \cong \mathbb{R}^2$. This is an injective linear map — the tangent line to S^1 sits inside the ambient plane. This is the prototype of an **immersion**: a map whose pushforward is injective at every point.

4. Inclusion $\iota : S^2 \hookrightarrow \mathbb{R}^3$. The same story works one dimension higher. Let $\iota : S^2 \to \mathbb{R}^3$ be the inclusion. At $p = (0, 0, 1)$ with the two basis vectors $v = \frac{\partial}{\partial\theta}\Big|_p$ and $w = \frac{\partial}{\partial\varphi}\Big|_p$, the pushforward embeds $T_pS^2 \cong \mathbb{R}^2$ as the horizontal plane $\{z = 1\} \subset T_p\mathbb{R}^3 \cong \mathbb{R}^3$. In standard coordinates:

$$d\iota_p(v) = \frac{\partial}{\partial z}\Big|_p \cdot (-1) = -\frac{\partial}{\partial z}\Big|_p, \qquad d\iota_p(w) = \frac{\partial}{\partial x}\Big|_p \cdot 0 + \frac{\partial}{\partial y}\Big|_p \cdot 1 = \frac{\partial}{\partial y}\Big|_p.$$

Again, $d\iota_p$ is injective: T_pS^2 sits as a 2-dimensional subspace of the 3-dimensional $T_p\mathbb{R}^3$.

5. The antipodal map on S^2. Define $A : S^2 \to S^2$ by $A(x, y, z) = (-x, -y, -z)$. Take $p = (0, 0, 1)$ and the curve $\gamma(t) = (\sin t, 0, \cos t)$. Then $A(\gamma(t)) = (-\sin t, 0, -\cos t)$, so

$$dA_p(v)(g) = \frac{d}{dt}\Big|_{t=0} g(-\sin t, 0, -\cos t).$$

Comparing with the curve $\sigma(t) = (-\sin t, 0, -\cos t)$, the pushforward sends $v \mapsto -v$ (with the sign coming from the chain rule). More precisely, $dA_p = -\mathrm{id}$ on T_pS^2. The antipodal map *reverses* all tangent vectors at every point.

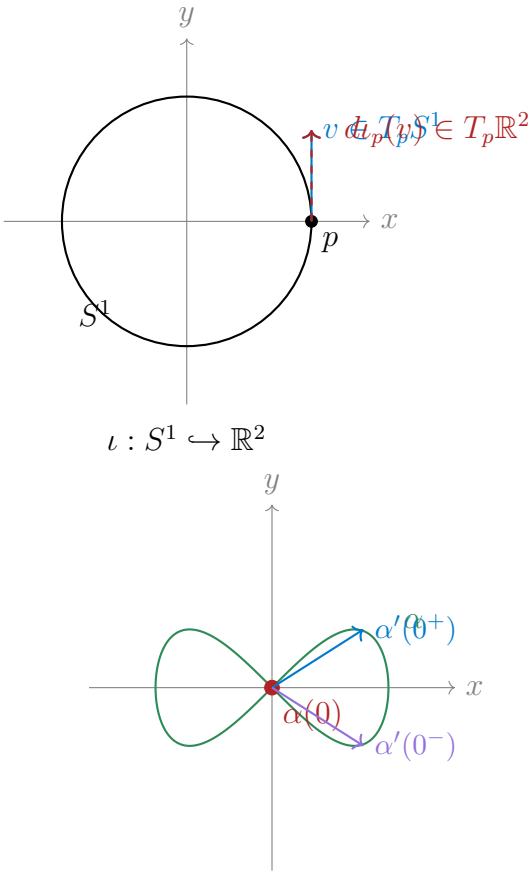


Figure-eight: immersion, not embedding

Figure 14: Left: the inclusion $\iota : S^1 \hookrightarrow \mathbb{R}^2$ is an embedding — the pushforward is injective and ι is globally injective. Right: the figure-eight is an immersion (pushforward injective everywhere) but not an embedding — the curve crosses itself.

6. A smooth map $f : \mathbb{R} \to S^1$ — winding. Define $f : \mathbb{R} \to S^1$ by $f(t) = (\cos t, \sin t)$, the universal cover of S^1. At any point $p = t_0 \in \mathbb{R}$, we have $T_{t_0}\mathbb{R} = \mathbb{R}$ with basis $\frac{d}{dt}\big|_{t_0}$. The pushforward is

$$df_{t_0}\left(\frac{d}{dt}\bigg|_{t_0}\right)(g) = \frac{d}{dt}\bigg|_{t=t_0} g(\cos t, \sin t) = \frac{\partial}{\partial\theta}\bigg|_{f(t_0)} g.$$

So df_{t_0} is an isomorphism $\mathbb{R} \to T_{f(t_0)}S^1$ at every point. The pushforward is always an isomorphism, yet f is not injective — it winds around S^1 infinitely many times. This previews the distinction between a **local diffeomorphism** (pushforward is an isomorphism everywhere) and a **global diffeomorphism** (also bijective).

7. Projection $\pi : S^1 \times \mathbb{R} \to S^1$ — a submersion. Let $\pi(a, b, z) = (a, b)$ be the projection from the cylinder onto the circle. At $p = (1, 0, 0)$ with basis $\{\frac{\partial}{\partial\theta}\big|_p, \frac{\partial}{\partial z}\big|_p\}$, the pushforward is

$$d\pi_p\left(\frac{\partial}{\partial\theta}\bigg|_p\right) = \frac{\partial}{\partial\theta}\bigg|_{\pi(p)}, \qquad d\pi_p\left(\frac{\partial}{\partial z}\bigg|_p\right) = 0.$$

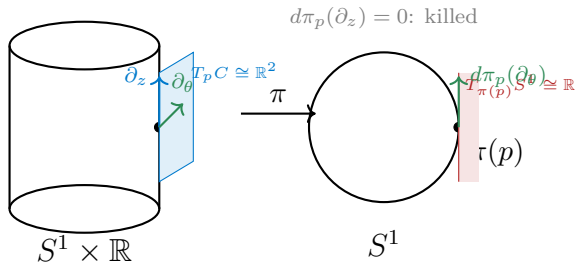


Figure 15: Projection $\pi : S^1 \times \mathbb{R} \to S^1$: the z-direction is killed by $d\pi_p$, the θ-direction survives. The pushforward is surjective — this is a submersion.

The z-direction gets *killed*: $\ker d\pi_p = \operatorname{span}\{\frac{\partial}{\partial z}\big|_p\}$. The pushforward is surjective but not injective — this is the prototype of a **submersion**. Every tangent vector on S^1 has many preimages (one for each value of the z-component).

Remark 10.

The examples above preview three fundamental types of smooth maps:

- **Immersion**: df_p is injective at every p (examples 3, 4 — inclusions of spheres).
- **Submersion**: df_p is surjective at every p (example 7 — projection from the cylinder).
- **Local diffeomorphism**: df_p is an isomorphism at every p (examples 1, 2, 6 — rotations and covering maps).

An **embedding** is an immersion that is also a homeomorphism onto its image (example 3, but not the figure-eight). We will return to all of these in detail later.

10.6 The tangent bundle and vector fields

Even though tangent spaces are local structures — each T_pM lives at a single point p — they fit together to form a global object called the *tangent bundle*. The tangent bundle is the natural setting for generalising one of the most important notions from Euclidean calculus: **vector fields**. In $\mathbb{R}^n$ a vector field is simply an assignment of a vector to each point. On a manifold, the vectors at different points live in different tangent spaces, so we need the tangent bundle to make sense of this assignment globally.

The tangent bundle

Definition 10.: Tangent bundle

Let M be a smooth d-dimensional manifold. The *tangent bundle* of M is the disjoint union of all tangent spaces:

$$TM = \bigsqcup_{p \in M} T_pM = \{(p, v) \mid p \in M,\ v \in T_pM\}.$$

It comes equipped with the *canonical projection*

$$\pi : TM \longrightarrow M, \qquad \pi(p, v) = p,$$

which sends each tangent vector to its base point.

Remark 10.

An element of TM is a pair (p, v) where $p \in M$ is a point and $v \in T_pM$ is a tangent vector at p. The projection π simply forgets the vector and remembers only the point. The *fiber* of π over a point p is exactly T_pM:

$$\pi^{-1}(p) = T_pM.$$

Proposition 10.: TM is a smooth manifold and a vector bundle

Let M be a smooth d-dimensional manifold. Then:

1. TM admits a natural smooth manifold structure of dimension $2d$.
2. The projection $\pi : TM \to M$ is a smooth surjective map.
3. Every point $p \in M$ has an open neighbourhood $U \subset M$ such that $\pi^{-1}(U) \cong U \times \mathbb{R}^d$ as smooth manifolds, with π corresponding to the projection onto the first factor.

In particular, TM is a *vector bundle of rank* d over M, with fiber $T_pM \cong \mathbb{R}^d$ over each point p.

Remark 10.

The dimension count $\dim TM = 2d$ is easy to understand: a point in TM is a pair (p, v), so it takes d coordinates to specify $p \in M$ and another d coordinates to specify the vector $v \in T_pM$. Hence $2d$ in total.
The local triviality condition in (3) says that TM *locally looks like a product* $U \times \mathbb{R}^d$ — but globally it may be twisted. Whether TM is globally a product $M \times \mathbb{R}^d$ depends on the topology of M. When it is, we say M is *parallelisable*.

Examples

Example 10.: $T\mathbb{R} \cong \mathbb{R} \times \mathbb{R}$

For $M = \mathbb{R}$, every tangent space $T_p\mathbb{R}$ is canonically identified with $\mathbb{R}$ via the basis vector $\frac{d}{dt}\Big|_p$. So

$$T\mathbb{R} \;=\; \bigsqcup_{p\in\mathbb{R}} T_p\mathbb{R} \;\cong\; \mathbb{R}\times\mathbb{R},$$

with a point $(p, v) \in T\mathbb{R}$ corresponding to the pair (position, velocity). This is the simplest example of a trivial bundle. More generally, $T\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$.

Example 10.: $TS^1 \cong S^1 \times \mathbb{R}$

For the circle, every tangent space T_pS^1 is a copy of $\mathbb{R}$ spanned by $\frac{\partial}{\partial\theta}\Big|_p$. As we move p around the circle, these lines fit together *without any twist*, so

$$TS^1 \;\cong\; S^1 \times \mathbb{R}.$$

S^1 is parallelisable: the tangent bundle is a trivial product. Geometrically, TS^1 looks like an infinite cylinder — the base circle with a real line attached at each point.

Example 10.: $T(S^1 \times \mathbb{R}) \cong (S^1 \times \mathbb{R}) \times \mathbb{R}^2$

The cylinder $C = S^1 \times \mathbb{R}$ is also parallelisable. At each point $p = (\theta, z)$ we have $T_pC = \operatorname{span}\left\{\frac{\partial}{\partial\theta}\Big|_p, \frac{\partial}{\partial z}\Big|_p\right\} \cong \mathbb{R}^2$, and these frames vary smoothly and without twisting, giving

$$TC \;\cong\; (S^1 \times \mathbb{R}) \times \mathbb{R}^2.$$

Example 10.: TS^2 **— a non-trivial bundle**

For the 2-sphere S^2, the tangent bundle TS^2 is *not* isomorphic to the trivial bundle $S^2 \times \mathbb{R}^2$. This is the content of the **hairy ball theorem**: there is no smooth non-vanishing vector field on S^2, which means there is no global frame for TS^2. Any attempt to consistently assign a basis to each tangent space T_pS^2 as p varies over S^2 must fail somewhere.

This is a fundamental difference between S^1 (parallelisable) and S^2 (not parallelisable), and it is a global topological obstruction — not something one can see in any single chart.

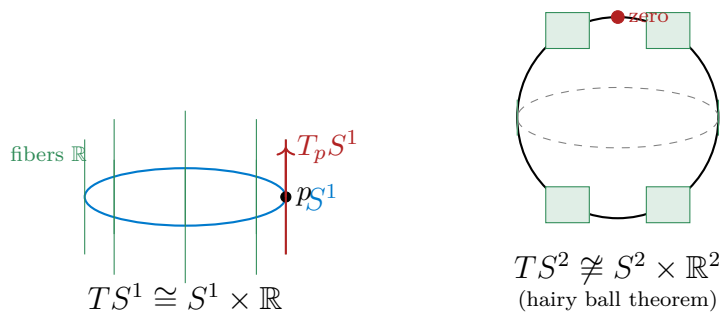


Figure 16: Left: TS^1 is a trivial bundle — a cylinder. The fibers are copies of $\mathbb{R}$ attached smoothly to each point of S^1. Right: TS^2 is non-trivial — any global vector field must vanish somewhere (marked in red).

Vector fields

Definition 10.: Vector field

Let M be a smooth manifold. A *vector field* on M is a smooth map

$$X : M \longrightarrow TM$$

such that $\pi \circ X = \mathrm{id}_M$, i.e. $X(p) \in T_pM$ for every $p \in M$. Such a map is called a *smooth section* of the tangent bundle $\pi : TM \to M$.

Remark 10.

The condition $\pi \circ X = \mathrm{id}_M$ is what makes X a *section* rather than just any smooth map $M \to TM$. Without it, X could send p to a tangent vector sitting at a *different* point $q \neq p$ — which is meaningless as a vector field. The composition with π sends the output back to M and demands it lands at p, enforcing that the vector $X(p)$ lives in the correct fiber T_pM.
In local coordinates $(x^1, \ldots, x^d)$ on an open set $U \subset M$, a vector field takes the form

$$X\big|_U = \sum_{i=1}^{d} X^i(x) \frac{\partial}{\partial x^i},$$

where $X^i : U \to \mathbb{R}$ are smooth functions called the *components* of X. Smoothness of X as a section is equivalent to smoothness of all the X^i.

Example 10.: Smooth vector field on S^1 — the angular field

The coordinate vector field

$$X = \frac{\partial}{\partial \theta}$$

assigns to each point $p = (\cos\theta, \sin\theta) \in S^1$ the tangent vector

$$X(p) = -\sin\theta \left.\frac{\partial}{\partial x}\right|_p + \cos\theta \left.\frac{\partial}{\partial y}\right|_p .$$

The component functions $X^x = -\sin\theta$ and $X^y = \cos\theta$ are smooth, so X is smooth vector field. It is nowhere-vanishing, confirming that S^1 is parallelisabl

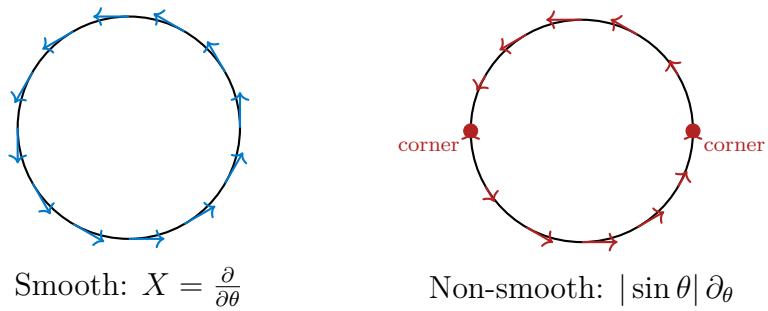
Vector fields on S^1.

Non-Example 10.: A non-smooth vector field on S^1

Define X by assigning to each point $p = (\cos\theta, \sin\theta)$ the vector

$$X(p) = |\sin\theta| \left.\frac{\partial}{\partial \theta}\right|_p .$$

The component function $|\sin\theta|$ is continuous but *not smooth* at $\theta = 0$ and $\theta = \pi$ (the points $(1, 0)$ and $(-1, 0)$), since $|\sin\theta|$ has a corner there. This is a well-defined section of TS^1 — the condition $\pi \circ X = \mathrm{id}$ holds — but it fails to be smooth.



Smooth: $X = \frac{\partial}{\partial \theta}$ Non-smooth: $|\sin\theta|\,\partial_\theta$

Figure 17: Left: smooth angular field — uniform arrows. Right: $|\sin\theta|\,\partial_\theta$ — vanishes at $(1, 0)$ and $(-1, 0)$ with a corner, continuous but not smooth.

Vector fields on the cylinder $S^1 \times \mathbb{R}$. The cylinder $C = S^1 \times \mathbb{R}$ has a 2-dimensional tangent space at each point, with basis $\{\partial_\theta, \partial_z\}$.

Example 10.: Purely vertical field

$$X_1 = \frac{\partial}{\partial z}$$

assigns the upward unit vector to every point — arrows all point along the axis.

Example 10.: Purely angular field

$$X_2 = \frac{\partial}{\partial \theta}$$

assigns the angular tangent vector — arrows circle the cylinder horizontally.

Example 10.: Helical field

$$X_3 = \frac{\partial}{\partial \theta} + \frac{\partial}{\partial z}$$

combines both. The integral curves are helices winding around the cylinder.

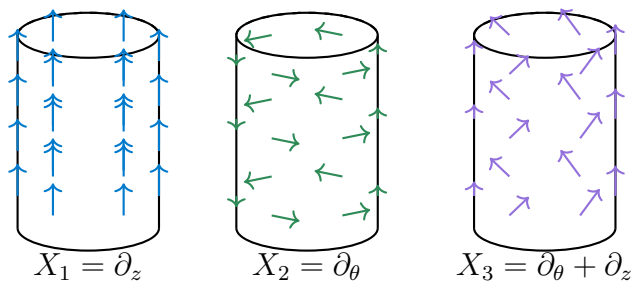


Figure 18: Three vector fields on the cylinder: vertical ∂_z (blue), angular ∂_θ (green), and helical $\partial_\theta + \partial_z$ (purple).

Vector fields on S^2.

Example 10.: Rotation field on S^2

The vector field generating rotation about the z-axis is

$$X(x, y, z) = -y \left.\frac{\partial}{\partial x}\right|_p + x \left.\frac{\partial}{\partial y}\right|_p .$$

It is smooth everywhere on S^2 but vanishes at the north and south poles $(0, 0, \pm$
where $x = y = 0$. The arrows circulate around latitude circles, shrinking to zero
both poles.

Non-Example 10.: The outward normal — not a vector field on S^2

At each point $p = (x, y, z) \in S^2$, the outward unit normal

$$N(p) = x \left.\frac{\partial}{\partial x}\right|_p + y \left.\frac{\partial}{\partial y}\right|_p + z \left.\frac{\partial}{\partial z}\right|_p$$

is a smooth, nowhere-vanishing map $S^2 \to T\mathbb{R}^3$. However, $N(p) \notin T_pS^2$ — it is normal to the sphere, not tangent to it. It violates the section condition: $N(p)$ does not live in the fiber T_pS^2. It is a vector field on $\mathbb{R}^3$ restricted to S^2, not a vector field *on* S^2.

Observation 10.

The outward normal never vanishes, yet it is not a section of TS^2. The hairy ball theorem is about *tangent* vector fields — genuine sections of TS^2 — and says any such section must vanish somewhere. There is no contradiction: the theorem does not apply to normal fields.

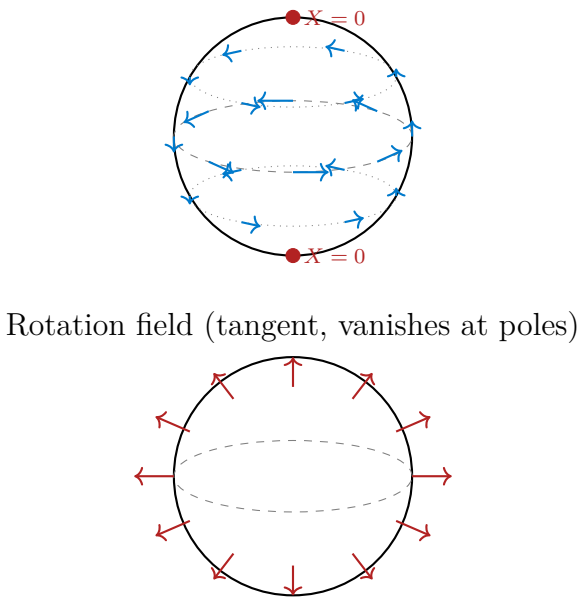


Figure 19: Left: rotation field on S^2 — tangent, vanishes at both poles. Right: outward normal — smooth and nowhere zero, but not a section of TS^2.

The algebra of vector fields

Local coordinate expression. In a chart (U, x) with local coordinates $(x^1, \ldots, x^d)$, every vector field $X \in \mathfrak{X}(M)$ restricts to

$$X\big|_U = \sum_{i=1}^{d} X^i(x)\, \frac{\partial}{\partial x^i},$$

where $X^i \in C^\infty(U)$ are the component functions. The action of X on a smooth function $f \in C^\infty(M)$ is then

$$X(f)\big|_U = \sum_{i=1}^{d} X^i(x)\, \frac{\partial f}{\partial x^i}.$$

This is simply directional differentiation in each chart, with the component functions telling us how much to differentiate in each coordinate direction.

Example 10.: Vector fields in local coordinates

- On $\mathbb{R}^2$ with coordinates (x, y): the rotation field $X = -y\,\partial_x + x\,\partial_y$ has components $X^x = -y$, $X^y = x$. It acts on f by $X(f) = -y\frac{\partial f}{\partial x} + x\frac{\partial f}{\partial y}$.
- On S^1 in the angle chart (θ): the field $X = \frac{\partial}{\partial \theta}$ has component $X^\theta = 1$. It acts on f by $X(f) = \frac{\partial f}{\partial \theta}$.
- On the cylinder in the chart (θ, z): the helical field $X = \partial_\theta + \partial_z$ has components $X^\theta = X^z = 1$. It acts by $X(f) = \frac{\partial f}{\partial \theta} + \frac{\partial f}{\partial z}$.

The $C^\infty(M)$-module $\mathfrak{X}(M)$. We denote the set of all smooth vector fields on M by

$$\mathfrak{X}(M) \;=\; \Gamma(TM).$$

This notation $\Gamma(TM)$ refers to the *smooth sections* of TM, and will generalise later to sections of other bundles.

Proposition 10.: $\mathfrak{X}(M)$ is a $C^\infty(M)$-module

Let $X, Y \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$. Define

$$(X + Y)(p) = X(p) + Y(p), \qquad (fX)(p) = f(p)\, X(p).$$

Then $X + Y$ and fX are again smooth vector fields. Moreover, each $X \in \mathfrak{X}(M)$ acts on smooth functions as a linear map $X : C^\infty(M) \to C^\infty(M)$ satisfying:

$$X(\alpha f + \beta g) = \alpha X(f) + \beta X(g), \qquad \alpha, \beta \in \mathbb{R},$$

$$X(fg) = f\, X(g) + g\, X(f) \qquad \text{(Leibniz rule)}.$$

Remark 10.

The Leibniz rule says X is a derivation of $C^\infty(M)$, consistent with our earlier definition of tangent vectors as derivations at a point. A vector field is simply a smoothly varying family of such derivations.

Why $\mathfrak{X}(M)$ is not an algebra under pointwise multiplication. A natural question: can we multiply two vector fields to get a third? The naive product XY defined by $XY(f) = X(Y(f))$ does *not* work. Expanding using the Leibniz rule twice:

$$XY(fg) = X\Big(fY(g) + gY(f)\Big) = fXY(g) + X(f)Y(g) + gXY(f) + X(g)Y(f).$$

For XY to be a vector field, it must satisfy the Leibniz rule: $XY(fg) = f \cdot XY(g) + g \cdot XY(f)$. But the expansion above contains the extra terms $X(f)Y(g) + X(g)Y(f)$, which do not vanish in general. So XY is *not* a derivation — it fails the Leibniz rule.

Observation 10.

The obstruction is exactly the symmetry of those extra terms. Notice that if we form the *antisymmetric combination*

$$[X, Y](f) \;=\; XY(f) - YX(f) \;=\; X(Y(f)) - Y(X(f)),$$

then:

$$[X, Y](fg) = X(Y(fg)) - Y(X(fg)) = \Big(fXY(g) + X(f)Y(g) + gXY(f) + X(g)Y(f)\Big)$$

$$-\Big(fYX(g) + Y(f)X(g) + gYX(f) + Y(g)X(f)\Big) = f[X, Y](g) + g[X, Y](f).$$

The unwanted terms cancel. So $[X, Y]$ *is* a derivation, hence a vector field. The antisymmetric product saves the Leibniz rule.

Definition 10.: Lie bracket

Let $X, Y \in \mathfrak{X}(M)$. The *Lie bracket* of X and Y is the vector field $[X, Y] \in \mathfrak{X}(M)$ defined by

$$[X, Y](f) \;=\; X(Y(f)) - Y(X(f)), \qquad f \in C^\infty(M).$$

In local coordinates $(x^1, \ldots, x^d)$, with $X = \sum_i X^i \partial_i$ and $Y = \sum_j Y^j \partial_j$:

$$[X, Y] \;=\; \sum_{i,j} \left(X^i \frac{\partial Y^j}{\partial x^i} - Y^i \frac{\partial X^j}{\partial x^i} \right) \frac{\partial}{\partial x^j}.$$

Proposition 10.: $(\mathfrak{X}(M), [\cdot,\cdot])$ is a Lie algebra

The Lie bracket satisfies:

1. **Antisymmetry**: $[X, Y] = -[Y, X]$.
2. **Bilinearity**: $[X, \alpha Y + \beta Z] = \alpha[X, Y] + \beta[X, Z]$ for $\alpha, \beta \in \mathbb{R}$.
3. **Jacobi identity**: $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

Together these make $\mathfrak{X}(M)$ into a **Lie algebra**.

Remark 10.

Note that the Lie bracket is *not* $C^\infty(M)$-linear in general:

$$[fX, Y] = f[X, Y] - Y(f)\, X.$$

This extra term $-Y(f)X$ is what distinguishes the Lie bracket from a purely algebraic product. It reflects the fact that $[X, Y]$ measures the failure of the flows of X and Y to commute — a deeply geometric statement we will revisit when we discuss Lie groups and Lie algebras.

Example 10.: Lie bracket on $\mathbb{R}^2$

Let $X = \partial_x$ and $Y = x\,\partial_y$ on $\mathbb{R}^2$. Then:

$$[X, Y](f) = X(Y(f)) - Y(X(f)) = \partial_x\left(x\frac{\partial f}{\partial y}\right) - x\,\partial_y\left(\frac{\partial f}{\partial x}\right) = \frac{\partial f}{\partial y} + x\frac{\partial^2 f}{\partial x \partial y} - x\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial f}{\partial y}.$$

So $[X, Y] = \partial_y$. The second-order terms cancel (as they must for $[X, Y]$ to be a vector field), leaving a first-order operator.

Example 10.: Commuting coordinate fields

On any chart (U, x), the coordinate vector fields commute:

$$\left[\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right] = 0, \qquad \forall\, i, j.$$

This follows since mixed partial derivatives commute for smooth functions. Geometrically, the flows along coordinate directions commute — moving first in the x^i direction then x^j gives the same result as the reverse order.

Example 10.: Non-commuting fields on S^2 — the rotation algebra

The three rotation vector fields on S^2 (and on $\mathbb{R}^3$),

$$X_1 = y\,\partial_z - z\,\partial_y, \quad X_2 = z\,\partial_x - x\,\partial_z, \quad X_3 = x\,\partial_y - y\,\partial_x,$$

satisfy the Lie bracket relations

$$[X_1, X_2] = X_3, \qquad [X_2, X_3] = X_1, \qquad [X_3, X_1] = X_2.$$

This is the Lie algebra $\mathfrak{so}(3)$ of the rotation group $SO(3)$. We will see this again when we study Lie groups and Lie algebras.

10.7 Integral curves and flows

Motivation. When we first defined a tangent vector at a point, we used a smooth curve

$$\gamma : (-\varepsilon, \varepsilon) \to M$$

through that point and extracted from it a derivation

$$v(f) = \left.\frac{d}{dt}\right|_{t=0} f(\gamma(t)).$$

So curves were our first way of encoding the idea of a *direction* on a manifold.

Now suppose instead of having just one tangent vector at one point, we are given a *vector field* X, that is, a smooth assignment of a tangent vector $X_p \in T_pM$ at every point $p \in M$. A natural question is:

> Can we recover curves whose velocity at each point is prescribed by the vector field?

Such curves describe the motion of a particle whose instantaneous velocity is determined by X. They are called *integral curves*. In physics, they represent trajectories generated by a velocity field, and later they will lead to the notion of a *flow*, which describes the continuous motion of all points of the manifold under the vector field.

Definition 10.: Integral curve

Let M be a smooth manifold and let $X \in \mathfrak{X}(M)$ be a smooth vector field. A smooth curve

$$\gamma : I \to M,$$

defined on an open interval $I \subseteq \mathbb{R}$, is called an *integral curve* of X if for every $t \in I$,

$$\gamma'(t) = X_{\gamma(t)}.$$

Equivalently, for every $f \in C^\infty(M)$,

$$\frac{d}{dt} f(\gamma(t)) = X(f)(\gamma(t)).$$

Remark 10.

The condition $\gamma'(t) = X_{\gamma(t)}$ means that at each point of the curve, its velocity is exactly the vector assigned by the field X. Thus an integral curve is a trajectory that "follows the arrows" of the vector field.

Proposition 10.: Existence and uniqueness of integral curves

Let $X \in \mathfrak{X}(M)$ and let $p \in M$. Then there exists an open interval $I \subseteq \mathbb{R}$ with $0 \in I$ and an integral curve

$$\gamma : I \to M$$

for X such that

$$\gamma(0) = p.$$

Moreover, if

$$\tilde{\gamma} : \tilde{I} \to M$$

is another integral curve for X with $\tilde{\gamma}(0) = p$, then

$$\tilde{\gamma} = \gamma \qquad \text{on } I \cap \tilde{I}.$$

Sketch of proof. Choose a chart (U, x) around p, with local coordinates

$$x = (x^1, \dots, x^n).$$

On U, the vector field X can be written as

$$X = \sum_{i=1}^{n} X^i \frac{\partial}{\partial x^i},$$

where the coefficients X^i are smooth functions.

If $\gamma(t)$ is an integral curve, then in coordinates

$$x(\gamma(t)) = (u^1(t), \dots, u^n(t))$$

must satisfy the system of ordinary differential equations

$$\frac{du^i}{dt} = X^i(u^1, \dots, u^n), \qquad i = 1, \dots, n,$$

with initial condition $u(0) = x(p)$. Since the functions X^i are smooth, the standard existence and uniqueness theorem for ODEs in $\mathbb{R}^n$ gives a unique local solution. Translating back through the chart gives the desired integral curve on the manifold. □

Definition 10.: Complete vector field

A vector field $X \in \mathfrak{X}(M)$ is called *complete* if every integral curve of X is defined for all $t \in \mathbb{R}$.

Proposition 10.: Compact manifolds imply completeness

If M is a compact smooth manifold, then every smooth vector field $X \in \mathfrak{X}(M)$ is complete.

Sketch of proof. Local existence gives an integral curve through each point, defined on some interval around $t = 0$. The only obstruction to extending such a solution is that it might "escape to infinity" or leave every compact set in finite time. But on a compact manifold this cannot happen. Hence every maximal integral curve extends to all $t \in \mathbb{R}$. □

Definition 10.: Flow

Let $X \in \mathfrak{X}(M)$ be a complete vector field. The *flow* generated by X is the smooth map

$$\Phi : \mathbb{R} \times M \to M, \qquad (t, p) \mapsto \Phi_t(p),$$

defined by the condition that for each $p \in M$, the curve

$$t \mapsto \Phi_t(p)$$

is the unique integral curve of X satisfying

$$\Phi_0(p) = p.$$

It satisfies

$$\Phi_{t+s}(p) = \Phi_t(\Phi_s(p)), \qquad \frac{d}{dt}\Phi_t(p) = X_{\Phi_t(p)}.$$

Remark 10.

The family $\{\Phi_t\}_{t\in\mathbb{R}}$ is called a one-parameter group of diffeomorphisms. The identity

$$\Phi_{t+s} = \Phi_t \circ \Phi_s$$

expresses the idea that flowing for time s and then for time t is the same as flowing once for time $t+s$.

Examples

Example 10.: Flow on S^1

Consider the vector field

$$X = \frac{\partial}{\partial\theta}$$

on S^1, written in the angle coordinate θ. The integral curves satisfy

$$\frac{d\theta}{dt} = 1,$$

so the solution through θ_0 is

$$\theta(t) = \theta_0 + t.$$

Thus the flow is

$$\Phi_t(e^{i\theta}) = e^{i(\theta+t)}.$$

Geometrically, this is uniform rotation of the circle.

Example 10.: Flow on the cylinder $S^1 \times \mathbb{R}$

Let (θ, z) be coordinates on the cylinder and consider the vector field

$$X = \frac{\partial}{\partial\theta} + \frac{\partial}{\partial z}.$$

Then an integral curve $(\theta(t), z(t))$ satisfies

$$\frac{d\theta}{dt} = 1, \qquad \frac{dz}{dt} = 1.$$

Hence the solution through (θ_0, z_0) is

$$\theta(t) = \theta_0 + t, \qquad z(t) = z_0 + t.$$

The flow is therefore

$$\Phi_t(\theta_0, z_0) = (\theta_0 + t,\ z_0 + t).$$

Geometrically, points move upward while simultaneously rotating around the cylinder, so the integral curves are helices.

Example 10.: Rotation flow on S^2

Consider the vector field on $S^2 \subset \mathbb{R}^3$ generated by rotation around the z-axis:

$$X_{(x,y,z)} = (-y, x, 0).$$

This vector field is tangent to S^2, since

$$(x, y, z) \cdot (-y, x, 0) = -xy + xy + 0 = 0.$$

Its integral curves satisfy

$$\frac{d}{dt}\begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = \begin{pmatrix} -y(t) \\ x(t) \\ 0 \end{pmatrix}.$$

Solving this system gives

$$x(t) = x_0 \cos t - y_0 \sin t, \qquad y(t) = x_0 \sin t + y_0 \cos t, \qquad z(t) = z_0.$$

Thus the flow is

$$\Phi_t(x_0, y_0, z_0) = (x_0 \cos t - y_0 \sin t,\ x_0 \sin t + y_0 \cos t,\ z_0).$$

Geometrically, each point moves along the horizontal circle of latitude through it.

Geometric illustrations

Remark 10.

These examples show the hierarchy clearly:

tangent vector at one point $\longrightarrow$ vector field on all points $\longrightarrow$ integral curves $\longrightarrow$ flow

So the notion of flow is the global dynamical version of the local idea of tangent direction.

11 Cotangent space, cotangent bundle and differential forms

11.1 The cotangent space

Motivation. At every point $p \in M$ we have the tangent space T_pM, a real vector space. Since T_pM is a finite-dimensional vector space, it is natural to study its *dual space*. The dual of T_pM consists of all linear maps from T_pM to $\mathbb{R}$ — objects that eat tangent vectors and produce real numbers.

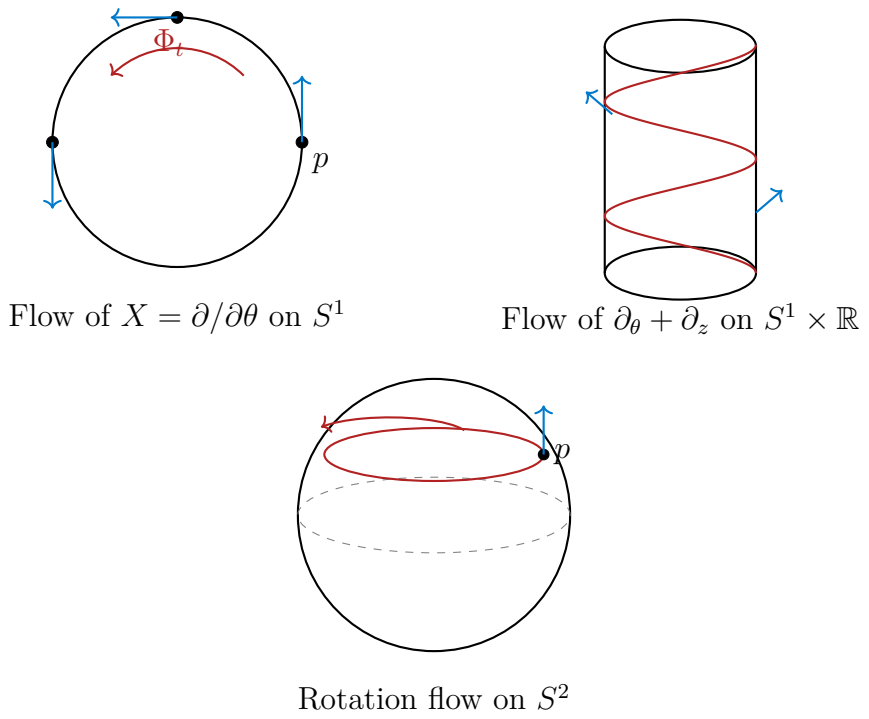


Figure 20: Integral curves and flows on the three running examples: uniform rotation on S^1, helical motion on the cylinder, and rotation about the z-axis on S^2.

This is not merely an abstract exercise. The objects physicists encounter most naturally — gradients, curls, divergences, integrands — are not vectors but *covectors*: they act on vectors to give numbers. The integral $\int_\gamma \omega$ of a 1-form along a curve, for instance, pairs a covector field ω with the velocity vector of γ at each point. The cotangent space is the correct setting for all of these notions.

Definition 11.: Cotangent space

Let M be a smooth manifold and $p \in M$. The *cotangent space at p* is the dual vector space of T_pM:

$$T_p^*M \;=\; (T_pM)^* \;=\; \mathrm{Hom}(T_pM,\, \mathbb{R}).$$

Elements of T_p^*M are called *cotangent vectors*, *covectors*, or *1-forms at p*. Given $\xi \in T_p^*M$ and $v \in T_pM$, we write the pairing as

$$\langle \xi,\, v \rangle \;\in\; \mathbb{R}.$$

Remark 11.

Since T_pM is a d-dimensional real vector space, $T_p^*M \cong \mathbb{R}^d$ as well — the cotangent space has the same dimension as the tangent space. However, there is no *canonical* (basis-independent) isomorphism between them. They become isomorphic only after choosing extra structure, such as a Riemannian metric.

Examples

Example 11.: Cotangent space of $\mathbb{R}^n$

At any point $p \in \mathbb{R}^n$, the tangent space $T_p\mathbb{R}^n$ has the standard basis $\{\partial_{x^1}|_p, \ldots, \partial_{x^n}|_p\}$. The cotangent space $T_p^*\mathbb{R}^n$ has the dual basis $\{dx^1|_p, \ldots, dx^n|_p\}$ defined by

$$\langle dx^i|_p,\, \partial_{x^j}|_p\rangle = \delta^i_j.$$

A general covector at p is $\xi = \sum_i \xi_i\, dx^i|_p$ for coefficients $\xi_i \in \mathbb{R}$, and it acts on $v = \sum_j v^j \partial_{x^j}|_p$ by

$$\langle \xi, v\rangle = \sum_i \xi_i v^i.$$

This is just the standard dot product in disguise — but written in a way that distinguishes vectors (upper indices) from covectors (lower indices).

Example 11.: Cotangent space of S^1

At $p = (1, 0) \in S^1$, the tangent space T_pS^1 is spanned by $\partial_\theta|_p$. The cotangent space $T_p^*S^1$ is therefore 1-dimensional, spanned by the dual basis element $d\theta|_p$ defined by

$$\langle d\theta|_p,\, \partial_\theta|_p\rangle = 1.$$

A general covector at p is $\xi = c\, d\theta|_p$ for some $c \in \mathbb{R}$. It acts on a tangent vector $v = a\, \partial_\theta|_p$ by $\langle \xi, v\rangle = ca$.

Example 11.: Cotangent space of S^2

At $p = (1, 0, 0) \in S^2$ using the spherical chart (θ, φ), the tangent space T_pS^2 has basis $\{\partial_\theta|_p, \partial_\varphi|_p\}$. The cotangent space $T_p^*S^2$ has the dual basis $\{d\theta|_p, d\varphi|_p\}$ defined by

$$\langle d\theta|_p, \partial_\theta|_p\rangle = 1, \quad \langle d\theta|_p, \partial_\varphi|_p\rangle = 0, \quad \langle d\varphi|_p, \partial_\theta|_p\rangle = 0, \quad \langle d\varphi|_p, \partial_\varphi|_p\rangle = 1.$$

A general covector is $\xi = a\, d\theta|_p + b\, d\varphi|_p$.

11.2 The coordinate basis for T_p^*M

Just as a chart (U, x) gives a basis $\{\partial_{x^i}|_p\}$ for T_pM, it gives a dual basis for T_p^*M.

T_pS^1 (blue) and $T_p^*S^1$ (red) $\qquad$ T_pS^2 (blue) and $T_p^*S^2$ (red)

Figure 21: Left: at $p \in S^1$, the tangent vector $\partial_\theta|_p$ (blue) and the dual covector $d\theta|_p$ (red, shown as equally-spaced level sets). The pairing $\langle d\theta, \partial_\theta \rangle = 1$ counts how many level sets the vector crosses. Right: at $p \in S^2$, the dual basis $\{d\theta, d\varphi\}$ pairs with $\{\partial_\theta, \partial_\varphi\}$ via the Kronecker delta.

Proposition 11.: Coordinate cobasis

Let (U, x) be a chart around p with coordinates $(x^1, \ldots, x^d)$. Define the covectors $dx^i|_p \in T_p^*M$ by

$$\left\langle dx^i|_p, \left.\frac{\partial}{\partial x^j}\right|_p \right\rangle = \delta^i_j = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Then $\{dx^1|_p, \ldots, dx^d|_p\}$ is a basis for T_p^*M, called the *coordinate cobasis* dual to $\{\partial_{x^i}|_p\}$. Every covector $\xi \in T_p^*M$ writes uniquely as

$$\xi = \sum_{i=1}^{d} \xi_i \, dx^i|_p, \qquad \xi_i = \left\langle \xi, \left.\frac{\partial}{\partial x^i}\right|_p \right\rangle.$$

Remark 11.

Notice the index convention: tangent vectors carry *upper* indices ($v = \sum v^i \partial_i$) while cotangent vectors carry *lower* indices ($\xi = \sum \xi_i \, dx^i$). This is not just a convention — it tracks the transformation behaviour under a change of coordinates. Tangent vectors transform with the Jacobian, cotangent vectors with the inverse Jacobian. The pairing $\langle \xi, v \rangle = \sum_i \xi_i v^i$ automatically contracts one upper and one lower index — this is *Einstein summation*.

11.3 The differential of a function

The most natural source of cotangent vectors is differentiation. Given a smooth function $f \in C^\infty(M)$, its differential at p is a covector that measures the directional rate of change of f.

Definition 11.: Differential of a function

Let $f \in C^\infty(M)$ and $p \in M$. The *differential of f at p* is the covector $df_p \in T_p^*M$ defined by

$$\langle df_p,\, v\rangle \;=\; v(f), \qquad v \in T_pM.$$

That is, df_p eats a tangent vector v and returns the directional derivative of f in the direction v.

Remark 11.

In a chart (U, x), the differential has the coordinate expression

$$df_p \;=\; \sum_{i=1}^{d} \frac{\partial f}{\partial x^i}\bigg|_p dx^i|_p.$$

This is the familiar gradient formula — the coefficients are the partial derivatives of f. The differential is the intrinsic, coordinate-free version of the gradient.
Note the notational consistency: applying df_p to the coordinate basis vector $\partial_{x^j}|_p$ gives

$$\langle df_p,\, \partial_{x^j}|_p\rangle = \partial_{x^j}|_p(f) = \frac{\partial f}{\partial x^j}\bigg|_p,$$

which correctly extracts the j-th coefficient.

Examples of differentials

Example 11.: Differential on $\mathbb{R}^2$

Let $f(x, y) = x^2 + y^2$ on $\mathbb{R}^2$. Then

$$df = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy = 2x\,dx + 2y\,dy.$$

At $p = (1, 0)$: $df_p = 2\,dx|_p$. Acting on $v = \partial_x|_p$: $\langle df_p, \partial_x|_p\rangle = 2$. Acting on $w = \partial_y|_p$: $\langle df_p, \partial_y|_p\rangle = 0$. The differential points in the direction of steepest ascent of f (outward from the origin), but it is a covector, not a vector.

Example 11.: Differential on S^1

Let $f(\cos\theta, \sin\theta) = \cos\theta$ on S^1. In the angle chart:

$$df = \frac{\partial \cos\theta}{\partial\theta}\, d\theta = -\sin\theta\, d\theta.$$

At $p = (1, 0)$ where $\theta = 0$: $df_p = 0$. The function $f = x$ has a critical point at $(1, 0)$ (a maximum restricted to S^1), so the differential vanishes there.
At $p = (0, 1)$ where $\theta = \pi/2$: $df_p = -d\theta|_p$. Acting on $\partial_\theta|_p$: $\langle df_p, \partial_\theta|_p\rangle = -1$, meaning f decreases as θ increases past $\pi/2$.

Example 11.: Differential on S^2

Let $f(x, y, z) = z$ on S^2 (the height function). In spherical coordinates (θ, φ) where $z = \cos\theta$:

$$df = \frac{\partial \cos\theta}{\partial\theta}\, d\theta + \frac{\partial \cos\theta}{\partial\varphi}\, d\varphi = -\sin\theta\, d\theta.$$

At the equator $\theta = \pi/2$: $df = -d\theta$, pointing in the direction of decreasing latitude. At the north pole $\theta = 0$: $df = 0$, a critical point. This is the height function on the sphere — its differential vanishes exactly at the poles (maximum and minimum of z on S^2).

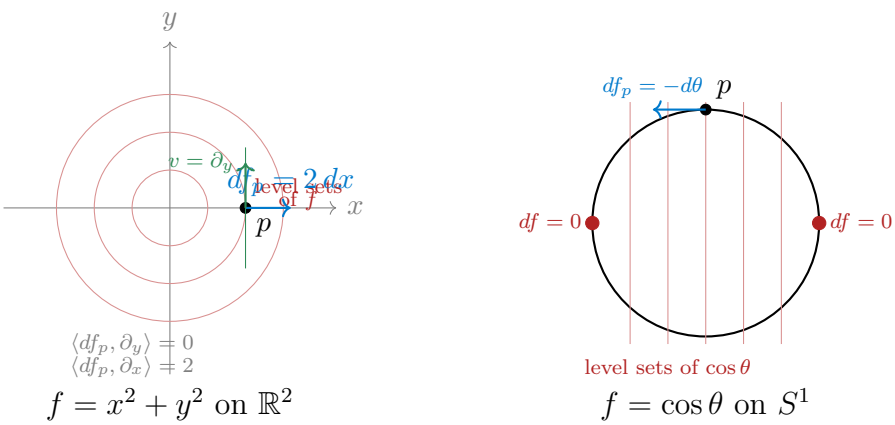


Figure 22: Left: the differential $df = 2x\,dx + 2y\,dy$ of $f = x^2 + y^2$ at $p = (1, 0)$ points outward (perpendicular to the level set through p). Right: the differential of $\cos\theta$ on S^1 vanishes at the two critical points $(\pm 1, 0)$ and is nonzero elsewhere.

11.4 The pullback

Motivation. In the tangent bundle section we defined the *pushforward* $df_p : T_pM \to T_{f(p)}N$, which sends tangent vectors *forward* along a smooth map $f : M \to N$. There is a dual operation going in the *opposite direction*: a smooth map $f : M \to N$ induces a map on cotangent spaces

$$f_p^* : T^*_{f(p)}N \longrightarrow T^*_pM$$

that pulls covectors *back* from N to M. This is the *pullback*. While the pushforward goes with f, the pullback goes against it.

Definition 11.: Pullback of a covector

Let $f : M \to N$ be smooth, $p \in M$, and $\xi \in T^*_{f(p)}N$. The *pullback* $f^*_p\,\xi \in T^*_pM$ is defined by

$$\langle f^*_p\,\xi,\ v\rangle \;=\; \langle \xi,\ df_p(v)\rangle, \qquad v \in T_pM.$$

That is, to evaluate $f^*\xi$ on a tangent vector $v \in T_pM$, first push v forward to $T_{f(p)}N$ using df_p, then pair with ξ.

Remark 11.

The definition says: *pull back by composing with the pushforward*. In local coordinates (x^i) on M and (y^j) on N, if $\xi = \sum_j \xi_j\, dy^j|_{f(p)}$ then

$$f^*_p\,\xi \;=\; \sum_{i,j} \xi_j \left.\frac{\partial(y^j \circ f)}{\partial x^i}\right|_p dx^i|_p.$$

The pullback is represented by the *transpose of the Jacobian* of f.

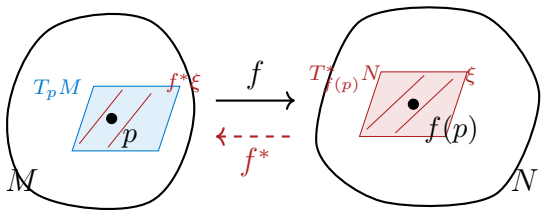


Figure 23: The pullback $f^* : T^*_{f(p)}N \to T^*_pM$ goes against the direction of f, while the pushforward $df_p : T_pM \to T_{f(p)}N$ goes with f. A covector ξ on N is pulled back to $f^*\xi$ on M by composing with the pushforward.

Examples of pullbacks

Example 11.: Pullback along inclusion $\iota : S^1 \hookrightarrow \mathbb{R}^2$

Let $\iota(\cos\theta, \sin\theta) = (\cos\theta, \sin\theta)$ be the inclusion. Consider the covector $\xi = x\,dx + y\,dy \in T^*_{\iota(p)}\mathbb{R}^2$. The pullback is

$$\iota^*\xi = \iota^*(x\,dx + y\,dy) = (\cos\theta)\,d(\cos\theta) + (\sin\theta)\,d(\sin\theta).$$

Since $d(\cos\theta) = -\sin\theta\,d\theta$ and $d(\sin\theta) = \cos\theta\,d\theta$:

$$\iota^*\xi = -\cos\theta\sin\theta\,d\theta + \sin\theta\cos\theta\,d\theta = 0.$$

The pullback of $x\,dx + y\,dy = \frac{1}{2}d(x^2 + y^2)$ vanishes on S^1 because $x^2 + y^2 = 1$ is constant on the circle — its differential restricts to zero.

Example 11.: Pullback of $d\theta$ along rotation $R_\alpha : S^1 \to S^1$

Let $R_\alpha(\cos\theta, \sin\theta) = (\cos(\theta + \alpha), \sin(\theta + \alpha))$. Then $\theta \circ R_\alpha = \theta + \alpha$, so

$$R_\alpha^*\,d\theta = d(\theta \circ R_\alpha) = d(\theta + \alpha) = d\theta.$$

Rotation preserves the angle form $d\theta$. This is consistent with the fact that dR_α is an isomorphism preserving ∂_θ.

Example 11.: Pullback of dz along projection $\pi : S^1 \times \mathbb{R} \to \mathbb{R}$

Let $\pi(\cos\theta, \sin\theta, z) = z$. Then

$$\pi^*\,dz = d(z \circ \pi) = dz,$$

but now dz lives on the cylinder. It acts on $\partial_z|_p$ by 1 and on $\partial_\theta|_p$ by 0. The pullback simply promotes dz from a covector on $\mathbb{R}$ to a covector on the cylinder, ignoring the angular direction.

11.5 Injectivity and surjectivity: pushforward vs. pullback

The relationship between injectivity/surjectivity of f, df_p, and f_p^* is summarised as follows.

Proposition 11.: Duality of pushforward and pullback

Let $f : M \to N$ be smooth and $p \in M$.

1. $df_p : T_pM \to T_{f(p)}N$ is injective $\iff$ $f_p^* : T^*_{f(p)}N \to T^*_pM$ is surjective.
2. $df_p : T_pM \to T_{f(p)}N$ is surjective $\iff$ $f_p^* : T^*_{f(p)}N \to T^*_pM$ is injective.
3. df_p is an isomorphism $\iff$ f_p^* is an isomorphism.

Remark 11.

This is just linear algebra: for a linear map $A : V \to W$ between finite-dimensional spaces, the transpose $A^T : W^* \to V^*$ satisfies $\ker A^T = (\operatorname{im} A)^\perp$ and $\operatorname{im} A^T = (\ker A)^\perp$. In particular A injective $\Leftrightarrow$ A^T surjective, and vice versa. The pushforward is $A = df_p$ and the pullback is $A^T = f_p^*$.

$$\begin{tikzcd}
T_pM \arrow[r, "df_p", "\text{immersion} \Rightarrow \text{injective}"'] \arrow[d, dash, "\text{dual pair}"'] \arrow[dr, phantom, "\langle f^*\xi, v\rangle = \langle \xi, df(v)\rangle"] & T_{f(p)}N \arrow[d, dash, "\text{dual pair}"] \\
T^*_pM & T^*_{f(p)}N \arrow[l, "f_p^*", "\text{immersion} \Rightarrow f^* \text{ surjective}"']
\end{tikzcd}$$

Figure 24: The pushforward df_p and pullback f_p^* are transpose to each other with respect to the natural pairings on T_pM, T_p^*M, $T_{f(p)}N$, $T^*_{f(p)}N$. Injectivity of one implies surjectivity of the other.

Example 11.: Inclusion $\iota : S^2 \hookrightarrow \mathbb{R}^3$ — immersion

The pushforward $d\iota_p : T_pS^2 \to T_p\mathbb{R}^3$ is injective (a 2-dimensional subspace inside a 3-dimensional space). By duality, the pullback $\iota_p^* : T_p^*\mathbb{R}^3 \to T_p^*S^2$ is surjective. Indeed, every covector on S^2 at p can be obtained by restricting some covector on $\mathbb{R}^3$ — but the restriction is not unique (different covectors on $\mathbb{R}^3$ restrict to the same covector on S^2, since the normal direction is killed).

Example 11.: Projection $\pi : S^1 \times \mathbb{R} \to S^1$ — submersion

The pushforward $d\pi_p$ is surjective (kills the z-direction). By duality, the pullback $\pi_p^* : T^*_{\pi(p)}S^1 \to T_p^*(S^1 \times \mathbb{R})$ is injective. Every covector ξ on S^1 pulls back to a unique covector $\pi^*\xi$ on the cylinder that vanishes on the z-direction.

11.6 The cotangent bundle

Just as the tangent spaces assemble into the tangent bundle TM, the cotangent spaces assemble into the *cotangent bundle*.

Definition 11.: Cotangent bundle

Let M be a smooth d-dimensional manifold. The *cotangent bundle* of M is the disjoint union of all cotangent spaces:

$$T^*M \;=\; \bigsqcup_{p\in M} T_p^*M \;=\; \{(p,\,\xi) \mid p \in M,\; \xi \in T_p^*M\}.$$

It carries the *canonical projection*

$$\pi : T^*M \longrightarrow M, \qquad \pi(p,\xi) = p,$$

whose fiber over p is $\pi^{-1}(p) = T_p^*M$.

Proposition 11.: T^*M is a smooth manifold and a vector bundle

Let M be a smooth d-dimensional manifold. Then:

1. T^*M admits a smooth manifold structure of dimension $2d$.
2. $\pi : T^*M \to M$ is a smooth surjective map.
3. Every $p \in M$ has a neighbourhood U such that $\pi^{-1}(U) \cong U \times \mathbb{R}^d$, with π corresponding to projection onto the first factor.

In particular, T^*M is a *vector bundle of rank d* over M, with fiber $T_p^*M \cong \mathbb{R}^d$ over each point p.

Remark 11.

A chart (U, x) on M gives a local trivialisation of T^*M over U: every $(p,\xi) \in \pi^{-1}(U)$ writes as

$$(p,\,\xi) \;=\; \left(p,\; \sum_{i=1}^{d} \xi_i\, dx^i|_p\right) \;\longleftrightarrow\; (x^1(p),\ldots,x^d(p),\;\xi_1,\ldots,\xi_d) \;\in\; \mathbb{R}^{2d}.$$

The coordinates $(x^1,\ldots,x^d,\xi_1,\ldots,\xi_d)$ on T^*M are called *canonical coordinates* or *Darboux coordinates* — they are the natural coordinates for Hamiltonian mechanics, where the x^i are positions and the ξ_i are momenta.

Examples

Example 11.: $T^*\mathbb{R}^n \cong \mathbb{R}^n \times \mathbb{R}^n$

For $M = \mathbb{R}^n$, the cotangent bundle is trivial:

$$T^*\mathbb{R}^n \;\cong\; \mathbb{R}^n \times \mathbb{R}^n.$$

A point $(p, \xi) \in T^*\mathbb{R}^n$ corresponds to a pair (position, momentum), which is exactly the phase space of classical mechanics on $\mathbb{R}^n$. The canonical coordinates are $(q^1, \ldots, q^n, p_1, \ldots, p_n)$.

Example 11.: $T^*S^1 \cong S^1 \times \mathbb{R}$

The cotangent bundle of S^1 is also trivial:

$$T^*S^1 \;\cong\; S^1 \times \mathbb{R}.$$

A point $(\theta, p_\theta) \in T^*S^1$ represents an angle and its conjugate momentum — this is the phase space of a particle moving on a circle, or of the planar rigid body. Since $TS^1 \cong S^1 \times \mathbb{R}$ as well, $T^*S^1 \cong TS^1$ in this case.

Example 11.: T^*S^2 — non-trivial bundle

The cotangent bundle T^*S^2 is non-trivial as a bundle: just as TS^2 has no global frame, T^*S^2 has no global cobasis. This is again the hairy ball theorem — a nowhere-vanishing 1-form on S^2 would give a nowhere-vanishing vector field via any Riemannian metric.

Example 11.: $T^*(S^1 \times \mathbb{R}) \cong (S^1 \times \mathbb{R}) \times \mathbb{R}^2$

The cylinder is parallelisable, so its cotangent bundle is trivial. Canonical coordinates are $(\theta, z, p_\theta, p_z)$, the phase space of a particle constrained to move on the cylinder.

11.7 Differential 1-forms

Just as a vector field is a smooth section of TM, a differential 1-form is a smooth section of the cotangent bundle.

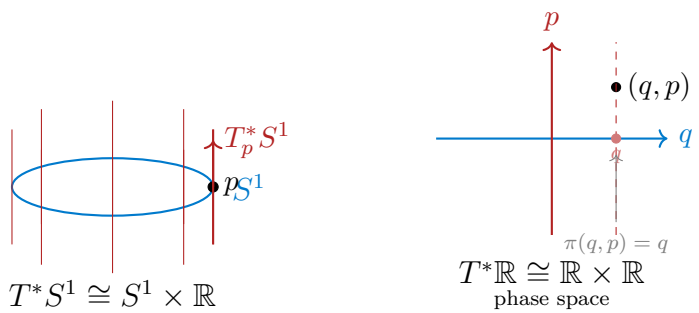


Figure 25: Left: T^*S^1 is a trivial bundle — a cylinder with $\mathbb{R}$-fibers (red) over the base circle (blue). Right: $T^*\mathbb{R}$ is the phase space of classical mechanics — position q on the base and momentum p in the fiber.

Definition 11.: Differential 1-form

Let M be a smooth manifold. A *differential 1-form* (or simply a *1-form*) on M is a smooth map

$$\omega : M \longrightarrow T^*M$$

such that $\pi \circ \omega = \mathrm{id}_M$, i.e. $\omega(p) \in T_p^*M$ for every $p \in M$. The set of all smooth 1-forms on M is denoted

$$\Omega^1(M) \;=\; \Gamma(T^*M).$$

Remark 11.

In a local chart (U, x), every 1-form $\omega \in \Omega^1(M)$ restricts to

$$\omega\Big|_U \;=\; \sum_{i=1}^{d} \omega_i(x)\, dx^i,$$

where $\omega_i : U \to \mathbb{R}$ are smooth functions called the *components* of ω. The component functions are

$$\omega_i(p) \;=\; \langle \omega(p),\, \partial_{x^i}|_p \rangle.$$

Smoothness of ω as a section is equivalent to smoothness of all ω_i. A 1-form acts on a vector field $X \in \mathfrak{X}(M)$ to give a smooth function:

$$\langle \omega, X \rangle : M \to \mathbb{R}, \qquad p \mapsto \langle \omega(p), X(p) \rangle.$$

Remark 11.

The space $\Omega^1(M)$ is a $C^\infty(M)$-module: for $\omega, \eta \in \Omega^1(M)$ and $f \in C^\infty(M)$,

$$(\omega + \eta)(p) = \omega(p) + \eta(p), \qquad (f\omega)(p) = f(p)\,\omega(p).$$

There is a natural pairing $\Omega^1(M) \times \mathfrak{X}(M) \to C^\infty(M)$ which is $C^\infty(M)$-bilinear. The modules $\Omega^1(M)$ and $\mathfrak{X}(M)$ are dual to each other over $C^\infty(M)$.

The differential as a 1-form. The most important source of 1-forms is the differential construction. For any $f \in C^\infty(M)$, the assignment $p \mapsto df_p$ defines a 1-form $df \in \Omega^1(M)$, called the *differential* of f. In coordinates:

$$df \;=\; \sum_{i=1}^{d} \frac{\partial f}{\partial x^i}\, dx^i.$$

A 1-form of the form df for some function f is called *exact*.

Examples of 1-forms

Example 11.: 1-forms on $\mathbb{R}^2$

- $\omega = dx$: the projection onto the x-component. $\langle dx, v^x\partial_x + v^y\partial_y\rangle = v^x$. This is exact: $dx = d(\mathrm{pr}_1)$ where $\mathrm{pr}_1(x, y) = x$.
- $\omega = x\,dy - y\,dx$: acts on $v = v^x\partial_x + v^y\partial_y$ by $\langle\omega, v\rangle = xv^y - yv^x$. This measures the angular momentum of v about the origin. It is *not exact* globally (though it is locally).
- $\omega = \frac{-y\,dx + x\,dy}{x^2+y^2}$: the angle form on $\mathbb{R}^2 \setminus \{0\}$. In polar coordinates, $\omega = d\theta$. It satisfies $d\omega = 0$ (closed) but is not exact on $\mathbb{R}^2 \setminus \{0\}$ — integrating around the origin gives $\int_{S^1} \omega = 2\pi \neq 0$. This is the simplest example of a closed but non-exact form, and its cohomology class generates $H^1(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{R}$.

Example 11.: The tautological 1-form on $T^*\mathbb{R}^n$

On the cotangent bundle $T^*\mathbb{R}^n$ with canonical coordinates $(q^1, \ldots, q^n, p_1, \ldots, p_n)$, the *tautological* (or *Liouville*) 1-form is

$$\theta \;=\; \sum_{i=1}^{n} p_i\, dq^i.$$

It has the defining property $\pi^*\omega = \theta$ for any 1-form ω on $\mathbb{R}^n$. Its exterior derivative $\omega_{can} = -d\theta = \sum dq^i \wedge dp_i$ is the canonical symplectic form — the foundation of Hamiltonian mechanics.

Example 11.: The angle form $d\theta$ on S^1

The 1-form $d\theta \in \Omega^1(S^1)$ assigns to each point $p \in S^1$ the covector $d\theta|_p$ that evaluates to 1 on $\partial_\theta|_p$. In ambient coordinates, at $p = (\cos\theta, \sin\theta)$:

$$\langle d\theta, v^x\partial_x + v^y\partial_y\rangle = -\sin\theta\, v^x + \cos\theta\, v^y.$$

It is the unique (up to scale) nowhere-vanishing 1-form on S^1, confirming that $T^*S^1 \cong S^1 \times \mathbb{R}$ is trivial. Its integral over S^1 gives $\int_{S^1} d\theta = 2\pi$.

Example 11.: Three 1-forms on the cylinder $S^1 \times \mathbb{R}$

Using coordinates (θ, z) on $C = S^1 \times \mathbb{R}$:

- $\omega_1 = d\theta$: pairs with the angular direction, kills the vertical. $\langle \omega_1, \partial_\theta \rangle = 1$, $\langle \omega_1, \partial_z \rangle = 0$.
- $\omega_2 = dz$: pairs with the vertical direction, kills the angular. $\langle \omega_2, \partial_\theta \rangle = 0$, $\langle \omega_2, \partial_z \rangle = 1$.
- $\omega_3 = \cos\theta\, d\theta + dz$: a mixed form that pairs non-trivially with both directions.

These are dual to the three vector fields $X_1 = \partial_\theta$, $X_2 = \partial_z$, $X_3 = \partial_\theta + \partial_z$ from the tangent bundle section.

Example 11.: 1-forms on S^2

Using spherical coordinates (θ, φ) on S^2:

- $\omega = \sin\theta\, d\varphi$: the area element's angular part. It vanishes at $\theta = 0$ (north pole) and $\theta = \pi$ (south pole), consistent with the hairy ball theorem — any 1-form on S^2 must vanish somewhere.
- $\omega = d\theta$: vanishes on the φ-direction everywhere. It also vanishes at the poles where θ is not a good coordinate.

There is no nowhere-vanishing 1-form on S^2, dual to the fact that T^*S^2 is a non-trivial bundle.

Pullback of 1-forms. The pullback acts naturally on 1-forms: for $f : M \to N$ and $\omega \in \Omega^1(N)$, the pullback $f^*\omega \in \Omega^1(M)$ is defined pointwise by $(f^*\omega)(p) = f_p^*(\omega(f(p)))$, or equivalently:

$$\langle f^*\omega,\, X \rangle \;=\; \langle \omega,\, df(X) \rangle, \qquad X \in \mathfrak{X}(M).$$

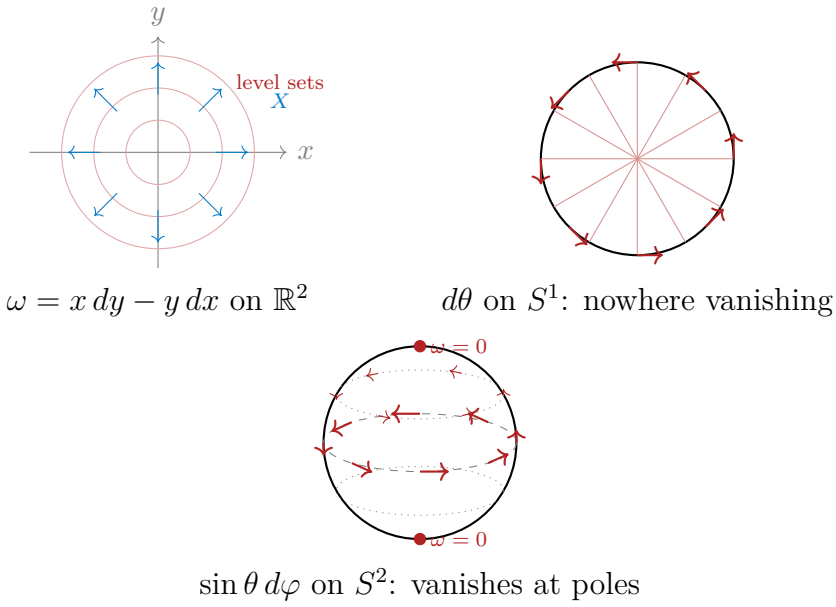
In coordinates, if $\omega = \sum_j \omega_j(y)\, dy^j$ then

$$f^*\omega \;=\; \sum_{i,j} \omega_j(f(x))\, \frac{\partial f^j}{\partial x^i}\, dx^i.$$

This is the chain rule for 1-forms. In particular, for an exact form:

$$f^*(dg) \;=\; d(g \circ f),$$

so the pullback commutes with the differential — a fundamental compatibility that will persist for all differential forms.



$\omega = x\,dy - y\,dx$ on $\mathbb{R}^2$ $\qquad$ $d\theta$ on S^1: nowhere vanishing

$\sin\theta\,d\varphi$ on S^2: vanishes at poles

Figure 26: Left: the 1-form $x\,dy - y\,dx$ on $\mathbb{R}^2$, whose level sets are circles centred at the origin. Middle: the angle form $d\theta$ on S^1 is nowhere vanishing. Right: $\sin\theta\,d\varphi$ on S^2 must vanish at both poles, consistent with the hairy ball theorem.

11.8 Geometric interpretation of 1-forms

A 1-form is a field of level sets. A tangent vector $v \in T_pM$ has a clear geometric picture: it is an arrow at p, pointing in some direction with some magnitude. What is the geometric picture of a covector $\xi \in T_p^*M$?

The answer is: a covector is a *family of equally-spaced hyperplanes* in T_pM, and the pairing $\langle \xi, v\rangle$ counts *how many hyperplanes the arrow v crosses* — with sign depending on direction.

More precisely, the covector ξ determines a linear functional on T_pM, whose level sets $\{\xi = 0\}$, $\{\xi = 1\}$, $\{\xi = -1\}$, ... are equally-spaced hyperplanes (lines in $2d$, planes in $3d$) in T_pM. The pairing $\langle \xi, v\rangle$ is simply how many of these hyperplanes the vector v crosses.

Remark 11.

This is why covectors are sometimes called *one-forms*: they measure something along a single direction. The key difference from a vector is that *a covector has no preferred direction* — only a family of level sets with a spacing and an orientation. Two covectors with the same level sets but different spacings are proportional; two with different orientations of level sets are genuinely different.

The coordinate 1-forms dx and dy on $\mathbb{R}^2$

The simplest examples make the picture clear. On $\mathbb{R}^2$ with coordinates (x, y), the two coordinate 1-forms are:

$$dx(v^x, v^y) = v^x, \qquad dy(v^x, v^y) = v^y.$$

- dx has level sets $\{x = c\}$ — equally spaced *vertical lines*. It counts how many vertical lines the vector crosses. A purely vertical vector $v = (0, 1)$ crosses none: $dx(v) = 0$. A horizontal vector $v = (2, 0)$ crosses two: $dx(v) = 2$.

- dy has level sets $\{y = c\}$ — equally spaced *horizontal lines*. It counts how many horizontal lines the vector crosses. A horizontal vector $v = (1, 0)$ crosses none: $dy(v) = 0$. A vertical vector $v = (0, 3)$ crosses three: $dy(v) = 3$.

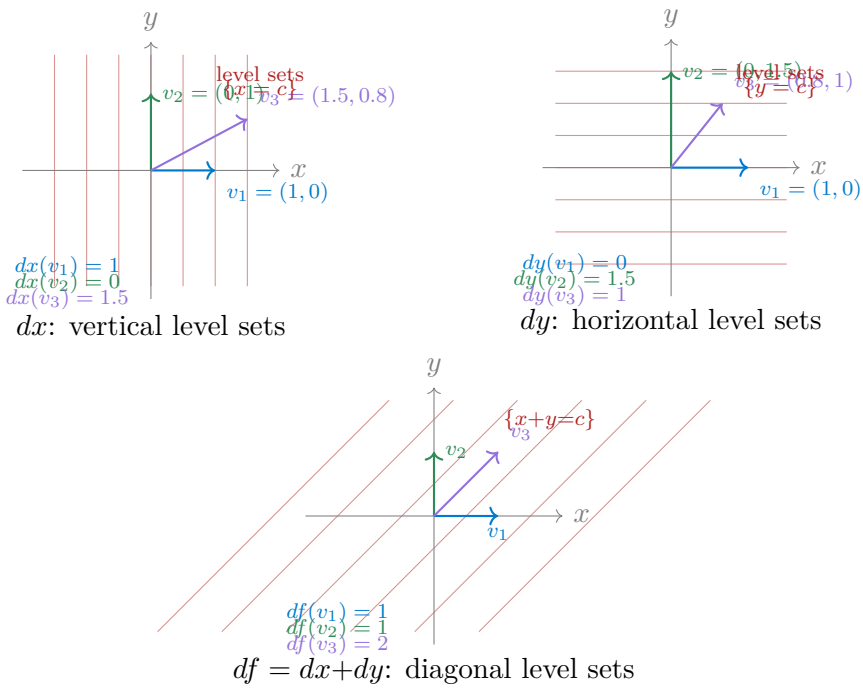


Figure 27: The coordinate 1-forms dx (left) and dy (middle) with their level sets and example vectors. Right: $df = dx + dy$ for $f = x + y$ has diagonal level sets — the vector $v_3 = (1, 1)$ crosses two diagonals, giving $df(v_3) = 2$.

The differential df as level sets of f

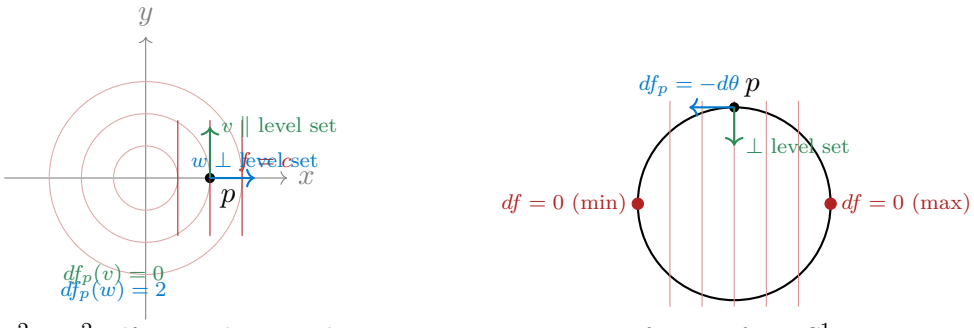
For a general function $f \in C^\infty(M)$, the differential df_p at a point p has a beautiful geometric meaning: its level sets in T_pM are the *tangent hyperplanes to the level sets of* f through p.

Equivalently, $\langle df_p, v\rangle$ measures the rate at which f changes as you move in the direction v — which is zero precisely when v is tangent to a level set of f.

Point-dependent spacing: dy vs. $y\,dy$

A 1-form field $\omega \in \Omega^1(M)$ assigns a covector — a family of level sets — to *each point separately*. The spacing of the level sets can vary from point to point.

- dy: at every point p, the level sets in T_pM are horizontal lines with *constant* unit spacing. The same vector $v = (0, 1)$ always crosses exactly one line, regardless of where p is.



$f = x^2 + y^2$, $df = 2x\,dx + 2y\,dy$ $\qquad$ $f = \cos\theta$ on S^1

Figure 28: Left: level sets of $f = x^2 + y^2$ (circles). At $p = (1, 0)$, a vector tangent to the level set gives $df_p = 0$; a vector crossing the level sets gives $df_p \neq 0$. The level sets of df_p in T_pM are the tangent lines to the circles. Right: $f = \cos\theta$ on S^1, vanishing at the critical points where v is tangent to the level set.

- $y\,dy = d(y^2/2)$: at a point $p = (x_0, y_0)$, the level sets have spacing $1/|y_0|$ — they are *denser* far from the x-axis and more spread out near it. The same vector $v = (0, 1)$ crosses $|y_0|$ level sets at height y_0.

dy: uniform, same value everywhere $\qquad$ $y\,dy = d(y^2/2)$: denser near $|y|$ large

Figure 29: Left: dy assigns uniformly-spaced horizontal level sets at every point — the same vector always gives the same value. Right: $y\,dy$ has level sets whose spacing varies with y — the same upward vector gives a larger value where $|y|$ is large (denser lines) and smaller where $|y| \approx 0$ (sparser lines).

Summary: vectors vs. covectors

Observation 11.

The geometric duality between vectors and covectors can be summarised as:

	Tangent vector $v \in T_pM$	**Covector** $\xi \in T_p^*M$
Picture	Arrow at p	Family of hyperplanes at p
Data	Direction + magnitude	Orientation + spacing
Pairing	$\langle \xi, v \rangle$ = number of hyperplanes v crosses	
Coordinate form	$v = \sum v^i \partial_i$ (upper indices)	$\xi = \sum \xi_i dx^i$ (lower indices)
Transforms	with Jacobian	with inverse Jacobian

Remark 11.

This picture extends to higher degrees: a 2-form assigns a family of *tubes* (intersections of two families of hyperplanes) at each point, and $\omega(v_1, v_2)$ counts how many tubes the parallelogram spanned by v_1, v_2 is pierced by. A 3-form assigns *boxes*, and $\omega(v_1, v_2, v_3)$ counts the volume in those boxes. This is the subject of the next section on k-forms.

12 Differential k-forms

12.1 Motivation

A 1-form at a point p eats a single tangent vector and returns a number — it measures change in one direction. But many natural quantities in geometry and physics involve *several* directions simultaneously:

- **Area**: to measure the area of a parallelogram spanned by two vectors v_1, v_2, you need something that eats both and returns a number.
- **Volume**: a volume element eats three vectors and returns the volume of the parallelepiped they span.
- **Flux**: the flux of a field through a surface requires pairing the field with two tangent vectors to the surface.
- **Curvature**: in differential geometry, curvature is naturally a 2-form — it measures the infinitesimal holonomy around a parallelogram.

The right framework for all of these is the theory of *differential k-forms*: fields of multilinear, antisymmetric maps that eat k tangent vectors and return a number. The antisymmetry condition is what gives them their oriented, signed character — the signed area/volume of a parallelepiped rather than just its magnitude.

12.2 Review of multilinear algebra

Multilinear maps

Definition 12.: Multilinear map

Let V be a real vector space. A map $T : V \times \cdots \times V \to \mathbb{R}$ (k factors) is *multilinear* (or *k-linear*) if it is linear in each argument separately: for each $i = 1, \ldots, k$ and all $v_1, \ldots, v_k \in V$, $\lambda \in \mathbb{R}$,

$$T(v_1, \ldots, \lambda u + \mu w, \ldots, v_k) = \lambda T(v_1, \ldots, u, \ldots, v_k) + \mu T(v_1, \ldots, w, \ldots, v_k).$$

Example 12.: Simple multilinear maps

- The dot product $(v, w) \mapsto v \cdot w$ on $\mathbb{R}^n$ is bilinear.
- The determinant $\det : \mathbb{R}^n \times \cdots \times \mathbb{R}^n \to \mathbb{R}$ (n factors) is multilinear.
- A 1-form $\xi \in T_p^* M$ is a 1-linear map $T_p M \to \mathbb{R}$.

Antisymmetry and alternating maps

Definition 12.: Alternating map

A multilinear map $T : V^k \to \mathbb{R}$ is *alternating* (or *antisymmetric*) if swapping any two arguments changes the sign:

$$T(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_k) = -T(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_k).$$

Equivalently, T vanishes whenever two arguments are equal: $T(\ldots, v, \ldots, v, \ldots) = 0$.

Remark 12.

Antisymmetry encodes *orientation*. The signed area of the parallelogram spanned by v_1, v_2 changes sign if you swap v_1 and v_2 — this is the same as reversing the orientation of the parallelogram. The determinant is the canonical example: $\det(v_1, v_2) = -\det(v_2, v_1)$.

Example 12.: Antisymmetry in $\mathbb{R}^2$

The map $T(v, w) = v^x w^y - v^y w^x$ (the 2×2 determinant) is bilinear and alternating:

$$T(w, v) = w^x v^y - w^y v^x = -(v^x w^y - v^y w^x) = -T(v, w).$$

Geometrically, $T(v, w)$ is the signed area of the parallelogram spanned by v and w: positive if w is counterclockwise from v, negative otherwise.

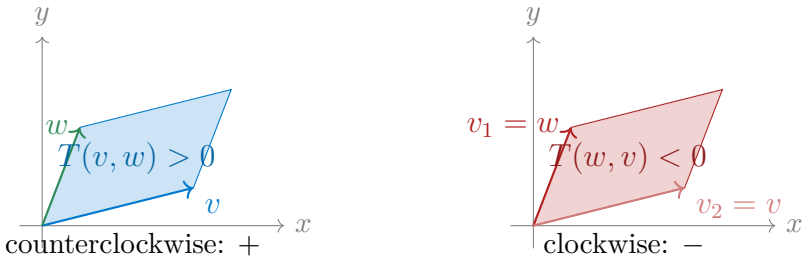


Figure 30: Antisymmetry as orientation. Swapping v and w gives the same parallelogram but with opposite orientation — the signed area changes sign.

The space of alternating k-linear maps

Definition 12.: Alternating k-linear forms

Let V be a d-dimensional real vector space. The space of all alternating k-linear maps $V^k \to \mathbb{R}$ is denoted $\Lambda^k(V^*)$. Its dimension is $\binom{d}{k}$. For $k > d$, $\Lambda^k(V^*) = 0$.

Remark 12.

The dimension count $\binom{d}{k}$ comes from the fact that an alternating k-linear map is completely determined by its values on ordered k-tuples of basis vectors, and antisymmetry means only the $\binom{d}{k}$ increasing index tuples matter. In particular:

$$\dim \Lambda^0(V^*) = 1, \quad \dim \Lambda^1(V^*) = d, \quad \dim \Lambda^2(V^*) = \tfrac{d(d-1)}{2}, \quad \dim \Lambda^d(V^*) = 1.$$

Example 12.: Dimensions in low degrees

For $V = \mathbb{R}^3$ $(d = 3)$:

- Λ^0: dimension 1 — just scalars.
- Λ^1: dimension 3 — covectors, spanned by dx, dy, dz.
- Λ^2: dimension 3 — spanned by $dx \wedge dy$, $dy \wedge dz$, $dx \wedge dz$.
- Λ^3: dimension 1 — spanned by $dx \wedge dy \wedge dz$ (the volume form).
- $\Lambda^k = 0$ for $k > 3$.

For $V = \mathbb{R}^2$ $(d = 2)$: dimensions $1, 2, 1, 0, 0, \ldots$

12.3 The wedge product and exterior algebra

Definition 12.: Wedge product

Let $\alpha \in \Lambda^j(V^*)$ and $\beta \in \Lambda^k(V^*)$. Their *wedge product* $\alpha \wedge \beta \in \Lambda^{j+k}(V^*)$ is defined by

$$(\alpha \wedge \beta)(v_1, \ldots, v_{j+k}) \;=\; \frac{1}{j!\,k!} \sum_{\sigma \in S_{j+k}} \operatorname{sgn}(\sigma)\; \alpha(v_{\sigma(1)}, \ldots, v_{\sigma(j)})\, \beta(v_{\sigma(j+1)}, \ldots, v_{\sigma(j+k)}),$$

where the sum is over all permutations σ of $\{1, \ldots, j+k\}$.

Remark 12.

The formula looks complicated but the key properties are simple:

1. **Associativity**: $(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma)$.
2. **Graded commutativity**: $\alpha \wedge \beta = (-1)^{jk}\, \beta \wedge \alpha$. In particular, $\alpha \wedge \alpha = 0$ for any odd-degree form α.
3. **Bilinearity**: linear in each factor.

For 1-forms α, β: $\alpha \wedge \beta = -\beta \wedge \alpha$, so $\alpha \wedge \alpha = 0$.

Wedge product of basis covectors

If $\{e^1, \ldots, e^d\}$ is the dual basis of V^*, then a basis for $\Lambda^k(V^*)$ is given by the wedge products

$$e^{i_1} \wedge e^{i_2} \wedge \cdots \wedge e^{i_k}, \qquad 1 \le i_1 < i_2 < \cdots < i_k \le d.$$

The action on basis vectors is:

$$(e^{i_1} \wedge \cdots \wedge e^{i_k})(e_{j_1}, \ldots, e_{j_k}) = \det\left[\delta^{i_a}_{j_b}\right]^k_{a,b=1} = \begin{cases} \operatorname{sgn}(\sigma) & \text{if } \{j_1, \ldots, j_k\} = \{i_1, \ldots, i_k\}, \\ 0 & \text{otherwise,} \end{cases}$$

where σ is the permutation relating the two orderings.

Example 12.: Wedge products in $\mathbb{R}^3$

With the standard dual basis $\{dx, dy, dz\}$:

$$dx \wedge dy\,(v_1, v_2) = \det \begin{pmatrix} v_1^x & v_2^x \\ v_1^y & v_2^y \end{pmatrix} = v_1^x v_2^y - v_1^y v_2^x.$$

This is the signed area of the projection of the parallelogram onto the xy-plane. Similarly:

$$dy \wedge dz\,(v_1, v_2) = v_1^y v_2^z - v_1^z v_2^y \quad \text{(area projected onto } yz\text{-plane)},$$

$$dx \wedge dz\,(v_1, v_2) = v_1^x v_2^z - v_1^z v_2^x \quad \text{(area projected onto } xz\text{-plane)}.$$

The 3-form:

$$dx \wedge dy \wedge dz\,(v_1, v_2, v_3) = \det \begin{pmatrix} v_1^x & v_2^x & v_3^x \\ v_1^y & v_2^y & v_3^y \\ v_1^z & v_2^z & v_3^z \end{pmatrix}.$$

This is the signed volume of the parallelepiped spanned by v_1, v_2, v_3.

Example 12.: Antisymmetry kills repeated indices

Since $dx \wedge dy = -dy \wedge dx$:

$$dx \wedge dx = 0, \quad dy \wedge dy = 0, \quad dz \wedge dz = 0.$$

A vector parallel to the x-axis spans no area in any plane containing the x-direction — the antisymmetry correctly encodes this.

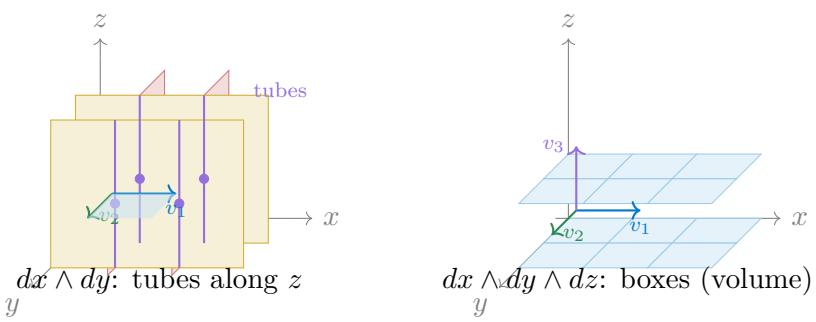


Figure 31: Left: $dx \wedge dy$ in $\mathbb{R}^3$ — red planes ($x = \text{const}$) and orange planes ($y = \text{const}$) intersect to form purple tubes running along z. The 2-form counts how many tubes the shaded parallelogram is pierced by. Right: $dx \wedge dy \wedge dz$ fills space with unit boxes — the 3-form counts the signed volume of the parallelepiped spanned by v_1, v_2, v_3.

Definition 12.: Exterior algebra

The *exterior algebra* of V^* is the direct sum

$$\Lambda^*(V^*) \;=\; \bigoplus_{k=0}^{d} \Lambda^k(V^*),$$

equipped with the wedge product. It is an associative, graded-commutative algebra of dimension 2^d.

12.4 Differential k-forms on a manifold

Definition 12.: Differential k-form

Let M be a smooth d-dimensional manifold. A *differential k-form* on M is a smooth section of the bundle $\Lambda^k T^*M$:

$$\omega : M \longrightarrow \Lambda^k T^*M, \qquad \pi \circ \omega = \mathrm{id}_M.$$

That is, ω assigns to each point $p \in M$ an alternating k-linear map

$$\omega_p : T_pM \times \cdots \times T_pM \;\longrightarrow\; \mathbb{R} \quad (k \text{ factors}),$$

varying smoothly with p. The space of all smooth k-forms is denoted $\Omega^k(M) = \Gamma(\Lambda^k T^*M)$.

Remark 12.

In local coordinates $(x^1, \ldots, x^d)$, every k-form writes as

$$\omega\Big|_U = \sum_{i_1 < i_2 < \cdots < i_k} \omega_{i_1 \cdots i_k}(x)\, dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k},$$

where $\omega_{i_1 \cdots i_k} \in C^\infty(U)$ are smooth component functions. We have:

$$\Omega^0(M) = C^\infty(M), \quad \Omega^1(M) = \Gamma(T^*M), \quad \Omega^k(M) = 0 \text{ for } k > d.$$

Examples of k-forms

Example 12.: 2-forms on $\mathbb{R}^3$

The general 2-form on $\mathbb{R}^3$ is

$$\omega = P\,dy \wedge dz + Q\,dz \wedge dx + R\,dx \wedge dy, \qquad P, Q, R \in C^\infty(\mathbb{R}^3).$$

This corresponds exactly to a vector field $F = (P, Q, R)$ via the identification

$$\omega(v_1, v_2) = F \cdot (v_1 \times v_2) \quad \text{(the flux of } F \text{ through the parallelogram } v_1, v_2\text{)}.$$

This is the starting point of the correspondence between vector calculus and differential forms.

Example 12.: The area form on S^2

The standard area form on the unit sphere S^2 is

$$\omega_{S^2} = x\,dy \wedge dz + y\,dz \wedge dx + z\,dx \wedge dy \;=\; \iota^*(r\,dr \wedge d\phi)$$

or in spherical coordinates (θ, φ):

$$\omega_{S^2} = \sin\theta\,d\theta \wedge d\varphi.$$

At any point $p \in S^2$, ω_{S^2} measures the area of the parallelogram spanned by two tangent vectors. Its integral gives the total area: $\int_{S^2} \omega_{S^2} = 4\pi$.

Example 12.: 2-form on the cylinder $S^1 \times \mathbb{R}$

On $C = S^1 \times \mathbb{R}$ with coordinates (θ, z), the general 2-form is

$$\omega = f(\theta, z)\,d\theta \wedge dz,$$

since $\Lambda^2(T_p^*C)$ is 1-dimensional ($\binom{2}{2} = 1$). The standard area form is $d\theta \wedge dz$. It satisfies:

$$d\theta \wedge dz\,(\partial_\theta, \partial_z) = d\theta(\partial_\theta)\,dz(\partial_z) - d\theta(\partial_z)\,dz(\partial_\theta) = 1 \cdot 1 - 0 \cdot 0 = 1.$$

Example 12.: The symplectic form on $T^*\mathbb{R}^n$

The cotangent bundle $T^*\mathbb{R}^n$ with canonical coordinates (q^i, p_i) carries the canonical *symplectic 2-form*

$$\omega = \sum_{i=1}^{n} dq^i \wedge dp_i.$$

It is closed ($d\omega = 0$), non-degenerate, and is the foundation of Hamiltonian mechanics. In 2-dimensions ($n = 1$): $\omega = dq \wedge dp$, the signed area element on phase space.

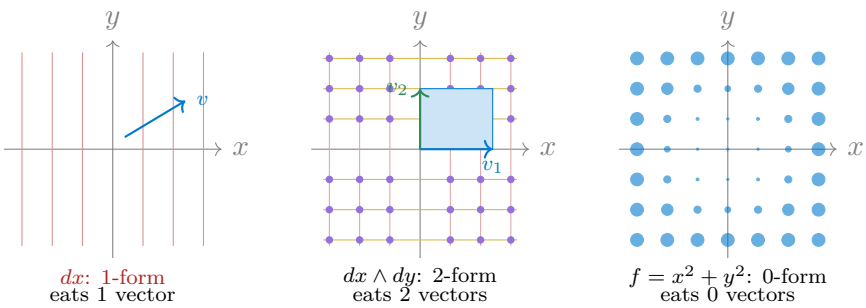


Figure 32: From left to right: a 0-form (function — a value at each point), a 1-form (dx — one family of lines, eats one vector), and a 2-form ($dx \wedge dy$ — a grid of points, eats two vectors and counts how many grid points lie inside the parallelogram).

12.5 The exterior derivative

The most important operation on differential forms is the *exterior derivative*, which generalises the gradient, curl, and divergence of vector calculus into a single unified operator.

Definition 12.: Exterior derivative

There exists a unique family of linear maps

$$d : \Omega^k(M) \longrightarrow \Omega^{k+1}(M), \qquad k = 0, 1, \ldots, d,$$

characterised by the following properties:

1. **On functions**: $d : \Omega^0(M) \to \Omega^1(M)$ is the differential $df_p(v) = v(f)$.
2. **Leibniz rule**: $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$ for $\alpha \in \Omega^k(M)$.
3. **Nilpotency**: $d \circ d = 0$, i.e. $d^2\omega = 0$ for all ω.

Remark 12.

In local coordinates, if $\omega = \sum_I \omega_I \, dx^I$ (where $I = (i_1 < \cdots < i_k)$ is a multi-index), then

$$d\omega \;=\; \sum_I \sum_{j=1}^{d} \frac{\partial \omega_I}{\partial x^j} \, dx^j \wedge dx^I.$$

The nilpotency $d^2 = 0$ follows from the equality of mixed partials $\frac{\partial^2 f}{\partial x^i \partial x^j} = \frac{\partial^2 f}{\partial x^j \partial x^i}$ combined with the antisymmetry $dx^i \wedge dx^j = -dx^j \wedge dx^i$.

The exterior derivative in coordinates

Example 12.: Exterior derivative of a function ($0 \to 1$)

For $f \in C^\infty(\mathbb{R}^3)$:

$$df = \frac{\partial f}{\partial x} \, dx + \frac{\partial f}{\partial y} \, dy + \frac{\partial f}{\partial z} \, dz.$$

This is the **gradient** of f written in the language of forms.

Example 12.: Exterior derivative of a 1-form ($1 \to 2$)

For $\omega = P \, dx + Q \, dy + R \, dz$ on $\mathbb{R}^3$:

$$d\omega = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy \wedge dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dz \wedge dx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \wedge dy.$$

The coefficients are exactly the components of $\operatorname{curl}(P, Q, R)$. So d on 1-forms in $\mathbb{R}^3$ is the **curl**.

Example 12.: Exterior derivative of a 2-form ($2 \to 3$)

For $\omega = P \, dy \wedge dz + Q \, dz \wedge dx + R \, dx \wedge dy$ on $\mathbb{R}^3$:

$$d\omega = \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx \wedge dy \wedge dz.$$

The coefficient is $\operatorname{div}(P, Q, R)$. So d on 2-forms in $\mathbb{R}^3$ is the **divergence**.

Observation 12.

The three classical operators of vector calculus are all instances of d:

$$\begin{array}{ccccccc} \Omega^0(\mathbb{R}^3) & \xrightarrow{d} & \Omega^1(\mathbb{R}^3) & \xrightarrow{d} & \Omega^2(\mathbb{R}^3) & \xrightarrow{d} & \Omega^3(\mathbb{R}^3) \\ \text{functions} & & \text{1-forms} & & \text{2-forms} & & \text{3-forms} \\ f & \xrightarrow{\text{grad}} & \nabla f & \xrightarrow{\text{curl}} & \nabla \times F & \xrightarrow{\text{div}} & \nabla \cdot G \end{array}$$

The identity $d^2 = 0$ encodes the classical identities $\operatorname{curl}(\nabla f) = 0$ and $\operatorname{div}(\operatorname{curl} F) = 0$.

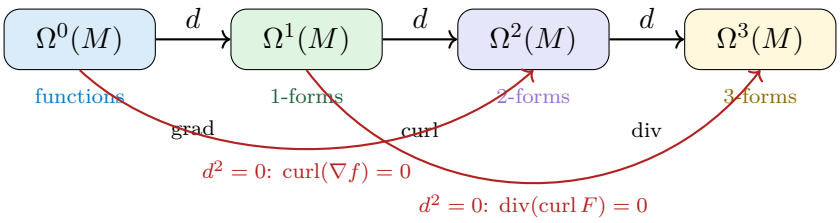


Figure 33: The exterior derivative d as a single operator unifying grad, curl, and div on $\mathbb{R}^3$. The nilpotency $d^2 = 0$ encodes the classical identities at once.

Examples on S^1 and S^2

Example 12.: Exterior derivative on S^1

On S^1 with coordinate θ, let $\omega = \sin\theta\, d\theta \in \Omega^1(S^1)$. Since $\Omega^2(S^1) = 0$ (a 1-manifold has no 2-forms), we have $d\omega = 0$ automatically. Every 1-form on S^1 is closed.
For a function $f(\theta) = \cos\theta$: $df = -\sin\theta\, d\theta$.

Example 12.: Exterior derivative of the area form on S^2

The area form $\omega = \sin\theta\, d\theta \wedge d\varphi \in \Omega^2(S^2)$. Since $\Omega^3(S^2) = 0$, we have $d\omega = 0$ automatically. Every top-degree form on a manifold is closed.
For $\alpha = \cos\theta\, d\varphi \in \Omega^1(S^2)$:

$$d\alpha = d(\cos\theta) \wedge d\varphi = -\sin\theta\, d\theta \wedge d\varphi = -\omega.$$

So $\omega = -d\alpha$ — the area form is exact on the chart $(0, \pi) \times (0, 2\pi)$, but *not* globally exact on S^2 (its integral $\int_{S^2} \omega = 4\pi \neq 0$).

12.6 The interior product

Motivation. The exterior derivative d raises the degree of a form by one. There is a complementary operation that *lowers* it by one: the *interior product* (or *contraction*), which plugs a vector field into the first slot of a form.

Definition 12.: Interior product

Let $X \in \mathfrak{X}(M)$ and $\omega \in \Omega^k(M)$. The *interior product* of ω with X is the $(k-1)$-form $\iota_X\omega \in \Omega^{k-1}(M)$ defined by

$$(\iota_X\omega)(v_1, \ldots, v_{k-1}) \;=\; \omega(X,\, v_1, \ldots, v_{k-1}).$$

For a function $f \in \Omega^0(M)$ we set $\iota_X f = 0$.

Remark 12.

The interior product is also written $X \lrcorner \omega$ or $i_X\omega$. It satisfies the graded Leibniz rule:

$$\iota_X(\alpha \wedge \beta) = (\iota_X\alpha) \wedge \beta + (-1)^k\, \alpha \wedge (\iota_X\beta), \qquad \alpha \in \Omega^k(M),$$

and is *nilpotent*: $\iota_X \circ \iota_X = 0$, since plugging the same vector twice into an alternating form gives zero.

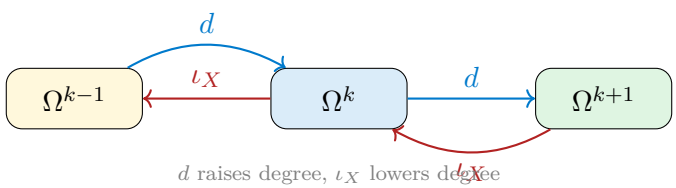


Figure 34: d and ι_X are complementary: d raises degree by 1, ι_X lowers it by 1.

Examples of the interior product

Example 12.: Interior product on $\mathbb{R}^3$ — 2-form

Let $\omega = dx \wedge dy \in \Omega^2(\mathbb{R}^3)$. Contract with coordinate fields:

$$\iota_{\partial_x}(dx \wedge dy) = dx(\partial_x)\, dy - dy(\partial_x)\, dx = 1 \cdot dy - 0 \cdot dx = dy.$$

$$\iota_{\partial_y}(dx \wedge dy) = dx(\partial_y)\, dy - dy(\partial_y)\, dx = 0 \cdot dy - 1 \cdot dx = -dx.$$

$$\iota_{\partial_z}(dx \wedge dy) = dx(\partial_z)\, dy - dy(\partial_z)\, dx = 0.$$

The z-direction is invisible to $dx \wedge dy$ — contracting with it gives zero, because ∂_z lies in the kernel of both dx and dy.

Example 12.: Interior product with the volume form

Let $\Omega = dx \wedge dy \wedge dz \in \Omega^3(\mathbb{R}^3)$ be the volume form and $X = P\partial_x + Q\partial_y + R\partial_z$ a vector field. Then:

$$\iota_X \Omega = P\, dy \wedge dz - Q\, dx \wedge dz + R\, dx \wedge dy = P\, dy \wedge dz + Q\, dz \wedge dx + R\, dx \wedge dy.$$

This is the 2-form corresponding to the vector field (P, Q, R) via the Hodge star. It measures the flux of X through oriented surface elements.

Example 12.: Interior product on S^2 — the area form

Let $\omega = \sin\theta\, d\theta \wedge d\varphi \in \Omega^2(S^2)$. Contract with $X = \partial_\theta$:

$$\iota_{\partial_\theta}(\sin\theta\, d\theta \wedge d\varphi) = \sin\theta\, d\varphi.$$

Contract with $Y = \partial_\varphi$:

$$\iota_{\partial_\varphi}(\sin\theta\, d\theta \wedge d\varphi) = -\sin\theta\, d\theta.$$

Example 12.: Nilpotency: $\iota_X^2 = 0$

Let $\omega = dx \wedge dy \wedge dz$ and $X = \partial_x$. Then:

$$\iota_X \omega = dy \wedge dz.$$

$$\iota_X(\iota_X \omega) = \iota_X(dy \wedge dz) = dx(\partial_x)\, dy \wedge dz\Big|_{x\text{-slot}}$$

But $\iota_X(dy \wedge dz) = dy(\partial_x)\, dz - dz(\partial_x)\, dy = 0$. Contracting twice with the same vector kills the form — antisymmetry forces this.

Example 12.: Interior product and flux in physics

In electromagnetism on Minkowski spacetime, the Faraday 2-form F encodes both $\mathbf{E}$ and $\mathbf{B}$. Given the 4-velocity u^μ, the contraction $\iota_u F$ gives the electric 1-form felt by an observer moving with velocity u. The magnetic part is recovered from $\iota_u(\star F)$, where $\star$ is the Hodge dual. This is how the interior product naturally separates the field tensor into electric and magnetic components depending on the observer.

12.7 The Lie derivative

Motivation. The interior product contracts a form with a vector. The exterior derivative differentiates a form. But neither answers the question: *how does a form change as you flow along a vector field?* This is the role of the Lie derivative.

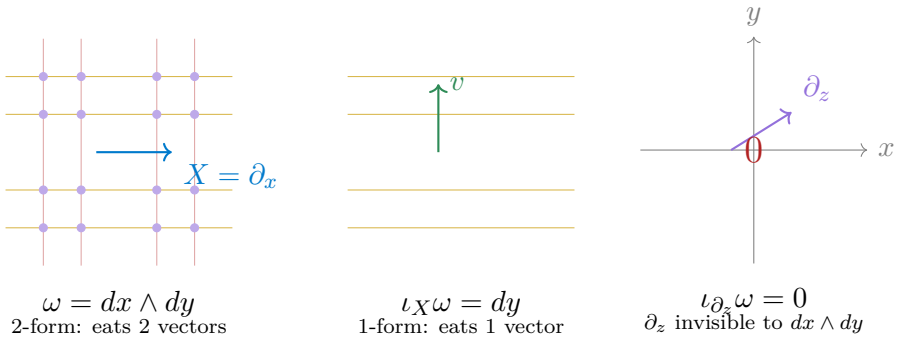


Figure 35: Interior product on $dx \wedge dy$. Left: the 2-form as a grid of purple dots. Middle: contracting with ∂_x plugs the horizontal direction in, leaving only the vertical level sets dy. Right: contracting with ∂_z gives zero — ∂_z is invisible to $dx \wedge dy$.

Definition 12.: Lie derivative of a form

Let $X \in \mathfrak{X}(M)$ with flow $\phi_t : M \to M$, and let $\omega \in \Omega^k(M)$. The *Lie derivative* of ω along X is

$$\mathcal{L}_X \omega \;=\; \lim_{t \to 0} \frac{\phi_t^* \omega - \omega}{t} \;=\; \left.\frac{d}{dt}\right|_{t=0} \phi_t^* \omega \;\in\; \Omega^k(M).$$

Remark 12.

Unlike d and ι_X, the Lie derivative *preserves degree*: $\mathcal{L}_X : \Omega^k(M) \to \Omega^k(M)$. It measures the infinitesimal change in ω as the manifold is "dragged" along the flow of X.

Operation	Symbol	Degree change	Meaning
Exterior derivative	d	$k \to k+1$	differentiate the form
Interior product	ι_X	$k \to k-1$	plug X into first slot
Lie derivative	$\mathcal{L}_X$	$k \to k$	drag along flow of X

Cartan's magic formula

Computing Lie derivatives from the definition requires computing flows, which is hard. Cartan's formula gives a purely algebraic shortcut.

Theorem 12.: Cartan's magic formula

For any $X \in \mathfrak{X}(M)$ and $\omega \in \Omega^k(M)$:

$$\mathcal{L}_X \omega \;=\; d(\iota_X \omega) + \iota_X (d\omega) \;=\; (d\iota_X + \iota_X d)\,\omega.$$

Remark 12.

This is the single most useful formula involving Lie derivatives. It expresses $\mathcal{L}_X$ entirely in terms of d and ι_X, so you never need to compute a flow to find $\mathcal{L}_X\omega$. The operator $d\iota_X + \iota_X d$ is sometimes called the *homotopy operator* associated to X.

An immediate consequence: if ω is closed ($d\omega = 0$), then

$$\mathcal{L}_X\omega = d(\iota_X\omega),$$

so the Lie derivative of a closed form is always exact. This is used constantly in symplectic geometry.

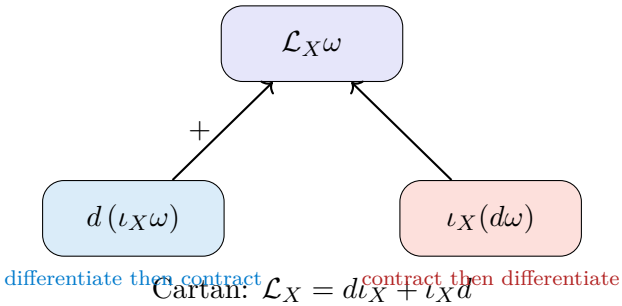


Figure 36: Cartan's magic formula: $\mathcal{L}_X\omega$ is the sum of "differentiate then contract" and "contract then differentiate".

Properties of the Lie derivative

The Lie derivative satisfies:

- $\mathcal{L}_X f = X(f)$ for $f \in C^\infty(M)$ (directional derivative of a function).
- $\mathcal{L}_X(\alpha \wedge \beta) = (\mathcal{L}_X\alpha) \wedge \beta + \alpha \wedge (\mathcal{L}_X\beta)$ (Leibniz rule — no sign!).
- $\mathcal{L}_X(d\omega) = d(\mathcal{L}_X\omega)$ ($\mathcal{L}_X$ commutes with d).
- $\mathcal{L}_{[X,Y]} = [\mathcal{L}_X, \mathcal{L}_Y] = \mathcal{L}_X\mathcal{L}_Y - \mathcal{L}_Y\mathcal{L}_X$.

Examples of the Lie derivative

Example 12.: Lie derivative of a function on $\mathbb{R}^2$

Let $X = x\,\partial_x + y\,\partial_y$ (the radial/dilation field) and $f = x^2 + y^2$. By definition:

$$\mathcal{L}_X f = X(f) = x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = x(2x) + y(2y) = 2(x^2 + y^2) = 2f.$$

The function f grows at rate $2f$ under dilation — consistent with the fact that the flow of X is $\phi_t(x, y) = (e^t x, e^t y)$, which scales f by e^{2t}.

Example 12.: Rotation preserves the area form: $\mathcal{L}_X(dx \wedge dy) = 0$

Let $X = -y\,\partial_x + x\,\partial_y$ (the rotation field on $\mathbb{R}^2$) and $\omega = dx \wedge dy$. Using Cartan's formula:

$$\iota_X\omega = \iota_X(dx \wedge dy) = dx(X)\,dy - dy(X)\,dx = (-y)\,dy - x\,dx = -y\,dy - x\,dx.$$

$$d(\iota_X\omega) = d(-y\,dy - x\,dx) = -dy \wedge dy - dx \wedge dx = 0.$$

$$\iota_X(d\omega) = \iota_X(0) = 0 \quad (\text{since } d(dx \wedge dy) = 0 \text{ on } \mathbb{R}^2).$$

Therefore $\mathcal{L}_X(dx \wedge dy) = 0 + 0 = 0$.
The area form is preserved by the rotation field — rotation is area-preserving.

Example 12.: Dilation stretches the area form: $\mathcal{L}_X(dx \wedge dy) = 2\,dx \wedge dy$

Now let $X = x\,\partial_x + y\,\partial_y$ (dilation) and $\omega = dx \wedge dy$.

$$\iota_X\omega = dx(X)\,dy - dy(X)\,dx = x\,dy - y\,dx.$$

$$d(\iota_X\omega) = d(x\,dy - y\,dx) = dx \wedge dy - dy \wedge dx = 2\,dx \wedge dy.$$

$$\iota_X(d\omega) = 0.$$

So $\mathcal{L}_X(dx \wedge dy) = 2\,dx \wedge dy$. The area form is scaled by 2 under dilation — the flow $\phi_t(x, y) = (e^t x, e^t y)$ scales areas by e^{2t}, whose derivative at $t = 0$ is indeed 2.

Example 12.: Lie derivative of $d\theta$ on S^1

Let $X = \partial_\theta$ (the angular field) and $\omega = d\theta$. Using Cartan:

$$\iota_X(d\theta) = d\theta(\partial_\theta) = 1 \in C^\infty(S^1).$$

$$d(\iota_X\,d\theta) = d(1) = 0.$$

$$\iota_X(d(d\theta)) = \iota_X(0) = 0.$$

So $\mathcal{L}_{\partial_\theta}(d\theta) = 0$. The angle form is preserved by the angular flow — rotating the circle doesn't change $d\theta$.

Example 12.: Lie derivative of the area form on S^2

Let $X_3 = x\,\partial_y - y\,\partial_x$ (rotation about the z-axis) and $\omega = \sin\theta\, d\theta \wedge d\varphi$. The flow of X_3 rotates S^2 about the z-axis, which clearly preserves the area form. By Cartan:

$$\iota_{X_3}\omega = \sin\theta\, d\varphi \qquad \text{(contracting the } \theta\text{-slot)}.$$

$$d(\iota_{X_3}\omega) = d(\sin\theta\, d\varphi) = \cos\theta\, d\theta \wedge d\varphi.$$

$$\iota_{X_3}(d\omega) = \iota_{X_3}(0) = 0.$$

Wait — this gives $\mathcal{L}_{X_3}\omega = \cos\theta\, d\theta \wedge d\varphi \neq 0$? Let us recheck: in spherical coordinates $X_3 = \partial_\varphi$, not $X_3 = x\partial_y - y\partial_x$ restricted to S^2. With $X = \partial_\varphi$:

$$\iota_{\partial_\varphi}(\sin\theta\, d\theta \wedge d\varphi) = -\sin\theta\, d\theta.$$

$$d(-\sin\theta\, d\theta) = -\cos\theta\, d\theta \wedge d\theta = 0.$$

$$\iota_{\partial_\varphi}(d\omega) = 0.$$

So $\mathcal{L}_{\partial_\varphi}\omega = 0$ — rotation about the z-axis preserves the area form, as expected.

Example 12.: Lie derivative and the symplectic form — Hamiltonian mechanics

On $T^*\mathbb{R}^n$ with symplectic form $\omega = \sum_i dq^i \wedge dp_i$, a vector field X_H is called *Hamiltonian* with Hamiltonian H if

$$\iota_{X_H}\omega = -dH.$$

By Cartan's formula:

$$\mathcal{L}_{X_H}\omega = d(\iota_{X_H}\omega) + \iota_{X_H}(d\omega) = d(-dH) + 0 = -d^2H = 0.$$

So the symplectic form is preserved by Hamiltonian flows — $\mathcal{L}_{X_H}\omega = 0$. This is Liouville's theorem: Hamiltonian flows preserve phase space volume.

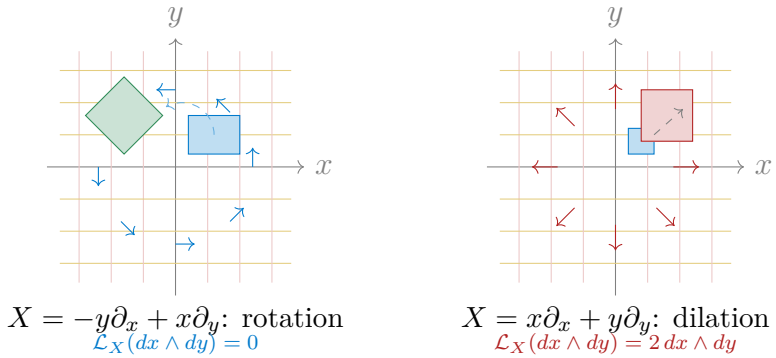


Figure 37: Left: the rotation field preserves the area form — the blue and green parallelograms have the same area ($\mathcal{L}_X\omega = 0$). Right: the dilation field stretches the area form — the red parallelogram has $4\times$ the area of the blue one after time $t = \ln 2$ ($\mathcal{L}_X\omega = 2\omega$).

12.8 The three operations together

Observation 12.

The three fundamental operations on differential forms satisfy a beautiful algebraic structure. Let $X \in \mathfrak{X}(M)$. Define operators d, ι_X, $\mathcal{L}_X$ on $\Omega^*(M)$. Then:

$$d^2 = 0, \qquad \iota_X^2 = 0, \qquad \mathcal{L}_X = d\iota_X + \iota_X d.$$

Further commutation relations:

$$[d, \mathcal{L}_X] = 0, \qquad [\iota_X, \mathcal{L}_X] = 0, \qquad [\mathcal{L}_X, \mathcal{L}_Y] = \mathcal{L}_{[X,Y]}.$$

The last identity says the Lie derivative is a Lie algebra homomorphism: the Lie bracket of vector fields corresponds to the commutator of Lie derivatives.

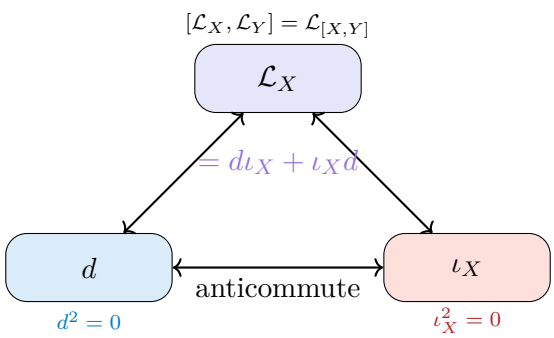


Figure 38: The three fundamental operations on differential forms and their relations. Cartan's formula $\mathcal{L}_X = d\iota_X + \iota_X d$ sits at the heart of the triangle.

Example 12.: Commutation: $\mathcal{L}_X$ commutes with d

Let $X = \partial_x$ on $\mathbb{R}^2$ and $f = xy$.

$$d(\mathcal{L}_X f) = d(X(f)) = d\left(\frac{\partial(xy)}{\partial x}\right) = d(y) = dy.$$

$$\mathcal{L}_X(df) = \mathcal{L}_X(y\,dx + x\,dy) = \mathcal{L}_X(y)\,dx + \mathcal{L}_X(x)\,dy = 0 \cdot dx + 1 \cdot dy = dy.$$

They agree: $d\mathcal{L}_X = \mathcal{L}_X d$.

13 Integration on Manifolds

13.1 Review: Integration on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^3$

Remark 13.

Before defining integration on manifolds, we isolate the common structure behind single, double, and triple integrals. The point is that in each case one integrates a *top-degree form* over an oriented domain of matching dimension.

Definition 13.Integration on $\mathbb{R}$: L

et $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be integrable. Then

$$\int_I f(x)\,dx$$

is the integral of the 1-form $f(x)\,dx$ over the oriented 1-dimensional domain I.

Example 13.: Examples

$$\int_0^1 x^2\,dx$$

is the integral of the 1-form $x^2\,dx$ over the interval $[0, 1]$.

Definition 13.Integration on $\mathbb{R}^2$: L

et $U \subseteq \mathbb{R}^2$ be measurable, and let $f : U \to \mathbb{R}$ be integrable. Then

$$\iint_U f(x, y)\,dx\,dy$$

is the integral of the 2-form

$$f(x, y)\,dx \wedge dy$$

over the oriented 2-dimensional domain U.

Example 13.: Examples

$$\iint_{[0,1]^2} (x+y)\,dx\,dy$$

is the integral of $(x+y)\,dx \wedge dy$ over the square $[0,1]^2$.

Definition 13.Integration on $\mathbb{R}^3$: L

et $V \subseteq \mathbb{R}^3$ be measurable, and let $f : V \to \mathbb{R}$ be integrable. Then

$$\iiint_V f(x,y,z)\,dx\,dy\,dz$$

is the integral of the 3-form

$$f(x,y,z)\,dx \wedge dy \wedge dz$$

over the oriented 3-dimensional domain V.

Example 13.: Examples

$$\iiint_{[0,1]^3} xyz\,dx\,dy\,dz$$

is the integral of $xyz\,dx \wedge dy \wedge dz$ over the cube $[0,1]^3$.

Non-Example 13.: Non-Examples

The expression

$$\int_{\mathbb{R}^2} x\,dy$$

is not a valid integral over the 2-dimensional manifold $\mathbb{R}^2$.

Failure: $x\,dy$ is a 1-form, not a 2-form. Its degree does not match the dimension of the domain.

Remark 13.

The unifying rule is:

a k-form integrates over an oriented k-dimensional domain.

This is the guiding principle for integration on manifolds.

13.2 Orientation

Definition 13.Orientation of a finite-dimensional real vector space: L

et V be a real vector space of dimension $n \geq 1$. An *orientation* on V is a choice of one of the two equivalence classes of ordered bases of V, where two ordered bases

$$(v_1, \ldots, v_n), \qquad (w_1, \ldots, w_n)$$

are equivalent if and only if the change-of-basis matrix $A \in GL(n, \mathbb{R})$ defined by

$$(w_1, \ldots, w_n) = (v_1, \ldots, v_n)A$$

satisfies

$$\det A > 0.$$

Example 13.: Examples

1. On $\mathbb{R}$, choosing an orientation is the same as choosing which direction is positive.
2. On $\mathbb{R}^2$, the standard basis (e_1, e_2) determines the standard orientation.
3. On $\mathbb{R}^3$, the basis (e_1, e_2, e_3) determines the standard right-handed orientation.

Theorem 13.: L

et V be a finite-dimensional real vector space. The relation above is an equivalence relation on the set of ordered bases of V, and there are exactly two equivalence classes.

Proof. Reflexivity holds because $\det(I) = 1 > 0$. Symmetry holds because if $\det(A) > 0$, then

$$\det(A^{-1}) = \frac{1}{\det(A)} > 0.$$

Transitivity holds because if $\det(A) > 0$ and $\det(B) > 0$, then

$$\det(AB) = \det(A)\det(B) > 0.$$

Thus the relation is an equivalence relation. Since $GL(n, \mathbb{R})$ has exactly two connected components, distinguished by the sign of the determinant, there are exactly two equivalence classes of ordered bases. $\square$

Definition 13.Orientable manifold: L

et M be a smooth manifold of dimension n. We say that M is *orientable* if there exists an atlas

$$\{(U_\alpha, \varphi_\alpha)\}$$

such that for every overlap $U_\alpha \cap U_\beta \neq \varnothing$, the transition map

$$\varphi_\beta \circ \varphi_\alpha^{-1}$$

has positive Jacobian determinant at every point of its domain.

Definition 13.Orientation on a manifold: A

n *orientation* on a smooth manifold M is a choice of an equivalence class of such atlases, called *oriented atlases*.

Example 13.: Examples

1. $\mathbb{R}^n$ is orientable, with standard orientation induced by

$$dx^1 \wedge \cdots \wedge dx^n.$$

2. S^1 is orientable; choosing an orientation corresponds to choosing clockwise or counterclockwise direction.

3. S^2 is orientable; one standard orientation is induced by the outward normal.

Non-Example 13.: Non-Examples

The Möbius strip is not orientable.

Failure: one cannot choose local positive frames consistently after going once around the strip.

Theorem 13.Volume form criterion: L

et M be a smooth manifold of dimension n. If there exists a nowhere-vanishing n-form

$$\omega \in \Omega^n(M),$$

then ω determines an orientation on M by declaring an ordered basis

$$(v_1, \ldots, v_n) \in T_pM$$

to be positively oriented if and only if

$$\omega_p(v_1, \ldots, v_n) > 0.$$

Proof. Since $\omega_p \neq 0$ for all $p \in M$, every ordered basis of T_pM is sent to a nonzero

real number. The sign of this number determines one of the two orientation classes. Smoothness of ω guarantees compatibility in neighborhoods, so this defines a smooth orientation on M. $\square$

Remark 13.

Orientation is exactly the datum needed to make sense of the sign of a top-degree volume element across coordinate charts.

13.3 Integration in One Oriented Chart

Definition 13.Local integral of a top form: L

et M be an oriented smooth manifold of dimension n. Let $\omega \in \Omega^n(M)$ satisfy

$$\operatorname{supp}(\omega) \subset U,$$

where (U, φ) is a positively oriented chart

$$\varphi : U \to \varphi(U) \subset \mathbb{R}^n.$$

Write

$$(\varphi^{-1})^*\omega = f(x)\, dx^1 \wedge \cdots \wedge dx^n.$$

Then we define

$$\int_M \omega := \int_{\varphi(U)} f(x)\, dx^1 \cdots dx^n.$$

Example 13.: Examples

Let $M = \mathbb{R}^2$ and

$$\omega = e^{-(x^2+y^2)}\, dx \wedge dy.$$

Using the identity chart,

$$\int_M \omega = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\, dx\, dy.$$

Non-Example 13.: Non-Examples

If

$$\alpha = x\, dy \in \Omega^1(\mathbb{R}^2),$$

then

$$\int_{\mathbb{R}^2} \alpha$$

is not defined.

Failure: degree mismatch. A 1-form cannot be integrated over a 2-dimensional manifold.

Theorem 13.Independence of positively oriented chart: L

et (U, φ) and (U, ψ) be positively oriented charts on an oriented smooth n-manifold M. If $\operatorname{supp}(\omega) \subset U$, then the local definition of

$$\int_M \omega$$

using φ agrees with the local definition using ψ.

Proof. Write

$$(\varphi^{-1})^*\omega = f(x)\, dx^1 \wedge \cdots \wedge dx^n, \qquad (\psi^{-1})^*\omega = g(y)\, dy^1 \wedge \cdots \wedge dy^n.$$

Let

$$y = \psi \circ \varphi^{-1}(x).$$

Then

$$dy^1 \wedge \cdots \wedge dy^n = \det\left(\frac{\partial y}{\partial x}\right) dx^1 \wedge \cdots \wedge dx^n.$$

Hence the coefficient functions satisfy the usual change-of-variables relation. Since the transition map is orientation-preserving, its Jacobian determinant is positive. Therefore the standard change-of-variables theorem on $\mathbb{R}^n$ shows that the two integrals agree. $\square$

Observation 13.Application

This theorem is the first step in defining integration intrinsically on manifolds. It shows that locally defined integrals do not depend on the choice of positively oriented coordinates.

13.4 Global Integration via Partition of Unity

Definition 13.Partition of unity: L

et M be a smooth manifold, and let $\{U_i\}_{i \in I}$ be an open cover of M. A family

$$\{\rho_i\}_{i \in I} \subset C^\infty(M)$$

is called a *partition of unity subordinate to* $\{U_i\}$ if:

1. $0 \leq \rho_i \leq 1$ for all i,
2. $\operatorname{supp}(\rho_i) \subset U_i$ for all i,
3. for every $p \in M$,
$$\sum_{i \in I} \rho_i(p) = 1,$$
4. the family is locally finite.

Definition 13.Global integral of a compactly supported top form: L

et M be an oriented smooth manifold of dimension n, and let $\omega \in \Omega_c^n(M)$. Let $\{\rho_i\}$ be a partition of unity subordinate to a positively oriented atlas $\{U_i\}$. Then define

$$\int_M \omega := \sum_i \int_M \rho_i \omega.$$

Each term is defined by the local construction above.

Example 13.: Examples

If $M = \mathbb{R}$ and $\omega = f(x)\,dx$ has compact support contained in the union of two overlapping intervals, then choosing a partition of unity $\rho_1 + \rho_2 = 1$ gives

$$\int_M \omega = \int \rho_1 f(x)\,dx + \int \rho_2 f(x)\,dx.$$

This recovers the usual integral.

Theorem 13.Well-definedness of global integration: T

he integral

$$\int_M \omega$$

is independent of the choice of positively oriented atlas and of the choice of subordinate partition of unity.

Proof. Let $\{\rho_i\}$ and $\{\sigma_j\}$ be two partitions of unity subordinate to oriented atlases. Then

$$\omega = \sum_{i,j} \rho_i \sigma_j \omega.$$

Each form $\rho_i \sigma_j \omega$ has support in a single oriented chart. By local finiteness, the sum is finite near each point. By the local chart-independence theorem, each piece has a well-defined integral independent of chart. Summing over all pieces gives the same value for either choice of partition of unity. □

Remark 13.

Partition of unity is the mechanism that turns local constructions into global ones.

13.5 Integration over Submanifolds

Definition 13.Integration over an oriented submanifold: L

et M be a smooth manifold, let $S \subset M$ be an oriented k-dimensional submanifold, let

$$\iota : S \hookrightarrow M$$

denote the inclusion, and let $\omega \in \Omega^k(M)$. Then define

$$\int_S \omega := \int_S \iota^* \omega.$$

Example 13.: Examples

1. If $S = \gamma([a, b])$ is an oriented curve in $\mathbb{R}^2$ and

$$\omega = P\,dx + Q\,dy,$$

then

$$\int_\gamma \omega = \int_a^b \Big(P(\gamma(t))x'(t) + Q(\gamma(t))y'(t)\Big)\,dt.$$

2. If S is an oriented surface in $\mathbb{R}^3$ and

$$\omega = f\,dy \wedge dz + g\,dz \wedge dx + h\,dx \wedge dy,$$

then

$$\int_S \omega$$

is the classical flux integral of the vector field (f, g, h) across S.

3. If $M = \mathbb{R}^3$ and

$$\omega = f\,dx \wedge dy \wedge dz,$$

then

$$\int_M \omega = \iiint_{\mathbb{R}^3} f\,dx\,dy\,dz$$

whenever the integral is defined.

Non-Example 13.: Non-Examples

If $\omega \in \Omega^2(M)$ and S is a curve, then

$$\int_S \omega$$

is not defined.

Failure: the degree of the form does not match the dimension of the submanifold.

Theorem 13.Parametrization formula: L

et $S \subset M$ be an oriented k-dimensional submanifold, let

$$\Phi : U \subset \mathbb{R}^k \to S$$

be an oriented parametrization, and let $\omega \in \Omega^k(M)$. Then

$$\int_S \omega = \int_U \Phi^* \omega.$$

Proof. By definition,

$$\int_S \omega = \int_S \iota^* \omega.$$

Pulling back further along Φ gives

$$\Phi^*(\iota^* \omega) = (\iota \circ \Phi)^* \omega.$$

Since Φ is a parametrization of S, the integral over S reduces to the usual integral of the pulled-back top-degree form on the open set $U \subset \mathbb{R}^k$. □

13.6 Manifolds with Boundary and Induced Orientation

Definition 13.Smooth manifold with boundary: A

topological space M is called a smooth n-manifold with boundary if each point $p \in M$ has a chart

$$\varphi : U \to \varphi(U) \subset \mathbb{H}^n, \qquad \mathbb{H}^n := \{x \in \mathbb{R}^n : x^n \geq 0\}.$$

Definition 13.Boundary: L

et M be a smooth manifold with boundary. The boundary of M is

$$\partial M := \{p \in M : \varphi(p)^n = 0 \text{ in some boundary chart}\}.$$

Example 13.: Examples

1. $\partial[a, b] = \{a, b\}$.
2. $\partial D^2 = S^1$.
3. $\partial B^3 = S^2$.

Definition 13.Induced orientation on the boundary: L

et M be an oriented smooth n-manifold with boundary. A basis

$$(v_1, \ldots, v_{n-1})$$

of $T_p(\partial M)$ is called positively oriented if and only if

$$(n_{\text{out}}, v_1, \ldots, v_{n-1})$$

is a positively oriented basis of T_pM, where n_{out} is an outward-pointing vector at $p \in \partial M$.

Example 13.: Examples

1. For $[a, b]$ with standard orientation, the induced boundary orientation is

$$\partial[a, b] = \{b\} - \{a\}.$$

2. For the oriented disk $D^2 \subset \mathbb{R}^2$, the induced orientation on $\partial D^2 = S^1$ is counterclockwise.

3. For the oriented ball $B^3 \subset \mathbb{R}^3$, the induced orientation on S^2 is the outward-pointing orientation.

Theorem 13.: I

f M is an oriented smooth manifold with boundary, then ∂M carries a natural induced orientation.

Proof. In a boundary chart

$$(x^1, \ldots, x^{n-1}, x^n), \qquad x^n \geq 0,$$

the boundary is locally given by $x^n = 0$. The outward transverse direction is represented by the outward normal, and the rule

$$(v_1, \ldots, v_{n-1}) \text{ positive on } \partial M \iff (n_{\text{out}}, v_1, \ldots, v_{n-1}) \text{ positive on } M$$

defines a consistent orientation on overlaps. □

13.7 Stokes' Theorem

Theorem 13.General Stokes' Theorem: L

et M be an oriented smooth n-manifold with boundary, and let

$$\omega \in \Omega_c^{n-1}(M).$$

Then

$$\int_M d\omega = \int_{\partial M} \omega.$$

Remark 13.Proof Strategy

The proof is local-to-global.

1. Use a partition of unity to reduce to the case where ω is supported in a single chart.
2. Reduce the statement to the corresponding theorem on a domain in $\mathbb{H}^n$.
3. In Euclidean coordinates, the result follows by repeated use of the Fundamental Theorem of Calculus and cancellation of interior boundary terms.
4. Sum the local identities.

Example 13.: Examples

1. **Fundamental Theorem of Calculus:** if $M = [a, b]$ and $\omega = f$, then
$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

2. **Green's Theorem:** if
$$\omega = P\,dx + Q\,dy$$
on a region $M \subset \mathbb{R}^2$, then
$$d\omega = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy,$$
so
$$\iint_M \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \oint_{\partial M} P\,dx + Q\,dy.$$

3. **Classical Stokes in $\mathbb{R}^3$:** for
$$\omega = P\,dx + Q\,dy + R\,dz,$$
one obtains
$$\iint_S (\nabla \times F) \cdot n\,dS = \oint_{\partial S} F \cdot dr.$$

4. **Divergence Theorem:** for
$$\omega = f\,dy \wedge dz + g\,dz \wedge dx + h\,dx \wedge dy,$$
one obtains
$$\iiint_M (\nabla \cdot F)\,dV = \iint_{\partial M} F \cdot n\,dS.$$

Theorem 13.Exact forms integrate to zero on closed manifolds: L

et M be an oriented smooth manifold without boundary, and let $\eta \in \Omega^{n-1}(M)$. Then
$$\int_M d\eta = 0.$$

Proof. Since $\partial M = \varnothing$, Stokes' theorem gives
$$\int_M d\eta = \int_{\partial M} \eta = 0.$$

□

Observation 13.Application

This theorem is the bridge to de Rham cohomology: on a closed manifold, the integral of a top-degree form depends only on its cohomology class.

Remark 13.

The slogan is:

$$\text{interior differential } d \quad \leftrightarrow \quad \text{boundary operator } \partial.$$

Stokes' theorem is the master identity that unifies the Fundamental Theorem of Calculus, Green's theorem, the classical Stokes theorem, and the Divergence theorem.

14 De Rham Cohomology

14.1 Motivation

The central question. You have learned that the exterior derivative satisfies $d^2 = 0$. This immediately implies:

$$\omega = d\eta \quad \Longrightarrow \quad d\omega = 0.$$

In words: *every exact form is closed.* But what about the converse? Given a closed form ω (i.e., $d\omega = 0$), can we always find η such that $\omega = d\eta$?

Why this matters. The failure of the converse is *topological.* When closed forms fail to be exact, they detect "holes" in the manifold. De Rham cohomology makes this precise: it measures the obstruction to solving $d\eta = \omega$.

Physical motivation. In electromagnetism, the Faraday 2-form F satisfies $dF = 0$ (Maxwell's equation). Locally, we can write $F = dA$ for a potential A. But globally? Only if the spacetime has trivial topology. Magnetic monopoles would obstruct this — they correspond to nontrivial cohomology classes.

14.2 Closed vs Exact Forms

Elementary examples first. Before the definitions, let us see the phenomenon in action.

Example 14.1: A form that is both closed and exact

On $\mathbb{R}^2$, consider the 1-form

$$\omega = y\,dx + x\,dy.$$

Closed? Compute:

$$d\omega = dy \wedge dx + dx \wedge dy = -dx \wedge dy + dx \wedge dy = 0. \quad \checkmark$$

Exact? Yes: $\omega = d(xy)$. $\checkmark$

Example 14.2: A form that is closed but NOT exact

On $\mathbb{R}^2 \setminus \{0\}$, consider the 1-form

$$\eta = \frac{-y\,dx + x\,dy}{x^2 + y^2}.$$

Closed? A computation shows $d\eta = 0$. $\checkmark$
Exact? Suppose $\eta = df$ for some smooth $f : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}$. Then by Stokes' theorem:

$$\oint_{S^1} \eta = \oint_{S^1} df = 0.$$

But direct computation gives $\oint_{S^1} \eta = 2\pi \neq 0$. Contradiction!
So η is closed but *not* exact.

Remark 14.1

The form η in Example 2 is often written $d\theta$ because in polar coordinates (r, θ), we have $\eta = d\theta$. But θ is not a globally defined function on $\mathbb{R}^2 \setminus \{0\}$ — it jumps by 2π around the origin. The notation $d\theta$ is suggestive but slightly misleading.

Definitions

Definition 14.1: Closed form

Let M be a smooth manifold. A k-form $\omega \in \Omega^k(M)$ is *closed* if

$$d\omega = 0.$$

Definition 14.2: Exact form

A k-form $\omega \in \Omega^k(M)$ is *exact* if there exists a $(k-1)$-form $\eta \in \Omega^{k-1}(M)$ such that

$$\omega = d\eta.$$

Proposition 14.1: Exact implies closed

Every exact form is closed. That is,

$$\omega = d\eta \quad \Longrightarrow \quad d\omega = 0.$$

Proof. If $\omega = d\eta$, then $d\omega = d(d\eta) = d^2\eta = 0$. □

Remark 14.2

The converse — "closed implies exact" — is **false** in general. The obstruction is topological.

More Examples

Example 14.3: Closed and exact forms on $\mathbb{R}^2$

Form	Closed?	Exact?
$x\,dy$	No ($d(x\,dy) = dx \wedge dy \neq 0$)	—
dx	Yes	Yes ($dx = d(x)$)
$dx + dy$	Yes	Yes ($= d(x+y)$)
$dx \wedge dy$	Yes (it's a top form)	Yes ($= d(x\,dy)$)

Example 14.4: The angle form on punctured domains

Domain	$d\theta$ closed?	$d\theta$ exact?
$\mathbb{R}^2 \setminus \{0\}$	Yes	**No**
$\mathbb{R}^2 \setminus \{(x,0) : x \leq 0\}$	Yes	Yes ($= d(\arctan(y/x))$)

Removing a ray (not just a point) makes the domain simply connected, so the obstruction vanishes.

Non-Example 14.1: Why topology matters

The form $d\theta = \frac{-y\,dx + x\,dy}{x^2+y^2}$ is:

- Exact on any simply connected subset of $\mathbb{R}^2 \setminus \{0\}$.
- **Not** exact on all of $\mathbb{R}^2 \setminus \{0\}$.

The "hole" at the origin is detected by the failure of exactness. This is exactly what cohomology measures.

14.3 The De Rham Cohomology Groups

Notation. Let M be a smooth manifold. We denote by $\Omega^k(M)$ the vector space of smooth k-forms on M.

Definition 14.3: Closed and exact forms as subspaces

Define the following subspaces of $\Omega^k(M)$:

$$Z^k(M) := \ker(d : \Omega^k(M) \to \Omega^{k+1}(M)) = \{\omega \in \Omega^k(M) : d\omega = 0\},$$
$$B^k(M) := \operatorname{im}(d : \Omega^{k-1}(M) \to \Omega^k(M)) = \{d\eta : \eta \in \Omega^{k-1}(M)\}.$$

By convention, $B^0(M) = \{0\}$ (there are no (-1)-forms).

Remark 14.3

Since $d^2 = 0$, we have $B^k(M) \subseteq Z^k(M)$. Every exact form is closed.

Definition 14.4: De Rham cohomology

The *k-th de Rham cohomology group* of M is the quotient vector space

$$\boxed{H^k_{\mathrm{dR}}(M) := \frac{Z^k(M)}{B^k(M)} = \frac{\text{closed } k\text{-forms}}{\text{exact } k\text{-forms}}.}$$

Elements of $H^k_{\mathrm{dR}}(M)$ are equivalence classes $[\omega]$, where

$$[\omega] = [\omega'] \quad \Longleftrightarrow \quad \omega - \omega' = d\eta \text{ for some } \eta \in \Omega^{k-1}(M).$$

Remark 14.4

The class $[\omega]$ is called a *cohomology class*. Two forms representing the same class are called *cohomologous*.

Interpretation

Observation 14.1

The de Rham cohomology measures the failure of closed forms to be exact:

$H^k_{\mathrm{dR}}(M) = 0$	Every closed k-form is exact (no obstruction).
$H^k_{\mathrm{dR}}(M) \neq 0$	Some closed k-forms are not exact (topology obstructs).
$\dim H^k_{\mathrm{dR}}(M)$	Counts independent "k-dimensional holes."

Example 14.5: Quick cohomology table

M	H^0_{dR}	H^1_{dR}	H^2_{dR}
$\mathbb{R}^n$	$\mathbb{R}$	0	0
S^1	$\mathbb{R}$	$\mathbb{R}$	0
S^2	$\mathbb{R}$	0	$\mathbb{R}$
T^2 (torus)	$\mathbb{R}$	$\mathbb{R}^2$	$\mathbb{R}$

Remark 14.5

$H^0_{\mathrm{dR}}(M) = \mathbb{R}$ for any connected manifold M. This is because a closed 0-form satisfies $df = 0$, hence f is locally constant. On a connected manifold, locally constant implies constant. There are no (-1)-forms, so $B^0 = 0$, and thus $H^0 = Z^0/0 = \{\text{constants}\} \cong \mathbb{R}$.

14.4 The Poincaré Lemma

Motivation. On $\mathbb{R}^n$, is every closed form exact? Since $\mathbb{R}^n$ has no holes, we expect: *yes*.

Theorem 14.1: Poincaré Lemma

For all $k \geq 1$:

$$H^k_{\mathrm{dR}}(\mathbb{R}^n) = 0.$$

Equivalently: every closed k-form on $\mathbb{R}^n$ is exact.

Proof idea: the homotopy operator. The key is to construct a linear map $h : \Omega^k(\mathbb{R}^n) \to \Omega^{k-1}(\mathbb{R}^n)$ satisfying

$$dh + hd = \mathrm{id}.$$

Then if $d\omega = 0$:

$$\omega = (dh + hd)\omega = d(h\omega) + h(\underbrace{d\omega}_{=0}) = d(h\omega).$$

So ω is exact. □

Example 14.6: Explicit computation

Let $\omega = dx \wedge dy$ on $\mathbb{R}^2$. Find η with $d\eta = \omega$.
Using the homotopy operator:

$$h(dx \wedge dy) = \int_0^1 t \cdot \iota_{(x,y)}(dx \wedge dy)\, dt = \int_0^1 t(x\, dy - y\, dx)\, dt = \frac{1}{2}(x\, dy - y\, dx).$$

Check:

$$d\left(\frac{1}{2}(x\, dy - y\, dx)\right) = \frac{1}{2}(dx \wedge dy + dy \wedge dx) = \frac{1}{2}(2\, dx \wedge dy) = dx \wedge dy. \quad \checkmark$$

Remark 14.6

$H^0_{\mathrm{dR}}(\mathbb{R}^n) = \mathbb{R}$, not 0. Constants are closed but not exact (no (-1)-forms exist).

Proposition 14.2: Local exactness

Every closed form is *locally* exact. That is, around any point $p \in M$, there exists a neighborhood U such that $\omega|_U = d\eta$ for some $\eta \in \Omega^{k-1}(U)$.

Proof. Work in a chart $U \cong \mathbb{R}^n$ and apply the Poincaré lemma. □

14.5 Homotopy Invariance

Motivation. If two manifolds are "topologically the same," they should have the same cohomology.

Definition 14.5: Smooth homotopy

Two smooth maps $f, g : M \to N$ are *smoothly homotopic*, written $f \simeq g$, if there exists a smooth map $H : M \times [0,1] \to N$ with

$$H(p,0) = f(p), \quad H(p,1) = g(p).$$

Definition 14.6: Homotopy equivalence

Manifolds M and N are *homotopy equivalent*, written $M \simeq N$, if there exist smooth maps $f : M \to N$ and $g : N \to M$ such that

$$g \circ f \simeq \mathrm{id}_M, \quad f \circ g \simeq \mathrm{id}_N.$$

Theorem 14.2: Homotopy invariance of de Rham cohomology

If $f \simeq g : M \to N$, then the induced maps on cohomology are equal:

$$f^* = g^* : H^k_{\mathrm{dR}}(N) \to H^k_{\mathrm{dR}}(M).$$

In particular, if $M \simeq N$, then

$$H^k_{\mathrm{dR}}(M) \cong H^k_{\mathrm{dR}}(N).$$

Example 14.7: Punctured plane and circle

The radial projection gives a homotopy equivalence $\mathbb{R}^2 \setminus \{0\} \simeq S^1$. Therefore:

$$H^k_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) \cong H^k_{\mathrm{dR}}(S^1).$$

In particular, $H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R}$, generated by $[d\theta]$.

Remark 14.7

This explains why $d\theta$ is not exact on $\mathbb{R}^2 \setminus \{0\}$: it represents a nontrivial cohomology class, detecting the "hole" at the origin.

14.6 Mayer–Vietoris Sequence

Motivation. How do we compute $H^k_{\mathrm{dR}}(M)$ for complicated manifolds? Strategy: decompose $M = U \cup V$ into simpler pieces and relate their cohomologies.

Setup. Let $M = U \cup V$ where $U, V \subseteq M$ are open. We have inclusion maps:

$$i_U : U \cap V \hookrightarrow U, \quad i_V : U \cap V \hookrightarrow V, \quad j_U : U \hookrightarrow M, \quad j_V : V \hookrightarrow M.$$

Theorem 14.3: Mayer–Vietoris sequence

There is a long exact sequence:

$$\cdots \to H^k(M) \xrightarrow{\alpha} H^k(U) \oplus H^k(V) \xrightarrow{\beta} H^k(U \cap V) \xrightarrow{\delta} H^{k+1}(M) \to \cdots$$

where:

- $\alpha([\omega]) = ([\omega|_U], [\omega|_V])$,
- $\beta([\sigma], [\tau]) = [\sigma|_{U\cap V}] - [\tau|_{U\cap V}]$,
- δ is the connecting homomorphism.

Example 14.8: Cohomology of S^1

Cover $S^1 = U \cup V$ with two overlapping arcs:

- $U, V \simeq \mathbb{R}$ (each arc is contractible).
- $U \cap V \simeq \mathbb{R} \sqcup \mathbb{R}$ (two disjoint intervals).

The Mayer–Vietoris sequence for $k = 0, 1$ gives:

$$0 \to H^0(S^1) \to \underbrace{H^0(U) \oplus H^0(V)}_{\mathbb{R}\oplus\mathbb{R}} \to \underbrace{H^0(U \cap V)}_{\mathbb{R}\oplus\mathbb{R}} \xrightarrow{\delta} H^1(S^1) \to \underbrace{H^1(U) \oplus H^1(V)}_{0} \to 0.$$

Chasing the sequence:

- $H^0(S^1) = \mathbb{R}$ (connected).
- $\dim H^1(S^1) = \dim \ker(\delta^{-1}\text{'s target}) = 2 + 1 - 2 = 1$, so $H^1(S^1) = \mathbb{R}$.

Example 14.9: Cohomology of S^n

By induction using Mayer–Vietoris with $U \cap V \simeq S^{n-1}$:

$$H^k_{\mathrm{dR}}(S^n) = \begin{cases} \mathbb{R} & k = 0 \text{ or } k = n, \\ 0 & \text{otherwise.} \end{cases}$$

14.7 De Rham's Theorem

Motivation. We have two ways to detect "holes" in a manifold:

Approach	Tool	Flavor
De Rham	Differential forms, d	Analytic/smooth
Singular	Chains, ∂	Topological/combinatorial

Are they the same?

The pairing. For $[\omega] \in H^k_{\mathrm{dR}}(M)$ and $[\sigma] \in H_k(M;\mathbb{R})$ (singular homology), define:

$$\langle [\omega], [\sigma] \rangle := \int_\sigma \omega.$$

This is well-defined by Stokes' theorem:

- $\omega \mapsto \omega + d\eta$: $\int_\sigma d\eta = \int_{\partial\sigma} \eta = 0$ (since $\partial\sigma = 0$).
- $\sigma \mapsto \sigma + \partial c$: $\int_{\partial c} \omega = \int_c d\omega = 0$ (since $d\omega = 0$).

Theorem 14.4: De Rham's theorem

For any smooth manifold M:

$$\boxed{H^k_{\mathrm{dR}}(M) \cong H^k(M;\mathbb{R}).}$$

The isomorphism is given by integration.

Remark 14.8

This is a deep result: it says that the purely analytic construction (differential forms and d) captures exactly the same information as the purely topological construction (singular cochains and δ).

Observation 14.2

Consequences of de Rham's theorem:

Consequence	Explanation
Computability	Use algebra (singular) or calculus (de Rham).
Topology $\leftrightarrow$ Analysis	Holes detected by forms = holes detected by chains.
Physics	EM, gauge theory: $dF = 0$ is topologically constrained.
Hodge theory	On compact Riemannian: harmonic forms $\cong H^k_{\mathrm{dR}}$.

14.8 Summary

Observation 14.3

What $H^k_{\mathrm{dR}}(M)$ counts:

k	$H^k_{\mathrm{dR}}(M)$ detects...
0	Connected components.
1	Independent loops (1-dimensional holes).
2	Enclosed voids (2-dimensional holes).
n	Orientation (if M is compact n-dimensional).

15 Lie Groups and Lie Algebras

15.1 Motivation

Why Lie groups? In gauge theory, physical fields transform under *symmetry groups*. These aren't just abstract groups — they're smooth manifolds where group operations are smooth. This is a Lie group.

Why Lie algebras? The full group G is often complicated (curved, global). Its Lie algebra $\mathfrak{g} = T_eG$ is a *linear* approximation at the identity — much easier to work with. Connections on principal bundles are $\mathfrak{g}$-valued, not G-valued.

The punchline for gauge theory.

Object	Role in gauge theory
Lie group G	Structure group (e.g., $U(1)$, $SU(2)$, $SU(3)$)
Lie algebra $\mathfrak{g}$	Where connections live
Adjoint rep Ad	How gauge transformations act on connections
Maurer–Cartan form	Prototype for connection 1-forms

15.2 Lie Groups

Definition 15.1: Lie group

A *Lie group* is a smooth manifold G equipped with a group structure such that:

1. Multiplication $\mu : G \times G \to G$, $(g, h) \mapsto gh$, is smooth.
2. Inversion $\iota : G \to G$, $g \mapsto g^{-1}$, is smooth.

Remark 15.1

Equivalently: G is a group object in the category of smooth manifolds.

The Matrix Lie Groups

Example 15.1: General linear group

$$GL(n, \mathbb{R}) = \{A \in M_{n\times n}(\mathbb{R}) : \det A \neq 0\}.$$

This is an open subset of $\mathbb{R}^{n^2}$, hence a smooth manifold of dimension n^2. Matrix multiplication and inversion are smooth (Cramer's rule).

Example 15.2: Special linear group

$$SL(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) : \det A = 1\}.$$

Dimension: $n^2 - 1$ (one constraint).

Example 15.3: Orthogonal and special orthogonal groups

$$O(n) = \{A \in GL(n, \mathbb{R}) : A^T A = I\}, \qquad SO(n) = \{A \in O(n) : \det A = 1\}.$$

Dimension: $\frac{n(n-1)}{2}$. These preserve the Euclidean inner product.

Example 15.4: Unitary and special unitary groups

$$U(n) = \{A \in GL(n, \mathbb{C}) : A^\dagger A = I\}, \qquad SU(n) = \{A \in U(n) : \det A = 1\}.$$

Dimensions: $\dim U(n) = n^2$, $\dim SU(n) = n^2 - 1$.

Observation 15.1

The gauge groups of the Standard Model:

Interaction	Gauge group	Dimension
Electromagnetism	$U(1)$	1
Weak force	$SU(2)$	3
Strong force	$SU(3)$	8

Group Actions

Definition 15.2: Left action

A *left action* of a Lie group G on a manifold M is a smooth map

$$\ell : G \times M \to M, \qquad (g, p) \mapsto g \cdot p,$$

such that:

1. $e \cdot p = p$ for all $p \in M$.
2. $(gh) \cdot p = g \cdot (h \cdot p)$ for all $g, h \in G$, $p \in M$.

Definition 15.3: Right action

A *right action* of G on M is a smooth map $r : M \times G \to M$, $(p, g) \mapsto p \cdot g$, with:

1. $p \cdot e = p$.
2. $p \cdot (gh) = (p \cdot g) \cdot h$.

Remark 15.2

Principal bundles use **right** actions. This is because we want the structure group to act on fibers without conflicting with the bundle projection.

Definition 15.4: Free and transitive actions

An action is:

- *Free* if $g \cdot p = p \Rightarrow g = e$ (no fixed points except by identity).
- *Transitive* if for any $p, q \in M$, there exists $g \in G$ with $g \cdot p = q$.

Remark 15.3

In a principal G-bundle, G acts **freely and transitively** on each fiber. This makes each fiber diffeomorphic to G.

15.3 Lie Algebras

Idea. The Lie algebra $\mathfrak{g}$ of a Lie group G is the tangent space at the identity, equipped with a bracket operation encoding the infinitesimal group structure.

Definition 15.5: Lie algebra of a Lie group

Let G be a Lie group with identity e. The *Lie algebra of* G is

$$\mathfrak{g} := T_e G,$$

equipped with the Lie bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ (defined below).

Definition 15.6: Abstract Lie algebra

A *Lie algebra* is a vector space $\mathfrak{g}$ with a bilinear map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ satisfying:

1. **Antisymmetry:** $[X, Y] = -[Y, X]$.
2. **Jacobi identity:** $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

Matrix Lie Algebras

Proposition 15.1: Lie algebra of matrix groups

For matrix Lie groups $G \subseteq GL(n, \mathbb{R})$ or $GL(n, \mathbb{C})$, the Lie bracket is the *matrix commutator*:

$$[X, Y] = XY - YX.$$

Example 15.5: $\mathfrak{gl}(n, \mathbb{R})$

$$\mathfrak{gl}(n, \mathbb{R}) = T_I\, GL(n, \mathbb{R}) = M_{n \times n}(\mathbb{R}) \quad \text{(all } n \times n \text{ matrices)}.$$

Dimension: n^2.

Example 15.6: $\mathfrak{sl}(n, \mathbb{R})$

$$\mathfrak{sl}(n, \mathbb{R}) = \{X \in \mathfrak{gl}(n, \mathbb{R}) : \operatorname{tr}(X) = 0\}.$$

Dimension: $n^2 - 1$.

Example 15.7: $\mathfrak{so}(n)$

$$\mathfrak{so}(n) = \{X \in \mathfrak{gl}(n, \mathbb{R}) : X^T = -X\} \quad \text{(skew-symmetric matrices)}.$$

Dimension: $\frac{n(n-1)}{2}$.

Example 15.8: $\mathfrak{u}(n)$ and $\mathfrak{su}(n)$

$$\begin{aligned}\mathfrak{u}(n) &= \{X \in \mathfrak{gl}(n, \mathbb{C}) : X^\dagger = -X\} \quad \text{(skew-Hermitian)},\\ \mathfrak{su}(n) &= \{X \in \mathfrak{u}(n) : \operatorname{tr}(X) = 0\}.\end{aligned}$$

Dimensions: n^2 and $n^2 - 1$ respectively.

Example 15.9: $\mathfrak{su}(2)$ explicitly

A basis for $\mathfrak{su}(2)$:

$$T_1 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad T_2 = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad T_3 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Or in terms of Pauli matrices: $T_a = \frac{i}{2}\sigma_a$. The bracket:

$$[T_a, T_b] = \epsilon_{abc} T_c.$$

This is the algebra of angular momentum in quantum mechanics.

Example 15.10: $\mathfrak{u}(1)$

$$\mathfrak{u}(1) = \{i\theta : \theta \in \mathbb{R}\} \cong i\mathbb{R}.$$

Dimension: 1. The bracket is trivial: $[X, Y] = 0$ (abelian).

15.4 The Exponential Map

Idea. The exponential map "integrates" infinitesimal generators (Lie algebra elements) into finite group elements.

Definition 15.7: Exponential map

The *exponential map* $\exp : \mathfrak{g} \to G$ is defined as follows: for $X \in \mathfrak{g}$, let $\gamma_X : \mathbb{R} \to G$ be the unique one-parameter subgroup with $\gamma_X'(0) = X$. Then

$$\exp(X) := \gamma_X(1).$$

Proposition 15.2: Properties of the exponential map

1. $\exp(0) = e$.
2. $\exp((s+t)X) = \exp(sX)\exp(tX)$.
3. $\exp(-X) = \exp(X)^{-1}$.
4. exp is a local diffeomorphism near $0 \in \mathfrak{g}$.
5. $\frac{d}{dt}\Big|_{t=0} \exp(tX) = X$.

Example 15.11: Matrix exponential

For matrix Lie groups, exp is the matrix exponential:

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!} = I + X + \frac{X^2}{2!} + \frac{X^3}{3!} + \cdots$$

Example 15.12: $U(1)$

For $X = i\theta \in \mathfrak{u}(1)$:

$$\exp(i\theta) = e^{i\theta} \in U(1) \cong S^1.$$

This wraps the real line around the circle — the exponential is surjective but not injective.

Example 15.13: $SO(2)$

For $X = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix} \in \mathfrak{so}(2)$:

$$\exp(X) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in SO(2).$$

Example 15.14: $SU(2)$

For $X = i\theta\, \hat{n} \cdot \vec{\sigma}/2 \in \mathfrak{su}(2)$ (where $|\hat{n}| = 1$):

$$\exp(X) = \cos(\theta/2)I + i\sin(\theta/2)\, \hat{n} \cdot \vec{\sigma}.$$

This gives rotations in the spin-$\frac{1}{2}$ representation.

Remark 15.4

The exponential is **not** always surjective. For $SL(2,\mathbb{R})$, the matrix $\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$ is not in the image of exp. However, for compact connected Lie groups, exp is surjective.

15.5 Adjoint Representations

Motivation. Gauge transformations act on connections. To understand this, we need to know how G acts on $\mathfrak{g}$. This is the adjoint representation.

The Adjoint Representation of G

Definition 15.8: Conjugation

For $g \in G$, the *conjugation map* is

$$C_g : G \to G, \qquad h \mapsto ghg^{-1}.$$

This is a Lie group automorphism: $C_g(h_1h_2) = C_g(h_1)C_g(h_2)$.

Definition 15.9: Adjoint representation Ad

The *adjoint representation* of G on $\mathfrak{g}$ is

$$\mathrm{Ad} : G \to GL(\mathfrak{g}), \qquad \mathrm{Ad}_g := (dC_g)_e : T_eG \to T_eG.$$

Explicitly, Ad_g is the derivative of conjugation at the identity.

Proposition 15.3: Ad is a representation

1. $\mathrm{Ad}_e = \mathrm{id}_{\mathfrak{g}}$.
2. $\mathrm{Ad}_{gh} = \mathrm{Ad}_g \circ \mathrm{Ad}_h$.
3. $\mathrm{Ad} : G \to GL(\mathfrak{g})$ is a smooth homomorphism.

Example 15.15: Ad for matrix groups

For $G \subseteq GL(n)$ and $g \in G$, $X \in \mathfrak{g}$:

$$\mathrm{Ad}_g(X) = gXg^{-1}.$$

Example 15.16: Ad for abelian groups

If G is abelian (e.g., $U(1)$, $\mathbb{R}^n$), then $ghg^{-1} = h$, so

$$\mathrm{Ad}_g = \mathrm{id}_{\mathfrak{g}} \quad \text{for all } g \in G.$$

The adjoint representation is trivial.

The Adjoint Representation of $\mathfrak{g}$

Definition 15.10: Adjoint representation ad

The *adjoint representation of* $\mathfrak{g}$ on itself is

$$\mathrm{ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}), \qquad \mathrm{ad}_X(Y) := [X, Y].$$

Proposition 15.4: ad is the derivative of Ad

$$\mathrm{ad} = d(\mathrm{Ad})_e : T_e G \to T_{\mathrm{id}} GL(\mathfrak{g}) = \mathfrak{gl}(\mathfrak{g}).$$

Equivalently:

$$\mathrm{ad}_X(Y) = \frac{d}{dt}\bigg|_{t=0} \mathrm{Ad}_{\exp(tX)}(Y) = [X, Y].$$

Proposition 15.5: ad is a Lie algebra homomorphism

$$\mathrm{ad}_{[X,Y]} = [\mathrm{ad}_X, \mathrm{ad}_Y] = \mathrm{ad}_X \circ \mathrm{ad}_Y - \mathrm{ad}_Y \circ \mathrm{ad}_X.$$

This is equivalent to the Jacobi identity.

Observation 15.2

Summary of adjoint representations:

Map	Domain $\to$ Codomain	Formula (matrix case)
Ad	$G \to GL(\mathfrak{g})$	$\mathrm{Ad}_g(X) = gXg^{-1}$
ad	$\mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$	$\mathrm{ad}_X(Y) = [X, Y] = XY - YX$

Remark 15.5

In gauge theory, under a gauge transformation $g : M \to G$:

- The connection transforms as $A \mapsto gAg^{-1} + g\,dg^{-1}$.
- The curvature transforms as $F \mapsto gFg^{-1} = \mathrm{Ad}_g(F)$.

The adjoint representation is the natural action on $\mathfrak{g}$-valued forms.

15.6 Left-Invariant Vector Fields and Forms

Motivation. To define the Maurer–Cartan form, we need to understand left-invariant structures on G.

Definition 15.11: Left translation

For $g \in G$, the *left translation* is

$$L_g : G \to G, \qquad h \mapsto gh.$$

This is a diffeomorphism with $(L_g)^{-1} = L_{g^{-1}}$.

Definition 15.12: Left-invariant vector field

A vector field $X \in \mathfrak{X}(G)$ is *left-invariant* if

$$(L_g)_* X = X \quad \text{for all } g \in G.$$

Equivalently: $(L_g)_{*h}(X_h) = X_{gh}$ for all $g, h \in G$.

Proposition 15.6: Left-invariant vector fields $\cong$ Lie algebra

The space of left-invariant vector fields on G is isomorphic to $\mathfrak{g} = T_e G$:

$$\mathfrak{g} \xrightarrow{\sim} \{\text{left-invariant vector fields}\}, \qquad X \mapsto \tilde{X},$$

where $\tilde{X}_g := (L_g)_{*e}(X)$.

Remark 15.6

The Lie bracket on $\mathfrak{g}$ corresponds to the Lie bracket of vector fields:

$$\widetilde{[X, Y]} = [\tilde{X}, \tilde{Y}].$$

This is how the bracket is often *defined*.

Definition 15.13: Left-invariant differential form

A differential form $\omega \in \Omega^k(G)$ is *left-invariant* if

$$L_g^* \omega = \omega \quad \text{for all } g \in G.$$

Proposition 15.7: Left-invariant forms $\cong \Lambda^k(\mathfrak{g}^*)$

The space of left-invariant k-forms on G is isomorphic to $\Lambda^k(\mathfrak{g}^*)$:

$$\Lambda^k(\mathfrak{g}^*) \xrightarrow{\sim} \{\text{left-invariant } k\text{-forms}\}, \qquad \alpha \mapsto \tilde{\alpha},$$

where $\tilde{\alpha}_g := (L_{g^{-1}})^* \alpha_e$.

15.7 The Maurer–Cartan Form

Motivation. The Maurer–Cartan form is a canonical $\mathfrak{g}$-valued 1-form on G. It is the *prototype* for connection 1-forms on principal bundles.

Definition 15.14: Maurer–Cartan form

The *Maurer–Cartan form* on a Lie group G is the $\mathfrak{g}$-valued 1-form $\theta \in \Omega^1(G; \mathfrak{g})$ defined by

$$\theta_g := (L_{g^{-1}})_{*g} : T_g G \to T_e G = \mathfrak{g}.$$

For $v \in T_g G$:

$$\theta_g(v) = (L_{g^{-1}})_* v \in \mathfrak{g}.$$

Remark 15.7

In words: θ takes a tangent vector at g and "left-translates it back to the identity" to get an element of $\mathfrak{g}$.

Proposition 15.8: Properties of the Maurer–Cartan form

1. **Left-invariance:** $L_g^* \theta = \theta$ for all $g \in G$.
2. **Right equivariance:** $R_g^* \theta = \mathrm{Ad}_{g^{-1}} \circ \theta$.
3. **Reproduces $\mathfrak{g}$:** For $X \in \mathfrak{g}$ and $\tilde{X}$ its left-invariant extension, $\theta(\tilde{X}) = X$.

Example 15.17: Maurer–Cartan form on matrix groups

For $G \subseteq GL(n)$, viewing $g \in G$ as a matrix and $v \in T_g G$ as a matrix:

$$\theta_g(v) = g^{-1} v.$$

In differential notation:

$$\theta = g^{-1} dg.$$

Example 15.18: θ on $U(1)$

Parametrize $U(1) = \{e^{i\phi} : \phi \in \mathbb{R}\}$. Then $g = e^{i\phi}$, $dg = ie^{i\phi}d\phi$, so:

$$\theta = g^{-1}dg = e^{-i\phi} \cdot ie^{i\phi}d\phi = i\,d\phi.$$

This is the canonical 1-form on the circle, valued in $\mathfrak{u}(1) = i\mathbb{R}$.

Example 15.19: θ on $SU(2)$

For $g \in SU(2)$:

$$\theta = g^{-1}dg = g^{\dagger}dg \in \Omega^1(SU(2); \mathfrak{su}(2)).$$

Expanding in a basis $\{T_a\}$ of $\mathfrak{su}(2)$: $\theta = \theta^a T_a$ where θ^a are ordinary 1-forms on $SU(2)$.

The Maurer–Cartan Equation

Theorem 15.1: Maurer–Cartan structure equation

The Maurer–Cartan form satisfies:

$$\boxed{d\theta + \frac{1}{2}[\theta, \theta] = 0.}$$

Here $[\theta, \theta]$ is the $\mathfrak{g}$-valued 2-form defined by

$$[\theta, \theta](X, Y) := 2[\theta(X), \theta(Y)].$$

Remark 15.8

Equivalently: $d\theta + \theta \wedge \theta = 0$, where

$$(\theta \wedge \theta)(X, Y) := [\theta(X), \theta(Y)].$$

Remark 15.9

For abelian groups (e.g., $U(1)$), $[\theta, \theta] = 0$, so $d\theta = 0$. The Maurer–Cartan form is closed.

Observation 15.3

Connection to gauge theory: On a principal G-bundle $P \xrightarrow{\pi} M$, a connection 1-form $\omega \in \Omega^1(P; \mathfrak{g})$ satisfies:

1. $R_g^* \omega = \mathrm{Ad}_{g^{-1}} \omega$ (same as Maurer–Cartan).
2. $\omega(\tilde{X}) = X$ for fundamental vector fields (analogue of reproducing $\mathfrak{g}$).

The curvature is $\Omega = d\omega + \frac{1}{2}[\omega, \omega]$. For the Maurer–Cartan form on G itself (trivial bundle), the curvature vanishes — this is the Maurer–Cartan equation. The Maurer–Cartan form is a *flat* connection.

15.8 Summary: Towards Principal Bundles

Observation 15.4

Here is how everything connects to gauge theory:

Lie theory concept	**Gauge theory role**
Lie group G	Structure group of principal bundle
Lie algebra $\mathfrak{g}$	Connections are $\mathfrak{g}$-valued 1-forms
$\exp : \mathfrak{g} \to G$	Generates gauge transformations
$\mathrm{Ad}_g : \mathfrak{g} \to \mathfrak{g}$	How connections transform under gauge
$\mathrm{ad}_X = [X, \cdot]$	Infinitesimal gauge transformations
Maurer–Cartan form θ	Prototype for connection 1-form
$d\theta + \frac{1}{2}[\theta, \theta] = 0$	Flat connection (curvature $= 0$)

16 Fiber Bundles and Vector Bundles

16.1 Motivation

Beyond product spaces. The simplest way to build a higher-dimensional space from two spaces F and M is the Cartesian product $M \times F$. But many natural spaces are *not* products globally, even though they *look like* products locally.

Example 16.1: The cylinder vs the Möbius strip

- The **cylinder** $S^1 \times I$ is a product: every point has a neighborhood of the form $U \times I$.
- The **Möbius strip** is *locally* a product $U \times I$, but globally it is *twisted* — it cannot be written as $S^1 \times I$.

Both are "I-bundles over S^1," but they are topologically distinct.

Example 16.2: The torus vs the Klein bottle

- The **torus** $T^2 = S^1 \times S^1$ is a product.
- The **Klein bottle** is locally $U \times S^1$, but globally twisted.

Both are "S^1-bundles over S^1."

Generalizing tangent bundles. You have seen that TM and T^*M are smooth manifolds where:

- The base is M.
- The fiber over each $p \in M$ is a vector space (T_pM or T_p^*M).
- Locally, $TM \cong U \times \mathbb{R}^n$.

We want a general framework capturing this structure: *vector bundles*.

Road map.

Concept	**Fiber type**
Fiber bundle	Any manifold F
Vector bundle	Vector space $\mathbb{R}^k$ or $\mathbb{C}^k$
Principal bundle	Lie group G (next section)

16.2 Fiber Bundles

Definition 16.1: Fiber bundle

A *fiber bundle* consists of:

- A smooth manifold E (the *total space*).
- A smooth manifold M (the *base space*).
- A smooth manifold F (the *typical fiber*).
- A smooth surjection $\pi : E \to M$ (the *projection*).

satisfying the **local triviality condition**: for every $p \in M$, there exists an open neighborhood $U \ni p$ and a diffeomorphism

$$\phi : \pi^{-1}(U) \xrightarrow{\sim} U \times F$$

such that the following diagram commutes:

$$\begin{tikzcd} \pi^{-1}(U) \arrow[rr, "\phi"] \arrow[dr, "\pi"'] & & U \times F \arrow[dl, "\mathrm{pr}_1"] \\ & U & \end{tikzcd}$$

We write $E \xrightarrow{\pi} M$ or (E, M, F, π) for a fiber bundle.

Definition 16.2: Fiber

The *fiber over* p is

$$E_p := \pi^{-1}(p) \cong F.$$

Local triviality ensures each fiber is diffeomorphic to F.

Definition 16.3: Trivial bundle

A fiber bundle is *trivial* if there exists a *global* trivialization:

$$\phi : E \xrightarrow{\sim} M \times F.$$

Examples

Example 16.3: Trivial bundle

$E = M \times F$ with $\pi = \mathrm{pr}_1$ is the trivial F-bundle over M.

Example 16.4: The cylinder

$E = S^1 \times [0,1]$, $\pi : E \to S^1$ is the trivial $[0,1]$-bundle over S^1.

Example 16.5: The Möbius strip

Let E be the Möbius strip, $M = S^1$, $F = (-1,1)$ (an open interval). The projection $\pi : E \to S^1$ sends each point to its "base circle" coordinate.
Locally, $E|_U \cong U \times (-1,1)$. But globally, $E \ncong S^1 \times (-1,1)$ — the bundle is *nontrivial*.

Example 16.6: The Hopf fibration

$$S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2.$$

The total space is S^3, base is S^2, fiber is S^1. This is a nontrivial S^1-bundle over S^2.
Explicitly: view $S^3 \subset \mathbb{C}^2$ as $\{(z_1, z_2) : |z_1|^2 + |z_2|^2 = 1\}$. Define $\pi(z_1, z_2) = [z_1 : z_2] \in \mathbb{C}P^1 \cong S^2$. Each fiber is $\{e^{i\theta}(z_1, z_2)\} \cong S^1$.

Non-Example 16.1: Not a fiber bundle

Let $E = \{(x,y) \in \mathbb{R}^2 : y^2 = x^2(x+1)\}$ (a nodal cubic) and $\pi(x,y) = x$.
Over $x = 0$, the fiber is a single point (the node). Over $x > 0$, the fiber is two points. The fiber is not constant — this is **not** a fiber bundle.

Local Trivializations and Transition Functions

Definition 16.4: Local trivialization

A *local trivialization* over $U \subseteq M$ is a diffeomorphism

$$\phi_U : \pi^{-1}(U) \xrightarrow{\sim} U \times F$$

commuting with projections: $\mathrm{pr}_1 \circ \phi_U = \pi$.

Definition 16.5: Transition functions

Given two local trivializations $\phi_U : \pi^{-1}(U) \to U \times F$ and $\phi_V : \pi^{-1}(V) \to V \times F$, the *transition function* on $U \cap V$ is

$$g_{UV} : U \cap V \to \mathrm{Diff}(F)$$

defined by

$$\phi_U \circ \phi_V^{-1}(p, f) = (p, g_{UV}(p)(f)).$$

Proposition 16.1: Cocycle condition

Transition functions satisfy:

1. $g_{UU}(p) = \mathrm{id}_F$.
2. $g_{UV}(p) = g_{VU}(p)^{-1}$.
3. $g_{UV}(p) \circ g_{VW}(p) \circ g_{WU}(p) = \mathrm{id}_F$ on $U \cap V \cap W$.

Remark 16.1

Transition functions encode the "twisting" of the bundle. A bundle is trivial if and only if all transition functions can be chosen to be the identity.

Example 16.7: Transition function for Möbius strip

Cover S^1 by $U = S^1 \setminus \{1\}$ and $V = S^1 \setminus \{-1\}$. Then $U \cap V$ has two components. On one component, $g_{UV} = \mathrm{id}$. On the other, $g_{UV}(t) = -t$ (reflection). This "twist" makes the bundle nontrivial.

Bundle Maps

Definition 16.6: Bundle map (morphism)

A *bundle map* from (E, M, F, π) to (E', M', F', π') is a pair of smooth maps $(\tilde{f}, f)$ making the diagram commute:

$$\begin{tikzcd} E \arrow[r, "\tilde{f}"] \arrow[d, "\pi"'] & E' \arrow[d, "\pi'"] \\ M \arrow[r, "f"'] & M' \end{tikzcd}$$

If $M = M'$ and $f = \mathrm{id}_M$, we call $\tilde{f}$ a *bundle map over* M.

Definition 16.7: Bundle isomorphism

A *bundle isomorphism* is a bundle map $(\tilde{f}, f)$ where both $\tilde{f}$ and f are diffeomorphisms.

Sections

Definition 16.8: Section

A *section* of a fiber bundle $\pi : E \to M$ is a smooth map $s : M \to E$ such that

$$\pi \circ s = \mathrm{id}_M.$$

Equivalently: $s(p) \in E_p$ for all $p \in M$. The space of sections is denoted $\Gamma(E)$ or $\Gamma(M, E)$.

Example 16.8: Sections of the trivial bundle

For $E = M \times F$, sections are functions $s : M \to F$:

$$s(p) = (p, f(p)) \quad \text{for some } f : M \to F.$$

Example 16.9: No global section of Möbius strip

The Möbius strip $E \to S^1$ has fiber $(-1, 1)$. A section would be a continuous "center line." But any such curve must cross the boundary — no global section exists.
More precisely: a *nonvanishing* section would trivialize the bundle.

16.3 Vector Bundles

Motivation. We want fiber bundles where fibers are vector spaces and the structure is compatible with linear algebra.

Definition 16.9: Vector bundle

A *(real) vector bundle of rank k* is a fiber bundle $\pi : E \to M$ with fiber $F = \mathbb{R}^k$, such that:

1. Each fiber $E_p = \pi^{-1}(p)$ has the structure of a k-dimensional real vector space.
2. Local trivializations are *linear on fibers*: for each local trivialization $\phi_U : \pi^{-1}(U) \to U \times \mathbb{R}^k$, the map

$$\phi_U|_{E_p} : E_p \xrightarrow{\sim} \{p\} \times \mathbb{R}^k \cong \mathbb{R}^k$$

is a linear isomorphism.

Remark 16.2

Equivalently: transition functions take values in $GL(k, \mathbb{R})$ rather than $\mathrm{Diff}(\mathbb{R}^k)$:

$$g_{UV} : U \cap V \to GL(k, \mathbb{R}).$$

Definition 16.10: Complex vector bundle

A *complex vector bundle of rank k* has fiber $\mathbb{C}^k$, with transition functions in $GL(k, \mathbb{C})$.

Examples

Example 16.10: Trivial vector bundle

$E = M \times \mathbb{R}^k$ with $\pi = \mathrm{pr}_1$ is the *trivial rank-k vector bundle.*

Example 16.11: Tangent bundle

The tangent bundle $\pi : TM \to M$ is a real vector bundle of rank $n = \dim M$.

- Fiber: $T_pM \cong \mathbb{R}^n$.
- Local trivialization: a chart (U, x) gives $\phi_U : TU \to U \times \mathbb{R}^n$ by $v \mapsto (p, v^1, \ldots, v^n)$.
- Transition functions: $g_{UV}(p) = D(x \circ y^{-1})|_{y(p)}$ (Jacobian).

Example 16.12: Cotangent bundle

The cotangent bundle $\pi : T^*M \to M$ is a real vector bundle of rank n.

- Fiber: $T_p^*M \cong \mathbb{R}^n$.
- Transition functions: $g_{UV}(p) = (D(y \circ x^{-1})|_{x(p)})^T$ (inverse transpose of Jacobian).

Example 16.13: Möbius strip as a line bundle

The Möbius strip is a rank-1 vector bundle (line bundle) over S^1.

- Fiber: $\mathbb{R}$.
- Transition function: $g_{UV}(p) = -1 \in GL(1, \mathbb{R}) = \mathbb{R}^*$ on one component of $U \cap V$.

This is the *nontrivial* line bundle over S^1.

Example 16.14: Tautological line bundle over $\mathbb{R}P^n$

Define

$$\gamma_n^1 := \{(\ell, v) \in \mathbb{R}P^n \times \mathbb{R}^{n+1} : v \in \ell\}.$$

Here $\ell \in \mathbb{R}P^n$ is a line through the origin in $\mathbb{R}^{n+1}$, and v is a vector on that line. Projection: $\pi(\ell, v) = \ell$. Fiber over ℓ: the line ℓ itself (a 1-dimensional vector space).
This is the *tautological* or *canonical* line bundle over $\mathbb{R}P^n$. It is nontrivial.

Example 16.15: Tautological line bundle over $\mathbb{C}P^n$

Similarly:

$$\mathcal{O}(-1) := \{(\ell, v) \in \mathbb{C}P^n \times \mathbb{C}^{n+1} : v \in \ell\}.$$

This is a complex line bundle, fundamental in algebraic geometry.

Non-Example 16.2: Not every $\mathbb{R}^k$-bundle is a vector bundle

Consider the Möbius strip with transition function $g(p)(t) = -t$ on one overlap. This *is* a vector bundle ($-1 \in GL(1, \mathbb{R})$).
But if we used $g(p)(t) = t^3$ (a diffeomorphism of $\mathbb{R}$ but not linear), the result would be a fiber bundle with fiber $\mathbb{R}$, but **not** a vector bundle.

Sections of Vector Bundles

Definition 16.11: Section of a vector bundle

A *section* of a vector bundle $E \to M$ is a smooth map $s : M \to E$ with $\pi \circ s = \mathrm{id}_M$.
The space of sections $\Gamma(E)$ is a $C^\infty(M)$-module: for $s, t \in \Gamma(E)$ and $f \in C^\infty(M)$:

$$(s + t)(p) := s(p) + t(p), \qquad (f \cdot s)(p) := f(p) \cdot s(p).$$

Example 16.16: Sections of TM

Sections of $TM \to M$ are *vector fields*:

$$\Gamma(TM) = \mathfrak{X}(M).$$

Example 16.17: Sections of T^*M

Sections of $T^*M \to M$ are 1-*forms*:

$$\Gamma(T^*M) = \Omega^1(M).$$

Example 16.18: Sections of trivial bundle

For $E = M \times \mathbb{R}^k$:

$$\Gamma(E) \cong C^\infty(M, \mathbb{R}^k) = \{(f^1, \ldots, f^k) : f^i \in C^\infty(M)\}.$$

Definition 16.12: Zero section

The *zero section* $s_0 : M \to E$ is defined by $s_0(p) = 0_p \in E_p$ (the zero vector in each fiber). This always exists.

Proposition 16.2: Nonvanishing section and triviality (rank 1)

A line bundle $L \to M$ is trivial if and only if it admits a *nowhere-vanishing* section.

Example 16.19: Möbius line bundle has no nonvanishing section

Any section of the Möbius bundle must vanish somewhere (it "twists"). Hence the bundle is nontrivial.

Vector Bundle Maps

Definition 16.13: Vector bundle homomorphism

A *vector bundle homomorphism* from $E \to M$ to $E' \to M$ (over the same base) is a smooth map $\phi : E \to E'$ such that:

1. $\pi' \circ \phi = \pi$ (preserves fibers).
2. $\phi|_{E_p} : E_p \to E'_p$ is linear for each $p \in M$.

Definition 16.14: Vector bundle isomorphism

A *vector bundle isomorphism* is a vector bundle homomorphism that is also a diffeomorphism (equivalently, linear isomorphism on each fiber).

16.4 Operations on Vector Bundles

Motivation. Vector spaces support operations like direct sum, tensor product, and duality. These extend to vector bundles fiberwise.

Definition 16.15: Direct sum (Whitney sum)

Given vector bundles $E \to M$ and $E' \to M$, the *direct sum* $E \oplus E'$ has:

- Total space: $E \oplus E' := \bigsqcup_{p \in M} (E_p \oplus E'_p)$.
- Fiber: $(E \oplus E')_p = E_p \oplus E'_p$.
- Rank: $\operatorname{rank}(E \oplus E') = \operatorname{rank}(E) + \operatorname{rank}(E')$.

Example 16.20: $TM \oplus T^*M$

This is a rank-$2n$ vector bundle over an n-manifold M.

Definition 16.16: Tensor product

The *tensor product* $E \otimes E'$ has:

- Fiber: $(E \otimes E')_p = E_p \otimes E'_p$.
- Rank: $\operatorname{rank}(E \otimes E') = \operatorname{rank}(E) \cdot \operatorname{rank}(E')$.

Definition 16.17: Dual bundle

The *dual bundle* E^* has:

- Fiber: $(E^*)_p = (E_p)^* = \operatorname{Hom}(E_p, \mathbb{R})$.
- Rank: same as E.

Example 16.21: $T^*M = (TM)^*$

The cotangent bundle is the dual of the tangent bundle.

Definition 16.18: Exterior powers

The *k-th exterior power* $\Lambda^k E$ has:

- Fiber: $(\Lambda^k E)_p = \Lambda^k(E_p)$.
- Rank: $\binom{\operatorname{rank}(E)}{k}$.

Example 16.22: Differential forms as sections

$$\Omega^k(M) = \Gamma(\Lambda^k T^* M).$$

A k-form is a section of the k-th exterior power of the cotangent bundle.

Definition 16.19: Determinant (top exterior power)

The *determinant line bundle* of a rank-k bundle E is

$$\det(E) := \Lambda^k E.$$

This is a line bundle (rank 1).

Example 16.23: Orientability

A manifold M is *orientable* if and only if $\det(TM) = \Lambda^n TM$ is trivial.

16.5 Transition Functions and Structure Groups

Definition 16.20: Structure group

The *structure group* of a vector bundle is the group in which transition functions take values:

- Real rank-k bundle: $GL(k, \mathbb{R})$.
- Complex rank-k bundle: $GL(k, \mathbb{C})$.

Definition 16.21: Reduction of structure group

A vector bundle has a *reduction of structure group* to $H \subseteq GL(k)$ if there exists an atlas with transition functions valued in H.

Example 16.24: Orientable bundles

A real vector bundle is *orientable* if its structure group reduces from $GL(k, \mathbb{R})$ to $GL^+(k, \mathbb{R})$ (matrices with positive determinant).

Example 16.25: Riemannian structure

A real vector bundle admits a *fiber metric* (inner product on each fiber varying smoothly) if and only if its structure group reduces to $O(k)$.
For the tangent bundle TM, this is a Riemannian metric on M.

Example 16.26: Hermitian structure

A complex vector bundle admits a Hermitian metric if its structure group reduces to $U(k)$.

Remark 16.3

Every real (resp. complex) vector bundle over a paracompact base admits an $O(k)$ (resp. $U(k)$) structure — use partitions of unity to patch local metrics.

16.6 Frame Bundles

Motivation. Given a vector bundle, we can extract a principal bundle whose fibers are the "frames" (bases) of the vector space fibers. This connects vector bundles to principal bundles.

Definition 16.22: Frame

A *frame* at p for a rank-k vector bundle $E \to M$ is an ordered basis $(e_1, \ldots, e_k)$ of E_p.

Definition 16.23: Frame bundle

The *frame bundle* of E is

$$\mathrm{Fr}(E) := \bigsqcup_{p \in M} \{\text{frames of } E_p\}.$$

This is a principal $GL(k, \mathbb{R})$-bundle over M:

- Projection: $\pi(e_1, \ldots, e_k) = p$ where $(e_1, \ldots, e_k) \in \mathrm{Fr}(E_p)$.
- Right action: $(e_1, \ldots, e_k) \cdot A = (e_1, \ldots, e_k)A$ (matrix multiplication).

Example 16.27: Frame bundle of TM

$$\mathrm{Fr}(TM) = \bigsqcup_{p \in M} \{\text{bases of } T_pM\}.$$

This is a principal $GL(n, \mathbb{R})$-bundle, often denoted LM or FM.

Observation 16.1

The relationship:

Vector bundle	**Associated principal bundle**
$E \to M$ (rank k)	$\mathrm{Fr}(E) \to M$ (structure group $GL(k)$)

Conversely, given a principal $GL(k)$-bundle P, we can reconstruct E as an *associated bundle*. This is the content of the next section.

16.7 Summary: Towards Principal Bundles

Observation 16.2

Bundle type	**Fiber**	**Transition functions**
Fiber bundle	Manifold F	$\mathrm{Diff}(F)$
Vector bundle (rank k)	$\mathbb{R}^k$ or $\mathbb{C}^k$	$GL(k, \mathbb{R})$ or $GL(k, \mathbb{C})$
Principal G-bundle	Lie group G	G (acting on itself)

Observation 16.3

Key constructions to remember:

- **Sections** $\Gamma(E)$: globalizations of fiber data.
- **Transition functions**: encode twisting, valued in structure group.
- **Operations**: $\oplus$, $\otimes$, E^*, $\Lambda^k E$ — all fiberwise.
- **Frame bundle**: extracts a principal bundle from a vector bundle.

17 Principal Bundles and Connections

17.1 Motivation

Why principal bundles? In gauge theory, matter fields (electrons, quarks) live in vector bundles. But the *gauge symmetry* itself — the group G acting on these fields — is encoded in a **principal G-bundle**.

Why connections? To differentiate sections of a vector bundle, we need to compare vectors in *different* fibers. A connection provides this comparison — it tells us how to "parallel transport" along curves. On a principal bundle, a connection is a $\mathfrak{g}$-valued 1-form.

Why Yang–Mills? The curvature of a connection measures its failure to be flat. Yang–Mills theory selects connections whose curvature is "as small as possible" — these are the physical gauge fields.

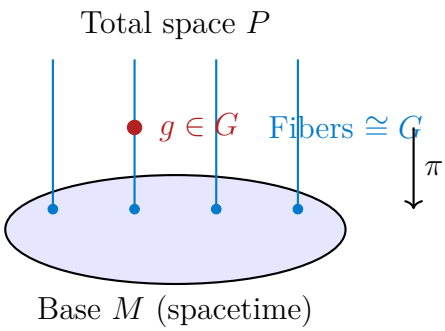


Figure 39: A principal G-bundle: fibers are copies of the Lie group G.

17.2 Group Actions on Principal Bundles

Definition 17.1: Right action (review)

A *right action* of a Lie group G on a manifold P is a smooth map

$$P \times G \to P, \qquad (u, g) \mapsto u \cdot g,$$

satisfying:

1. $u \cdot e = u$ for all $u \in P$.
2. $(u \cdot g) \cdot h = u \cdot (gh)$ for all $u \in P$, $g, h \in G$.

Definition 17.2: Free action

The action is *free* if

$$u \cdot g = u \quad \Longrightarrow \quad g = e.$$

No element except the identity has fixed points.

Definition 17.3: Transitive action

The action is *transitive on a set* S if for any $u, v \in S$, there exists $g \in G$ with $u \cdot g = v$.

Remark 17.1

In a principal bundle, G acts **freely and transitively on each fiber**. This makes each fiber diffeomorphic to G (non-canonically).

17.3 Principal Bundles

Definition 17.4: Principal G-bundle

A *principal G-bundle* over a manifold M is a fiber bundle $\pi : P \to M$ together with a right action of G on P such that:

1. G **acts freely:** $u \cdot g = u \Rightarrow g = e$.
2. **Fibers are orbits:** $\pi^{-1}(p) = \{u \cdot g : g \in G\}$ for any $u \in \pi^{-1}(p)$.
3. **Local triviality:** For each $p \in M$, there exists $U \ni p$ and a G-equivariant diffeomorphism
$$\phi_U : \pi^{-1}(U) \xrightarrow{\sim} U \times G$$
with $\mathrm{pr}_1 \circ \phi_U = \pi$ and $\phi_U(u \cdot g) = \phi_U(u) \cdot g$.

We write $P \xrightarrow{G} M$ or (P, M, G, π).

Remark 17.2

The key difference from a general fiber bundle: the structure group G is the *same* as the fiber G, and G acts on itself by right multiplication.

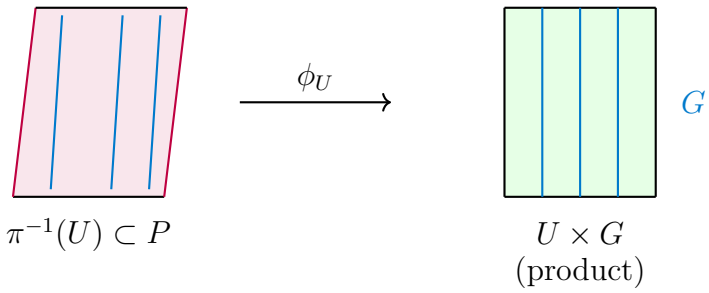


Figure 40: Local trivialization: $\pi^{-1}(U)$ is diffeomorphic to $U \times G$.

Examples

Example 17.1: Trivial principal bundle

$P = M \times G$ with $\pi = \mathrm{pr}_1$ and right action $(p, h) \cdot g = (p, hg)$.

Example 17.2: Frame bundle

For a vector bundle $E \to M$ of rank k:

$$\mathrm{Fr}(E) = \bigsqcup_{p \in M} \{\text{ordered bases of } E_p\}$$

is a principal $GL(k, \mathbb{R})$-bundle. The action: $(e_1, \ldots, e_k) \cdot A = (e_1, \ldots, e_k)A$.

Example 17.3: Orthonormal frame bundle

If E has a fiber metric, the *orthonormal frame bundle*

$$\mathrm{Fr}_O(E) = \{\text{orthonormal bases of } E_p\}$$

is a principal $O(k)$-bundle (or $SO(k)$ if oriented).

Example 17.4: Hopf fibration as principal bundle

$$U(1) \hookrightarrow S^3 \xrightarrow{\pi} S^2$$

is a principal $U(1)$-bundle. The action: $z \cdot e^{i\theta} = e^{i\theta} z$ for $z \in S^3 \subset \mathbb{C}^2$.

Example 17.5: Universal cover

The universal cover $\tilde{M} \to M$ is a principal $\pi_1(M)$-bundle (with discrete structure group).

Example 17.6: $SU(2) \to SO(3)$

The double cover $SU(2) \to SO(3)$ is a principal $\mathbb{Z}_2$-bundle. Fiber: $\{\pm I\}$.

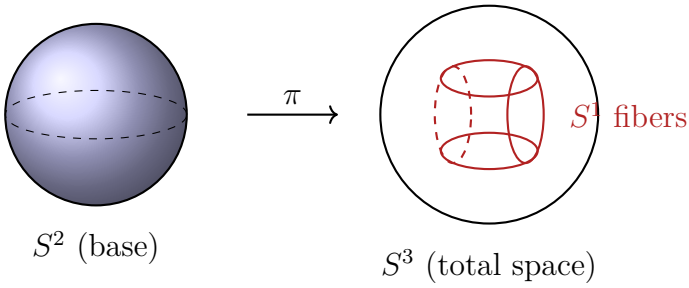


Figure 41: Hopf fibration: S^3 fibers over S^2 with S^1 fibers.

Transition Functions

Proposition 17.1: Transition functions for principal bundles

Given local trivializations ϕ_U and ϕ_V over $U, V \subseteq M$, the transition function

$$g_{UV} : U \cap V \to G$$

is defined by

$$\phi_U \circ \phi_V^{-1}(p, h) = (p, g_{UV}(p) \cdot h).$$

These satisfy the cocycle conditions:

1. $g_{UU} = e$.
2. $g_{UV} = g_{VU}^{-1}$.
3. $g_{UV} \cdot g_{VW} \cdot g_{WU} = e$ on $U \cap V \cap W$.

Theorem 17.1: Reconstruction from cocycles

Given an open cover $\{U_\alpha\}$ of M and smooth maps $g_{\alpha\beta} : U_\alpha \cap U_\beta \to G$ satisfying the cocycle conditions, there exists a principal G-bundle $P \to M$ with these transition functions, unique up to isomorphism.

Fundamental Vector Fields

Definition 17.5: Fundamental vector field

For $X \in \mathfrak{g}$, the *fundamental vector field* $X^\sharp \in \mathfrak{X}(P)$ is

$$X^\sharp_u := \frac{d}{dt}\bigg|_{t=0} u \cdot \exp(tX).$$

This is the infinitesimal generator of the G-action in the direction X.

Proposition 17.2: Properties of fundamental vector fields

1. The map $\mathfrak{g} \to \mathfrak{X}(P)$, $X \mapsto X^\sharp$, is a Lie algebra anti-homomorphism:

$$[X, Y]^\sharp = -[X^\sharp, Y^\sharp].$$

2. $X^\sharp$ is *vertical*: $\pi_* X^\sharp = 0$.
3. $(R_g)_* X^\sharp = (\mathrm{Ad}_{g^{-1}} X)^\sharp$.

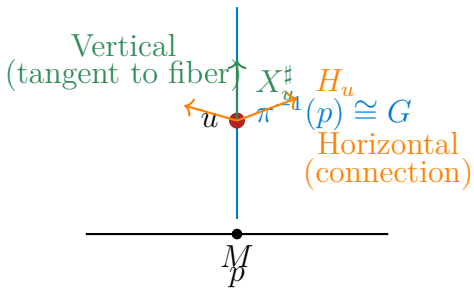


Figure 42: Vertical vectors are tangent to fibers. A connection chooses horizontal complements.

17.4 Connections on Principal Bundles

The problem. At each $u \in P$, the tangent space $T_u P$ contains the *vertical subspace*

$$V_u := \ker(\pi_{*u}) = T_u(\pi^{-1}(\pi(u))) \cong \mathfrak{g}.$$

But there is no canonical complement. A **connection** chooses one.

Definition 17.6: Connection (horizontal distribution)

A *connection* on a principal G-bundle $P \to M$ is a smooth distribution $H \subset TP$ such that:

1. **Complement:** $T_uP = H_u \oplus V_u$ for all $u \in P$.
2. **G-invariance:** $(R_g)_* H_u = H_{u \cdot g}$ for all $g \in G$.

The subspace H_u is called the *horizontal subspace* at u.

Definition 17.7: Connection 1-form

A *connection 1-form* on P is a $\mathfrak{g}$-valued 1-form $\omega \in \Omega^1(P; \mathfrak{g})$ satisfying:

1. **Reproduces generators:** $\omega(X^\sharp) = X$ for all $X \in \mathfrak{g}$.
2. **Equivariance:** $R_g^* \omega = \mathrm{Ad}_{g^{-1}} \circ \omega$ for all $g \in G$.

Proposition 17.3: Equivalence of definitions

There is a bijection between connections (horizontal distributions) and connection 1-forms:

$$H_u = \ker(\omega_u).$$

Proof. Given ω, set $H_u = \ker \omega_u$. Given H, define $\omega_u(v) = X$ where $v - X_u^\sharp \in H_u$. □

Remark 17.3

The connection form ω generalizes the Maurer–Cartan form $\theta = g^{-1}dg$. On the trivial bundle $M \times G$, the Maurer–Cartan form pulled back from G is a (flat) connection.

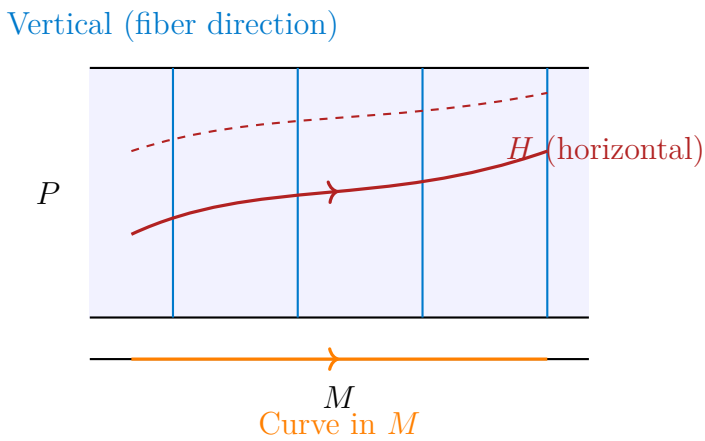


Figure 43: A connection: horizontal subspaces complement vertical fibers. Horizontal curves project to curves in M.

Local Connection Forms (Gauge Potentials)

Definition 17.8: Local section

A *local section* of $P \to M$ over $U \subseteq M$ is a smooth map $\sigma : U \to P$ with $\pi \circ \sigma = \mathrm{id}_U$.

Definition 17.9: Local connection form (gauge potential)

Given a connection ω on P and a local section $\sigma : U \to P$, the *local connection form* or *gauge potential* is

$$A := \sigma^* \omega \in \Omega^1(U; \mathfrak{g}).$$

Remark 17.4

In physics notation:

- $A = A^a_\mu T_a \, dx^\mu$ is the gauge potential.
- A^a_μ are the component functions.
- $\{T_a\}$ is a basis of $\mathfrak{g}$.

Proposition 17.4: Reconstruction of ω from local data

On a trivializing neighborhood U with section σ, every $u \in \pi^{-1}(U)$ can be written uniquely as $u = \sigma(p) \cdot g$ for $p = \pi(u)$ and $g \in G$. Then:

$$\omega_u = \mathrm{Ad}_{g^{-1}}(\pi^* A)_u + \theta_g,$$

where $\theta = g^{-1} dg$ is the Maurer–Cartan form on G.

Gauge Transformations

Definition 17.10: Gauge transformation

A *gauge transformation* is a G-equivariant diffeomorphism $\Phi : P \to P$ covering the identity:

$$\pi \circ \Phi = \pi, \qquad \Phi(u \cdot g) = \Phi(u) \cdot g.$$

Equivalently, it is a smooth map $g : M \to G$ (via local sections).

Theorem 17.2: Transformation of gauge potential

Under a gauge transformation $g : U \to G$, the local connection form transforms as:

$$\boxed{A \mapsto A^g = gAg^{-1} + g\, dg^{-1} = \mathrm{Ad}_g(A) - dg \cdot g^{-1}.}$$

For matrix groups: $A^g = gAg^{-1} - (dg)g^{-1}$.

Example 17.7: $U(1)$ gauge transformation

For $G = U(1)$ (abelian), $g = e^{i\lambda}$ for $\lambda : U \to \mathbb{R}$:

$$A \mapsto A + i\, d\lambda.$$

This is the familiar gauge transformation of electromagnetism.

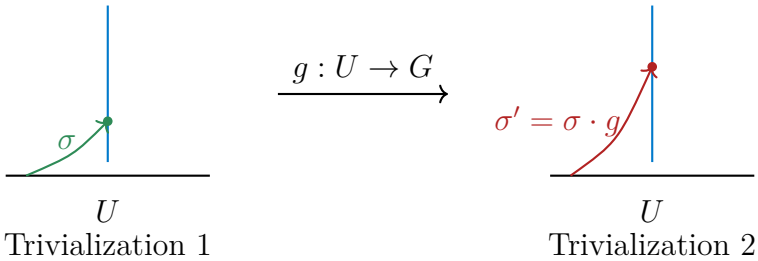


Figure 44: Gauge transformation: changing the local section from σ to $\sigma' = \sigma \cdot g$.

17.5 Curvature

Definition 17.11: Curvature 2-form

The *curvature* of a connection ω is the $\mathfrak{g}$-valued 2-form $\Omega \in \Omega^2(P; \mathfrak{g})$ defined by:

$$\boxed{\Omega = d\omega + \frac{1}{2}[\omega, \omega].}$$

Here $[\omega, \omega](X, Y) = 2[\omega(X), \omega(Y)]$.

Remark 17.5

Equivalently: $\Omega = d\omega + \omega \wedge \omega$, where $(\omega \wedge \omega)(X, Y) = [\omega(X), \omega(Y)]$.

Theorem 17.3: Structure equation (Cartan)

The curvature satisfies:

$$\Omega(X, Y) = d\omega(X^h, Y^h),$$

where X^h, Y^h denote horizontal projections. In other words, Ω measures the failure of horizontal distribution to be integrable.

Proposition 17.5: Properties of curvature

1. **Horizontal:** $\Omega(X, Y) = 0$ if X or Y is vertical.
2. **Equivariant:** $R_g^* \Omega = \mathrm{Ad}_{g^{-1}} \circ \Omega$.
3. **Bianchi identity:** $d\Omega + [\omega, \Omega] = 0$ (equivalently, $D\Omega = 0$).

Local Curvature (Field Strength)

Definition 17.12: Local curvature (field strength)

Given a local section $\sigma : U \to P$, the *local curvature* or *field strength* is

$$F := \sigma^* \Omega \in \Omega^2(U; \mathfrak{g}).$$

Proposition 17.6: Field strength formula

$$\boxed{F = dA + \frac{1}{2}[A, A] = dA + A \wedge A.}$$

In components: $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^a_{bc} A^b_\mu A^c_\nu$, where $[T_b, T_c] = f^a_{bc} T_a$.

Example 17.8: $U(1)$ field strength

For $G = U(1)$, $[A, A] = 0$ (abelian), so:

$$F = dA.$$

This is the electromagnetic field tensor $F = dA$, i.e., $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

Example 17.9: $SU(2)$ field strength

For $G = SU(2)$, with $A = A^a T_a$:

$$F = dA + A \wedge A = \left(dA^a + \frac{1}{2} \epsilon^a_{bc} A^b \wedge A^c \right) T_a.$$

The non-abelian term $A \wedge A$ gives self-interaction of gauge bosons.

Theorem 17.4: Gauge transformation of curvature

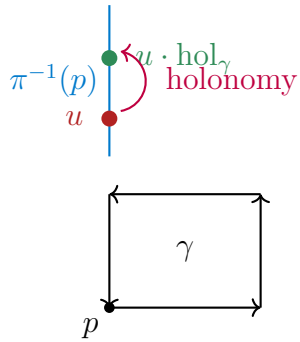
Under $g : U \to G$:

$$\boxed{F \mapsto F^g = g F g^{-1} = \mathrm{Ad}_g(F).}$$

The curvature transforms *covariantly* (no inhomogeneous term).

Remark 17.6

This is why F (not A) is the physical observable. For $U(1)$: $F^g = F$ (gauge invariant). For non-abelian: F transforms, but $\mathrm{tr}(F \wedge F)$ is gauge invariant.



Curvature $\Leftrightarrow$ holonomy around loops

Figure 45: Parallel transport around a closed loop returns to a different point in the fiber. The failure is measured by the holonomy, which is computed from the curvature.

Flat Connections

Definition 17.13: Flat connection

A connection is *flat* if its curvature vanishes: $\Omega = 0$ (equivalently, $F = 0$ locally).

Proposition 17.7: Flat $\Leftrightarrow$ integrable

A connection is flat if and only if the horizontal distribution H is integrable (Frobenius).

Example 17.10: Maurer–Cartan connection is flat

On the trivial bundle $M \times G$, pull back the Maurer–Cartan form θ from G. The curvature is:

$$d\theta + \frac{1}{2}[\theta, \theta] = 0$$

by the Maurer–Cartan equation. This connection is flat.

17.6 Yang–Mills Theory

Setup. Let $P \to M$ be a principal G-bundle over an oriented Riemannian (or Lorentzian) n-manifold (M, g). Assume G is compact with an Ad-invariant inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{g}$ (e.g., $\langle X, Y \rangle = -\mathrm{tr}(XY)$ for $\mathfrak{su}(n)$).

Definition 17.14: Yang–Mills action

The *Yang–Mills action* of a connection A (with curvature F) is:

$$\boxed{S_{YM}[A] = \frac{1}{2}\int_M \langle F \wedge *F \rangle = \frac{1}{4}\int_M F^a_{\mu\nu} F^{a\,\mu\nu}\,\mathrm{vol}_g.}$$

Here $*$ is the Hodge star on M.

Remark 17.7

The action is:

- Gauge invariant (since $F \mapsto gFg^{-1}$ and $\langle \cdot, \cdot \rangle$ is Ad-invariant).
- Non-negative (for Riemannian signature).
- Quadratic in F.

Definition 17.15: Yang–Mills equations

The *Yang–Mills equations* are the Euler–Lagrange equations for S_{YM}:

$$\boxed{D_A * F = d * F + [A, *F] = 0.}$$

Together with the Bianchi identity $D_A F = 0$, these are the equations of motion.

Example 17.11: Maxwell's equations

For $G = U(1)$ (electromagnetism), $[A, *F] = 0$, so Yang–Mills reduces to:

$$d * F = 0, \qquad dF = 0.$$

These are Maxwell's equations in vacuum: $dF = 0$ (Faraday/no monopoles), $d*F = 0$ (Ampère/Gauss).

Electromagnetism		**Yang–Mills**
$G = U(1)$		G non-abelian
$A \in \Omega^1(M; i\mathbb{R})$		$A \in \Omega^1(M; \mathfrak{g})$
$F = dA$		$F = dA + A \wedge A$
$d * F = 0$	generalize $\longrightarrow$	$D_A * F = 0$
(linear)		(nonlinear!)

Figure 46: Yang–Mills generalizes Maxwell to non-abelian gauge groups.

Yang–Mills in 4 Dimensions

Proposition 17.8: Decomposition of 2-forms in 4D

On an oriented 4-manifold, $\Omega^2(M)$ splits into self-dual and anti-self-dual parts:

$$\Omega^2 = \Omega^2_+ \oplus \Omega^2_-, \qquad *\omega_\pm = \pm\omega_\pm.$$

Correspondingly: $F = F^+ + F^-$.

Proposition 17.9: Topological bound

For a compact 4-manifold:

$$S_{YM}[A] = \frac{1}{2}\int_M |F|^2 = \frac{1}{2}\int_M (|F^+|^2 + |F^-|^2) \geq \left|\frac{1}{2}\int_M (|F^+|^2 - |F^-|^2)\right| = |k| \cdot 8\pi^2,$$

where $k = \frac{1}{8\pi^2}\int_M \operatorname{tr}(F \wedge F)$ is the *instanton number* (second Chern number).

Definition 17.16: Instantons (self-dual connections)

A connection is an *instanton* if $F = F^+$ (self-dual: $*F = F$) or an *anti-instanton* if $F = F^-$ (anti-self-dual: $*F = -F$).

Theorem 17.5: Instantons are Yang–Mills

Self-dual and anti-self-dual connections automatically satisfy the Yang–Mills equations. They are absolute minima of S_{YM} within their topological class.

Proof. $D_A * F = D_A(\pm F) = \pm D_A F = 0$ by Bianchi. □

Physical Interpretation

Observation 17.1

Yang–Mills theory describes:

Gauge Group	Physics	Gauge Bosons
$U(1)$	Electromagnetism	Photon (γ)
$SU(2)$	Weak interaction	$W^\pm$, Z^0
$SU(3)$	Strong interaction (QCD)	Gluons (g)
$SU(3) \times SU(2) \times U(1)$	Standard Model	All of the above

Observation 17.2

Key features of non-abelian Yang–Mills:

- **Self-interaction:** $F = dA + A \wedge A$ means gauge bosons interact with themselves (unlike photons).
- **Asymptotic freedom:** Coupling decreases at high energies (QCD).
- **Confinement:** Quarks cannot be isolated (QCD).
- **Instantons:** Topologically nontrivial solutions; contribute to path integral.

17.7 Summary

Observation 17.3

The complete picture:

Mathematical Object	**Physical Interpretation**
Principal G-bundle $P \to M$	Gauge symmetry
Connection ω / gauge potential A	Gauge field
Curvature Ω / field strength F	Field tensor
Gauge transformation	Change of local trivialization
$F = dA + A \wedge A$	Non-abelian field strength
$D_A * F = 0$	Equations of motion
$\mathrm{tr}(F \wedge F)$	Topological charge (instanton number)

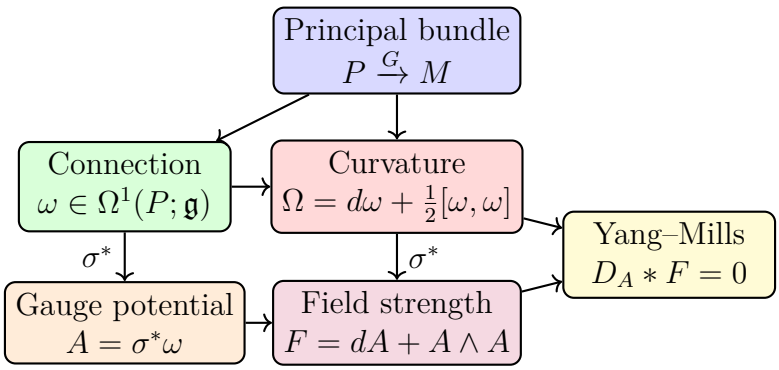


Figure 47: Summary: from principal bundles to Yang–Mills equations.

18 Riemannian Geometry

18.1 Motivation

The problem. On a smooth manifold M, you can talk about smooth functions, tangent vectors, and differential forms. But you **cannot** talk about:

- the *length* of a tangent vector,
- the *angle* between two vectors,
- the *distance* between two points,
- the *volume* of a region.

Why? Because T_pM is just a vector space — no inner product is given.

The solution. Equip each T_pM with an inner product, varying smoothly with p. This is a **Riemannian metric**. From it, everything else follows: lengths, angles, volumes, curvature, geodesics, and ultimately Einstein's equations of general relativity.

18.2 Riemannian Metrics

Elementary examples first.

Example 18.1: $\mathbb{R}^n$ with the Euclidean metric

At each $p \in \mathbb{R}^n$, we have $T_p\mathbb{R}^n \cong \mathbb{R}^n$. The usual dot product:

$$g_p(v, w) = v^1 w^1 + v^2 w^2 + \cdots + v^n w^n.$$

This is so natural we forget it's a *choice*.

Example 18.2: A different metric on $\mathbb{R}^2$

Same manifold $\mathbb{R}^2$, but define:

$$g_p(v, w) = 4v^1 w^1 + v^2 w^2.$$

Now the "unit circle" $\{v : g(v, v) = 1\}$ is an ellipse, not a circle. **Same manifold, different geometry.**

Example 18.3: The sphere $S^2 \subset \mathbb{R}^3$

S^2 inherits a metric from $\mathbb{R}^3$: for $v, w \in T_pS^2 \subset T_p\mathbb{R}^3$:

$$g_p(v, w) = v \cdot w \quad \text{(Euclidean dot product)}.$$

In spherical coordinates (θ, ϕ):

$$g = d\theta^2 + \sin^2\theta \, d\phi^2.$$

The coefficient $\sin^2\theta$ varies with position — geometry is *curved*.

Example 18.4: The hyperbolic plane $\mathbb{H}^2$

Upper half-plane $\{(x, y) : y > 0\}$ with:

$$g = \frac{dx^2 + dy^2}{y^2}.$$

Near $y = 0$: metric blows up. Near $y \to \infty$: metric shrinks. This has **constant negative curvature**.

Definition

Definition 18.1: Riemannian metric

A *Riemannian metric* on a smooth manifold M is a smooth assignment of an inner product g_p on each tangent space T_pM. That is:

- $g_p : T_pM \times T_pM \to \mathbb{R}$,
- **Symmetric:** $g_p(v, w) = g_p(w, v)$,
- **Positive definite:** $g_p(v, v) > 0$ for $v \neq 0$,
- **Smooth:** In any chart, the components $g_{ij}(x)$ are smooth.

In coordinates:

$$g = g_{ij}(x)\, dx^i \otimes dx^j.$$

A manifold equipped with a Riemannian metric is called a *Riemannian manifold*.

Remark 18.1

The metric is a $(0, 2)$-tensor. Under coordinate change $x \to x'$:

$$g'_{kl} = \frac{\partial x^i}{\partial x'^k} \frac{\partial x^j}{\partial x'^l} g_{ij}.$$

What a Metric Gives You

Proposition 18.1: Geometric quantities from the metric

Concept	Formula
Length of vector	$\|v\| = \sqrt{g(v, v)}$
Angle between vectors	$\cos\theta = \dfrac{g(v, w)}{\|v\|\|w\|}$
Length of curve $\gamma : [a, b] \to M$	$L(\gamma) = \displaystyle\int_a^b \sqrt{g(\dot\gamma, \dot\gamma)}\, dt$
Distance	$d(p, q) = \inf\{L(\gamma) : \gamma \text{ from } p \text{ to } q\}$
Volume form	$\mathrm{vol}_g = \sqrt{\det(g_{ij})}\, dx^1 \wedge \cdots \wedge dx^n$

Example 18.5: Volume of S^2

With $g = d\theta^2 + \sin^2\theta\, d\phi^2$:

$$\det(g_{ij}) = \sin^2\theta, \quad \mathrm{vol}_g = \sin\theta\, d\theta \wedge d\phi.$$

$$\mathrm{Vol}(S^2) = \int_{S^2} \mathrm{vol}_g = \int_0^\pi \int_0^{2\pi} \sin\theta\, d\phi\, d\theta = 4\pi.$$

Musical Isomorphisms

The problem. Tangent vectors $v \in T_pM$ and covectors $\omega \in T_p^*M$ are different objects. But with a metric, we can identify them.

Definition 18.2: Lowering indices: $\flat$ ("flat")

Given $v \in T_pM$, define $v^\flat \in T_p^*M$ by:

$$v^\flat(w) := g(v, w) \quad \text{for all } w \in T_pM.$$

In components:

$$(v^\flat)_i = g_{ij}v^j.$$

Definition 18.3: Raising indices: $\sharp$ ("sharp")

Given $\omega \in T_p^*M$, define $\omega^\sharp \in T_pM$ by:

$$g(\omega^\sharp, w) = \omega(w) \quad \text{for all } w \in T_pM.$$

In components:

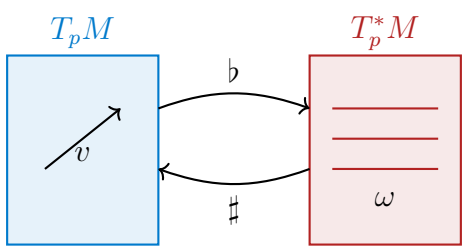
$$(\omega^\sharp)^i = g^{ij}\omega_j,$$

where g^{ij} is the inverse matrix: $g^{ik}g_{kj} = \delta^i_j$.

Example 18.6: Gradient

For $f \in C^\infty(M)$, the differential df is a 1-form. The *gradient* is:

$$\text{grad}(f) := (df)^\sharp = g^{ij}\frac{\partial f}{\partial x^j}\partial_i.$$



The metric identifies vectors and covectors

Figure 48: Musical isomorphisms: $\flat$ lowers, $\sharp$ raises.

18.3 The Hodge Star

Motivation. The musical isomorphisms convert vectors $\leftrightarrow$ covectors (1-forms). The **Hodge star** extends this to all differential forms:

$$\star : \Omega^k(M) \xrightarrow{\sim} \Omega^{n-k}(M).$$

Definition 18.4: Hodge star

Let (M, g) be an oriented Riemannian n-manifold with volume form vol_g. The *Hodge star* is the linear map $\star : \Omega^k(M) \to \Omega^{n-k}(M)$ defined by:

$$\alpha \wedge \star\beta = g(\alpha, \beta)\,\mathrm{vol}_g$$

for all $\alpha, \beta \in \Omega^k(M)$.

Example 18.7: Hodge star on $\mathbb{R}^3$

With standard orientation and Euclidean metric:

$$\begin{aligned}
&\star 1 = dx \wedge dy \wedge dz, \\
&\star\, dx = dy \wedge dz, \quad \star dy = dz \wedge dx, \quad \star dz = dx \wedge dy, \\
&\star\,(dy \wedge dz) = dx, \quad \star(dz \wedge dx) = dy, \quad \star(dx \wedge dy) = dz, \\
&\star\,(dx \wedge dy \wedge dz) = 1.
\end{aligned}$$

Example 18.8: Hodge star on S^2

With $g = d\theta^2 + \sin^2\theta\, d\phi^2$ and $\mathrm{vol} = \sin\theta\, d\theta \wedge d\phi$:

$$\star 1 = \sin\theta\, d\theta \wedge d\phi, \quad \star d\theta = \sin\theta\, d\phi, \quad \star d\phi = -\csc\theta\, d\theta.$$

Proposition 18.2: Double Hodge star

On a Riemannian n-manifold:

$$\star\star\,\alpha = (-1)^{k(n-k)}\alpha \quad \text{for } \alpha \in \Omega^k.$$

18.4 Codifferential and Laplacian

Motivation. The exterior derivative $d : \Omega^k \to \Omega^{k+1}$ goes "up" in degree. Is there something going "down"?

Definition 18.5: Inner product on forms

For $\alpha, \beta \in \Omega^k(M)$ on a compact manifold:

$$\langle \alpha, \beta \rangle := \int_M \alpha \wedge \star\beta = \int_M g(\alpha, \beta)\, \mathrm{vol}_g.$$

Definition 18.6: Codifferential

The *codifferential* $\delta : \Omega^k(M) \to \Omega^{k-1}(M)$ is the formal adjoint of d:

$$\langle d\alpha, \beta \rangle = \langle \alpha, \delta\beta \rangle.$$

Explicitly:

$$\delta = (-1)^{n(k+1)+1} \star d \star .$$

Example 18.9: Codifferential on $\mathbb{R}^3$

For $\omega = f\,dx + g\,dy + h\,dz$ (corresponding to vector field $\vec{F} = (f, g, h)$):

$$\delta\omega = -\left(\frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}\right) = -\mathrm{div}(\vec{F}).$$

Definition 18.7: Hodge Laplacian

The *Hodge Laplacian* (or Laplace–de Rham operator) is:

$$\Delta = d\delta + \delta d : \Omega^k(M) \to \Omega^k(M).$$

Example 18.10: Laplacian on functions

For $f \in \Omega^0(M)$:

$$\Delta f = \delta(df) = -\mathrm{div}(\mathrm{grad}(f)).$$

On $\mathbb{R}^n$: $\Delta f = -\sum_i \partial_i^2 f$ (geometer's convention, negative).

Definition 18.8: Harmonic form

A form $\alpha \in \Omega^k(M)$ is *harmonic* if $\Delta\alpha = 0$.

Proposition 18.3: Characterization of harmonic forms

On a compact manifold, $\Delta\alpha = 0$ if and only if $d\alpha = 0$ and $\delta\alpha = 0$.

18.5 Hodge Decomposition

Theorem 18.1: Hodge Decomposition

Let (M, g) be a compact oriented Riemannian manifold. Then:

$$\boxed{\Omega^k(M) = \mathcal{H}^k(M) \oplus d\Omega^{k-1}(M) \oplus \delta\Omega^{k+1}(M)}$$

where $\mathcal{H}^k(M) = \ker(\Delta)$ is the space of harmonic k-forms. This is an orthogonal direct sum.

Theorem 18.2: Hodge Isomorphism

Every de Rham cohomology class contains a unique harmonic representative:

$$\boxed{H^k_{dR}(M) \cong \mathcal{H}^k(M).}$$

Observation 18.1

This is remarkable: a **topological** invariant (cohomology) equals an **analytic** object (kernel of the Laplacian PDE). The dimension $\dim \mathcal{H}^k = b_k$ (Betti number) is independent of the metric!

Example 18.11: Harmonic forms on S^2

k	$\mathcal{H}^k(S^2)$	Generators
0	$\mathbb{R}$	constant functions
1	0	(none)
2	$\mathbb{R}$	volume form

18.6 The Levi-Civita Connection

The problem. How do you differentiate a vector field on a curved manifold? On $\mathbb{R}^n$, just differentiate components. But on a manifold, the basis vectors ∂_i themselves change from point to point.

The solution. A *connection* ∇ tells us how to compare vectors at different points. Among all connections, there is a unique one compatible with the metric.

Definition 18.9: Connection (covariant derivative)

A *connection* on M is a map $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$, written $(X, Y) \mapsto \nabla_X Y$, satisfying:

1. $\nabla_{fX+Z} Y = f\nabla_X Y + \nabla_Z Y$,
2. $\nabla_X (Y + Z) = \nabla_X Y + \nabla_X Z$,
3. $\nabla_X (fY) = (Xf)Y + f\nabla_X Y$ (Leibniz rule).

Definition 18.10: Christoffel symbols

In coordinates, define $\nabla_{\partial_i} \partial_j = \Gamma^k_{ij} \partial_k$. Then:

$$\nabla_i Y^k = \partial_i Y^k + \Gamma^k_{ij} Y^j.$$

Definition 18.11: Metric compatibility

A connection is *metric compatible* if $\nabla g = 0$, equivalently:

$$X \cdot g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

Definition 18.12: Torsion-free

A connection is *torsion-free* if:

$$\nabla_X Y - \nabla_Y X = [X, Y], \quad \text{i.e., } \Gamma^k_{ij} = \Gamma^k_{ji}.$$

Theorem 18.3: Fundamental Theorem of Riemannian Geometry

On a Riemannian manifold (M, g), there exists a unique connection ∇ that is:

1. metric compatible: $\nabla g = 0$,
2. torsion-free: $\Gamma^k_{ij} = \Gamma^k_{ji}$.

This is called the *Levi-Civita connection*. Its Christoffel symbols are:

$$\boxed{\Gamma^k_{ij} = \frac{1}{2} g^{kl} \left(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij} \right).}$$

Example 18.12: Christoffel symbols on $\mathbb{R}^n$

With $g_{ij} = \delta_{ij}$ (constant): $\Gamma^k_{ij} = 0$. Covariant derivative = ordinary derivative.

Example 18.13: Christoffel symbols on S^2

With $g = d\theta^2 + \sin^2\theta\, d\phi^2$:

$$\Gamma^\theta_{\phi\phi} = -\sin\theta\cos\theta, \quad \Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot\theta, \quad \text{others } = 0.$$

18.7 Parallel Transport

Motivation. Given a curve $\gamma(t)$ and a vector $v_0 \in T_{\gamma(0)}M$, how do we "carry" v_0 along γ to other points?

We want a vector field $V(t) \in T_{\gamma(t)}M$ that doesn't change "relative to the curve" — it has zero covariant derivative along γ.

Definition 18.13: Covariant derivative along a curve

Let $\gamma : [a, b] \to M$ be a curve and V a vector field along γ. The *covariant derivative of V along γ* is:

$$\frac{DV}{dt} := \nabla_{\dot\gamma} V.$$

In coordinates, if $\gamma(t) = (x^1(t), \ldots, x^n(t))$ and $V = V^k \partial_k$:

$$\frac{DV^k}{dt} = \frac{dV^k}{dt} + \Gamma^k_{ij}\frac{dx^i}{dt}V^j.$$

Definition 18.14: Parallel transport

A vector field V along γ is *parallel* (or *parallel transported*) if:

$$\frac{DV}{dt} = 0.$$

Proposition 18.4: Existence and uniqueness

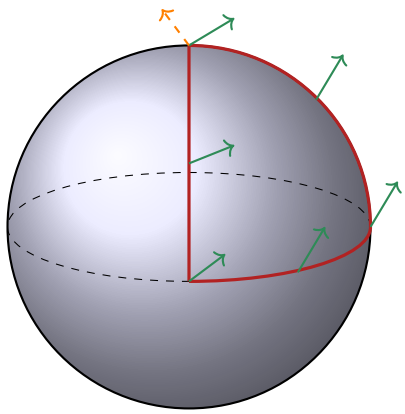
Given any $v_0 \in T_{\gamma(0)}M$, there exists a unique parallel vector field $V(t)$ along γ with $V(0) = v_0$.

Example 18.14: Parallel transport on $\mathbb{R}^n$

$\Gamma^k_{ij} = 0$, so $\frac{DV}{dt} = \frac{dV}{dt}$. Parallel means $\frac{dV}{dt} = 0$, i.e., V is constant. No surprises.

Example 18.15: Parallel transport on S^2

Transport a vector around a closed loop. It comes back **rotated**! The rotation angle equals the solid angle enclosed by the loop (Gauss–Bonnet in disguise).



Parallel transport around a closed loop
returns a **rotated** vector!

Figure 49: On S^2: parallel transporting a vector around a triangle. It comes back rotated — this is curvature!

Observation 18.2

Parallel transport depends on the path, not just the endpoints. The failure to return to the same vector around a closed loop measures **curvature**.

18.8 Geodesics

Motivation. What are the "straight lines" on a curved manifold?

Two equivalent answers:

1. **Locally shortest paths** (variational definition).
2. **Curves that parallel transport their own velocity** (autoparallel).

Definition 18.15: Geodesic

A curve $\gamma : [a, b] \to M$ is a *geodesic* if its velocity is parallel transported along itself:

$$\frac{D\dot{\gamma}}{dt} = \nabla_{\dot{\gamma}}\dot{\gamma} = 0.$$

Proposition 18.5: Geodesic equation

In coordinates, $\gamma(t) = (x^1(t), \ldots, x^n(t))$ is a geodesic iff:

$$\boxed{\frac{d^2x^k}{dt^2} + \Gamma^k_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt} = 0.}$$

This is a system of n second-order ODEs.

Example 18.16: Geodesics on $\mathbb{R}^n$

$\Gamma^k_{ij} = 0$, so $\ddot{x}^k = 0$. Solutions: $x(t) = at + b$ — straight lines!

Example 18.17: Geodesics on S^2

Great circles! These are the intersections of S^2 with planes through the origin. The equator $\theta = \pi/2$, $\phi = t$ is a geodesic. Lines of latitude (except the equator) are *not* geodesics — you have to keep turning to stay on them.

Example 18.18: Geodesics on $\mathbb{H}^2$

Vertical lines $x = \text{const}$ and semicircles centered on the x-axis.

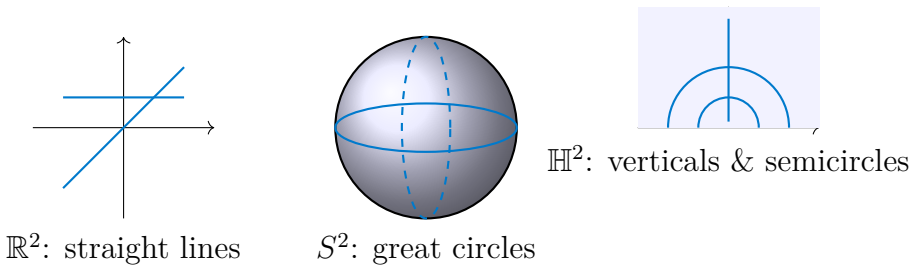


Figure 50: Geodesics on different spaces.

The Exponential Map

Definition 18.16: Exponential map

For $p \in M$ and $v \in T_pM$, let $\gamma_v(t)$ be the geodesic with $\gamma_v(0) = p$ and $\dot{\gamma}_v(0) = v$. Define:

$$\exp_p(v) := \gamma_v(1)$$

when it exists. This is the *exponential map* $\exp_p : T_pM \supset U \to M$.

Remark 18.2

The exponential map "shoots out" from p in direction v for unit time. It's a local diffeomorphism near $0 \in T_pM$.

Example 18.19: Exponential map on $\mathbb{R}^n$

$\exp_p(v) = p + v$ — just translation.

Example 18.20: Exponential map on S^2

$\exp_p(v)$ follows the great circle from p in direction v for distance $|v|$. At distance π, you hit the antipodal point.

18.9 Riemann Curvature

Motivation. Parallel transport around a closed loop doesn't return to the same vector — this is **curvature**.

Equivalently: covariant derivatives don't commute.

$$\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z \neq \nabla_{[X,Y]} Z \quad \text{in general.}$$

Definition 18.17: Riemann curvature tensor

The *Riemann curvature tensor* is the $(1,3)$-tensor:

$$\boxed{R(X,Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.}$$

It measures the failure of covariant derivatives to commute.

Proposition 18.6: Components of Riemann tensor

In coordinates, $R(\partial_i, \partial_j)\partial_k = R^l{}_{kij}\partial_l$, where:

$$\boxed{R^l{}_{kij} = \partial_i \Gamma^l_{jk} - \partial_j \Gamma^l_{ik} + \Gamma^l_{im}\Gamma^m_{jk} - \Gamma^l_{jm}\Gamma^m_{ik}.}$$

Proposition 18.7: Symmetries of Riemann tensor

Let $R_{ijkl} = g_{lm} R^m{}_{kij}$. Then:

1. $R_{ijkl} = -R_{jikl}$ (antisymmetric in first pair),
2. $R_{ijkl} = -R_{ijlk}$ (antisymmetric in second pair),
3. $R_{ijkl} = R_{klij}$ (swap pairs),
4. $R_{ijkl} + R_{iklj} + R_{iljk} = 0$ (first Bianchi identity).

Remark 18.3

These symmetries reduce the number of independent components from n^4 to $\frac{n^2(n^2-1)}{12}$. For $n = 4$ (spacetime): 20 components.

Contractions: Ricci and Scalar

Definition 18.18: Ricci tensor

The *Ricci tensor* is the contraction:

$$R_{ij} := R^k{}_{ikj} = g^{kl} R_{kilj}.$$

It's symmetric: $R_{ij} = R_{ji}$.

Definition 18.19: Scalar curvature

The *scalar curvature* is:

$$R := g^{ij} R_{ij}.$$

A single function measuring "total curvature" at each point.

Observation 18.3

The hierarchy:

Tensor	Components	Information
Riemann $R^l{}_{kij}$	$\frac{n^2(n^2-1)}{12}$	Full curvature
Ricci R_{ij}	$\frac{n(n+1)}{2}$	"Volume distortion"
Scalar R	1	Average curvature

Curvature in 2D: Gaussian Curvature

Definition 18.20: Gaussian curvature

For a 2-dimensional Riemannian manifold:

$$K = \frac{R}{2} = \frac{R_{1212}}{g_{11}g_{22} - g_{12}^2} = \frac{R_{1212}}{\det(g)}.$$

This single number captures all curvature in 2D.

Example 18.21: Gaussian curvature examples

Surface	K	Meaning
Plane	0	Flat
Sphere S^2 (radius r)	$1/r^2$	Positive (elliptic)
Hyperbolic plane $\mathbb{H}^2$	-1	Negative (hyperbolic)
Cylinder	0	Flat! (locally isometric to plane)
Saddle	< 0	Negative

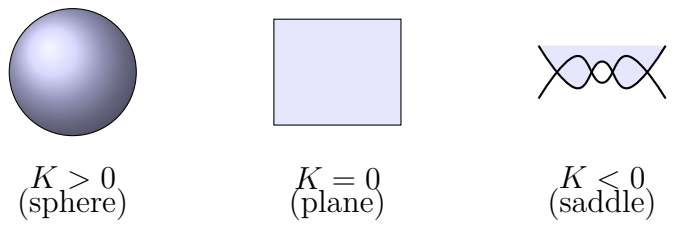


Figure 51: Sign of Gaussian curvature.

18.10 The Gauss–Bonnet Theorem

Motivation. Is there a relationship between curvature (geometry) and topology?

Yes! The total Gaussian curvature of a closed surface is determined by its topology.

Theorem 18.4: Gauss–Bonnet Theorem (2D)

Let (M, g) be a compact oriented 2-dimensional Riemannian manifold without boundary. Then:

$$\boxed{\int_M K\, dA = 2\pi\chi(M),}$$

where K is the Gaussian curvature and $\chi(M) = 2 - 2\mathfrak{g}$ is the Euler characteristic ($\mathfrak{g}$ = genus = number of handles).

Example 18.22: Gauss–Bonnet for S^2

$K = 1/r^2$ (constant), $\text{Area}(S^2) = 4\pi r^2$:

$$\int_{S^2} K\, dA = \frac{1}{r^2} \cdot 4\pi r^2 = 4\pi = 2\pi \cdot 2 = 2\pi\chi(S^2).$$

Check: $\chi(S^2) = 2$ (genus 0). ✓

Example 18.23: Gauss–Bonnet for T^2

Torus has $\chi(T^2) = 0$ (genus 1). So:

$$\int_{T^2} K\, dA = 0.$$

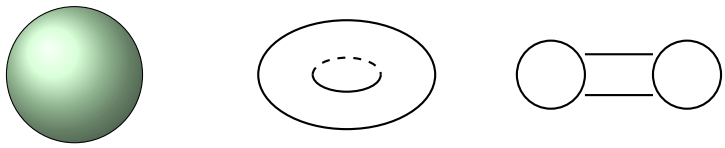
The positive curvature on the outside exactly cancels the negative curvature on the inside!

Example 18.24: Gauss–Bonnet for double torus

$\chi = 2 - 2 \cdot 2 = -2$, so $\int K\, dA = -4\pi$. Must have net negative curvature.

Remark 18.4

The left side depends on the metric. The right side is purely topological. The theorem says: no matter how you bend and stretch the surface (keeping it smooth), the total curvature is fixed by topology!



$\chi = 2$ $\int K = 4\pi$ | $\chi = 0$ $\int K = 0$ | $\chi = -2$ $\int K = -4\pi$

Figure 52: Gauss–Bonnet: total curvature is topological.

Gauss–Bonnet with Boundary

Theorem 18.5: Gauss–Bonnet with boundary

If M has boundary ∂M, then:

$$\int_M K\, dA + \int_{\partial M} \kappa_g\, ds = 2\pi\chi(M),$$

where κ_g is the geodesic curvature of the boundary.

Higher Dimensions: Chern–Gauss–Bonnet

Theorem 18.6: Chern–Gauss–Bonnet (even dimensions)

For a compact oriented $2n$-dimensional Riemannian manifold:

$$\int_M \mathrm{Pf}(\Omega) = (2\pi)^n \chi(M),$$

where $\mathrm{Pf}(\Omega)$ is the Pfaffian of the curvature 2-form.

Remark 18.5

In 4D, this involves the *Euler density*:

$$e_4 = \frac{1}{32\pi^2} \epsilon^{ijkl} R_{ijab} R_{kl}{}^{ab}.$$

18.11 Einstein's Equations

Motivation. In general relativity, gravity is not a force — it's the curvature of spacetime. Matter tells spacetime how to curve; spacetime tells matter how to move.

Setup. Spacetime is a 4-dimensional *Lorentzian* manifold (M, g) with signature $(-, +, +, +)$. Everything we did carries over, with some sign changes.

The Einstein Tensor

Definition 18.21: Einstein tensor

The *Einstein tensor* is:

$$G_{\mu\nu} := R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu}.$$

Proposition 18.8: Bianchi identity for Einstein tensor

The Einstein tensor is divergence-free:

$$\nabla^\mu G_{\mu\nu} = 0.$$

This follows from the contracted Bianchi identity.

Remark 18.6

This is crucial! It means Einstein's equations are compatible with conservation of energy-momentum.

Stress-Energy Tensor

Definition 18.22: Stress-energy tensor

The *stress-energy tensor* $T_{\mu\nu}$ encodes the distribution of matter and energy:

- T_{00}: energy density,
- T_{0i}: momentum density / energy flux,
- T_{ij}: stress (pressure, shear).

It satisfies $\nabla^\mu T_{\mu\nu} = 0$ (conservation of energy-momentum).

Einstein's Field Equations

Theorem 18.7: Einstein's Field Equations

The curvature of spacetime is determined by matter:

$$\boxed{G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}}$$

or equivalently:

$$\boxed{R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.}$$

In units where $G = c = 1$:

$$R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = 8\pi T_{\mu\nu}.$$

Remark 18.7

Both sides are symmetric, divergence-free $(0, 2)$-tensors. The equation is 10 independent PDEs for the 10 components of the metric.

Special Cases

Example 18.25: Vacuum: $T_{\mu\nu} = 0$

Einstein's equations become:

$$R_{\mu\nu} = 0.$$

Ricci-flat, but *not* necessarily flat! The Schwarzschild black hole is a vacuum solution with $R_{\mu\nu} = 0$ but $R^{\alpha}{}_{\beta\gamma\delta} \neq 0$.

Example 18.26: Cosmological constant

Add a term:

$$R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}.$$

$\Lambda > 0$: dark energy, accelerating expansion.

Example 18.27: Weak field / Newtonian limit

For slow motion, weak gravity, $g_{00} \approx -(1 + 2\Phi/c^2)$ where Φ is Newtonian potential. Einstein's equations reduce to:

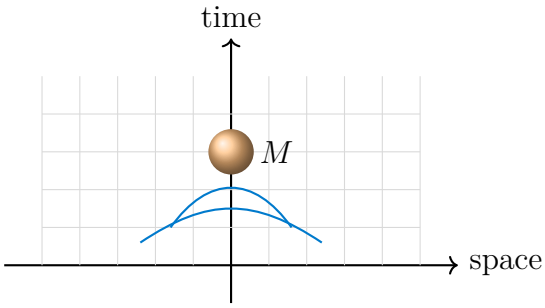
$$\nabla^2 \Phi = 4\pi G \rho,$$

Poisson's equation!

Famous Solutions

Observation 18.4

Solution	Description	Features
Minkowski	Flat spacetime	$R^{\alpha}{}_{\beta\gamma\delta} = 0$
Schwarzschild	Spherical black hole	Event horizon at $r = 2GM/c^2$
Kerr	Rotating black hole	Ergosphere, frame dragging
FLRW	Cosmology	Expanding universe
de Sitter	$\Lambda > 0$, empty	Exponential expansion



Mass curves spacetime

Figure 53: Einstein's insight: gravity is geometry.

Geodesics as Free Fall

Proposition 18.9: Free particles follow geodesics

In general relativity, a freely falling particle (no forces except gravity) follows a geodesic:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0,$$

where τ is proper time.

Remark 18.8

This is the "spacetime tells matter how to move" part. The geodesic equation replaces Newton's $F = ma$ — but now "gravity" is encoded in $\Gamma^\mu_{\alpha\beta}$, not a force.

18.12 Summary

Observation 18.5

The complete picture of Riemannian geometry:

Structure	**What it gives**
Metric g_{ij}	Lengths, angles, volume
Musical isomorphisms $\flat$, $\sharp$	Vectors $\leftrightarrow$ covectors
Hodge star $\star$	k-forms $\leftrightarrow$ $(n-k)$-forms
Laplacian Δ	Harmonic forms, Hodge theory
Levi-Civita ∇	Covariant derivative
Parallel transport	Move vectors along curves
Geodesics	"Straight lines," shortest paths
Riemann tensor $R^l{}_{kij}$	Full curvature
Ricci tensor R_{ij}	Contracted curvature
Scalar curvature R	Average curvature
Gauss–Bonnet	Curvature $\leftrightarrow$ topology
Einstein equations	Curvature = matter (gravity!)

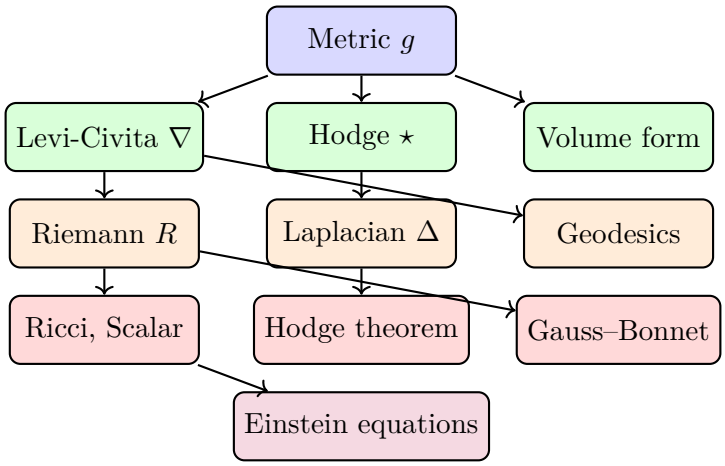


Figure 54: From the metric, everything flows.

19 Symplectic Geometry

19.1 Motivation

What is symplectic geometry? Riemannian geometry is about *lengths* and *angles* — symmetric inner products. Symplectic geometry is about *areas* and *volumes* — anti-symmetric pairings.

Why care? Symplectic geometry is the natural language of **classical mechanics**. Phase space — the space of positions and momenta — carries a symplectic structure, not

a metric.

Key contrast:

	Riemannian	**Symplectic**
Structure	g (symmetric)	ω (antisymmetric)
Measures	Lengths, angles	Areas
Local invariants	Curvature	**None!** (Darboux)
Physics	General relativity	Classical mechanics

19.2 Hamiltonian Formulation of Classical Mechanics

The problem. You know Lagrangian mechanics: start with $L(q, \dot{q}, t)$, get equations of motion from $\delta \int L\, dt = 0$.

But the Lagrangian formulation has drawbacks:

- Second-order ODEs (harder to analyze).
- Symmetries and conservation laws not manifest.
- Doesn't generalize well to field theory and quantization.

The solution. Pass to the **Hamiltonian formulation**: trade $\dot{q}$ for momentum p, get first-order equations on **phase space**.

From Lagrangian to Hamiltonian

Definition 19.1: Conjugate momentum

Given a Lagrangian $L(q, \dot{q}, t)$, the *conjugate momentum* to q^i is:

$$p_i := \frac{\partial L}{\partial \dot{q}^i}.$$

Definition 19.2: Legendre transform

The *Hamiltonian* is defined by the Legendre transform:

$$H(q, p, t) := p_i \dot{q}^i - L(q, \dot{q}, t),$$

where $\dot{q}$ is expressed in terms of (q, p) by inverting $p_i = \frac{\partial L}{\partial \dot{q}^i}$.

Example 19.1: Free particle

$L = \frac{1}{2}m\dot{q}^2$. Then $p = m\dot{q}$, so $\dot{q} = p/m$.

$$H = p \cdot \frac{p}{m} - \frac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{m} - \frac{p^2}{2m} = \frac{p^2}{2m}.$$

The Hamiltonian is kinetic energy (in terms of momentum).

Example 19.2: Harmonic oscillator

$L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2$. Then $p = m\dot{q}$.

$$H = \frac{p^2}{2m} + \frac{1}{2}kq^2.$$

Total energy = kinetic + potential.

Example 19.3: Particle in a potential

$L = \frac{1}{2}m|\dot{\vec{q}}|^2 - V(\vec{q})$. Then $\vec{p} = m\dot{\vec{q}}$.

$$H = \frac{|\vec{p}|^2}{2m} + V(\vec{q}).$$

Hamilton's Equations

Theorem 19.1: Hamilton's equations

The Euler–Lagrange equations $\frac{d}{dt}\frac{\partial L}{\partial \dot{q}^i} = \frac{\partial L}{\partial q^i}$ are equivalent to:

$$\boxed{\dot{q}^i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q^i}.}$$

These are *Hamilton's equations* — $2n$ first-order ODEs.

Remark 19.1

The antisymmetry is striking: q evolves via $\partial H/\partial p$, while p evolves via $-\partial H/\partial q$. This antisymmetry is the heart of symplectic geometry.

Example 19.4: Harmonic oscillator dynamics

$H = \frac{p^2}{2m} + \frac{1}{2}kq^2$.

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{p}{m}, \qquad \dot{p} = -\frac{\partial H}{\partial q} = -kq.$$

Combine: $\ddot{q} = \dot{p}/m = -kq/m$, giving $\ddot{q} + \omega^2 q = 0$ with $\omega = \sqrt{k/m}$.

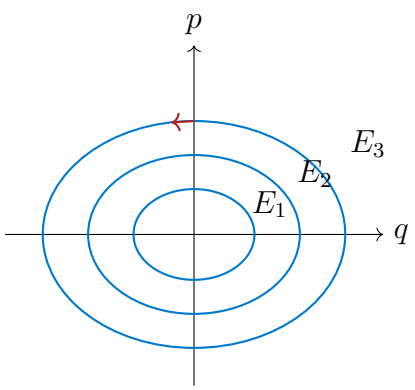
Phase Space

Definition 19.3: Phase space

The *phase space* of a mechanical system with n degrees of freedom is the $2n$-dimensional manifold:

$$\mathcal{M} = \{(q^1, \ldots, q^n, p_1, \ldots, p_n)\} \cong T^*Q,$$

where Q is the configuration space. Points in phase space represent complete states of the system.



Phase portrait of harmonic oscillator

Figure 55: Orbits in phase space are level sets of the Hamiltonian (energy).

Remark 19.2

Why phase space and not configuration space?

- In configuration space Q, you need $(q, \dot{q})$ to specify the state — but $\dot{q}$ is not a point in Q.
- In phase space T^*Q, the state is a single point (q, p) — much cleaner!

19.3 Symplectic Manifolds

The key observation. Hamilton's equations have a hidden structure. Define the matrix:

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Then Hamilton's equations become:

$$\begin{pmatrix} \dot{q} \\ \dot{p} \end{pmatrix} = J \begin{pmatrix} \partial H/\partial q \\ \partial H/\partial p \end{pmatrix} = J \nabla H.$$

The matrix J defines a **symplectic structure**.

Symplectic Vector Spaces

Definition 19.4: Symplectic form on a vector space

A *symplectic form* on a vector space V is a bilinear map $\omega : V \times V \to \mathbb{R}$ that is:

1. **Antisymmetric:** $\omega(u, v) = -\omega(v, u)$.
2. **Nondegenerate:** $\omega(u, v) = 0$ for all v implies $u = 0$.

The pair (V, ω) is a *symplectic vector space*.

Proposition 19.1: Symplectic vector spaces are even-dimensional

If (V, ω) is a symplectic vector space, then $\dim V = 2n$ for some n.

Proof. Nondegeneracy gives an isomorphism $V \to V^*$. But antisymmetry means $\omega(v, v) = 0$ for all v. In odd dimensions, this forces degeneracy. □

Example 19.5: Standard symplectic form on $\mathbb{R}^{2n}$

On $\mathbb{R}^{2n}$ with coordinates $(q^1, \ldots, q^n, p_1, \ldots, p_n)$:

$$\omega_0 = \sum_{i=1}^{n} dq^i \wedge dp_i.$$

In matrix form: $\omega_0(u, v) = u^T J v$ where $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$.

Definition 19.5: Symplectic basis

A basis $\{e_1, \ldots, e_n, f_1, \ldots, f_n\}$ of (V, ω) is *symplectic* (or *canonical*) if:

$$\omega(e_i, e_j) = 0, \quad \omega(f_i, f_j) = 0, \quad \omega(e_i, f_j) = \delta_{ij}.$$

Proposition 19.2: Existence of symplectic basis

Every symplectic vector space admits a symplectic basis.

Symplectic Manifolds

Definition 19.6: Symplectic manifold

A *symplectic manifold* is a smooth manifold M equipped with a 2-form $\omega \in \Omega^2(M)$ that is:

1. **Closed:** $d\omega = 0$.
2. **Nondegenerate:** $\omega_p : T_pM \times T_pM \to \mathbb{R}$ is nondegenerate at each p.

Remark 19.3

Nondegeneracy implies $\dim M = 2n$. Closedness is the "integrability condition" — it ensures the structure is consistent globally.

Example 19.6: Cotangent bundle T^*Q

For any manifold Q, the cotangent bundle T^*Q is symplectic. In local coordinates (q^i, p_i):

$$\omega = \sum_i dq^i \wedge dp_i = -d\theta,$$

where $\theta = \sum_i p_i \, dq^i$ is the *tautological 1-form*.

Example 19.7: $\mathbb{R}^{2n}$ with standard form

$(\mathbb{R}^{2n}, \omega_0)$ with $\omega_0 = \sum_i dq^i \wedge dp_i$ is symplectic.

Example 19.8: S^2 with area form

The sphere S^2 with the area form $\omega = \sin\theta \, d\theta \wedge d\phi$ is symplectic.
Check: $d\omega = \cos\theta \, d\theta \wedge d\theta \wedge d\phi = 0$. ✓
Nondegenerate since it's a volume form on a 2-manifold.

Example 19.9: Kähler manifolds

Any Kähler manifold (complex manifold with compatible Riemannian metric) is symplectic. Examples: $\mathbb{C}^n$, $\mathbb{C}P^n$, complex tori.

Non-Example 19.1: S^4 is not symplectic

S^4 admits no symplectic structure! If it did, $\omega \wedge \omega$ would be a volume form, so $\int_{S^4} \omega \wedge \omega \neq 0$. But $[\omega \wedge \omega] = [\omega]^2 \in H^4(S^4)$, and $[\omega] \in H^2(S^4) = 0$.

The Darboux Theorem

Motivation. In Riemannian geometry, the curvature tensor provides local invariants — you can distinguish S^2 from $\mathbb{R}^2$ locally.

In symplectic geometry, there are **no local invariants**!

Theorem 19.2: Darboux's Theorem

Let (M, ω) be a $2n$-dimensional symplectic manifold. Around any point $p \in M$, there exist local coordinates $(q^1, \ldots, q^n, p_1, \ldots, p_n)$ such that:

$$\boxed{\omega = \sum_{i=1}^{n} dq^i \wedge dp_i.}$$

These are called *Darboux coordinates* or *canonical coordinates*.

Remark 19.4

This is remarkable: **all symplectic manifolds of the same dimension look identical locally**. The only invariant is dimension. All the interesting structure is global.

Observation 19.1

Comparison:

Geometry	**Local normal form**	**Local invariants**
Riemannian	$g = \delta_{ij}$ only at a point	Curvature
Symplectic	$\omega = \sum dq^i \wedge dp_i$ in a neighborhood	**None!**

Symplectic Volume

Definition 19.7: Liouville volume

On a $2n$-dimensional symplectic manifold (M, ω), the *Liouville volume form* is:

$$\mathrm{vol} = \frac{\omega^n}{n!} = \frac{1}{n!} \underbrace{\omega \wedge \cdots \wedge \omega}_{n \text{ times}}.$$

Example 19.10: Volume on $\mathbb{R}^{2n}$

With $\omega = \sum dq^i \wedge dp_i$:

$$\omega^n = n!\, dq^1 \wedge dp_1 \wedge \cdots \wedge dq^n \wedge dp_n,$$

so $\mathrm{vol} = dq^1 \wedge dp_1 \wedge \cdots \wedge dq^n \wedge dp_n$.

19.4 Hamiltonian Systems

Hamiltonian Vector Fields

The idea. A function $H : M \to \mathbb{R}$ on a symplectic manifold generates a vector field X_H via the symplectic form.

Definition 19.8: Hamiltonian vector field

Given $H \in C^\infty(M)$, the *Hamiltonian vector field* X_H is defined by:

$$\boxed{\iota_{X_H}\omega = dH,}$$

or equivalently: $\omega(X_H, Y) = dH(Y) = Y(H)$ for all vector fields Y.

Remark 19.5

This uses nondegeneracy: dH is a 1-form, and ω provides an isomorphism $TM \to T^*M$, so we can find a unique X_H with $\omega(X_H, \cdot) = dH$.

Example 19.11: Hamiltonian vector field in Darboux coordinates

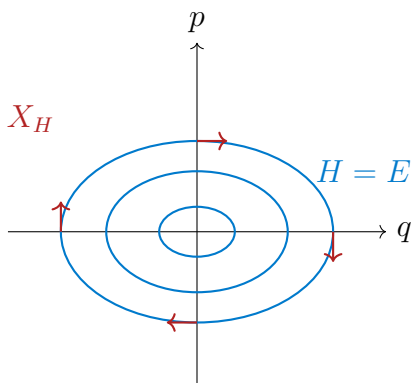
If $\omega = \sum dq^i \wedge dp_i$ and $H = H(q, p)$:

$$X_H = \sum_i \left(\frac{\partial H}{\partial p_i} \frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i} \frac{\partial}{\partial p_i} \right).$$

The flow of X_H is exactly Hamilton's equations!

Definition 19.9: Hamiltonian system

A *Hamiltonian system* is a triple (M, ω, H) where (M, ω) is symplectic and $H \in C^\infty(M)$ is the Hamiltonian. The dynamics is the flow of X_H.



X_H is tangent to level sets of H

Figure 56: Hamiltonian vector field X_H flows along energy surfaces.

Conservation of Energy

Proposition 19.3: Energy is conserved

Along the flow of X_H:

$$\frac{dH}{dt} = X_H(H) = \omega(X_H, X_H) = 0.$$

The Hamiltonian is a constant of motion.

Proof. $X_H(H) = dH(X_H) = \omega(X_H, X_H) = 0$ by antisymmetry. □

Poisson Brackets

Definition 19.10: Poisson bracket

For $f, g \in C^\infty(M)$, the *Poisson bracket* is:

$$\boxed{\{f, g\} := \omega(X_f, X_g) = X_g(f) = -X_f(g).}$$

Proposition 19.4: Poisson bracket in Darboux coordinates

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q^i} \right).$$

Example 19.12: Canonical Poisson brackets

$$\{q^i, q^j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q^i, p_j\} = \delta^i_j.$$

Proposition 19.5: Properties of Poisson brackets

1. **Antisymmetry:** $\{f, g\} = -\{g, f\}$.
2. **Bilinearity:** $\{af + bg, h\} = a\{f, h\} + b\{g, h\}$.
3. **Leibniz:** $\{fg, h\} = f\{g, h\} + g\{f, h\}$.
4. **Jacobi identity:** $\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$.

Remark 19.6

The Jacobi identity is equivalent to $d\omega = 0$. The Poisson bracket makes $C^\infty(M)$ into a *Lie algebra*!

Proposition 19.6: Time evolution via Poisson bracket

For any observable f:

$$\boxed{\frac{df}{dt} = \{f, H\}.}$$

In particular, f is conserved iff $\{f, H\} = 0$.

Example 19.13: Hamilton's equations via Poisson brackets

$$\dot{q}^i = \{q^i, H\} = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = \{p_i, H\} = -\frac{\partial H}{\partial q^i}.$$

Symplectomorphisms

Definition 19.11: Symplectomorphism

A diffeomorphism $\phi : (M_1, \omega_1) \to (M_2, \omega_2)$ is a *symplectomorphism* if:

$$\phi^* \omega_2 = \omega_1.$$

Example 19.14: Hamiltonian flow is a symplectomorphism

The time-t flow ϕ_t of a Hamiltonian vector field X_H satisfies $\phi_t^* \omega = \omega$.

Remark 19.7

In physics, symplectomorphisms are called *canonical transformations*. They preserve the form of Hamilton's equations.

Liouville's Theorem

Theorem 19.3: Liouville's Theorem

Hamiltonian flow preserves the Liouville volume:

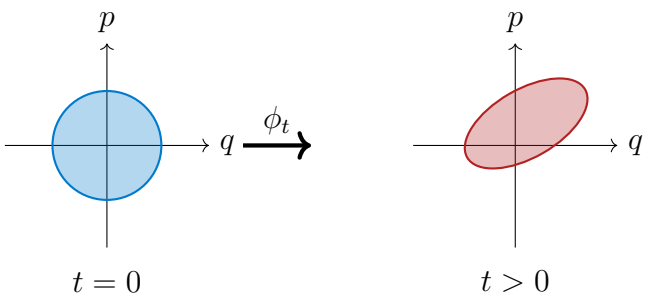
$$\phi_t^* \left(\frac{\omega^n}{n!} \right) = \frac{\omega^n}{n!}.$$

Equivalently: phase space volume is conserved under time evolution.

Proof. Hamiltonian flow preserves ω, hence preserves ω^n. □

Remark 19.8

Physical consequence: if you start with a cloud of initial conditions in phase space, the cloud may stretch and deform, but its volume stays constant. This is foundational for statistical mechanics.



Shape changes, but **area is preserved**

Figure 57: Liouville's theorem: Hamiltonian flow preserves phase space volume.

19.5 Arnold–Liouville Theorem

Motivation. Most Hamiltonian systems are chaotic and cannot be solved exactly. But some special systems — *integrable systems* — have enough conserved quantities that they can be solved completely.

Integrability

Definition 19.12: Poisson-commuting functions

Functions $f_1, \ldots, f_k \in C^\infty(M)$ are *in involution* (or *Poisson-commute*) if:

$$\{f_i, f_j\} = 0 \quad \text{for all } i, j.$$

Definition 19.13: Independent functions

Functions $f_1, \ldots, f_k$ are *independent* on M if $df_1, \ldots, df_k$ are linearly independent almost everywhere.

Definition 19.14: Completely integrable system

A Hamiltonian system (M^{2n}, ω, H) is *completely integrable* if there exist n functions $f_1 = H, f_2, \ldots, f_n$ that are:

1. Independent.
2. In involution: $\{f_i, f_j\} = 0$ for all i, j.

These are called *integrals of motion* or *first integrals*.

Remark 19.9

Why n functions? On a $2n$-dimensional phase space:

- Each conserved quantity constrains motion to a $(2n-1)$-dimensional surface.
- n independent conserved quantities $\Rightarrow$ motion on n-dimensional surface.
- Involution ensures these surfaces are "nice" (Lagrangian).

The Arnold–Liouville Theorem

Theorem 19.4: Arnold–Liouville Theorem

Let (M^{2n}, ω, H) be completely integrable with integrals $f_1 = H, f_2, \ldots, f_n$ in involution. Let

$$M_c := \{p \in M : f_i(p) = c_i,\ i = 1, \ldots, n\}$$

be a level set (assumed compact and connected). Then:

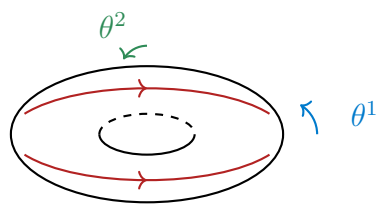
1. M_c is diffeomorphic to the n-torus T^n.
2. There exist *action-angle coordinates* $(I_1, \ldots, I_n, \theta^1, \ldots, \theta^n)$ near M_c such that:

$$\omega = \sum_i dI_i \wedge d\theta^i,$$

   and $f_1, \ldots, f_n$ depend only on $I_1, \ldots, I_n$.
3. The flow is linear on the torus:

$$\dot{I}_i = 0, \qquad \dot{\theta}^i = \nu^i(I),$$

   where $\nu^i = \frac{\partial H}{\partial I_i}$ are the *frequencies*.



Motion is quasi-periodic on torus T^n

Figure 58: Arnold–Liouville: integrable systems have motion on tori.

Observation 19.2

The motion is *quasi-periodic*: each angle θ^i advances at constant rate ν^i. If the frequencies are rationally related, the orbit closes. If not, it densely fills the torus.

Examples of Integrable Systems

Example 19.15: Harmonic oscillator (1D)

$H = \frac{p^2}{2m} + \frac{1}{2}kq^2$. One degree of freedom, one integral (H itself). Level sets are ellipses $\cong S^1 = T^1$. Action-angle: $I = E/\omega$, $\theta =$ phase.

Example 19.16: n uncoupled harmonic oscillators

$H = \sum_{i=1}^n \left(\frac{p_i^2}{2m_i} + \frac{1}{2}k_i q_i^2\right)$. Integrals: $H_i = \frac{p_i^2}{2m_i} + \frac{1}{2}k_i q_i^2$ for each i. All Poisson-commute since they depend on different variables.

Example 19.17: Kepler problem

$H = \frac{|\vec{p}|^2}{2m} - \frac{k}{|\vec{q}|}$ in 3D. Integrals: H, $\vec{L}$ (angular momentum, 3 components but only 2 independent on H-surface), plus Runge–Lenz vector. The system is "super-integrable."

Example 19.18: Geodesic flow on ellipsoid

The geodesic flow on a triaxial ellipsoid in $\mathbb{R}^3$ is integrable (Jacobi).

19.6 Calogero–Moser System

Motivation. The Calogero–Moser system is a beautiful example of an integrable many-body system. It describes n particles on a line with pairwise inverse-square interactions.

Definition

Definition 19.15: Calogero–Moser Hamiltonian

The (rational) *Calogero–Moser system* consists of n particles on $\mathbb{R}$ with positions $q_1, \ldots, q_n$ and momenta $p_1, \ldots, p_n$. The Hamiltonian is:

$$\boxed{H = \frac{1}{2}\sum_{i=1}^n p_i^2 + g^2 \sum_{i<j} \frac{1}{(q_i - q_j)^2},}$$

where $g > 0$ is the coupling constant.

Remark 19.10

The potential $\frac{1}{(q_i-q_j)^2}$ is repulsive and singular when particles collide. For $g > 0$, particles cannot pass through each other — they maintain their ordering.

Example 19.19: $n = 2$ case

Let $Q = q_1 - q_2$ (relative position), $P = (p_1 - p_2)/2$ (relative momentum). The center of mass decouples. The relative motion has:

$$H_{\text{rel}} = P^2 + \frac{g^2}{Q^2}.$$

This is exactly solvable: particles approach, bounce off the $1/Q^2$ barrier, and separate.

Lax Pair and Integrability

The key to integrability. Calogero–Moser has a *Lax pair* representation: matrices L, M such that Hamilton's equations are equivalent to:

$$\frac{dL}{dt} = [M, L] = ML - LM.$$

Definition 19.16: Lax pair for Calogero–Moser

Define $n \times n$ matrices:

$$L_{ij} = \begin{cases} p_i & i = j \\ \dfrac{ig}{q_i - q_j} & i \neq j \end{cases}, \qquad M_{ij} = \begin{cases} ig \displaystyle\sum_{k \neq i} \frac{1}{(q_i - q_k)^2} & i = j \\ \dfrac{-ig}{(q_i - q_j)^2} & i \neq j \end{cases}.$$

Proposition 19.7: Lax equation

The equations of motion $\dot{q}_i = p_i$, $\dot{p}_i = 2g^2 \sum_{j \neq i} \frac{1}{(q_i - q_j)^3}$ are equivalent to:

$$\frac{dL}{dt} = [M, L].$$

Theorem 19.5: Integrals from Lax pair

The eigenvalues of L are conserved. Equivalently, the functions

$$I_k := \frac{1}{k} \operatorname{tr}(L^k), \qquad k = 1, 2, \ldots, n$$

are integrals of motion in involution.

Proof sketch. If $\dot{L} = [M, L]$, then:

$$\frac{d}{dt}\mathrm{tr}(L^k) = k\,\mathrm{tr}(L^{k-1}\dot{L}) = k\,\mathrm{tr}(L^{k-1}[M, L]) = k\,\mathrm{tr}([L^{k-1}, M]L) = 0.$$

The vanishing of Poisson brackets requires more work. □

Example 19.20: First integrals

$$I_1 = \mathrm{tr}(L) = \sum_i p_i = P_{\text{total}} \quad \text{(total momentum)},$$
$$I_2 = \frac{1}{2}\mathrm{tr}(L^2) = \frac{1}{2}\sum_i p_i^2 - g^2 \sum_{i<j} \frac{1}{(q_i - q_j)^2} = H - g^2 \cdot (\text{const}).$$

(Actually $I_2 = H$ up to a constant depending on conventions.)

Scattering and Asymptotic Behavior

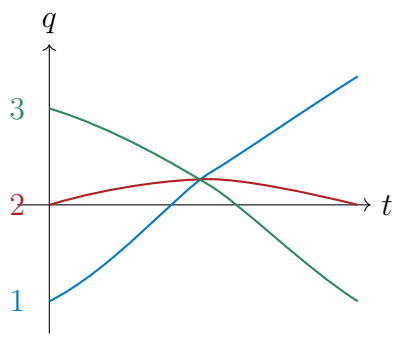
Proposition 19.8: Scattering behavior

As $t \to \pm\infty$:

- Particles separate: $q_i - q_j \to \pm\infty$ for $i \neq j$.
- Momenta approach constant values: $p_i \to p_i^{\pm}$.
- The asymptotic momenta $\{p_1^-, \ldots, p_n^-\}$ and $\{p_1^+, \ldots, p_n^+\}$ are permutations of each other.

Remark 19.11

This is remarkable: the particles interact, exchange momenta, but the *set* of asymptotic momenta is conserved. The scattering is "factorized" — the n-body scattering reduces to pairwise 2-body scattering.



Particles repel but cannot pass through each other

Figure 59: Calogero–Moser scattering: particles exchange momenta.

Variants

Observation 19.3

The Calogero–Moser family includes several variants:

Type	Potential
Rational	$\sum_{i<j} \frac{1}{(q_i - q_j)^2}$
Trigonometric (Sutherland)	$\sum_{i<j} \frac{1}{\sin^2(q_i - q_j)}$
Hyperbolic	$\sum_{i<j} \frac{1}{\sinh^2(q_i - q_j)}$
Elliptic	$\sum_{i<j} \wp(q_i - q_j)$ (Weierstrass $\wp$)

All are completely integrable with Lax pairs!

Remark 19.12

The Calogero–Moser system has deep connections to:

- Representation theory (Cherednik algebras).
- Quantum groups.
- Seiberg–Witten theory.
- Random matrix theory.

19.7 Summary

Observation 19.4

The structure of symplectic geometry:

Concept	Role
Symplectic form ω	Replaces metric; gives phase space structure
Darboux theorem	No local invariants; all symplectic manifolds locally same
Hamiltonian H	Energy; generates time evolution
X_H	Hamiltonian vector field; flow = dynamics
$\{f, g\}$	Poisson bracket; Lie algebra of observables
Symplectomorphism	Canonical transformation; preserves ω
Liouville theorem	Phase space volume conserved
Integrability	n integrals in involution $\Rightarrow$ solvable
Arnold–Liouville	Integrable $\Rightarrow$ motion on tori
Lax pair	$\dot{L} = [M, L] \Rightarrow$ spectral integrals

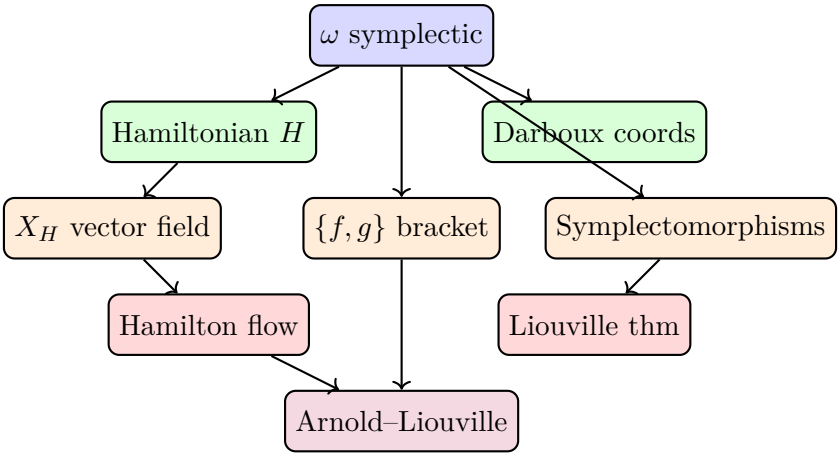


Figure 60: From symplectic structure to integrable systems.

20 Complex Geometry

20.1 Motivation

What is complex geometry? Complex geometry studies manifolds with a notion of "multiplication by i" on tangent spaces. This seemingly simple structure has profound consequences.

The grand unification. We've seen three types of geometry:

Geometry	**Structure**	**Measures**
Riemannian	Metric g	Lengths, angles
Symplectic	2-form ω	Areas
Complex	J with $J^2 = -1$	"Rotation by 90°"

Kähler geometry unifies all three: g, ω, and J compatible. This is the natural setting for much of modern physics.

Physics connections.

- Kähler manifolds: supersymmetric sigma models, moduli spaces
- Calabi–Yau manifolds: string compactifications, mirror symmetry
- Chern classes: anomalies, topological field theory

20.2 Almost Complex Structures

Motivation. On $\mathbb{C}^n$, multiplication by i rotates tangent vectors by 90ř. We want to abstract this to manifolds.

Definition 20.1: Almost complex structure

An *almost complex structure* on a smooth manifold M is a smooth bundle map $J : TM \to TM$ such that:

$$J^2 = -\mathrm{id}.$$

That is, $J_p : T_pM \to T_pM$ satisfies $J_p^2 = -\mathrm{id}$ for all $p \in M$.

Proposition 20.1: Dimension must be even

If M admits an almost complex structure, then $\dim M = 2n$ is even.

Proof. $J^2 = -\mathrm{id}$ has eigenvalues $\pm i$. Over $\mathbb{R}$, such a linear map exists only in even dimensions. □

Example 20.1: $\mathbb{C}^n$ with standard J

Identify $\mathbb{C}^n \cong \mathbb{R}^{2n}$ via $z^j = x^j + iy^j$. Define:

$$J\left(\frac{\partial}{\partial x^j}\right) = \frac{\partial}{\partial y^j}, \qquad J\left(\frac{\partial}{\partial y^j}\right) = -\frac{\partial}{\partial x^j}.$$

This is "multiplication by i" on tangent vectors.

Example 20.2: S^2 is almost complex

$S^2 = \mathbb{C}P^1$ inherits a complex structure from $\mathbb{C}$. At each point, J rotates tangent vectors by 90ř (using the round metric to define "90ř").

Non-Example 20.1: S^4 is not almost complex

S^4 does not admit an almost complex structure. This is a topological obstruction (related to characteristic classes).

Non-Example 20.2: S^6 is almost complex but...

S^6 admits an almost complex structure (from octonions), but it is *not integrable* — S^6 is not a complex manifold. (This is the famous "S^6 problem.")

Remark 20.1

Which spheres are almost complex? Only S^2 and S^6. Which are actually complex manifolds? Only $S^2 = \mathbb{C}P^1$.

Complexified Tangent Bundle

Idea. To work with J, it's useful to complexify and diagonalize.

Definition 20.2: Complexified tangent bundle

The *complexified tangent bundle* is:

$$TM \otimes \mathbb{C} = TM \oplus iTM.$$

Sections are complex vector fields $X + iY$ where $X, Y \in \mathfrak{X}(M)$.

Definition 20.3: $(1,0)$ and $(0,1)$ vectors

The almost complex structure J extends $\mathbb{C}$-linearly to $TM \otimes \mathbb{C}$. Its eigenspaces are:

$$\begin{aligned} T^{1,0}M &= \{v \in TM \otimes \mathbb{C} : Jv = iv\} \quad \text{(eigenvalue } +i\text{)}, \\ T^{0,1}M &= \{v \in TM \otimes \mathbb{C} : Jv = -iv\} \quad \text{(eigenvalue } -i\text{)}. \end{aligned}$$

We have $TM \otimes \mathbb{C} = T^{1,0}M \oplus T^{0,1}M$.

Example 20.3: $(1,0)$ and $(0,1)$ vectors on $\mathbb{C}^n$

Define:

$$\frac{\partial}{\partial z^j} := \frac{1}{2}\left(\frac{\partial}{\partial x^j} - i\frac{\partial}{\partial y^j}\right), \qquad \frac{\partial}{\partial \bar{z}^j} := \frac{1}{2}\left(\frac{\partial}{\partial x^j} + i\frac{\partial}{\partial y^j}\right).$$

Then:

$$T^{1,0}\mathbb{C}^n = \operatorname{span}_{\mathbb{C}}\left\{\frac{\partial}{\partial z^j}\right\}, \qquad T^{0,1}\mathbb{C}^n = \operatorname{span}_{\mathbb{C}}\left\{\frac{\partial}{\partial \bar{z}^j}\right\}.$$

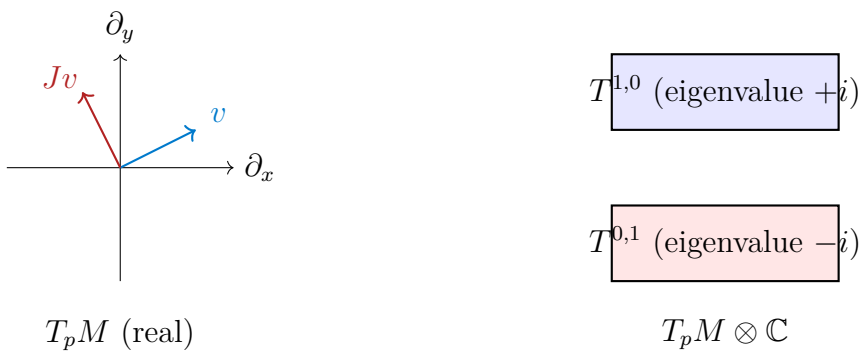


Figure 61: J acts as rotation on real tangent space; diagonalizes over $\mathbb{C}$.

20.3 Complex Manifolds

Motivation. An almost complex structure is "pointwise" data. When does it come from actual complex coordinates?

Definition 20.4: Complex manifold

A *complex manifold* of complex dimension n is a smooth manifold M with an atlas $\{(U_\alpha, \phi_\alpha)\}$ where:

1. Each $\phi_\alpha : U_\alpha \to \mathbb{C}^n$ is a homeomorphism onto an open set.
2. Transition functions $\phi_\beta \circ \phi_\alpha^{-1}$ are *holomorphic*.

Remark 20.2

A complex manifold has real dimension $2n$ and a natural almost complex structure J induced by the complex charts.

Example 20.4: $\mathbb{C}^n$

The model complex manifold. One chart, identity map.

Example 20.5: Complex projective space $\mathbb{C}P^n$

$$\mathbb{C}P^n = (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^* = \{[z_0 : \cdots : z_n]\}.$$

Charts: $U_j = \{z_j \neq 0\}$ with coordinates $(z_0/z_j, \ldots, \widehat{z_j/z_j}, \ldots, z_n/z_j)$.
Transition functions are rational, hence holomorphic where defined.

Example 20.6: Riemann surfaces

Complex manifolds of dimension 1: $\mathbb{C}$, $\mathbb{C}P^1 = S^2$, tori $\mathbb{C}/\Lambda$, higher genus surfaces.

Example 20.7: Complex tori

$T^{2n}_{\mathbb{C}} = \mathbb{C}^n/\Lambda$ where $\Lambda \cong \mathbb{Z}^{2n}$ is a lattice. These are compact complex manifolds.

Integrability: Newlander–Nirenberg

The question. Given an almost complex structure J, when does it come from a complex atlas?

Definition 20.5: Nijenhuis tensor

The *Nijenhuis tensor* of an almost complex structure J is:

$$N_J(X, Y) := [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y].$$

Definition 20.6: Integrable almost complex structure

An almost complex structure J is *integrable* if $N_J = 0$.

Theorem 20.1: Newlander–Nirenberg Theorem

An almost complex structure J on M comes from a complex atlas (i.e., M is a complex manifold with induced J) if and only if J is integrable:

$$N_J = 0.$$

Remark 20.3

In real dimension 2, every almost complex structure is automatically integrable (no room for N_J to be nonzero). This is why every oriented surface is a Riemann surface.

Example 20.8: S^6 revisited

S^6 has an almost complex structure from the octonions, but $N_J \neq 0$. So S^6 is not a complex manifold.

20.4 Hermitian Manifolds

Motivation. Combine complex structure J with a Riemannian metric g in a compatible way.

Definition 20.7: Hermitian metric

Let (M, J) be an almost complex manifold. A Riemannian metric g is *Hermitian* (or *J-compatible*) if:

$$g(JX, JY) = g(X, Y) \quad \text{for all } X, Y \in TM.$$

That is, J is an isometry.

Definition 20.8: Hermitian manifold

A *Hermitian manifold* is a complex manifold (M, J) with a Hermitian metric g.

Proposition 20.2: Existence of Hermitian metrics

Every almost complex manifold admits a Hermitian metric.

Proof. Start with any metric $\tilde{g}$. Define $g(X, Y) = \tilde{g}(X, Y) + \tilde{g}(JX, JY)$. Then $g(JX, JY) = g(X, Y)$. $\square$

The Fundamental 2-Form

Definition 20.9: Fundamental 2-form (Kähler form)

Given a Hermitian manifold (M, J, g), the *fundamental 2-form* is:

$$\omega(X, Y) := g(JX, Y).$$

Proposition 20.3: Properties of ω

1. ω is antisymmetric: $\omega(X, Y) = -\omega(Y, X)$.
2. ω is J-invariant: $\omega(JX, JY) = \omega(X, Y)$.
3. ω is nondegenerate (since g is).

Proof of (1). $\omega(Y, X) = g(JY, X) = g(J^2Y, JX) = g(-Y, JX) = -g(JX, Y) = -\omega(X, Y)$. □

Remark 20.4

The fundamental 2-form ω is always nondegenerate. But it need not be closed! When $d\omega = 0$, we get something special.

20.5 Kähler Manifolds

The magic. When ω is also closed, the three structures — complex, Riemannian, symplectic — are perfectly compatible.

Definition 20.10: Kähler manifold

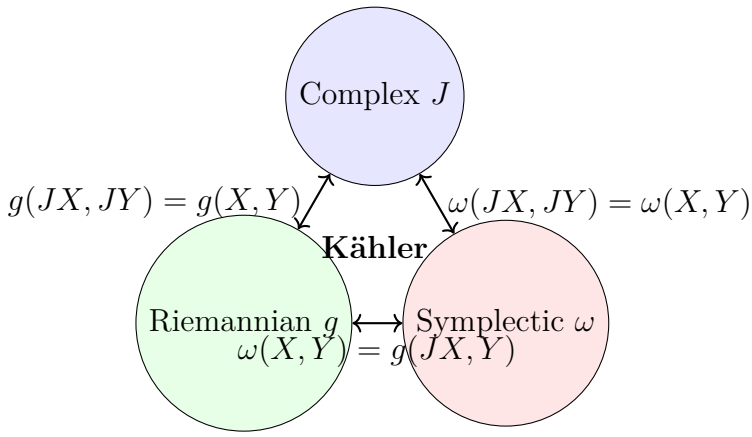
A *Kähler manifold* is a Hermitian manifold (M, J, g) such that the fundamental 2-form ω is closed:

$$d\omega = 0.$$

Equivalently: ω is a symplectic form compatible with J and g.

Observation 20.1

The Kähler condition intertwines three geometries:



Theorem 20.2: Equivalent characterizations of Kähler

For a Hermitian manifold (M, J, g, ω), the following are equivalent:

1. $d\omega = 0$.
2. $\nabla J = 0$ (where ∇ is the Levi-Civita connection).
3. Locally, $\omega = i\partial\bar{\partial}\phi$ for some real function ϕ (Kähler potential).
4. Parallel transport preserves J.

Examples

Example 20.9: $\mathbb{C}^n$ with flat metric

$g = \sum_j (dx^j)^2 + (dy^j)^2$, $\omega = \sum_j dx^j \wedge dy^j = \frac{i}{2}\sum_j dz^j \wedge d\bar{z}^j$.
Kähler potential: $\phi = \sum_j |z^j|^2$.

Example 20.10: $\mathbb{C}P^n$ with Fubini–Study metric

The *Fubini–Study metric* makes $\mathbb{C}P^n$ Kähler. In homogeneous coordinates $[z_0 : \cdots : z_n]$:

$$\omega_{FS} = \frac{i}{2\pi}\partial\bar{\partial}\log|z|^2.$$

This is the unique (up to scale) $U(n+1)$-invariant Kähler metric.

Example 20.11: Complex submanifolds of Kähler

Any complex submanifold of a Kähler manifold is Kähler (restrict g, ω, J).
In particular: smooth projective varieties (submanifolds of $\mathbb{C}P^n$ cut out by homogeneous polynomials) are Kähler.

Example 20.12: Complex tori

$T^{2n}_{\mathbb{C}} = \mathbb{C}^n/\Lambda$ with the flat metric inherited from $\mathbb{C}^n$ is Kähler.

Non-Example 20.3: Hopf surface

The Hopf surface $S^3 \times S^1 \cong (\mathbb{C}^2 \setminus \{0\})/\mathbb{Z}$ is a complex manifold but **not** Kähler. Why? $H^2(S^3 \times S^1) = 0$, so no nonzero closed 2-form exists. But Kähler requires $\omega \neq 0$ with $d\omega = 0$. Contradiction.

Kähler Potential

Definition 20.11: Kähler potential

A *Kähler potential* is a real-valued function ϕ such that locally:

$$\omega = i\partial\bar{\partial}\phi = \frac{i}{2}\frac{\partial^2\phi}{\partial z^j \partial \bar{z}^k} dz^j \wedge d\bar{z}^k.$$

Remark 20.5

The metric can be recovered: $g_{j\bar{k}} = \frac{\partial^2\phi}{\partial z^j \partial \bar{z}^k}$. So the entire Kähler geometry is encoded in one function!

Example 20.13: Kähler potential for $\mathbb{C}^n$

$\phi = |z|^2 = \sum_j z^j \bar{z}^j$. Then:

$$i\partial\bar{\partial}\phi = i\sum_j dz^j \wedge d\bar{z}^j = \omega_{\text{flat}}.$$

20.6 Dolbeault Cohomology

Motivation. On a complex manifold, differential forms split by type. This leads to refined cohomology theories.

(p, q)-Forms

Definition 20.12: (p, q)-form

On a complex manifold M, a (p, q)-*form* is a section of $\Lambda^{p,q}T^*M := \Lambda^p(T^{1,0}M)^* \otimes \Lambda^q(T^{0,1}M)^*$.
In local coordinates $(z^1, \ldots, z^n)$:

$$\alpha = \sum_{|I|=p, |J|=q} \alpha_{I\bar{J}}\, dz^I \wedge d\bar{z}^J,$$

where $dz^I = dz^{i_1} \wedge \cdots \wedge dz^{i_p}$ and $d\bar{z}^J = d\bar{z}^{j_1} \wedge \cdots \wedge d\bar{z}^{j_q}$.

Example 20.14: Types of forms on $\mathbb{C}$

- $(0,0)$: functions f
- $(1,0)$: $f\,dz$
- $(0,1)$: $f\,d\bar{z}$
- $(1,1)$: $f\,dz \wedge d\bar{z}$

Proposition 20.4: Decomposition of forms

On a complex n-manifold:

$$\Omega^k(M;\mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}(M).$$

The ∂ and $\bar{\partial}$ Operators

Definition 20.13: Dolbeault operators

The exterior derivative d splits as:

$$d = \partial + \bar{\partial},$$

where:

$$\partial : \Omega^{p,q} \to \Omega^{p+1,q},$$
$$\bar{\partial} : \Omega^{p,q} \to \Omega^{p,q+1}.$$

Example 20.15: ∂ and $\bar{\partial}$ on $\mathbb{C}$

For $f : \mathbb{C} \to \mathbb{C}$:

$$df = \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z} = \partial f + \bar{\partial} f.$$

where $\frac{\partial}{\partial z} = \frac{1}{2}(\partial_x - i\partial_y)$ and $\frac{\partial}{\partial \bar{z}} = \frac{1}{2}(\partial_x + i\partial_y)$.

Proposition 20.5: Properties of ∂ and $\bar{\partial}$

1. $\partial^2 = 0$, $\bar{\partial}^2 = 0$.
2. $\partial\bar{\partial} + \bar{\partial}\partial = 0$.
3. $d^2 = 0$ follows from (1) and (2).

Definition 20.14: Holomorphic function

f is *holomorphic* if $\bar{\partial} f = 0$, i.e., $\frac{\partial f}{\partial \bar{z}^j} = 0$ for all j.

Dolbeault Cohomology

Definition 20.15: Dolbeault cohomology

The *Dolbeault cohomology groups* are:

$$H^{p,q}_{\bar{\partial}}(M) := \frac{\ker(\bar{\partial} : \Omega^{p,q} \to \Omega^{p,q+1})}{\operatorname{im}(\bar{\partial} : \Omega^{p,q-1} \to \Omega^{p,q})}.$$

Remark 20.6

This is analogous to de Rham cohomology, but using $\bar{\partial}$ instead of d.

Definition 20.16: Hodge numbers

The *Hodge numbers* of a compact complex manifold are:

$$h^{p,q} := \dim H^{p,q}_{\bar{\partial}}(M).$$

Theorem 20.3: Dolbeault theorem

For a compact complex manifold:

$$H^{p,q}_{\bar{\partial}}(M) \cong H^q(M, \Omega^p),$$

where Ω^p is the sheaf of holomorphic p-forms.

Hodge Decomposition for Kähler

Theorem 20.4: Hodge decomposition (Kähler case)

On a compact Kähler manifold:

$$H^k(M; \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}_{\bar{\partial}}(M).$$

Moreover:

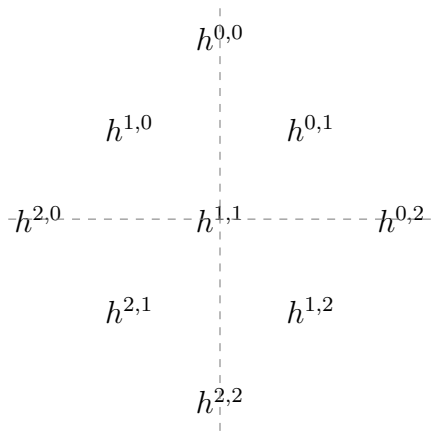
1. $H^{p,q} \cong \overline{H^{q,p}}$ (complex conjugation).
2. $h^{p,q} = h^{q,p}$.
3. $b_k = \sum_{p+q=k} h^{p,q}$ (Betti numbers).

Example 20.16: Hodge diamond of $\mathbb{C}P^2$

For $\mathbb{C}P^2$ $(n = 2)$:

$$\begin{array}{ccccc} & & h^{0,0} & & \\ & h^{1,0} & & h^{0,1} & \\ h^{2,0} & & h^{1,1} & & h^{0,2} \\ & h^{2,1} & & h^{1,2} & \\ & & h^{2,2} & & \end{array} = \begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 0 & & 1 & & 0 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

Betti numbers: $b_0 = 1$, $b_1 = 0$, $b_2 = 1$, $b_3 = 0$, $b_4 = 1$.



Hodge diamond (symmetric about both axes for Kähler)

Figure 62: The Hodge diamond encodes Hodge numbers $h^{p,q}$.

20.7 Chern Classes

Motivation. Chern classes are topological invariants of complex vector bundles. They measure "twisting" and obstruct triviality.

First Chern Class

Definition 20.17: First Chern class (line bundle)

Let $L \to M$ be a complex line bundle with connection ∇ and curvature F. The *first Chern class* is:

$$c_1(L) = \left[\frac{i}{2\pi} F\right] \in H^2(M; \mathbb{Z}).$$

Proposition 20.6: c_1 is independent of connection

Different connections give cohomologous curvature forms, so $c_1(L)$ is well-defined.

Example 20.17: Tautological bundle over $\mathbb{C}P^1$

The tautological line bundle $\mathcal{O}(-1) \to \mathbb{C}P^1$ has $c_1 = -1$ (generates $H^2(\mathbb{C}P^1; \mathbb{Z}) \cong \mathbb{Z}$).

Example 20.18: Hyperplane bundle over $\mathbb{C}P^n$

The hyperplane bundle $\mathcal{O}(1) \to \mathbb{C}P^n$ has $c_1 = 1$.

Total Chern Class

Definition 20.18: Total Chern class

For a rank-r complex vector bundle $E \to M$, the *total Chern class* is:

$$c(E) = 1 + c_1(E) + c_2(E) + \cdots + c_r(E) \in H^*(M; \mathbb{Z}),$$

where $c_k(E) \in H^{2k}(M; \mathbb{Z})$.

Proposition 20.7: Chern–Weil theory

With curvature F (a $\mathfrak{u}(r)$-valued 2-form):

$$c(E) = \det\left(I + \frac{iF}{2\pi}\right).$$

Expanding: $c_k(E) = \frac{1}{(2\pi i)^k}\sigma_k(F)$, where σ_k is the k-th symmetric polynomial in eigenvalues of F.

Proposition 20.8: Properties of Chern classes

1. **Naturality:** $c(f^*E) = f^*c(E)$.
2. **Whitney sum:** $c(E \oplus F) = c(E) \smile c(F)$.
3. **Normalization:** $c_1(\mathcal{O}(1)) =$ generator of $H^2(\mathbb{C}P^n; \mathbb{Z})$.

First Chern Class of a Manifold

Definition 20.19: First Chern class of M

For a complex manifold M, define:

$$c_1(M) := c_1(TM) = -c_1(K_M),$$

where $K_M = \Lambda^n T^*M$ is the *canonical bundle*.

Example 20.19: $c_1(\mathbb{C}P^n)$

$c_1(\mathbb{C}P^n) = (n+1)[\omega_{FS}]$, where $[\omega_{FS}]$ is the class of the Fubini–Study form.

Remark 20.7

$c_1(M) = 0$ means the canonical bundle is (topologically) trivial. This is the first condition for Calabi–Yau!

20.8 Calabi–Yau Manifolds

Motivation. Calabi–Yau manifolds are Kähler manifolds with vanishing Ricci curvature. They're central to string theory: extra dimensions are often compactified on Calabi–Yau 3-folds.

Definition 20.20: Calabi–Yau manifold

A *Calabi–Yau manifold* is a compact Kähler manifold (M, g, ω, J) satisfying:

1. $c_1(M) = 0$ (trivial canonical bundle).
2. $\mathrm{Ric}(g) = 0$ (Ricci-flat).

Remark 20.8

Condition (1) is topological, condition (2) is geometric. The Calabi conjecture (proved by Yau) says: given (1), there exists a unique Kähler metric satisfying (2) in each Kähler class.

Theorem 20.5: Calabi–Yau theorem (Yau)

Let (M, ω) be a compact Kähler manifold with $c_1(M) = 0$. Then there exists a unique Ricci-flat Kähler metric in the cohomology class $[\omega]$.

Equivalent Characterizations

Proposition 20.9: Equivalent conditions for Calabi–Yau

For a compact Kähler manifold M of dimension n, the following are equivalent:

1. $c_1(M) = 0$.
2. K_M is trivial (there exists a nowhere-vanishing holomorphic n-form Ω).
3. M admits a Ricci-flat Kähler metric.
4. Holonomy $\mathrm{Hol}(g) \subseteq SU(n)$.

Remark 20.9

The holomorphic volume form $\Omega \in H^0(M, K_M)$ is a key player. It's used in mirror symmetry and defines special Lagrangian submanifolds.

Examples

Example 20.20: Complex tori

$T^{2n} = \mathbb{C}^n/\Lambda$ is Calabi–Yau. The flat metric is Ricci-flat. $\Omega = dz^1 \wedge \cdots \wedge dz^n$. Holonomy: trivial (flat!), so $\mathrm{Hol} = \{1\} \subset SU(n)$.

Example 20.21: K3 surfaces

A *K3 surface* is a simply connected compact complex surface with $c_1 = 0$. Examples:

- Quartic hypersurface in $\mathbb{C}P^3$: $\{z_0^4 + z_1^4 + z_2^4 + z_3^4 = 0\}$.
- Kummer surface: resolution of $T^4/\mathbb{Z}_2$.

Hodge numbers: $h^{0,0} = h^{2,0} = h^{0,2} = h^{2,2} = 1$, $h^{1,1} = 20$.

Example 20.22: Quintic threefold

The hypersurface in $\mathbb{C}P^4$:

$$M = \{z_0^5 + z_1^5 + z_2^5 + z_3^5 + z_4^5 = 0\} \subset \mathbb{C}P^4.$$

This is a Calabi–Yau 3-fold (complex dimension 3, real dimension 6).
Hodge numbers: $h^{1,1} = 1$, $h^{2,1} = 101$.

$$\begin{tikzcd}
\boxed{\text{Kähler}} \arrow[d, "+\text{ topological condition}"] & \mathbb{C}P^n\text{, projective varieties} \\
\boxed{c_1 = 0} \arrow[d, "+\text{ Ricci-flat metric (Yau)}"] & \text{quintic in } \mathbb{C}P^4\text{, K3} \\
\boxed{\text{Calabi–Yau}} & \text{same, with Ricci-flat metric}
\end{tikzcd}$$

Figure 63: Hierarchy: Kähler $\supset c_1 = 0 \supset$ Calabi–Yau.

Physics: String Compactification

Observation 20.2

In string theory, spacetime is 10-dimensional. To get 4D physics:

$$M^{10} = M^4 \times X^6,$$

where X^6 is a compact 6-manifold.
For $\mathcal{N} = 1$ supersymmetry in 4D, X^6 must be Calabi–Yau (complex dimension 3).

Observation 20.3

The Hodge numbers of X determine physics:

- $h^{1,1}$: number of Kähler moduli (sizes of 2-cycles)
- $h^{2,1}$: number of complex structure moduli (shapes)
- Total moduli: $h^{1,1} + h^{2,1}$ massless scalars in 4D

For the quintic: $1 + 101 = 102$ moduli.

Mirror Symmetry

Observation 20.4

Mirror symmetry is a duality between pairs of Calabi–Yau manifolds $(X, \tilde{X})$ where:

$$h^{1,1}(X) = h^{2,1}(\tilde{X}), \qquad h^{2,1}(X) = h^{1,1}(\tilde{X}).$$

The Hodge diamond is reflected!
This has profound consequences:

- Hard enumerative geometry on $X \leftrightarrow$ easy classical geometry on $\tilde{X}$.
- Counting rational curves on the quintic: famously solved via mirror symmetry.

20.9 Summary

Observation 20.5

The hierarchy of complex geometry:

Structure	Definition	Example
Almost complex	$J^2 = -1$	S^6
Complex manifold	J integrable	$\mathbb{C}P^n$, Riemann surfaces
Hermitian	+ compatible metric	Any complex manifold
Kähler	$d\omega = 0$	$\mathbb{C}P^n$, smooth projective varieties
Calabi–Yau	$c_1 = 0$, Ricci-flat	K3, quintic 3-fold

Observation 20.6

Key tools and results:

Tool	Purpose
∂, $\bar{\partial}$ operators	Split d by type
Dolbeault cohomology $H^{p,q}$	Refined cohomology
Hodge numbers $h^{p,q}$	Numerical invariants
Chern classes $c_k(E)$	Topological invariants of bundles
Calabi–Yau theorem	$c_1 = 0 \Rightarrow$ Ricci-flat metric exists
Mirror symmetry	Duality swapping $h^{1,1} \leftrightarrow h^{2,1}$

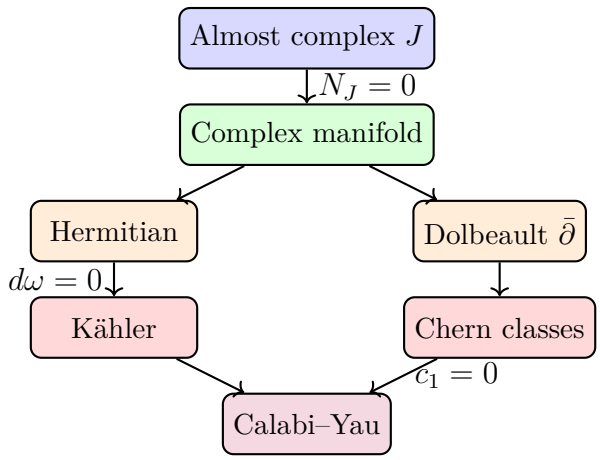


Figure 64: The landscape of complex geometry.

21 Characteristic Classes

21.1 Motivation

The question. Given a vector bundle $E \to M$, how can we detect whether it's trivial? And if not, how "twisted" is it?

The answer. *Characteristic classes* are cohomology classes associated to bundles that measure their non-triviality. If a characteristic class is nonzero, the bundle is nontrivial.

The hierarchy.

Class	Bundle type	Lives in
Chern classes c_k	Complex	$H^{2k}(M;\mathbb{Z})$
Pontryagin classes p_k	Real	$H^{4k}(M;\mathbb{Z})$
Euler class e	Oriented real	$H^n(M;\mathbb{Z})$
Stiefel–Whitney w_k	Real	$H^k(M;\mathbb{Z}_2)$

21.2 Chern Classes (Review and Extension)

Definition 21.1: Chern classes via Chern–Weil

Let $E \to M$ be a complex vector bundle of rank r with connection ∇ and curvature $F \in \Omega^2(M;\mathrm{End}(E))$. The *total Chern class* is:

$$c(E) = \det\left(I + \frac{iF}{2\pi}\right) = 1 + c_1(E) + c_2(E) + \cdots + c_r(E),$$

where $c_k(E) \in H^{2k}(M;\mathbb{Z})$.

Proposition 21.1: Explicit formulas

$$\begin{aligned} c_1(E) &= \frac{i}{2\pi}\mathrm{tr}(F), \\ c_2(E) &= \frac{1}{8\pi^2}\left[\mathrm{tr}(F\wedge F) - \mathrm{tr}(F)\wedge \mathrm{tr}(F)\right]. \end{aligned}$$

For a line bundle: $c(L) = 1 + c_1(L)$.

Proposition 21.2: Axiomatic characterization

Chern classes are uniquely determined by:

1. **Naturality:** $c(f^*E) = f^*c(E)$.
2. **Whitney sum:** $c(E \oplus F) = c(E) \smile c(F)$.
3. **Normalization:** $c_1(\mathcal{O}(1)) =$ generator of $H^2(\mathbb{C}P^n)$.
4. **Dimension:** $c_k(E) = 0$ for $k > \mathrm{rank}(E)$.

Example 21.1: Chern classes of $\mathbb{C}P^n$

The tangent bundle $T\mathbb{C}P^n$ has:

$$c(T\mathbb{C}P^n) = (1+h)^{n+1},$$

where $h \in H^2(\mathbb{C}P^n)$ is the hyperplane class. Thus:

$$c_1 = (n+1)h, \quad c_2 = \binom{n+1}{2} h^2, \quad \text{etc.}$$

21.3 Pontryagin Classes

Motivation. For *real* vector bundles, Chern classes don't directly apply. But we can complexify!

Definition 21.2: Complexification

For a real vector bundle $E \to M$, the *complexification* is:

$$E_{\mathbb{C}} := E \otimes_{\mathbb{R}} \mathbb{C}.$$

This is a complex vector bundle of the same rank.

Definition 21.3: Pontryagin classes

The *k-th Pontryagin class* of a real vector bundle E is:

$$p_k(E) := (-1)^k c_{2k}(E_{\mathbb{C}}) \in H^{4k}(M; \mathbb{Z}).$$

The *total Pontryagin class* is:

$$p(E) = 1 + p_1(E) + p_2(E) + \cdots$$

Remark 21.1

Why c_{2k} and not c_{2k+1}? Because $\overline{E_{\mathbb{C}}} \cong E_{\mathbb{C}}$, so $c_k(E_{\mathbb{C}}) = (-1)^k c_k(\overline{E_{\mathbb{C}}}) = (-1)^k \overline{c_k(E_{\mathbb{C}})}$. The odd Chern classes are 2-torsion (and vanish in $H^*(M; \mathbb{R})$).

Proposition 21.3: Chern–Weil for Pontryagin

Using a connection with curvature F:

$$p(E) = \det\left(I + \frac{F}{2\pi}\right)^{1/2} \det\left(I - \frac{F}{2\pi}\right)^{1/2} = \det\left(I - \frac{F^2}{4\pi^2}\right)^{1/2}.$$

Explicitly:

$$p_1(E) = -\frac{1}{8\pi^2}\mathrm{tr}(F \wedge F).$$

Example 21.2: Pontryagin class of S^4

For the tangent bundle TS^4:

$$p_1(TS^4) = 0 \quad \text{(since } S^4 \text{ is parallelizable after removing a point).}$$

But more interestingly, for bundles over S^4, p_1 classifies them up to stable equivalence.

Example 21.3: Pontryagin classes and cobordism

The signature of a 4-manifold M^4 is:

$$\sigma(M) = \frac{1}{3}\int_M p_1(TM).$$

This is the *Hirzebruch signature theorem* in dimension 4.

21.4 Euler Class

Motivation. The Euler class is the "top" characteristic class for oriented real bundles. It generalizes the Euler characteristic.

Definition 21.4: Euler class

Let $E \to M$ be an oriented real vector bundle of rank n. The *Euler class* is:

$$e(E) \in H^n(M; \mathbb{Z}).$$

Proposition 21.4: Chern–Weil for Euler class

Using a connection with curvature F:

$$e(E) = \mathrm{Pf}\left(\frac{F}{2\pi}\right),$$

where Pf is the Pfaffian (square root of determinant for antisymmetric matrices).

Proposition 21.5: Properties of Euler class

1. $e(E)^2 = p_{n/2}(E)$ when n is even.
2. $e(E \oplus F) = e(E) \smile e(F)$.
3. Reversing orientation: $e(-E) = -e(E)$.
4. For complex bundles: $e(E_{\mathbb{R}}) = c_n(E)$ (top Chern class).

Example 21.4: Euler class of S^n

For the tangent bundle TS^n:

$$\int_{S^n} e(TS^n) = \chi(S^n) = 1 + (-1)^n = \begin{cases} 2 & n \text{ even} \\ 0 & n \text{ odd} \end{cases}.$$

Theorem 21.1: Gauss–Bonnet–Chern

For a compact oriented Riemannian n-manifold:

$$\int_M e(TM) = \chi(M).$$

This generalizes Gauss–Bonnet to all dimensions.

Proposition 21.6: Euler class and sections

$e(E) = 0$ if and only if E admits a nowhere-vanishing section.

21.5 Todd, L-, and $\hat{A}$-Classes

Motivation. These are multiplicative characteristic classes appearing in index theorems. They're defined via formal power series in Chern/Pontryagin classes.

Definition 21.5: Todd class

For a complex vector bundle E, the *Todd class* is:

$$\mathrm{Td}(E) = \prod_i \frac{x_i}{1 - e^{-x_i}},$$

where x_i are the Chern roots (formal eigenvalues of $\frac{iF}{2\pi}$).

Proposition 21.7: Todd class expansion

$$\mathrm{Td}(E) = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \frac{1}{24}c_1c_2 + \cdots$$

Definition 21.6: L-class (Hirzebruch)

For a real vector bundle E, the *L-class* is:

$$L(E) = \prod_i \frac{x_i}{\tanh x_i},$$

where x_i^2 are related to Pontryagin classes.

Proposition 21.8: L-class expansion

$$L(E) = 1 + \frac{1}{3}p_1 + \frac{1}{45}(7p_2 - p_1^2) + \cdots$$

Definition 21.7: $\hat{A}$-class (A-roof / Dirac genus)

For a real vector bundle E, the *$\hat{A}$-class* is:

$$\hat{A}(E) = \prod_i \frac{x_i/2}{\sinh(x_i/2)}.$$

Proposition 21.9: $\hat{A}$-class expansion

$$\hat{A}(E) = 1 - \frac{1}{24}p_1 + \frac{1}{5760}(7p_1^2 - 4p_2) + \cdots$$

Observation 21.1

These classes appear in index theorems:

Class	Operator	Index theorem
Todd	$\bar{\partial}$	Hirzebruch–Riemann–Roch
L	Signature operator	Hirzebruch signature
$\hat{A}$	Dirac operator	Atiyah–Singer

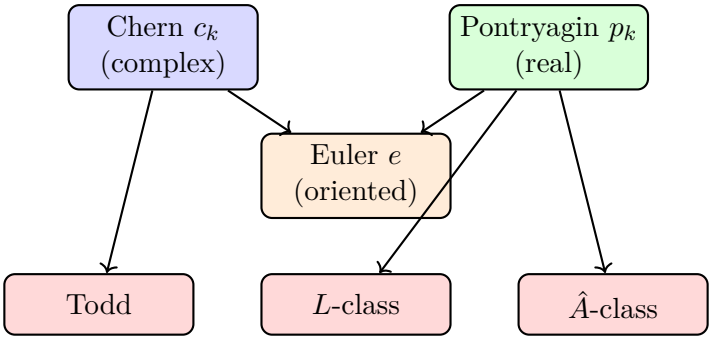
22 Index Theorem

22.1 Motivation

The question. Given a differential operator D, what is its *index*?

$$\operatorname{ind}(D) = \dim\ker(D) - \dim\operatorname{coker}(D).$$

The miracle. For elliptic operators, the index is a *topological invariant* — it depends only on the symbol, not on lower-order terms. The Atiyah–Singer theorem computes it via characteristic classes.



Characteristic classes hierarchy

Figure 65: Relations between characteristic classes.

Why physicists care.

- Zero modes of Dirac operator $\to$ fermion content
- Anomalies $\to$ index of Dirac operator in background gauge field
- SUSY $\to$ Witten index = index of supercharge

22.2 Symbol and Elliptic Operators

Definition 22.8: Differential operator

A *differential operator of order* k between vector bundles $E, F \to M$ is a linear map $D : \Gamma(E) \to \Gamma(F)$ that in local coordinates has the form:

$$D = \sum_{|\alpha| \leq k} A_\alpha(x) \partial^\alpha,$$

where $\alpha = (\alpha_1, \ldots, \alpha_n)$ is a multi-index and A_α are bundle maps.

Definition 22.9: Principal symbol

The *principal symbol* of D is the map $\sigma_D : T^*M \to \mathrm{Hom}(E, F)$ given by the leading-order terms:

$$\sigma_D(x, \xi) = \sum_{|\alpha| = k} A_\alpha(x) \xi^\alpha,$$

where $\xi = \xi_i dx^i \in T_x^* M$.

Example 22.5: Symbol of Laplacian

For $\Delta = -\sum_i \partial_i^2$ on $\mathbb{R}^n$:

$$\sigma_\Delta(x, \xi) = |\xi|^2 = \xi_1^2 + \cdots + \xi_n^2.$$

Example 22.6: Symbol of Dirac operator

For the Dirac operator $D = i\gamma^\mu \nabla_\mu$:

$$\sigma_D(x, \xi) = i\gamma^\mu \xi_\mu = i\not{\xi}.$$

Definition 22.10: Elliptic operator

D is *elliptic* if $\sigma_D(x, \xi)$ is invertible for all $\xi \neq 0$.

Example 22.7: Elliptic examples

- Laplacian: $\sigma = |\xi|^2 \neq 0$ for $\xi \neq 0$. ✓
- Dirac: $\sigma = i\not{\xi}$, $\sigma^2 = |\xi|^2$. ✓
- Wave operator $\partial_t^2 - \nabla^2$: $\sigma = \tau^2 - |\xi|^2$. No (vanishes on light cone)

Analytic Index

Definition 22.11: Fredholm operator

An operator $D : H_1 \to H_2$ between Hilbert spaces is *Fredholm* if $\ker D$ and $\operatorname{coker}(D) = H_2/\operatorname{im}(D)$ are finite-dimensional.

Theorem 22.2: Elliptic implies Fredholm

If D is an elliptic operator on a compact manifold, then D is Fredholm.

Definition 22.12: Analytic index

For an elliptic operator D:

$$\operatorname{ind}_a(D) := \dim\ker(D) - \dim\ker(D^*) = \dim\ker(D) - \dim\operatorname{coker}(D).$$

Proposition 22.10: Index is stable

1. $\operatorname{ind}(D)$ is invariant under continuous deformations of D (within elliptic operators).
2. $\operatorname{ind}(D)$ depends only on the principal symbol.

22.3 The de Rham Complex

Definition 22.13: de Rham complex

$$0 \to \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \cdots \xrightarrow{d} \Omega^n(M) \to 0.$$

Definition 22.14: Euler characteristic as index

Define $D = d + d^*$ on $\Omega^{\text{even}}(M) \to \Omega^{\text{odd}}(M)$. Then:

$$\operatorname{ind}(d + d^*) = \sum_k (-1)^k \dim H^k(M) = \chi(M).$$

Remark 22.2

$\ker(d + d^*) = \ker(d) \cap \ker(d^*) =$ harmonic forms. By Hodge, $\dim \mathcal{H}^k = b_k$.

Theorem 22.3: Index theorem for de Rham

$$\chi(M) = \int_M e(TM).$$

This is Gauss–Bonnet–Chern.

22.4 The Dolbeault Complex

Definition 22.15: Dolbeault complex

On a complex manifold M:

$$0 \to \Omega^{p,0}(M) \xrightarrow{\bar\partial} \Omega^{p,1}(M) \xrightarrow{\bar\partial} \cdots \xrightarrow{\bar\partial} \Omega^{p,n}(M) \to 0.$$

Definition 22.16: Arithmetic genus

The *arithmetic genus* (or Todd genus) is:

$$\chi(M, \mathcal{O}) = \sum_q (-1)^q h^{0,q} = \sum_q (-1)^q \dim H^q(M, \mathcal{O}).$$

Theorem 22.4: Hirzebruch–Riemann–Roch

For a compact complex manifold M of dimension n:

$$\chi(M, \mathcal{O}) = \int_M \operatorname{Td}(TM).$$

More generally, for a holomorphic vector bundle E:

$$\chi(M, E) = \int_M \operatorname{ch}(E) \wedge \operatorname{Td}(TM),$$

where $\operatorname{ch}(E) = \operatorname{tr}(e^{iF/2\pi})$ is the Chern character.

Example 22.8: Riemann–Roch for curves

For a Riemann surface Σ of genus g and line bundle L:

$$h^0(L) - h^1(L) = \deg(L) + 1 - g.$$

Classical Riemann–Roch!

22.5 The Dirac Operator

Setup. Let (M, g) be a Riemannian spin manifold with spinor bundle S. The Dirac operator $D : \Gamma(S) \to \Gamma(S)$.

Definition 22.17: Dirac operator

In local coordinates with orthonormal frame $\{e_\mu\}$:

$$D = i\gamma^\mu \nabla_{e_\mu} = i\gamma^\mu(\partial_\mu + \omega_\mu),$$

where γ^μ are gamma matrices and ω_μ is the spin connection.

Proposition 22.11: Properties of Dirac operator

1. D is self-adjoint: $D^* = D$.
2. $D^2 = \nabla^*\nabla + \frac{R}{4}$ (Lichnerowicz formula).
3. D is elliptic.

Definition 22.18: Chiral Dirac operator

On an even-dimensional manifold, $S = S^+ \oplus S^-$ (positive/negative chirality). The Dirac operator splits:

$$D = \begin{pmatrix} 0 & D^- \\ D^+ & 0 \end{pmatrix}, \quad D^+ : \Gamma(S^+) \to \Gamma(S^-).$$

Theorem 22.5: Atiyah–Singer Index Theorem (Dirac)

For the chiral Dirac operator:

$$\operatorname{ind}(D^+) = \dim\ker(D^+) - \dim\ker(D^-) = \int_M \hat{A}(TM).$$

Example 22.9: Index on S^{4k}

$\hat{A}(S^{4k})$ can be computed. For S^4, $\hat{A} = 0$ (no harmonic spinors).

22.6 Anomalies

Physics context. A classical symmetry may fail at the quantum level — this is an *anomaly*. Chiral anomalies are related to the index of the Dirac operator.

Definition 22.19: Chiral anomaly

Consider a Dirac fermion coupled to a gauge field A. The classical $U(1)$ axial symmetry has anomalous divergence:

$$\partial_\mu j_5^\mu = \frac{1}{16\pi^2}\mathrm{tr}(F \wedge F).$$

Theorem 22.6: Index and anomaly

The integrated anomaly equals the index:

$$\int_M \partial_\mu j_5^\mu = \mathrm{ind}(D_A) = \frac{1}{8\pi^2}\int_M \mathrm{tr}(F \wedge F).$$

This is the second Chern number.

Remark 22.3

For $SU(2)$ instantons on S^4: $\mathrm{ind}(D_A) = k$ (instanton number). The Dirac operator has k zero modes.

Observation 22.2

Anomaly matching:

- Anomalies are RG invariants
- Must match in UV and IR
- Constrains possible IR phases

22.7 Supersymmetric Quantum Mechanics

Motivation. SUSY QM provides a beautiful proof of the index theorem (Witten, Alvarez-Gaumé). It also connects to Morse theory.

Definition 22.20: SUSY quantum mechanics

A SUSY QM system has:

- Hilbert space $\mathcal{H} = \mathcal{H}^+ \oplus \mathcal{H}^-$ (bosonic/fermionic)
- Supercharges $Q, Q^\dagger$ with $Q : \mathcal{H}^+ \to \mathcal{H}^-$
- Hamiltonian $H = \{Q, Q^\dagger\} = QQ^\dagger + Q^\dagger Q$
- Fermion number $(-1)^F$ with $(-1)^F|_{\mathcal{H}^\pm} = \pm 1$

Proposition 22.12: SUSY structure

1. $H \geq 0$ (positivity).
2. $[H, Q] = [H, Q^\dagger] = 0$.
3. $E = 0$ states satisfy $Q|\psi\rangle = Q^\dagger|\psi\rangle = 0$.
4. Nonzero energy states come in boson-fermion pairs.

Definition 22.21: Witten index

The *Witten index* is:

$$\mathrm{Tr}((-1)^F e^{-\beta H}) = n_B^0 - n_F^0 = \dim\ker Q|_{\mathcal{H}^+} - \dim\ker Q|_{\mathcal{H}^-}.$$

Proposition 22.13: Witten index = topological

The Witten index is independent of β (and continuous parameters) — it's a topological invariant.

Example 22.10: SUSY QM on a manifold

Take $\mathcal{H} = \Omega^*(M)$, $Q = d + d^*$. Then:

- $H = \Delta$ (Hodge Laplacian)
- Zero modes = harmonic forms
- Witten index = $\chi(M)$

Theorem 22.7: SUSY proof of index theorem

Using path integral localization:

$$\mathrm{ind}(D) = \int_M \hat{A}(TM).$$

The path integral localizes to fixed points of the supercharge — this gives the characteristic class integral.

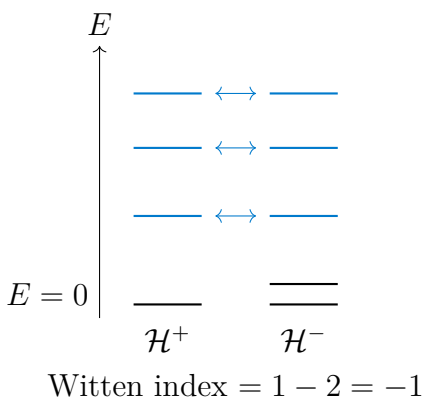


Figure 66: SUSY spectrum: $E > 0$ states are paired; index counts unpaired ground states.

23 Chern–Simons Theory

23.1 Motivation

The setup. Chern–Simons theory is a topological quantum field theory (TQFT) in 3 dimensions. It has no local degrees of freedom — all observables are topological.

Why it matters.

- Knot invariants (Jones polynomial)
- 3-manifold invariants
- Boundary theory of 4D topological insulators
- Quantum Hall effect

23.2 Flat Connections and Holonomy

Definition 23.22: Flat connection

A connection A on a principal G-bundle $P \to M$ is *flat* if its curvature vanishes:

$$F = dA + A \wedge A = 0.$$

Proposition 23.14: Flat connections and parallel transport

A is flat if and only if parallel transport around any contractible loop is trivial.

Definition 23.23: Holonomy representation

For a flat connection A on M, the *holonomy* defines a representation:

$$\rho : \pi_1(M) \to G.$$

The holonomy around a loop γ is $\mathrm{Hol}_\gamma(A) = \mathcal{P}\exp\left(-\oint_\gamma A\right)$.

Theorem 23.8: Flat connections $\leftrightarrow$ representations

There is a bijection:

$$\frac{\{\text{flat connections}\}}{\text{gauge equivalence}} \cong \frac{\mathrm{Hom}(\pi_1(M), G)}{G},$$

where G acts by conjugation.

Example 23.11: Flat connections on T^2

$\pi_1(T^2) = \mathbb{Z}^2$. A flat $U(1)$ connection is determined by holonomies $(e^{i\theta_1}, e^{i\theta_2})$ around the two cycles. Moduli space: T^2 itself.

Example 23.12: Flat connections on a Riemann surface

For Σ_g of genus g and $G = SU(2)$:

$$\dim \mathcal{M}_{\text{flat}} = (2g-2)\dim G = 6g - 6 \quad (g \geq 2).$$

23.3 Chern–Simons Theory

The Chern–Simons Form

Definition 23.24: Chern–Simons 3-form

For a connection A with curvature F:

$$\mathrm{CS}(A) = \mathrm{tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right).$$

Proposition 23.15: Chern–Simons and Chern class

$$d\,\mathrm{CS}(A) = \mathrm{tr}(F \wedge F).$$

The Chern–Simons form is a "potential" for the second Chern class.

Definition 23.25: Chern–Simons action

On a 3-manifold M^3 with G-bundle P:

$$S_{\mathrm{CS}}[A] = \frac{k}{4\pi} \int_{M^3} \mathrm{tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A\right),$$

where $k \in \mathbb{Z}$ is the *level*.

Properties

Proposition 23.16: Gauge invariance

Under a gauge transformation $g : M \to G$:

$$S_{\mathrm{CS}}[A^g] = S_{\mathrm{CS}}[A] + 2\pi k \cdot w(g),$$

where $w(g) \in \mathbb{Z}$ is the winding number of g. For $k \in \mathbb{Z}$, $e^{iS_{\mathrm{CS}}}$ is gauge invariant.

Proposition 23.17: Equations of motion

$$\frac{\delta S_{\mathrm{CS}}}{\delta A} = 0 \quad \Leftrightarrow \quad F = 0.$$

Classical solutions are flat connections!

Remark 23.4

No metric appears in S_{CS} — the theory is topological. It's a TQFT.

Quantization and Knot Invariants

Theorem 23.9: Witten (1989)

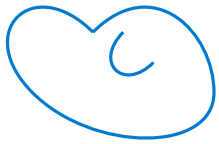
The partition function of Chern–Simons theory with gauge group G at level k on S^3 with a Wilson loop in representation R along knot K gives:

$$Z(S^3, K) = \langle W_R(K) \rangle = \text{Jones polynomial (or generalizations)}.$$

Observation 23.3

Knot invariants from Chern–Simons:

G	Level	Invariant
$SU(2)$	k	Jones polynomial at $q = e^{2\pi i/(k+2)}$
$SU(N)$	k	HOMFLY polynomial
$SO(N)$	k	Kauffman polynomial



Wilson loop $W_R(K) = \mathrm{tr}_R\, \mathcal{P} e^{-\oint_K A}$
$\langle W_R(K) \rangle_{\mathrm{CS}} =$ knot invariant

Figure 67: Chern–Simons computes knot invariants via Wilson loops.

Chern–Simons as Boundary Theory

Theorem 23.10: CS as boundary of 4D theory

On a 4-manifold M^4 with boundary $\partial M^4 = M^3$:

$$\int_{M^4} \mathrm{tr}(F \wedge F) = \int_{M^3} \mathrm{CS}(A) \pmod{\mathbb{Z}}.$$

Chern–Simons is the boundary term needed to make the 4D topological term well-defined.

Observation 23.4

Applications:

- **Topological insulators:** Surface states described by CS theory
- **Quantum Hall:** CS effective theory captures Hall conductance
- **Gravity:** 3D gravity is equivalent to CS theory with $G = SL(2, \mathbb{R})$

24 Moduli Spaces

24.1 Motivation

What is a moduli space? A *moduli space* is a space whose points parametrize some class of geometric objects. Instead of studying one object, we study the "space of all such objects."

Why moduli spaces?

- Understand how objects vary in families
- Often carry rich geometric structure (Kähler, symplectic)
- Count solutions to equations
- Central in string theory (moduli = massless fields)

24.2 Toy Examples

Example 24.13: Lines through the origin in $\mathbb{R}^2$

Each line through 0 is determined by its slope $m \in \mathbb{R}$, plus the vertical line.
Moduli space: $\mathbb{R}P^1 \cong S^1$.

Example 24.14: Lines through the origin in $\mathbb{R}^{n+1}$

Moduli space: $\mathbb{R}P^n$.

Example 24.15: Complex lines through origin in $\mathbb{C}^{n+1}$

Moduli space: $\mathbb{C}P^n$.

Example 24.16: k-planes in $\mathbb{R}^n$

Moduli space: Grassmannian $\mathrm{Gr}(k, n)$.

$$\dim \mathrm{Gr}(k, n) = k(n - k).$$

Example 24.17: Flat connections on M

Moduli space: $\mathcal{M}_{\text{flat}} = \mathrm{Hom}(\pi_1(M), G)/G$.

Remark 24.5

Notice: moduli spaces themselves are often interesting manifolds with geometric structure.

24.3 Moduli Space of Triangles

Setup. Consider triangles in the plane. What is the "space of all triangles"?

Definition 24.26: Triangle moduli problem

Two triangles are *equivalent* if related by:

- Translation
- Rotation
- Scaling (optionally)
- Reflection (optionally)

The moduli space is the set of equivalence classes.

Example 24.18: Triangles up to similarity

A triangle (up to similarity) is determined by its angles (α, β, γ) with $\alpha+\beta+\gamma = \pi$ and $\alpha, \beta, \gamma > 0$.
Moduli space: interior of a 2-simplex Δ^2.

$$\mathcal{M} = \{(\alpha, \beta, \gamma) : \alpha + \beta + \gamma = \pi,\ \alpha, \beta, \gamma > 0\}.$$

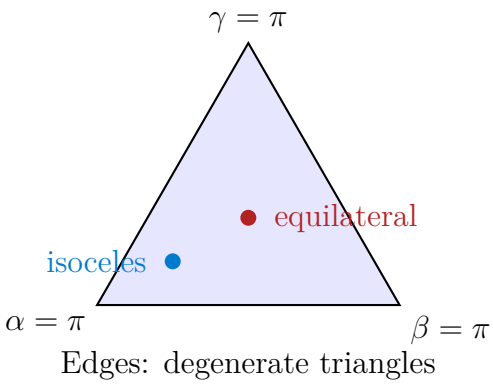


Figure 68: Moduli space of triangles (up to similarity) = interior of simplex.

Example 24.19: Labeled vs unlabeled

If we don't label vertices, we quotient by S_3 (permutations).

$$\mathcal{M}_{\text{unlabeled}} = \Delta^2/S_3.$$

This is an orbifold with singularities at isoceles and equilateral triangles.

24.4 Moduli Space of Riemann Surfaces

Setup. Fix genus $g \geq 2$. What is the space of all complex structures on a surface of genus g?

Definition 24.27: Moduli space of Riemann surfaces

The *moduli space* $\mathcal{M}_g$ parametrizes isomorphism classes of Riemann surfaces of genus g.

Definition 24.28: Teichmüller space

The *Teichmüller space* $\mathcal{T}_g$ parametrizes marked Riemann surfaces (surface + choice of generators for π_1).

$$\mathcal{M}_g = \mathcal{T}_g/\mathrm{MCG}_g,$$

where MCG_g is the mapping class group.

Proposition 24.18: Dimension of moduli space

$$\dim_{\mathbb{C}} \mathcal{M}_g = 3g - 3 \quad (g \geq 2).$$

Equivalently, $\dim_{\mathbb{R}} = 6g - 6$.

Remark 24.6

Why $3g - 3$? By Riemann–Roch, holomorphic quadratic differentials on Σ_g have dimension $3g - 3$. These are cotangent vectors to $\mathcal{M}_g$.

Example 24.20: Low genus

- $g = 0$: $\mathcal{M}_0 = \{\text{pt}\}$ (unique: $\mathbb{C}P^1$)
- $g = 1$: $\mathcal{M}_1 = \mathbb{H}/SL(2, \mathbb{Z})$ (modular curve)
- $g = 2$: $\dim \mathcal{M}_2 = 3$

The Genus 1 Case: Elliptic Curves

Definition 24.29: Elliptic curve

An *elliptic curve* is a Riemann surface of genus 1, equivalently a complex torus $\mathbb{C}/\Lambda$ where $\Lambda = \mathbb{Z} + \tau\mathbb{Z}$ for $\tau \in \mathbb{H}$ (upper half-plane).

Proposition 24.19: Moduli of elliptic curves

Two tori $\mathbb{C}/\Lambda$ and $\mathbb{C}/\Lambda'$ are isomorphic iff $\tau' = \frac{a\tau+b}{c\tau+d}$ for $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z})$.

Thus:

$$\mathcal{M}_1 = \mathbb{H}/SL(2, \mathbb{Z}).$$

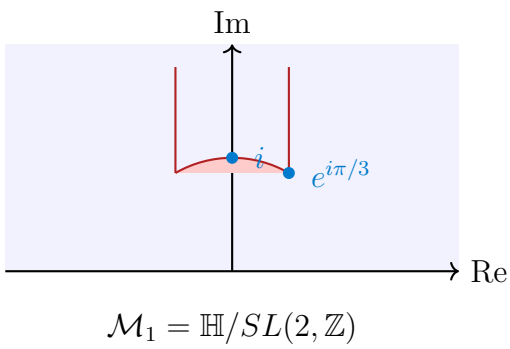


Figure 69: Moduli space of elliptic curves: fundamental domain in upper half-plane.

Compactification: Deligne–Mumford

Definition 24.30: Stable curve

A *stable curve* is a connected curve with at worst nodal singularities, such that each component has finitely many automorphisms.

Definition 24.31: Deligne–Mumford compactification

$\overline{\mathcal{M}}_g$ is the moduli space of stable curves of genus g. It compactifies $\mathcal{M}_g$ by adding "boundary divisors" corresponding to nodal curves.

Observation 24.5

Boundary of $\overline{\mathcal{M}}_g$:

- Curves with one node: $\mathcal{M}_{g-1,2}$ (pinch a handle) or $\mathcal{M}_{g_1,1} \times \mathcal{M}_{g_2,1}$ (separate into two pieces)
- Higher codimension: more nodes

Weil–Petersson Geometry

Definition 24.32: Weil–Petersson metric

The *Weil–Petersson metric* on $\mathcal{M}_g$ is a natural Kähler metric. At a point $[\Sigma]$:

$$\langle \phi, \psi \rangle_{WP} = \int_\Sigma \phi \bar{\psi}\, \rho^{-2},$$

where ϕ, ψ are holomorphic quadratic differentials and ρ is the hyperbolic metric on Σ.

Proposition 24.20: Properties of WP metric

1. Kähler and incomplete (geodesics can reach boundary in finite time).
2. Negative sectional curvature (like hyperbolic space).
3. The WP volume of $\mathcal{M}_g$ is finite.

24.5 Summary

Observation 24.6

Key moduli spaces in geometry and physics:

Moduli space	Objects parametrized	Structure
$\mathrm{Gr}(k, n)$	k-planes in $\mathbb{R}^n$	Compact, smooth
$\mathcal{M}_{\text{flat}}$	Flat connections	Symplectic
$\mathcal{M}_g$	Riemann surfaces of genus g	Kähler (WP)
$\mathcal{M}_H$	Yang–Mills instantons	Hyperkähler
$\mathcal{M}_{\text{Higgs}}$	Higgs bundles	Hyperkähler

Observation 24.7

Why moduli spaces are interesting:

1. **Geometry:** Often have rich structures (Kähler, symplectic, hyperkähler).
2. **Topology:** Their cohomology encodes information about the objects they parametrize.
3. **Physics:** Massless fields = coordinates on moduli space. Dynamics = geodesics.
4. **Counting:** Intersection theory on moduli spaces → enumerative invariants.

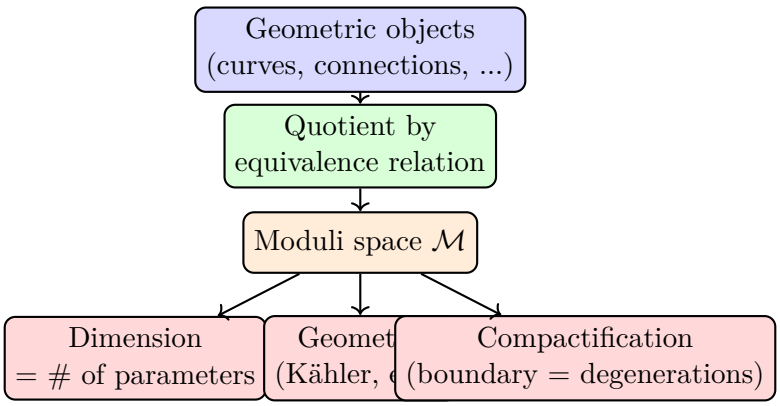


Figure 70: The moduli space philosophy.

A Smooth Manifolds: Fundamental Examples

In this section we build every important manifold from scratch: we write down the set, construct an explicit atlas, give the chart maps and their inverses, draw the geometry, and verify smoothness of every transition function. Multiple atlases are given for S^1, S^2,

the cylinder, and the torus to emphasise that a smooth structure is an equivalence class of atlases, not a single one.

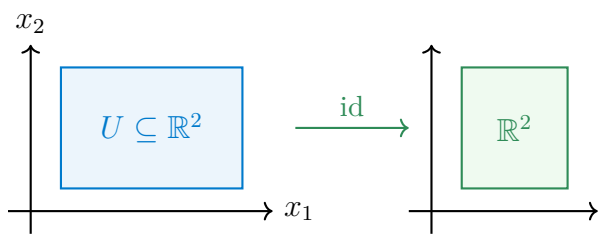
A.1 Euclidean Space $\mathbb{R}^n$

Definition A.1: Euclidean Space as a Manifold

The set $\mathbb{R}^n$ is an n-dimensional smooth manifold. Its topology is the standard metric topology, and its smooth structure is generated by the single global chart

$$\mathcal{A} = \left\{(\mathbb{R}^n, \mathrm{id}_{\mathbb{R}^n})\right\}, \qquad \mathrm{id}(x_1, \ldots, x_n) = (x_1, \ldots, x_n).$$

Since there is only one chart, no transition function needs to be checked. Every open subset $U \subseteq \mathbb{R}^n$ is itself a smooth n-manifold with chart $(U, \mathrm{id}|_U)$. This trivial case is the local model for *every* manifold: by definition, each point of an n-manifold has a neighbourhood that looks like an open subset of $\mathbb{R}^n$.



A.2 The Circle S^1

Definition A.1: The Unit Circle

The **unit circle** is

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

It is a compact, connected, 1-dimensional smooth manifold without boundary.

Non-Example A.1: One chart is not enough

No single chart $\varphi : S^1 \to \mathbb{R}$ can be a homeomorphism: S^1 is compact but $\mathbb{R}$ is not. Hence we need *at least* two charts.

We now present two distinct smooth atlases.

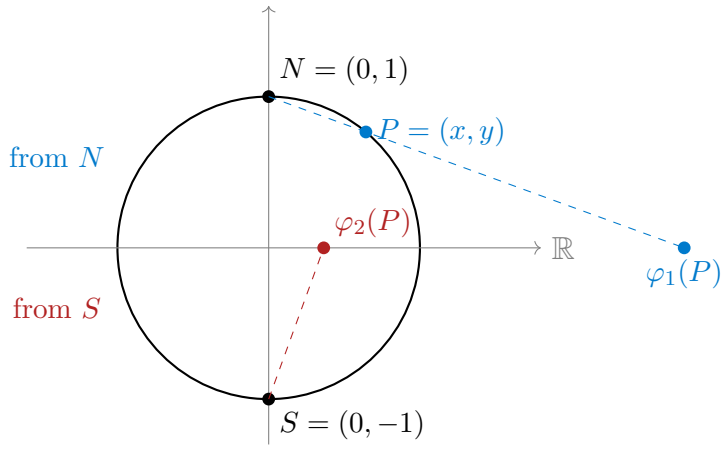
A.2.1 Atlas I on S^1: Stereographic Projection

Let $N = (0, 1)$ (*north pole*) and $S = (0, -1)$ (*south pole*).

Definition A.2: Stereographic Atlas on S^1

$$U_1 = S^1 \setminus \{N\}, \quad \varphi_1 : U_1 \to \mathbb{R}, \quad \varphi_1(x, y) = \frac{x}{1-y}.$$
$$U_2 = S^1 \setminus \{S\}, \quad \varphi_2 : U_2 \to \mathbb{R}, \quad \varphi_2(x, y) = \frac{x}{1+y}.$$

Geometric picture. $\varphi_1(P)$ is the x-intercept of the line through N and P.



Proposition A.1: Inverse Charts

$$\varphi_1^{-1}(t) = \left(\frac{2t}{t^2+1}, \frac{t^2-1}{t^2+1}\right), \qquad \varphi_2^{-1}(t) = \left(\frac{2t}{t^2+1}, \frac{1-t^2}{1+t^2}\right).$$

Proof. For φ_1^{-1}: set $t = x/(1-y)$, so $x = t(1-y)$. Substituting into $x^2 + y^2 = 1$:

$$t^2(1-y)^2 + y^2 = 1 \implies (t^2+1)y^2 - 2t^2 y + (t^2-1) = 0.$$

Since $y \neq 1$, the quadratic gives $y = (t^2-1)/(t^2+1)$ and then $x = 2t/(t^2+1)$. The computation for φ_2^{-1} is analogous. □

Theorem A.1: Transition Function: Stereographic Atlas on S^1

The overlap is $U_1 \cap U_2 = S^1 \setminus \{N, S\}$, with $\varphi_1(U_1 \cap U_2) = \mathbb{R} \setminus \{0\}$. The transition function is:

$$\tau_{12} := \varphi_2 \circ \varphi_1^{-1} : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R} \setminus \{0\}, \qquad \tau_{12}(t) = \frac{1}{t}.$$

This is a C^∞ diffeomorphism.

Proof.

$$\varphi_2\left(\varphi_1^{-1}(t)\right) = \varphi_2\left(\frac{2t}{t^2+1}, \frac{t^2-1}{t^2+1}\right) = \frac{2t/(t^2+1)}{1 + (t^2-1)/(t^2+1)} = \frac{2t/(t^2+1)}{2t^2/(t^2+1)} = \frac{1}{t}.$$

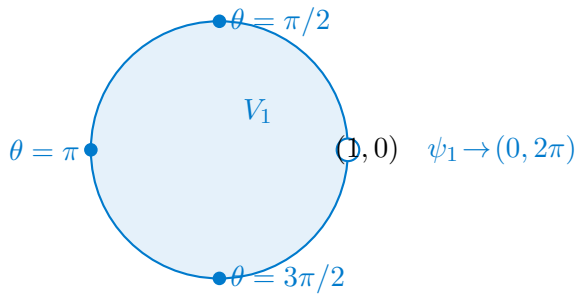
The map $t \mapsto 1/t$ is C^∞ on $\mathbb{R} \setminus \{0\}$ with C^∞ inverse $t \mapsto 1/t$. □

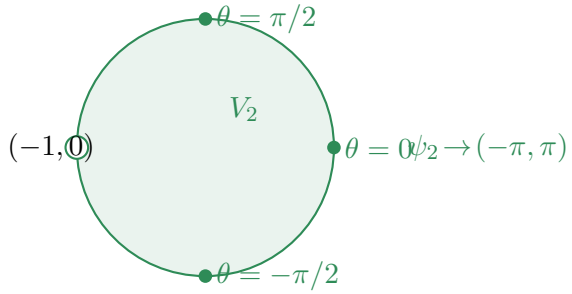
A.2.2 Atlas II on S^1: Angle (Arc) Charts

Definition A.3: Angle Atlas on S^1

Write points of S^1 as $(\cos\theta, \sin\theta)$ for $\theta \in \mathbb{R}$. Define:

$$V_1 = S^1 \setminus \{(1,0)\}, \qquad \psi_1 : V_1 \to (0, 2\pi), \quad \psi_1(\cos\theta, \sin\theta) = \theta \text{ for } \theta \in (0, 2\pi),$$
$$V_2 = S^1 \setminus \{(-1,0)\}, \quad \psi_2 : V_2 \to (-\pi, \pi), \quad \psi_2(\cos\theta, \sin\theta) = \theta \text{ for } \theta \in (-\pi, \pi).$$





Theorem A.2: Transition Function: Angle Atlas on S^1

The overlap $V_1 \cap V_2 = S^1 \setminus \{(1,0), (-1,0)\}$ has two connected components:

$$A_+ = \{\theta \in (0, \pi)\}, \qquad A_- = \{\theta \in (\pi, 2\pi)\}.$$

The transition $\psi_2 \circ \psi_1^{-1} : (0, \pi) \cup (\pi, 2\pi) \to (-\pi, 0) \cup (0, \pi)$ is:

$$(\psi_2 \circ \psi_1^{-1})(\theta) = \begin{cases} \theta & \theta \in (0, \pi), \\ \theta - 2\pi & \theta \in (\pi, 2\pi). \end{cases}$$

Both branches are C^∞.

Observation A.

The two atlases $\mathcal{A}_1 = \{(U_i, \varphi_i)\}$ and $\mathcal{A}_2 = \{(V_j, \psi_j)\}$ are *compatible*: all four cross-transitions $\varphi_i \circ \psi_j^{-1}$ are smooth. They therefore generate the same maximal smooth atlas on S^1.

A.3 The 2-Sphere S^2

Definition A.1: The 2-Sphere

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}.$$

It is a compact, connected, orientable 2-dimensional smooth manifold without boundary.

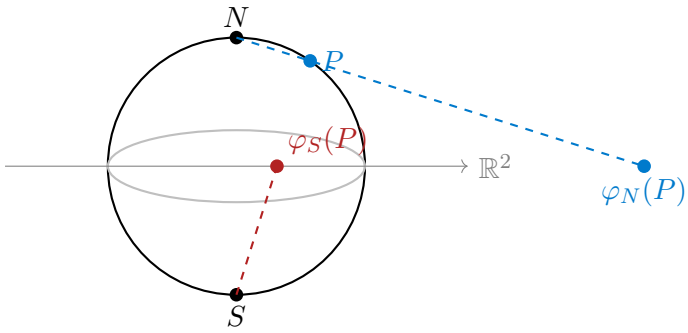
A.3.1 Atlas I on S^2: Stereographic Projection

Let $N = (0, 0, 1)$ and $S = (0, 0, -1)$.

Definition A.2: Stereographic Atlas on S^2

$$U_N = S^2 \setminus \{N\}, \qquad \varphi_N(x, y, z) = \left(\frac{x}{1-z}, \frac{y}{1-z}\right) \in \mathbb{R}^2,$$

$$U_S = S^2 \setminus \{S\}, \qquad \varphi_S(x, y, z) = \left(\frac{x}{1+z}, \frac{y}{1+z}\right) \in \mathbb{R}^2.$$



Proposition A.2: Inverse of φ_N

For $(u, v) \in \mathbb{R}^2$, writing $r^2 = u^2 + v^2$:

$$\varphi_N^{-1}(u, v) = \left(\frac{2u}{r^2+1}, \frac{2v}{r^2+1}, \frac{r^2-1}{r^2+1}\right).$$

Proof. Set $u = x/(1-z)$, $v = y/(1-z)$. Then $x = u(1-z)$, $y = v(1-z)$. Substituting into $x^2 + y^2 + z^2 = 1$: $(u^2 + v^2)(1-z)^2 + z^2 = 1$, i.e. $(r^2+1)z^2 - 2r^2 z + (r^2 - 1) = 0$. Since $z \neq 1$, we get $z = (r^2 - 1)/(r^2 + 1)$, then $1 - z = 2/(r^2 + 1)$. $\square$

Theorem A.1: Transition Function: Stereographic Atlas on S^2

The transition function $\varphi_S \circ \varphi_N^{-1} : \mathbb{R}^2 \setminus \{0\} \to \mathbb{R}^2 \setminus \{0\}$ is *inversion*:

$$\varphi_S \circ \varphi_N^{-1}(u, v) = \left(\frac{u}{u^2 + v^2}, \frac{v}{u^2 + v^2} \right).$$

This is a C^∞ diffeomorphism on $\mathbb{R}^2 \setminus \{0\}$.

Proof. Apply φ_S to $\varphi_N^{-1}(u, v) = \left(\frac{2u}{r^2+1}, \frac{2v}{r^2+1}, \frac{r^2-1}{r^2+1} \right)$:

$$\begin{aligned} \varphi_S\left(\frac{2u}{r^2+1}, \frac{2v}{r^2+1}, \frac{r^2-1}{r^2+1} \right) &= \frac{1}{1 + \frac{r^2-1}{r^2+1}} \cdot \left(\frac{2u}{r^2+1}, \frac{2v}{r^2+1} \right) \\ &= \frac{r^2+1}{2r^2} \cdot \frac{(2u, 2v)}{r^2+1} = \left(\frac{u}{r^2}, \frac{v}{r^2} \right). \end{aligned}$$

All partial derivatives of all orders exist and are continuous on $\mathbb{R}^2 \setminus \{0\}$. The map is its own inverse, confirming it is a diffeomorphism. □

A.3.2 Atlas II on S^2: Six Hemisphere Charts

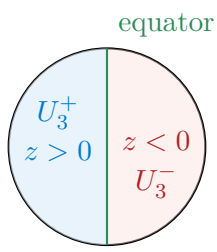
Definition A.3: Hemisphere Atlas on S^2

For $i \in \{1, 2, 3\}$ and $\varepsilon \in \{+, -\}$, define open hemispheres:

$$U_i^\varepsilon = \{(x_1, x_2, x_3) \in S^2 : \varepsilon x_i > 0\}.$$

The chart $\varphi_i^\varepsilon : U_i^\varepsilon \to \mathbb{D}^2 = \{(u, v) : u^2 + v^2 < 1\}$ is orthogonal projection onto the remaining two coordinates:

$$\varphi_1^\pm(x, y, z) = (y, z), \quad \varphi_2^\pm(x, y, z) = (x, z), \quad \varphi_3^\pm(x, y, z) = (x, y).$$



Two of the six hemisphere charts

Example A.1: Transition Between φ_3^+ and φ_1^+

On $U_3^+ \cap U_1^+ = \{x > 0, z > 0\}$, the transition function is:

$$\varphi_3^+ \circ (\varphi_1^+)^{-1}(u, v) = \varphi_3^+\left(\sqrt{1 - u^2 - v^2},\ u,\ v \right) = \left(\sqrt{1 - u^2 - v^2},\ u \right).$$

This is C^∞ on $\{(u, v) : u^2 + v^2 < 1,\ v > 0\}$ since $1 - u^2 - v^2 > 0$ there.

Observation A.

The two atlases (stereographic and hemispheric) are compatible and define the same smooth structure on S^2.

A.4 The Cylinder

Definition A.1: The Cylinder

The **cylinder** is the smooth manifold $\mathrm{Cyl} = S^1 \times \mathbb{R}$. It is non-compact, connected, orientable, 2-dimensional, and has no boundary.

A.4.1 Construction I: As a Product Manifold

Proposition A.1: Product Atlas on the Cylinder

Let $\{(U_i, \varphi_i)\}_{i=1,2}$ be the stereographic atlas on S^1 from §2. The product atlas on $S^1 \times \mathbb{R}$ consists of:

$$\Phi_i : U_i \times \mathbb{R} \to \mathbb{R}^2, \qquad \Phi_i\Big((x,y),\, t\Big) = \Big(\varphi_i(x,y),\ t\Big).$$

Explicitly:

$$\Phi_1\Big((x,y),t\Big) = \left(\frac{x}{1-y},\ t\right), \qquad \Phi_2\Big((x,y),t\Big) = \left(\frac{x}{1+y},\ t\right).$$

Transition function. On $(U_1 \cap U_2) \times \mathbb{R}$:

$$\Phi_2 \circ \Phi_1^{-1}(s,t) = \left(\frac{1}{s},\ t\right), \qquad C^\infty \text{ on } (\mathbb{R} \setminus \{0\}) \times \mathbb{R}. \checkmark$$

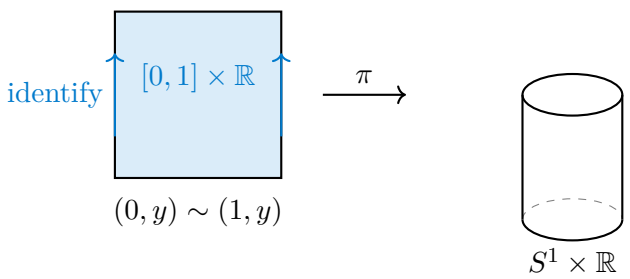
A.4.2 Construction II: As a Quotient Manifold

Definition A.2: Cylinder as Quotient

Define the equivalence relation on $\mathbb{R}^2$:

$$(x,y) \sim (x+n,\ y) \qquad \forall\, n \in \mathbb{Z}.$$

The quotient $\mathbb{R}^2/\!\sim$ is diffeomorphic to $S^1 \times \mathbb{R}$ via $[(x,y)] \mapsto (e^{2\pi i x}, y)$.



Quotient charts. Let $\pi : \mathbb{R}^2 \to \mathbb{R}^2/\sim$ be the quotient map. For $x_0 \in \mathbb{R}$ define:

$$W_{x_0} = \pi\Big(\big(x_0 - \tfrac{1}{2}, x_0 + \tfrac{1}{2}\big) \times \mathbb{R}\Big), \qquad \chi_{x_0}([x, y]) = (x \bmod 1,\ y).$$

These are homeomorphisms onto open subsets of $\mathbb{R}^2$.

Theorem A.1: Smoothness of Quotient Cylinder Atlas

Two charts χ_{x_0} and $\chi_{x_0'}$ overlap when $|x_0 - x_0'| < 1$. On the overlap, the transition is:

$$\chi_{x_0'} \circ \chi_{x_0}^{-1}(x, y) = (x + m,\ y), \quad m \in \mathbb{Z} \text{ fixed on each component.}$$

This is a smooth (in fact affine) diffeomorphism. □

A.5 The Torus T^2

Definition A.1: The Torus

The **torus** is $T^2 = S^1 \times S^1$. It is compact, connected, orientable, and 2-dimensional.

A.5.1 Construction I: As a Product Manifold

Proposition A.1: Product Atlas on T^2

Using stereographic atlases $\{(U_i, \varphi_i)\}$ and $\{(V_j, \psi_j)\}$ on the two S^1 factors, the product atlas on T^2 has four charts:

$$\Phi_{ij} : U_i \times V_j \to \mathbb{R}^2, \qquad \Phi_{ij}\Big((x_1, y_1), (x_2, y_2)\Big) = \Big(\varphi_i(x_1, y_1),\ \psi_j(x_2, y_2)\Big).$$

Transition functions. On $(U_1 \cap U_2) \times (V_1 \cap V_2)$:

$$\Phi_{22} \circ \Phi_{11}^{-1}(s, t) = \left(\frac{1}{s}, \frac{1}{t}\right), \qquad C^\infty \text{ on } (\mathbb{R} \setminus \{0\})^2. \checkmark$$

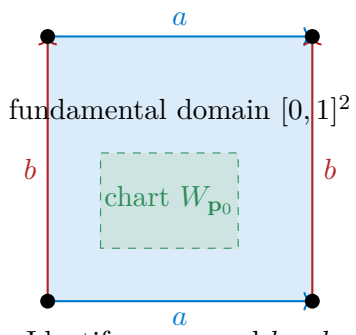
A.5.2 Construction II: As a Quotient Manifold

Definition A.2: Torus as Quotient of $\mathbb{R}^2$

Define the equivalence relation on $\mathbb{R}^2$:

$$(x, y) \sim (x + m,\ y + n) \qquad \forall\, m, n \in \mathbb{Z}.$$

The quotient $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ is a smooth manifold diffeomorphic to $S^1 \times S^1$.



Identify $a \sim a$ and $b \sim b$

Quotient charts. Let $\pi : \mathbb{R}^2 \to T^2 = \mathbb{R}^2/\mathbb{Z}^2$. For $\mathbf{p}_0 = (x_0, y_0) \in \mathbb{R}^2$:

$$W_{\mathbf{p}_0} = \pi\Big(\big(x_0 - \tfrac{1}{2}, x_0 + \tfrac{1}{2}\big) \times \big(y_0 - \tfrac{1}{2}, y_0 + \tfrac{1}{2}\big)\Big), \quad \chi_{\mathbf{p}_0}([x,y]) = (x \bmod 1,\ y \bmod 1) \in \Big(-\tfrac{1}{2}, \tfrac{1}{2}\Big)^2.$$

Theorem A.2: Smoothness of Quotient Torus Atlas

Any two overlapping quotient charts $\chi_{\mathbf{p}_0}$, $\chi_{\mathbf{p}_0'}$ have transition function:

$$\chi_{\mathbf{p}_0'} \circ \chi_{\mathbf{p}_0}^{-1}(x,y) = (x+m,\ y+n), \quad m, n \in \mathbb{Z} \text{ fixed on each component.}$$

This is a C^∞ (affine) diffeomorphism. □

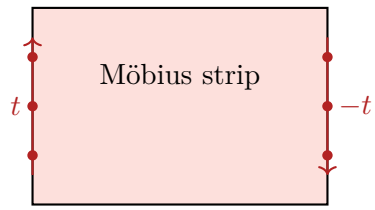
A.6 The Möbius Strip

Definition A.1: The Möbius Strip

The **Möbius strip** is the quotient:

$$\mathrm{Mob} = [0,1] \times (-1,1) \big/ \sim, \qquad (0,t) \sim (1,-t).$$

It is a non-compact, connected, *non-orientable* 2-dimensional smooth manifold *with boundary*.



$(0,t) \sim (1,-t)$ **twisted identification**

A.6.1 As a Fiber Bundle over S^1

Definition A.2: Möbius Strip as a Line Bundle

The Möbius strip is the total space of a smooth rank-1 vector bundle ξ over S^1:

- **Base:** $B = S^1$.
- **Fiber:** $F = (-1, 1) \subset \mathbb{R}$.
- **Structure group:** $G = \mathbb{Z}/2\mathbb{Z} = \{+1, -1\} \subset GL(1, \mathbb{R})$.
- **Projection:** $\pi : \mathrm{Mob} \to S^1$, $\pi([s, t]) = e^{2\pi i s}$.

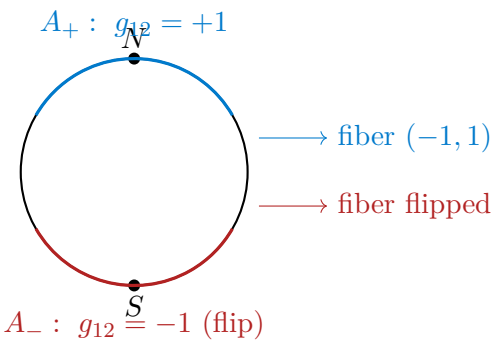
Local trivializations. Cover S^1 with:

$$\begin{aligned} V_1 &= S^1 \setminus \{S\}, \quad h_1 : V_1 \times (-1, 1) \xrightarrow{\sim} \pi^{-1}(V_1), \\ V_2 &= S^1 \setminus \{N\}, \quad h_2 : V_2 \times (-1, 1) \xrightarrow{\sim} \pi^{-1}(V_2). \end{aligned}$$

Transition (cocycle). On $V_1 \cap V_2 = S^1 \setminus \{N, S\}$, which has two components A_+ (upper arc) and A_- (lower arc):

$$g_{12}(\theta) = h_2^{-1} \circ h_1\Big|_{\{\theta\} \times (-1,1)} = \begin{cases} +1 & \theta \in A_+, \\ -1 & \theta \in A_-. \end{cases}$$

The fiber is **flipped** on the lower arc A_-. This is what makes the bundle *non-trivial* and the total space non-orientable. If we used $g_{12} \equiv +1$ everywhere, we would get the trivial bundle $S^1 \times (-1, 1)$, i.e. the cylinder.



Smooth atlas on Mob**.** The quotient map $\pi : [0, 1] \times (-1, 1) \to \mathrm{Mob}$ restricts to a homeomorphism on any open strip not crossing the seam $\{0\} \times (-1, 1)$. Two chart types suffice:

$$W_1 = \pi\Big(\big(\tfrac{1}{4}, \tfrac{3}{4}\big) \times (-1, 1)\Big), \qquad W_2 = \pi\Big(\big([0, \tfrac{1}{4}) \cup (\tfrac{3}{4}, 1]\big) \times (-1, 1)\Big).$$

The transition on $W_1 \cap W_2$ involves the flip $(s, t) \mapsto (s, -t)$ near the seam, which is smooth.

A.7 Real Projective Spaces $\mathbb{R}P^1$ and $\mathbb{R}P^2$

Definition A.1: Real Projective Space

The **real projective space** $\mathbb{R}P^n$ is the set of lines through the origin in $\mathbb{R}^{n+1}$:

$$\mathbb{R}P^n = \left(\mathbb{R}^{n+1} \setminus \{0\}\right) \Big/ \sim, \qquad \mathbf{x} \sim \lambda \mathbf{x} \text{ for all } \lambda \in \mathbb{R} \setminus \{0\}.$$

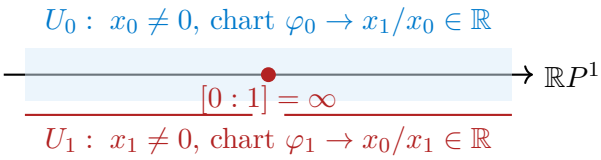
Points are denoted by *homogeneous coordinates* $[x_0 : x_1 : \cdots : x_n]$.

A.7.1 $\mathbb{R}P^1$: The Projective Line

Definition A.2: Standard Atlas on $\mathbb{R}P^1$

$$U_0 = \{[x_0 : x_1] : x_0 \neq 0\}, \quad \varphi_0([x_0 : x_1]) = \frac{x_1}{x_0} \in \mathbb{R}.$$

$$U_1 = \{[x_0 : x_1] : x_1 \neq 0\}, \quad \varphi_1([x_0 : x_1]) = \frac{x_0}{x_1} \in \mathbb{R}.$$



Theorem A.1: Transition Function on $\mathbb{R}P^1$

On $U_0 \cap U_1 = \{[x_0 : x_1] : x_0 \neq 0, x_1 \neq 0\}$, with $\varphi_0(U_0 \cap U_1) = \mathbb{R} \setminus \{0\}$:

$$\varphi_1 \circ \varphi_0^{-1}(t) = \varphi_1([1 : t]) = \frac{1}{t}, \qquad C^\infty \text{ on } \mathbb{R} \setminus \{0\}. \checkmark$$

This is the same transition as the stereographic atlas on S^1, confirming $\mathbb{R}P^1 \cong S^1$.

A.7.2 $\mathbb{R}P^2$: The Projective Plane

Definition A.3: Standard Atlas on $\mathbb{R}P^2$

For $i \in \{0, 1, 2\}$:

$$U_i = \{[x_0 : x_1 : x_2] : x_i \neq 0\}.$$

With $\{j, k\} = \{0, 1, 2\} \setminus \{i\}$:

$$\varphi_i([x_0 : x_1 : x_2]) = \left(\frac{x_j}{x_i}, \frac{x_k}{x_i}\right) \in \mathbb{R}^2.$$

Explicitly:

$$\varphi_0 = \left(\frac{x_1}{x_0}, \frac{x_2}{x_0}\right), \quad \varphi_1 = \left(\frac{x_0}{x_1}, \frac{x_2}{x_1}\right), \quad \varphi_2 = \left(\frac{x_0}{x_2}, \frac{x_1}{x_2}\right).$$

Theorem A.2: Transition Functions on $\mathbb{R}P^2$

On $U_0 \cap U_1$ (where $x_0 \neq 0, x_1 \neq 0$), let $(s,t) = \varphi_0([x_0 : x_1 : x_2]) = (x_1/x_0, x_2/x_0)$, so $s \neq 0$:

$$\varphi_1 \circ \varphi_0^{-1}(s,t) = \varphi_1([1 : s : t]) = \left(\frac{1}{s}, \frac{t}{s}\right), \qquad C^\infty \text{ on } \{s \neq 0\}. \checkmark$$

On $U_0 \cap U_2$ (where $x_0 \neq 0, x_2 \neq 0$), let $(s,t) = \varphi_0$, so $t \neq 0$:

$$\varphi_2 \circ \varphi_0^{-1}(s,t) = \varphi_2([1 : s : t]) = \left(\frac{1}{t}, \frac{s}{t}\right), \qquad C^\infty \text{ on } \{t \neq 0\}. \checkmark$$

On $U_1 \cap U_2$ (where $x_1 \neq 0, x_2 \neq 0$), let $(s,t) = \varphi_1([x_0 : x_1 : x_2]) = (x_0/x_1, x_2/x_1)$, so $t \neq 0$:

$$\varphi_2 \circ \varphi_1^{-1}(s,t) = \varphi_2([s : 1 : t]) = \left(\frac{s}{t}, \frac{1}{t}\right), \qquad C^\infty \text{ on } \{t \neq 0\}. \checkmark$$

A.8 The General Linear Group $GL(n, \mathbb{R})$

Definition A.1: $GL(n, \mathbb{R})$ as an Open Submanifold

The **general linear group** is:

$$GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det(A) \neq 0\} = \overset{-1}{\det}(\mathbb{R} \setminus \{0\}).$$

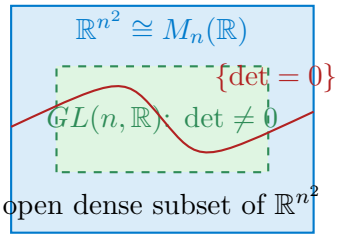
Since $\det : M_n(\mathbb{R}) \cong \mathbb{R}^{n^2} \to \mathbb{R}$ is a polynomial (hence continuous), and $\mathbb{R} \setminus \{0\}$ is open, $GL(n, \mathbb{R})$ is an **open submanifold** of $\mathbb{R}^{n^2}$: an n^2-dimensional smooth manifold.

Proposition A.1: Global Chart

The single chart:

$$\varphi : GL(n, \mathbb{R}) \hookrightarrow \mathbb{R}^{n^2}, \qquad \varphi(A) = (a_{11}, a_{12}, \ldots, a_{1n}, a_{21}, \ldots, a_{nn}),$$

reads off the n^2 matrix entries in row-major order. No transition functions are needed.



Observation A.

$GL(n,\mathbb{R})$ has two connected components: $GL^+(n,\mathbb{R}) = \{\det > 0\}$ and $GL^-(n,\mathbb{R}) = \{\det < 0\}$. The submanifold $SL(n,\mathbb{R}) = \{\det = 1\}$ has dimension $n^2 - 1$ by the Regular Value Theorem.

A.9 The Special Orthogonal Group $SO(3)$

Definition A.1: $SO(3)$

$$SO(3) = \{A \in M_3(\mathbb{R}) : A^\top A = I_3,\ \det(A) = 1\}.$$

This is a compact, connected, non-simply-connected Lie group of dimension 3.

Proposition A.1: Dimension of $SO(3)$

The Lie algebra $\mathfrak{so}(3) = \{X \in M_3(\mathbb{R}) : X^\top = -X\}$ consists of 3×3 skew-symmetric matrices and has dimension $\binom{3}{2} = 3$. Hence $SO(3)$ is a 3-manifold.

Proof. A skew-symmetric 3×3 matrix has the form

$$X = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix}, \qquad a, b, c \in \mathbb{R},$$

giving a 3-dimensional vector space. □

A.9.1 Atlas via Axis-Angle (Rodrigues) Parametrization

By Euler's theorem, every rotation $R \in SO(3) \setminus \{I\}$ is a rotation by angle $\theta \in (0, \pi]$ about a unit axis $\hat{n} \in S^2$, with the identification $(\hat{n}, \pi) \sim (-\hat{n}, \pi)$ for $\theta = \pi$.

Definition A.2: Axis-Angle Chart: Rodrigues Formula

The **rotation vector** is $\mathbf{v} = \theta\hat{n} \in \mathbb{R}^3$, $\|\mathbf{v}\| \le \pi$. The Rodrigues formula gives:

$$R(\mathbf{v}) = I + \frac{\sin\theta}{\theta}[\mathbf{v}]_\times + \frac{1-\cos\theta}{\theta^2}[\mathbf{v}]_\times^2,$$

where the skew matrix $[\mathbf{v}]_\times$ encodes the cross product with $\mathbf{v}$:

$$[\mathbf{v}]_\times = \begin{pmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{pmatrix}, \qquad [\mathbf{v}]_\times \mathbf{u} = \mathbf{v} \times \mathbf{u}.$$

Example A.1: Chart 1 — Interior Ball

$$U_1 = \left\{ R(\mathbf{v}) : \mathbf{v} \in B_\pi^\circ(0) \subset \mathbb{R}^3 \right\}, \qquad \varphi_1(R) = R^{-1}(R) \in B_\pi^\circ(0).$$

This open ball covers all rotations with angle $\theta \in [0, \pi)$. Rotations with $\theta = \pi$ (the antipodal boundary) are excluded.

Example A.2: Chart 2 — Translated Ball

Fix $R_0 \in SO(3)$ to be the rotation by π about the x-axis. Define:

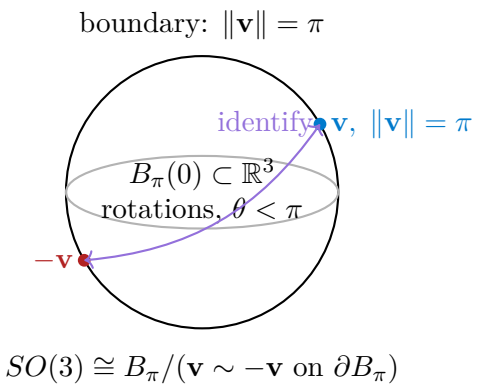
$$U_2 = \left\{ R \in SO(3) : R_0^{-1} R \in U_1 \right\}, \qquad \varphi_2(R) = \varphi_1(R_0^{-1} R).$$

This choice ensures $U_1 \cup U_2 = SO(3)$.

Transition function. On $U_1 \cap U_2$:

$$\varphi_2 \circ \varphi_1^{-1}(\mathbf{v}) = \text{axis-angle coordinates of } R_0^{-1} R(\mathbf{v}),$$

which is C^∞ since group multiplication in $SO(3) \subset \mathbb{R}^9$ is smooth (matrix multiplication is polynomial).



Theorem A.1: $SO(3) \cong \mathbb{R}P^3$

The axis-angle parametrization identifies:

$$SO(3) \;\cong\; \overline{B_\pi(0)} \Big/ (\mathbf{v} \sim -\mathbf{v} \text{ for } \|\mathbf{v}\| = \pi) \;\cong\; \mathbb{R}P^3.$$

In particular $\pi_1(SO(3)) \cong \mathbb{Z}/2\mathbb{Z}$: $SO(3)$ is **not** simply connected.

A.10 The Special Unitary Group $SU(2)$

Definition A.1: $SU(2)$

$$SU(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} : \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}.$$

It is a compact, connected, **simply connected** Lie group of dimension 3.

Theorem A.1: $SU(2) \cong S^3$

Write $\alpha = a_1 + ia_2$, $\beta = a_3 + ia_4$ with $a_i \in \mathbb{R}$. The condition $|\alpha|^2 + |\beta|^2 = 1$ becomes:

$$a_1^2 + a_2^2 + a_3^2 + a_4^2 = 1,$$

which is the equation of $S^3 \subset \mathbb{R}^4 \cong \mathbb{C}^2$. The identification

$$(a_1, a_2, a_3, a_4) \longleftrightarrow \begin{pmatrix} a_1 + ia_2 & -(a_3 - ia_4) \\ a_3 + ia_4 & a_1 - ia_2 \end{pmatrix}$$

is a diffeomorphism $SU(2) \xrightarrow{\sim} S^3$.

A.10.1 Atlas on $SU(2)$ Inherited from S^3

Definition A.2: Stereographic Atlas on S^3 (and $SU(2)$)

Let $N = (0, 0, 0, 1) \in S^3$ and $S = (0, 0, 0, -1)$.

$$U_N = S^3 \setminus \{N\}, \qquad \varphi_N(a_1, a_2, a_3, a_4) = \frac{(a_1, a_2, a_3)}{1 - a_4} \in \mathbb{R}^3,$$

$$U_S = S^3 \setminus \{S\}, \qquad \varphi_S(a_1, a_2, a_3, a_4) = \frac{(a_1, a_2, a_3)}{1 + a_4} \in \mathbb{R}^3.$$

Theorem A.2: Transition Function on $SU(2) \cong S^3$

With $\mathbf{u} = (u_1, u_2, u_3) \in \mathbb{R}^3 \setminus \{0\}$ and $r^2 = |\mathbf{u}|^2$:

$$\varphi_S \circ \varphi_N^{-1}(\mathbf{u}) = \frac{\mathbf{u}}{|\mathbf{u}|^2} = \frac{\mathbf{u}}{r^2}, \qquad C^\infty \text{ on } \mathbb{R}^3 \setminus \{0\}. \checkmark$$

This is inversion in $\mathbb{R}^3$, a diffeomorphism on $\mathbb{R}^3 \setminus \{0\}$.

Proof. The inverse is $\varphi_N^{-1}(\mathbf{u}) = \left(\frac{2\mathbf{u}}{r^2+1}, \frac{r^2-1}{r^2+1}\right) \in S^3$. Applying φ_S:

$$\varphi_S\left(\frac{2\mathbf{u}}{r^2+1}, \frac{r^2-1}{r^2+1}\right) = \frac{2\mathbf{u}/(r^2+1)}{1 + (r^2-1)/(r^2+1)} = \frac{2\mathbf{u}/(r^2+1)}{2r^2/(r^2+1)} = \frac{\mathbf{u}}{r^2}. \qquad \square$$

A.10.2 The Double Cover $SU(2) \to SO(3)$

Theorem A.3: $SU(2)$ is the Double Cover of $SO(3)$

There is a smooth surjective Lie group homomorphism

$$\Phi : SU(2) \longrightarrow SO(3), \qquad \ker \Phi = \{\pm I\} \cong \mathbb{Z}/2\mathbb{Z}.$$

Hence $SO(3) \cong SU(2)/\{\pm I\} \cong S^3/\{\pm 1\} \cong \mathbb{R}P^3$.

Example A.3: Explicit Formula for Φ

Identify $\mathbb{R}^3$ with the space of traceless skew-Hermitian matrices:

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} it & -\bar{z} \\ z & -it \end{pmatrix} : t \in \mathbb{R},\ z \in \mathbb{C} \right\} \cong \mathbb{R}^3.$$

For $U \in SU(2)$, define $\Phi(U) \in GL(\mathfrak{su}(2)) \cong GL(3, \mathbb{R})$ by conjugation:

$$\Phi(U)(X) = UXU^{-1} = UXU^{\dagger}.$$

One verifies: (i) $UXU^{\dagger}$ is again skew-Hermitian traceless; (ii) the map is an isometry of $\mathbb{R}^3$, so $\Phi(U) \in O(3)$; (iii) $\det \Phi(U) = +1$ by connectivity; (iv) $\ker \Phi = \{\pm I\}$.

$$\begin{tikzcd}
SU(2) \cong S^3 \arrow[rr, "\Phi,\ \text{2-to-1}"] \arrow[dr, "\text{quotient by } \pm I"'] & & SO(3) \cong \mathbb{R}P^3 \\
& S^3/\{\pm 1\} \arrow[ur, dashed, "\cong"'] &
\end{tikzcd}$$

Remark A.

This double cover is fundamental in quantum mechanics: the states of a spin-$\frac{1}{2}$ particle transform under $SU(2)$, not $SO(3)$. A rotation by 2π gives $-I \in SU(2)$ (a sign change), whereas a rotation by 4π returns to $+I$.

Summary: Relationships Between the Manifolds

Remark A.

The key identifications and relationships proved in this section are:

$$S^1 \cong U(1) \cong \mathbb{R}P^1, \qquad S^3 \cong SU(2) \xrightarrow{2:1} SO(3) \cong \mathbb{R}P^3,$$

$$\mathrm{Cyl} = S^1 \times \mathbb{R} \cong \mathbb{R}^2/\mathbb{Z} \quad \text{(product and quotient)},$$

$$T^2 = S^1 \times S^1 \cong \mathbb{R}^2/\mathbb{Z}^2 \quad \text{(product and quotient)},$$

$$\mathrm{Mob} \to S^1 \quad \text{(non-trivial line bundle; non-orientable total space)}.$$

The cylinder and torus illustrate that product manifolds can equally be described as quotients by free, proper, discrete group actions. The Möbius strip shows that a non-trivial sign flip in the transition cocycle is precisely what produces non-orientability.